USA

THE NORTH AMERICAN HEATING AND COOLING PRODUCT MARKETS

- 2024 Edition -

BRG BUILDING SOLUTIONS



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1400 Old Country Road, Suite 403N, Westbury, NY, USA 11590

Tel: +1 973-347-5300

Email: us@brgbuildingsolutions.com

Website: www.brgbuildingsolutions.com

Table of Contents

USA Heating

The US Boilers & Water Heaters Market

Introduction : Heating 7

Background : USA - Heating & Cooling

A.0 US - Background 11

A.1 Background : Overview 12

A.2 Key Demographic Figures 33

A.3 Macroeconomic Trends and Forecasts 33

A.4 Non-Residential Construction Put in Place History 34

A.5 New Dwelling Construction 35

A.5.1 Housing Stock by State 36

A.5.2 Dwelling Stock by Type and Tenure 37

A.6.1.a Domestic Space Heating Park 38

A.6.1.b Domestic Space Heating Park (Cont.) 39

A.6.2 Domestic Water Heating Park 40

A.6.3 Air-Conditioning Park 41

A.7.1/2 Top 10 Largest House Builders and Firms 42

A.7.3 US Climate Map 43

A.8 Boiler & Water Heater Standards 44

A.8.3 Residential Central A/C & HP Standards 45

A.8.4.1/2 Commercial Central A/C & HP Standards 46

A.8.4.3/4 Commercial Central A/C & HP Standards (Cont.) 47

A.9.1 Residential Energy Code Adoption by State 48

A.9.2 Commercial Energy Code Adoption by State 49

A.10 Historical and Current Exchange Rates 50

Distribution : US - Heating & Cooling

B.0 US - Distribution 51

B.1 Distribution : Structural Overview 52

B.2 Summary of Leading Distributors 62

Boilers : USA

1.0	US - Boilers	63
1.1	Market Sector Overview : Boilers	64
1.2.1	Historical Trends and Forecasts : All Boilers	74
1.3.1	Prices and Market Values : Boilers	75
1.3.2	Prices & Market Values by Capacity: GFSB Cond	76
1.3.3	Prices & Market Values by Capacity: GFSB Non Cond	77
1.3.4	Prices & Market Values by Capacity: GWHB Cond	78
1.3.5	Prices & Market Values by Capacity: GWHB Non Cond	79
1.3.6	Prices & Market Values by Capacity: JBB Cond	80
1.3.7	Prices & Market Values by Capacity: JBB Non Cond	81
1.3.8	Prices & Market Values by Capacity: Solid Fuel	82
1.4.1	End Use : Base Year	83
1.4.2	End Use : Historical and Forecasts	84
1.4.3	Technical Segmentation : Gas FS Boilers	85
1.4.4	Technical Segmentation : Gas Wall Hung Boilers	85
1.4.5	Technical Segmentation : Jet Burner Boilers	86
1.4.6	Technical Segmentation : Solid Fuel Boilers	86
1.5.1	Market Shares : Electric	87
1.5.2	Market Shares : Floor Standing Gas Condensing	88
1.5.3	Market Shares : Floor Standing Gas Non-Condensing	89
1.5.4	Market Shares : Wall Hung Gas Condensing	90
1.5.5	Market Shares : Wall Hung Gas-Combi vs. Heat Only	91
1.5.6	Market Shares : Total Gas Condensing (Res vs. Com)	92
1.5.7	Market Shares : Oil Non-Condensing	93
1.6.1/2	Imports-Exports : Central Heating Boilers	94
1.7.1/2	Distribution Flow : Boilers	95
1.8	Market Revision : Boilers	96

Water Heaters : USA

3.0	US - Water Heaters	97
3.1	Market Sector Overview : Water Heaters	98

3.2 Historical Trends and Forecasts : Water Heaters....109
3.3.1 Prices and Market Values : Water Heaters....110
3.3.2 Prices & Market Values by Capacity: Indirect Tanks....112
3.3.3 Prices & Market Values by Output: Elec. Tankless....112
3.3.4 Prices & Market Values by Capacity: ESWH Res....113
3.3.5 Prices & Market Values by Capacity: ESWH Com....114
3.3.6 Prices & Market Values by Capacity: GIWH Cond....115
3.3.7 Prices & Market Values by Capacity: GIWH N-Cond....116
3.3.8 Prices & Market Values by Capacity: Gas Storage....117
3.3.9 Prices & Market Values by Capacity: HPWH....118
3.4.1 End Use : Base Year....119
3.4.2 End Use : Historical and Forecasts....120
3.4.3 Sales by Type : Water Heaters....121
3.4.4 Sales by Type of Flue and Type of Material : WH....122
3.4.5 Sales by Storage Capacity : Electric Storage....122
3.5.1 Market Shares : Indirect Tanks....123
3.5.2 Market Shares : Gas Storage Condensing Commercial....123
3.5.3 Market Shares : Gas Storage Condensing Residential....124
3.5.4 Market Shares : Gas Storage Non-Cond. Commercial....124
3.5.5 Market Shares : Gas Storage Non-Cond. Residential....125
3.5.6 Market Shares : Gas Tankless Condensing....125
3.5.7 Market Shares : Gas Tankless Non-Condensing....126
3.5.8 Market Shares : Electric Storage Residential....126
3.5.9 Market Shares : Electric Storage Commercial....127
3.5.10 Market Shares : Electric Instantaneous....127
3.5.11 Market Shares : Electric Instantaneous by Type....128
3.5.12 Market Shares : Solar Storage Tanks....129
3.5.13 Market Shares : HP Water Heater....129
3.6.1/2 Imports & Exports : Other Non-Electric....130
3.6.3/4 Imports & Exports : Gas Instantaneous....131
3.6.5/6 Imports & Exports : Electric Storage....132

3.7.1/2 Distribution Flow : Water Heaters.....133
3.8 Market Revision : Water Heaters.....134

Overview

BRG Building Solutions is a specialized market research consultancy serving the global building technologies industry. Combining unparalleled industry experience with expert research and analysis, our team delivers lasting results to hundreds of companies worldwide.

Our core expertise lies in our coverage of the HVAC (heating, ventilation and air-conditioning), bathroom and kitchen, and other building products/ technologies. Throughout a history spanning over 50 years, we have assisted industry players in identifying and realizing growth opportunities through a combination of regular market monitoring and ad-hoc consultancy services.

BRG's unique Enterprise Database is at the center of our offering. This is a comprehensive tool and very detailed source of historic, current and forecast market trends and market share data (by holding and brand) on a country comparable basis. The database provides an essential tool to our clients as a platform to support their strategic decision making.

BRG Building Solutions is a genuinely global company, staffed with individuals from varied backgrounds and nationalities, working collaboratively from four offices across four countries. Combining a high level of industry experience with expert research and analysis, our team delivers lasting results to hundreds of companies worldwide.

Our market coverage is also global, reflecting the nature of the industries we serve. Currently, 75 individual markets are covered annually in detail across the following key geographical regions (and number of countries):

- Europe (37)
- Middle East (8)
- Africa (7)
- Australasia (2)
- Central Asia (4)
- East Asia (1)
- South Asia (3)
- Southeast Asia (5)
- North America (3)
- South America (5)

BRG has its own in-house dedicated modelling and forecasting unit. In cooperation with clients, we have since extended our offering to include market sizing for countries outside the traditional geographic reach of our main research program, via global market models.

Methodology

The methodology behind the data remains true to our origins in that the industry participates in the research process, in many cases by providing BRG Building Solutions with annualized volume sales by product category per country and contacts for detailed expert interviews in all countries covered.

Our approach follows a multi-pronged strategy, gathering data inputs and information from many different sources. A further USP of BRG is the fact that we live and breathe the markets in which we specialize (bathroom, kitchen, plumbing & HVAC) and are in constant contact with manufacturers, distributors and other key influencers all of the time during our repeat annual research program. The markets are built via a bottom-up approach. Extensive primary and secondary research complement the picture and provide all necessary insight for many additional segmentations shown in the reports, like e.g. price segmentations, technical and end-use segmentations.

The reports are constantly evolving and refined based on industry feedback, and always reflect the latest developments in the industries.

All research is conducted in-house with a team of around 70 individuals who speak 35 different languages between them. As a result, we can transfer that knowledge and experience into services designed to assist stakeholders in understanding the opportunities available to them, at a fraction of that charged by the larger and less nimble global consultancies. BRG operates a policy of direct management of the team required to conduct the regular research work and avoids subcontracting its market studies to outside agencies.

The analysis has grown in complexity as the number of drivers affecting the markets grows, like e.g. economic uncertainty, construction cycles, new technologies, legislation. As a result, BRG Building Solutions considers all drivers through a holistic approach, recognizing that products, end use segments and countries are all inter-linked.

Product Coverage

Since the 1980s, BRG's heating &cooling program has provided industry players with insightful analysis and evidence-based information. In addition to detailed data on trends, values, end use, technical segmentations, market shares and distribution, our program also includes perceptive text commentary, enabling our clients to easily identify and understand market characteristics and the latest industry developments.

BRG's program has evolved steadily in terms of both geographic scope and product coverage and continues to do so each year. Each country report follows a uniform structure, to ensure that the data can be readily compared across regions and countries.

The product coverage has also expanded to cover virtually all heating related products, used in both residential and non-residential buildings. Although BRG's product coverage varies slightly according to the industry structure in each country, the studies generally distinguish between the following main product types, covering both the residential and non-residential markets:

- **Boilers** : gas and oil-fired, solid fuel, including biomass, and electric boilers.
- **Jet Burners** : fitted to new units, solo and replacement.
- **Heat Pumps** : space heating and sanitary hot water heat pumps.
- **Heat Emitters** : radiators, convectors and electric fixed heat emitters for selected countries
- **Water Heaters** : dedicated water heaters and water heaters linked to boilers.
- **Solar Thermal** : pressurized and unpressurized solar thermal systems.
- **Furnaces** (North America only).
- **Air Conditioners**.

In-depth product definitions are included in a separate document and available upon request.

Definitions

Volume Trends

Time series show historical market developments as well as forecasts up to 2028. Market sizes are quantified in terms of volume sales (units), with the exception of solar collectors, which is quantified in square meters (m^2).

Values

Manufacturers' selling prices (MSPs) are provided for all heating products in euro and local currency. All MSPs exclude VAT (value added tax) and are net of all discounts/rebates. MSPs are calculated on the basis of our market shares, e.g., if manufacturer X sells a water heater to manufacturer Y who then in turn sells it on under his own brand, the market share table as well as the MSP is based on the level of manufacturer Y.

End Use

BRG's end use data details the installation location of the researched products. BRG analyses the end-use for the heating and cooling markets at product type level (historical trends and forecasts) and also provides a more detailed analysis of the end use for the base year (at product level).

BRG provides end use data for historical trends and a five-year forecast for the following segments:

- **Residential**
- New Housing
- Replacement
- First Time Installations
- **Non-Residential**

Market Shares

All market shares show both the holding and the brand (in brackets). Small markets in volume terms and markets dominated by the DIY channels may show a significant number of sales from “Others”, usually referring to imports and also possibly including private labelling specialist sales. BRG’s market shares focus on branded sales and not OEM sales to other manufacturers.

For further information, please contact us:

BRG Building Solutions Inc.

1400 Old Country Road, Suite 403N

Westbury, New York

USA 11590

Website : www.brgbuildingsolutions.com

Email : us@brgbuildingsolutions.com

USA

BACKGROUND

THE NORTH AMERICAN HEATING & COOLING PRODUCT MARKETS

BRG BUILDING SOLUTIONS

A.1 Background : Overview

Geography and Demographics

The United States of America (also known as the United States, the US, the USA, or America) has a land area of 3.79 million square miles (9.83 million square kilometers) and is the third largest country in terms of land area and population. The United States, after Russia and Canada, is the world's third or fourth largest nation by total area, ranking just above or below China, depending on how the country's total size is calculated and how the two territories disputed by China and India are counted.

The US Census Bureau’s Population Clock estimated 335.9 million as of January 1, 2024, making it the third-most populous country in the world behind China and India. It is also one of the world’s most ethnically diverse and multicultural populations, the product of large-scale immigration from many countries.

The United States is a federal constitutional republic comprising fifty states, a federal district, and various dependent territories. The country is situated almost entirely in the Western Hemisphere. Its contiguous states stretch from the Pacific Ocean in the west to the Atlantic Ocean in the east, with the Gulf of Mexico to the southeast; it is bordered by Canada to the north and Mexico to the south. Alaska, the largest state in terms of area, is separated from the contiguous states by Canada and touches the Pacific Ocean to the south and the Arctic Ocean to the north. Hawaii, the other noncontiguous state, occupies an archipelago in the central Pacific Ocean over 2,000 miles west-southwest of the country. The US territories are scattered around the West Indies (Puerto Rico, US Virgin Islands, Bajo Nuevo Bank, and Serranilla Bank) and the Pacific (Guam, American Samoa, Northern Mariana Islands, Baker Island, Howland Island, Jarvis Island, Johnston Atoll, Midway Islands, Wake Island, Palmyra Atoll, Kingman Reef, and Navassa Island).

The climate in the US varies greatly. West of the 100th meridian, semi-arid and desert conditions exist, while humid continental climate prevails in the northern areas, humid temperate on the central and Atlantic coasts, and humid subtropical in the Gulf and South Atlantic regions. Like many countries, the US has seen an increase in the number of climate-related natural disasters that ultimately lead to increased infrastructure repair expenditures.

Article I, Section 2, of the US Constitution calls for a census of the nation's population every 10 years to apportion the House of Representative seats among the states. The first census in 1790 recorded 4 million Americans. The 2020 Census represented the 24th official count of the entire population in the country's history, recording 331.4 million people. The five most populous states—California, Texas, Florida, New York, and Pennsylvania—accounted for 37.2%.

The US population grew by 0.53% in 2023, or 1.76 million people, according to US Census Bureau projections. This was an increase over 2022, and a notable increase over the historically low 2021 increase. Roughly two-thirds of 2023 growth came from international immigration, the second year in a row it fueled population gains.

Since 2000, the national median age (the point at which half the population is older and half younger) has increased by 3.4%. The US adult population increased faster than both the under-18 population and the population as a whole in the last decade, according to the 2020 Census data. Roughly 258.3 million, or 77.9%, were adults. The aging of Baby Boomers, those born from 1946 to 1964, is helping drive the growth in the adult population. In comparison, those under 18 numbered 73.1 million, or 22.1%. The younger population's slow decline can be attributed to a general decrease in fertility observed since 2007. As the nation ages, it also becomes more diverse. Nationally, all race groups experienced population increases except for whites, which declined by 0.03%.

The number of Millennials, or Gen Y, (those born from 1981 to 1996) became America’s largest generation in 2019 with an estimated 73 million members, while the Baby Boomer population declined to 72 million, per Census Bureau estimates. While Millennials may be the future, the Baby Boomers are the wealthiest age group and continue to dominate consumer spending in the US.

Macroeconomic Overview

According to the International Monetary Fund, the United States has the world's largest and most technologically powerful economy, accounting for an estimated 15.4% of world GDP in 2023. The World Bank estimated the per capita GDP for 2022 at $76,330. Going into 2020, the United States was experiencing the longest economic expansion in history, breaking the 120-month record of economic growth that occurred from March 1991 to March 2001. Low unemployment rates, low gas prices, and low interest rates bolstered the economy, giving consumers more money to spend on housing and other goods and services. However, 2020 provided multiple economic and social challenges that greatly affected growth.

The US economy grew at a 4% annual rate in the first three months of 2020, then shrank by the largest amount in 74 years due to the COVID-19 pandemic—a record-shattering 31% plunge in Q2. The IRS began depositing stimulus checks in April of 2020 to help ease the devastation, and Americans went back to work and shopping by June. Between two congressional stimulus packages and presidential executive actions, the US government spent over $2.6 trillion on COVID relief in 2020. Still, the US economy shrank by 3.5% that year—the first GDP contraction since 2009.

The US economy ended 2021 with GDP growth of 5.7%, the strongest calendar-year growth since 1984. According to the US Bureau of Economic Analysis, real GDP increased in all 50 states and the District of Columbia. The percent change in real GDP ranged from 0.3% in Alaska to 8.6% in Tennessee. The state of

California alone achieved a $3.36 trillion GDP. It would have been the world's 5th largest economy as a separate country, ahead of the UK ($2.76 trillion), India ($2.66 trillion), and France ($2.63 trillion). The next largest state, Texas, with $1.99 trillion of output in 2021, would be the 9th largest economy in the world, behind France but bigger than Italy ($1.89 trillion) and Canada ($1.64 trillion).

The strength of the US economy is reflected in the dollar's relative value compared to most other currencies in the world. A strong US dollar hurts the manufacturing sector as American goods become more expensive for other countries to buy. In 2021, US trade with foreign countries totaled $6.97 trillion, with $3.01 trillion in exports and $3.96 trillion in imports of both goods and services. Exports increased by $453.1 billion in 2021, while imports increased by $556.1 billion. Exports of goods alone increased $324.2 billion to $2,085.6 billion, while imports of goods increased $425.7 billion to $3,277.3 billion, both of which set records.

Canada was the top destination market for US goods exports, hitting $356.1 billion. Mexico finished third behind the European Union with $324.4 billion. The Americas region accounted for 43.4 percent of US exports. When services are included in a single market, the European Union remains the US' top trading partner. China is the top source of US goods imports, reaching $536.8 billion in 2022.

The COVID-19 pandemic created quite a supply shock, which is responsible for a portion of the price increases consumers experienced afterward. Inflation is mostly the result of monetary policy, i.e., the control of the money supply and the channels by which new money is provided. The Federal Reserve oversees monetary policy in the US and kicked off its pandemic relief efforts in March 2020 with large-scale asset purchases to prop up financial markets. By mid-2021, it had doubled its balance sheet to $8 trillion. The money supply spiked during this effort, as did consumer demand, which helped push prices up across the board.

The US Bureau of Labor Statistics releases the monthly Consumer Price Index (CPI), which measures the change in prices paid by consumers for goods and services. The CPI climbed 7% from December 2020 to December 2021, the largest 12-month gain since June 1982. Higher prices for shelter and used automobiles led to the increase, but food costs were also a major contributor. Excluding the more volatile food and energy components, "core" prices accelerated 5.5% from December 2020 to December 2021—the biggest jump since 1991. This extreme inflation forced the US Federal Reserve to change its view of it as transitory and instead embark on an interest rate hike and quantitative tightening campaign to halt rising prices and achieve its desired 2% inflation rate. The initial rate hike of this cycle took place in March 2022 and, as of April 2024, appears to have ended in July 2023 at an effective federal fund rate of 5.33%.

The Federal Reserve appears to have adopted a wait-and-see approach to watch the effects of its actions on the economy, but it has also recently signaled three 25-basis-point interest rate cuts in 2024. However, some in the Federal Reserve have recently stated that cuts may not be necessary at all in 2024 due to economic momentum. Supporting this possibility was March CPI coming in above

expectations for the third consecutive month, driven by energy prices and shelter; the new data point led markets to hypothesize there would only be two rate cuts as opposed to three occurring late in the year. At this stage, it is clear that time is needed to see how conditions develop, as elevated rates tend to constrict growth, and cutting too soon risks inflation re-accelerating.

The US Conference Board Leading Economic Index (LEI) had been signaling the US economy was headed for a recession since July 2022 and repeated that forecast monthly until January 2024. Data was released showing a 0.4% decline, but six of its ten components were positive contributors, and the remainder was poised to turn higher. As predicted, February 2024 data crossed over into positive territory with a 0.1% increase.

Every economic cycle is different, but ultimately the depth of any US downturn at least partially depends on the amount of leverage built up in the system during the cycle, which is currently more than twice the amount it carried heading into the global financial crisis (national debt of $10.0 trillion in 2008, or 68% of GDP, versus $33.2 trillion in 2023, or 120%).

The pace of this hiking cycle is the fastest in decades, and many businesses were unprepared for how it would affect them. Silicon Valley Bank, the 16th-largest bank in the US at the end of 2022, grew rich with deposits from tech companies and locked them for 10 years into low-yielding government bonds and securities. Once rates started spiking, these interest-rate-sensitive customers began laying off workers and drawing down their bank deposits to stay afloat, but the bank was unable to exit its hold-to-maturity positions. To save depositors, the US government had to step in, not by lowering interest rates or pumping money into the system as it had done many times before, but by protecting all uninsured deposits.

An unemployment rate near 5% is considered full employment. The country had been at full employment since September 2015, even reaching a 50-year low of 3.5% in 2019. In January 2020, before COVID-19 lockdowns, the unemployment rate was 3.6%. By April, the rate hit a high of 14.8% before ending the year at 6.7%. According to the US Bureau of Labor and Statistics, the 2021 US unemployment rate came in at 5.4%, a significant improvement versus the previous year. 2022 and 2023 were the lowest annual rates seen in decades, both at 3.6%, according to Federal Reserve Economic Data. The March 2024 unemployment rate was 3.8%.

Bottom line: recession risk still exists in 2024, but there is much more of a bull case.

Residential Construction

New Housing

Residential construction showed consistent growth from 1995 through 2005. By mid-2006, when new residential construction permits peaked at 2.273 million, the

housing bubble had burst, and the market entered a deep and long correction. In 2007, residential construction expenditures dropped 17.9%, and the market corrected further until 2011. The market turned around in 2012, when it grew 14.8% to reach $282 billion in value. It has been on a roll since then, but it has yet to reach the level of activity seen in 2006 or even 2007, when the decline began.

The residential real estate market was anything but typical in 2021. Home sales reached the highest level in 15 years, with roughly 6 million homes sold. Home prices rose by about 20% over 2020, the largest annual increase in the history of the Federal Housing Finance Agency House Price Index. Homes sold quickly and often over the asking price, with sellers frequently receiving multiple offers, including all cash offers. The stimulus payments boosted record consumer savings, and some of the lowest home loan interest rates on record helped fuel demand, quickly driving inventory to record lows. This remained one of the factors affecting the housing market for half of 2022.

According to the 2024 National Association of Realtors' Home Buyers and Sellers Generational Trends Report, Millennials were the largest group of home buyers last year, accounting for 38% of home buyers (both younger aged 25 to 33 at 17% and older aged 34 to 43 subgroups at 21%). This was up from 28% in 2023, but still below 43% in 2021. Boomers represented 31% (younger adults aged 59 to 68 at 19% and older adults aged 69 to 77 at 12%). Generation X (aged 44 to 58) made up 24% of home buyers, the Silent Generation (aged 78 to 98) 4%, and Generation Z (aged 18 to 24) just 3%. First-time home buyers accounted for 32%, below the 38% average but up from an historic low of 26% the year before. The sales market was dominated by boomers, who accounted for 45% of all sellers.

On a monthly basis, the US Census Bureau tracks housing in four ways: the number of building permits issued, the number of units started, the number of units completed, and the value of put-in-place construction. The data provided in units only includes homes that were built with private dollars; therefore, this information does not take into account any public housing projects that were permitted, started, or completed during the calendar year.

An estimated 1,470,600 housing units were authorized by building permits in 2023, an 11.7% drop versus the 1,665,100 in 2022.

In 2023, there were 1,420,000 housing units started in the US, down 8.5% from the 1,552,600 of 2022. New single-family home starts were down 5.7%, and multifamily home starts were down 13.7%. All regions were down. The South had the most starts, at 792,100 units, but decreased 2.9% compared to 2021. The West saw 328,200 units start, down 11.5% from 2022. The Midwest and North came in with 184,900 and 114,800 starts, down 9.0% and 3.5%, respectively.

For the tenth year in a row, residential new construction unit completions rose. About 1,448,800 units were completed in 2023, an increase of 4.2% over 2022. Single-family home completions fell 2.3% to 998,900 units, which represented 68.9% of all new dwelling units completed. Multifamily units, completed in buildings

containing two or more units, came in at 449,900 units, up 22.1%. All regions of the US but the Midwest saw more new housing units completed, which was down 2.6%.

The value of residential construction put in place includes construction work done each month on new structures or improvements to existing structures for the private and public sectors. In 2023, total residential spending was $872.9 billion, a decrease of 5.9% versus the $927.4 billion in 2022 but up from the $802.9 billion in 2021.

According to the US Census Bureau's Characteristics of New Housing (published in June of each year), the median square footage of newly completed single-family homes in 2023 was 2,233 square feet (207 square meters). The median size of multifamily units completed in 2023 were 998 square feet (93 square meters).

The US Census also tracks shipments of manufactured housing; these homes are not accounted for in the new construction numbers above. Manufactured homes are factory-built and can be up to 95% complete when leaving the factory; they are delivered to the building site equipped with all lighting, appliances, cabinets, and fixtures already installed. Homes are typically built in one to three large sections, up to 20 feet (6.1 meters) in width and 76 feet (23.2 meters) in length, and can be ready for occupancy in as little as one day after delivery. Manufactured homes used to be called "mobile homes" or "trailers," but these are terms that only correctly refer to homes built before 1976. In 1976, a new HUD (Housing and Urban Development) code set stricter standards for the construction, design, and performance of manufactured homes, which made them different from and superior to mobile homes. Many of the manufactured homes made today are almost indistinguishable from site-built homes, but they cost an average of $50,000 less than a similar site-built home.

According to the 2021 American Housing Survey, over 6.7 million mobile and manufactured homes exist in the US, of which almost 1.3 million are set on a masonry foundation and another 1.2 million rest on a concrete pad (the majority are up on blocks but not on a concrete pad). Manufactured homes have historically been most popular in the South and West. For example, the share of manufactured homes in New Mexico, South Carolina, Mississippi, and West Virginia is more than twice the national average. On the other hand, less than 2% of homes in Northeastern states like Rhode Island, Massachusetts, New Jersey, and Connecticut are manufactured.

According to the US Census Bureau's Manufactured Housing Survey, 89,169 new manufactured homes were shipped in the US in 2023, down from 112,882 in 2022 and 105,772 in 2021, but consistent with the number of shipments from 2017 through 2020.

CLAYTON HOMES is the largest builder of manufactured housing and modular homes in the US. The company is owned by Warren Buffett's BERKSHIRE HATHAWAY and headquartered in Maryville, Tennessee.

Housing RMI

Home repair, maintenance, and improvement spending occurs for many reasons, but generally, the three primary drivers are to make necessary repairs caused by failure, remodel an aging or outdated home, prepare a home for sale, or customize it after purchase.

With respect to the first of the three drivers, the aging US housing stock is a primary reason that sales in the repair and remodeling market have been increasing. According to the US Census 2021 American Housing Survey, the median age of occupied homes is 46 years old, effectively built in 1978. The 2022 1-Year American Community Survey Estimates detail that 35% of homes were built in 1969 or earlier. In total, 60% of US housing units were built before 1990.

During the COVID-19 pandemic, more time spent in the home meant more product use, which accelerated failure. For example, plumbing fixtures that may have only been used a couple times per day before the pandemic saw dramatically increased usage with work-from-home and home-schooling families, which caused more repair issues to surface and become unbearable.

In retrospect, the COVID-19 pandemic has been a boon for the remodeling industry because the money homeowners were able to save by not going out or traveling was spent on home improvements. Now that this chapter is over, it is natural that there will be somewhat of a spending hangover.

Regarding the third driver of RMI spending, the National Association of Realtors (NAR) reports that existing-home sales declined 19% to the lowest amount in almost 30 years in 2023, at 4.09 million, a significant drop from 5.03 million in 2022 and 6.12 million in 2021. The lack of homes for sale due to elevated mortgage rates was the primary driver behind the decline. The median existing home sales price was $389,300, up <1% from 386,400 in 2022 and well above $350,700 in 2021. The NAR also estimates that the average selling price for an existing home sold in 2023 will be $510,000.

The NAR estimates that there were 990,000 existing homes for sale at the end of December 2023, or 3.1 months' supply. This compares favorably to the 2.7 months in 2022 and the 2.3 months in 2021. Their year-end Monthly Housing Trends Report estimated active housing inventory still remained 36% below typical 2017 to 2019 levels, and, in the 40 largest metro areas of the US, inventory of homes actively for sale was down between 24% and 49% from pre-pandemic levels.

Homeowners are also spending more to repair failures caused by natural disasters. The National Oceanic and Atmospheric Administration's (NOAA's) National Centers for Environmental Information (NCEI) releases an annual update to its billion-dollar disaster report each January. In 2023, there were 28 weather and climate disaster events exceeding $1 billion in losses, which included: 1 drought, 4 floods, 19 severe storms, 2 hurricanes and tropical cyclones, 1 wildfire, and 1

winter storm. In total, these events cost the nation $93.7 billion in damages. Harvard's Joint Center for Housing Studies estimates that repair spending on owner-occupied homes following a disaster is typically spread over three years.

The Joint Center for Housing Studies (JCHS) at Harvard monitors US remodeling statistics and trends and publishes the Leading Indicatory or Remodeling Activity (LIRA), which forecasts year-over-year growth in homeowner remodeling expenditure. The research center estimates home remodeling activity (excluding rental improvements and maintenance spending) reached $481 billion in 2023, a modest 1.9% gain over the $472 of 2022. The LIRA predicts that annual owner improvement and repair spending will plummet to $450 billion in 2024.

Non-Residential Construction

Non-residential construction activity increased from 1995 to 2001, followed by a two-year correction with a combined market adjustment of -7.4%. The segment emerged from this correction and grew robustly until 2008, with a total combined growth of 62% from 2003 through 2008. The market clearly overheated and started to correct at the beginning of 2009, with an adjustment of almost -25% until 2011. By 2012, America began to spend money on non-residential projects again, and the value of non-residential construction increased every year for five years straight. However, in 2017, the value of non-residential spending put in place declined slightly, only to bounce back the following year and gain again in 2019.

The US Census Bureau publishes Annual Value of Construction Put In Place statistics for the US as a whole. In 2023, total construction reached $1,979.7 billion, up 7.1% versus the prior year. Of this total, 77.8% was private construction and 22.2% was public. Further, 55.9% was nonresidential construction and 44.1% was residential.

In 2023, all 16 types of nonresidential construction tracked grew compared to 2022. The largest sectors in 2023 by value were manufacturing ($196.4 billion, +71.3%), highway and street ($134.7 billion, +8.0%), and commercial ($132.12 billion, +8.9%). The more relevant categories to this publication were educational ($116.4 billion, +14.0%), office ($98.9 billion, +7.9%), health care ($63.2 billion, +15.5%), and lodging ($23.6 billion, +19.6%).

The American Hotel &Lodging Association's 2024 State of the Hotel Industry Report shows domestic tourism has effectively returned to normal. Revenue per room in 2023 exceeded the 2019 pre-pandemic level by 13.0%, while nationwide occupancy came within 4.4%. Occupancy is expected to reach 63.6% in 2024, just above the 62.9% of 2023 but short of the record 65.8% posted in 2019. The average daily rate hit $155,47 in 2023 and is projected to grow 3% to $160.16 in 2024, well above the $131.42 rate of 2019.

According to the year-end 2023 Global Hotel Construction Pipeline Trend Report from Lodging Econometrics, the US hotel construction pipeline stood at 5,964 projects, equating to 693,963 rooms at the end of 2023. Compared to 2022,

projects were up 9% and rooms 7%.

Domestic Space Heating Park

According to the US Energy Information Administration's 2020 (latest available) Residential Energy Consumption Survey (RECS), 95.1% of the 123.5 million primary-occupied housing units in the US used some form of space heating equipment. 58.3% of US homes used a central warm-air furnace, 14.4% used a heat pump, 7.5% used a steam or hot water system, and 6.7% used built-in electric units.

Single-family detached homes accounted for 77.1 million of the 123.5 million homes in 2020, or 62.4%, according to the RECS. Almost exactly two-thirds of these relied on a central warm-air furnace as the main heating equipment, followed by 14.7% heat pump and 5.6% steam or hot water systems. Single-family attached homes accounted for just 6.0%, but furnaces accounted for effectively the same share as their detached counterparts, while heat pumps and steam or hot water systems were 10.6% and 7.4%, respectively. Shares of apartment heating systems differed from single-family dwellings. Apartment buildings with 5 or more units accounted for 18.5% of homes in 2020, or 22.8 million dwellings, but only 37.0% utilized a warm-air furnace. These types of homes utilized comparatively more heat pumps and steam or water systems, accounting for 13.0% and 12.3% of total usage, respectively. In buildings with 2-4 apartments, which accounted for 7.6% of homes at 9.3 million dwellings, the penetration rates are 43.1% for warm-air furnaces, 11.0% for heat pumps, and 16.7% for steam or hot water systems.

Different forms of space heating are more popular in owner-occupied properties than rental accommodations, according to the 2020 RECS. In the former, 64.9% of the 82.9 million homes rely on a central warm-air furnace as the primary heating equipment. In the latter, the share drops to 44.7% of the 40.6 million homes. Heat pump share between owned and rented homes is more comparable at 14.8% and 13.5%, respectively. Homes relying on steam and hot water systems as the main heating equipment differ more between the two, coming in at 6.6% for owned homes versus 9.5% for rentals.

According to the RECS, warm-air furnaces are by far the most popular type of primary heating equipment across all census regions, with the largest market penetration in the Midwest at 77.9% of homes. In all regions, these appliances make up at least half of the market, with the South having the lowest share at 50.1% of homes. In this region, heat pumps accounted for 29.0% of the main heating equipment in 2020. Approximately 58.5% of homes in the West and 51.2% of homes in the Northeast utilized furnaces.

Steam or hot water systems made up the most significant share of homes in the Northeast, accounting for 27.5% of the 21.9 million homes in 2020. The next closest would be the Midwest, with 7.3% of the 27.0 million homes. The more notable market penetration in the Northeast is linked to the relatively old age of the housing stock as well as tradition. However, the Census Bureau's Characteristics

of New Housing data reveals a consistent rise in the furnace share in the region, at the expense of other systems, since the early 1990s, when new single-family houses were completed. In 1992, 50% of these dwelling completions utilized a warm-air furnace as the main heating system and 36% steam or hot water systems; in 2023, those shares jump to 79% for the former and a mere 4% for the latter.

According to new housing characteristics, 45% of single-family homes completed in the US during 2023 use a heat pump as their primary heating source. The warmer regional climates of the South and the Southern West had the highest heat pump market penetration, primarily the air-source type. An estimated 61% of single-family homes completed in the South used a heat pump to warm the space. The share has remained between 59% and 62% since 2009. In the West, the heat pump share of single-family homes completed in 2023 was 29%—the highest recorded share and a significant jump from the 16% reported in 2019-2021.

The warm climates in the South and West are more likely to have specific room heaters than a system that heats the entire house. Because the days tend to be hot and the nights are often cold, people typically only provide heating in the bedroom and possibly the living room.

The market dominance of warm-air furnaces may be changing. According to data provided in the Characteristics of New Housing, forced-air furnaces fell from 60% of heating systems used in new single-family houses completed in 2021 to 53% in 2023. Heat pumps were the primary beneficiaries of the decline, particularly air-source, which jumped from 37% of homes completed to 44%—an all-time high.

Domestic Air Conditioning Park

According to the 2020 Residential Energy Consumption Survey (RECS) conducted by the US Energy Information Administration, 88.2% of the 123.5 million primary-occupied housing units in the US used some form of air-conditioning equipment. An estimated 66.5% of US homes used central air-conditioning equipment; 17.3% used a window or wall air conditioner; 2.2% used a portable air conditioner; and a mere 1.4% used a ductless heat pump (aka “mini-split”).

According to the RECS, an estimated 74.6% of single-family detached homes relied on central air-conditioning equipment (including a central heat pump) as the primary type of air conditioning, followed by 11.6% using window or wall air conditioners. Those shares for single-family attached homes were 68.5% and 14.4%, respectively. As was the case with heating, apartment heating system shares differed from single-family dwellings. According to the RECS, 52.6% of apartment buildings with 5 or more units used central air-conditioning equipment (including a central heat pump) as the main type of air conditioning, while 27.1% used window or wall air conditioners. In buildings with 2–4 apartments, those penetration rates are more evenly split at 39.3% and 36.5%, respectively.

The forms of air conditioning vary in owner-occupied properties versus rental accommodations, according to the 2020 RECS. In the former, 74.1% of the 82.9 million homes rely on central air-conditioning equipment (including a central heat pump) as the main type of air conditioning, while 12.6% rely on window or wall air conditioners. In the latter, those shares jump to 50.9% and 26.9%, respectively, for the 40.6 million homes.

According to the RECS, central air-conditioning equipment (including central heat pumps) is the most popular type of primary air-conditioning equipment across all census regions, with the largest market penetration in the South at 82.9% of homes, followed by the Midwest with 73.4%. Window or wall air conditioners accounted for 9.3% and 17.4% of homes in these two regions, respectively. The Northeast sees the lowest share of central systems at 38.8% of homes, likely a result of the relatively higher share of homes utilizing a steam or hot water heating system that lacks ductwork for a central air-conditioning system. In this region, window or wall air conditioners accounted for 41.6% of the main air-conditioning equipment in 2020. Approximately 53.9% of homes in the West used central systems, and 11.6% used window or wall units.

According to the Census Bureau's Characteristics of New Housing data, 98% of new single-family homes completed in 2023 had air conditioning. This share has been steadily increasing from 88% in 2009.

Domestic Water Heating Park

According to the 2021 American Housing Survey, conducted by the US Census Bureau, 99% of occupied housing units in the US have hot piped water. Of these homes, 47.5% use electricity, and 45.7% use piped gas. The 2020 Residential Energy Consumption Survey (RECS), conducted by the US Energy Information Administration, agrees with these numbers, and based on its estimated number of main and secondary water heaters, there are 1.05 water heaters per dwelling. Larger homes would be the most likely to have two water heaters, but these are not typical homes.

The RECS estimated that 5.6% of houses utilized propane, fuel oil, or kerosene as the main water heater fuel. A very small number of homes reported using a wood stove and an indirect coil, which translated to having water heating equipment but not a water heater appliance.

The 2020 RECS estimated 52.1% of single-family homes used a main water heater powered by natural gas and 41.1% electricity. For multifamily apartment buildings, these shares were 39.3% and 57.0%, respectively. While apartment buildings with 5 or more units broke as 36.3% and 60.0%, buildings with 2-4 apartments were more evenly split at 46.5% and 49.5%. Tankless units accounted for 7.4% of single-family housing, 6.0% of multifamily buildings with 5 or more units, and just 2.7% of those with 2-4 apartments.

Water heaters using natural gas as the primary fuel are by far the most popular type of unit in the West, Midwest, and Northeast census regions. These units accounted for 61.7%, 57.5%, and 50.5% of homes in these areas, compared to 33.8%, 37.6%, and 34.0%, respectively. However, this regional dominance does not translate into national dominance, as electricity dominates the relatively larger housing market in the South. Accounting for 37.9% of homes in the country, 66.7% of water heaters in the South utilize electricity as the primary source of energy, and just 30.03% use natural gas. All totaled, electric water heaters were found in 47.1% of homes in 2020, and natural gas at 46.9%.

District Heating

According to the American Council for an Energy-Efficient Economy, 5,800 district energy systems were in operation in 2012 and served more than 6.5% of commercial buildings, downtown districts, campuses, military bases, research facilities, and even some residential locations in the United States. Most district heating systems in the United States are in northern cities like Minneapolis, Milwaukee, Chicago, and Detroit, and they are especially popular heating options on college campuses; however, systems can be found in all 50 states. Steam systems are the most common.

New York City's district heating system is the world's largest steam system, with 105 miles of steam pipe running beneath the city's streets. It was first developed in the 1880s. Denver's commercial district steam system is the oldest continuously operating system in the world, serving 135 customers.

The United States currently has 20 geothermal district heating systems (GDHS) operational. Six of the existing plants are in Idaho, where the first GDHS in the US was designed and operational in 1892.

In residential areas, district heating penetration is very low.

Trade Organizations

There are several US non-profit organizations that lobby state and federal legislatures to get favorable laws and financial incentives passed for their supporting members. The following are some examples of the types of organizations that lobby for HVAC products:

The largest of these organizations is the AIR-CONDITIONING, HEATING, AND REFRIGERATION INSTITUTE (AHRI). AHRI is a trade association representing 300+ manufacturers of HVACR and water heating equipment that account for more than 90% of all the HVACR and water heating residential and commercial equipment manufactured and sold in North America. The association has three stated purposes: “AHRI advocates on behalf of its members at all levels of government, working to ensure that members’ interests are included in final drafts of legislation” and “a certification program that is relied heavily upon by regulators for accurate and unbiased evaluation of heating, water heating, ventilation, air

conditioning, and commercial refrigeration equipment." Lastly, AHRI “develops industry-recognized performance standards for industry equipment." More information may be found at www.ahrinet.org.

The GEOTHERMAL EXCHANGE ORGANIZATION (GEO) calls itself “The Voice of the Geothermal Heat Pump Industry in the United States." Like other non-profit trade associations, it promotes the manufacture, design, and installation of its members’ geothermal systems. GEO’s government affairs program has achieved successes for its industry, including increasing tax credits for residential geothermal heat pump installations from $300 to 30% of system costs. Going forward, legislative and regulatory initiatives include the qualification of geothermal heat pumps under energy and climate legislation, which could result in consumer benefits and electric utility promotion programs. More information about the GeoExchange can be found here: www.geoexchange.org.

The SOLAR ENERGY INDUSTRIES ASSOCIATION (SEIA), the national trade association of the US solar energy industry (including PV manufacturers), champions the US solar thermal industry. The organization represents approximately 1,000 member companies, which employ more than 250,000 Americans, “to build jobs and diversity, champion the use of cost-competitive solar in America, remove market barriers, and educate the public on the benefits of solar energy." More information can be found at www.SEIA.org.

To encourage local markets to favor energy-efficient products and services, six of the Northwest’s investor-owned utilities and the Bonneville Power Administration joined forces to form the non-profit corporation NORTHWEST ENERGY EFFICIENCY ALLIANCE (NEEA). The organization has grown since its foundation and now focuses on encouraging incentives for heat pump water heaters, among other residential products. More information may be found at www.NEEA.org.

Building Codes

Building codes are regulations governing the design, construction, alteration, and maintenance of structures. They detail minimum requirements to adequately safeguard the health, safety, and welfare of building occupants.

In the US, the power to regulate the safety of buildings is reserved by the states, not the federal government. This regulation may be done at the state level, but some or all authority for the adoption, maintenance, and enforcement of building codes is delegated to counties, cities, or other local governments.

The International Code Council (ICC) is a non-profit organization dedicated to the development of model codes and standards used in the design, construction, and compliance processes. The International Codes are a widely accepted, comprehensive set of model codes used in dozens of countries to help ensure safe, sustainable, affordable, and resilient structures. All fifty states, as well as the District of Columbia, have adopted the International Codes at the state or jurisdictional level.

All codes are coordinated, compatible with each other, and developed using the same ICC consensus code development process that draws on the expertise of plumbing, building, and safety experts from across North America.

The International Building Code (IBC) is a set of regulations for commercial construction. It addresses the design and installation of materials that meet or exceed public health and safety goals. In general, the IBC prioritizes the design and installation of materials that meet or exceed public health and safety goals.

The IBC is in use or adopted in 50 states: the District of Columbia, Guam, the Northern Marianas Islands, New York City, the U.S. Virgin Islands, and Puerto Rico. It also serves as the foundation for laws and regulations in other countries' communities. It is also used in a variety of nonregulatory settings, including voluntary compliance programs, the insurance industry, certification of individuals and buildings, U.S. federal agencies, best practices benchmarks, and reference works.

The IBC also includes acoustic and noise requirements. IBC sound transmission requirements apply to occupants of residential buildings (i.e., houses, apartments, condominiums) or places offering overnight accommodations, such as hotels and motels. The IBC does not have acoustic requirements for non-residential buildings. However, the ICC has developed a separate set of recommendations, which are detailed in the ICC G2–2010 Guideline for Acoustics. This guideline sets targets for sound isolation performance for construction systems that separate occupied spaces in commercial and multiple-family buildings.

The International Energy Conservation Code (IECC) is a comprehensive energy conservation code that establishes minimum requirements for energy-efficient buildings. This code addresses the design of energy-efficient building envelopes, as well as the installation of energy-efficient mechanical, lighting, and power systems. The goal is to create optimal utilization of fossil fuels and nondepletable resources. Many cities and states often look to this national model energy code as the starting point for their own.

Nonresidential building HVAC systems must comply with the 2015 International Energy Conservation Code (IECC). Section C401.2 states that these buildings must comply with ANSI/ASHRAE/IESNA 90.1, or sections C403.2, "Provisions Applicable to All Mechanical Systems," C403.3, "Economizers," and C403.4, "Hydronic and Multiple-Zone HVAC Systems Controls and Equipment."

This code contains separate provisions for commercial buildings and for low-rise residential buildings (three stories or less in height above grade). The IECC Commercial Provision contains chapters on scope and administration, definitions, general requirements, energy efficiency requirements, and existing building provisions.

The IECC is revised on a three-year cycle, and the 2021 update process began in 2019. The ASHRAE 90.1-2019, another national model energy code, was released this year and can track into the 2021 IECC process.

The International Mechanical Code (IMC) is intended to provide comprehensive regulations for mechanical systems and equipment, including HVAC, exhaust systems, chimneys and vents, ducts, appliances, boilers, water heaters, refrigerators, hydronic piping, and solar systems. The IMC is founded on broad-based principles that make the use of new materials and new plumbing designs possible. The IMC is in use or adopted in 46 states (as of August 2019), the District of Columbia, Guam, Puerto Rico, and the U.S. Virgin Islands. The latest version of the IMC is the 2018 edition; the next version is 2021.

A chapter of the IBC covered the Existing Building Code (IEBC), which was previously covered in another chapter of the IBC. The stand-alone IEBC applies to existing structures repair, alteration, change of occupancy, addition, and relocation. The IEBC's goal is to encourage the use and reuse of existing buildings while also maintaining minimum safety requirements. The IEBC's idea is that any additions or alterations must comply with new codes, but existing buildings only need to be maintained to their current level of safety.

Legislation

Boilers

Standards for residential boilers were included in the Energy Independence and Security Act (EISA) of 2007 included standards for residential boilers, which apply to gas-fired hot water boilers, gas-fired steam boilers, oil-fired hot water boilers, oil-fired steam boilers, and electric hot water boilers with an input of less than 300,000 BTU per hour manufactured after September 2012. The standards for each type of boiler include a requirement for annual fuel utilization efficiency (AFUE) and/or design requirements. AFUE is the ratio of the heat output of the boiler compared to the total energy consumption of the boiler.

In 2016, the Department of Energy (DOE) finalized new standards for residential boilers that raised minimum efficiency levels from 82% to 84% AFUE for gas-fired boilers and 84% to 86% for oil-fired hot water boilers beginning January 15, 2021.

A second regulation, finalized under the Obama administration in December 2016, raised minimum efficiency levels and cut energy waste for four product categories, including commercial boilers. On January 10, 2020, the DOE announced the first new national appliance efficiency standards since 2017:

https://www.federalregister.gov/documents/2020/01/10/2019-26356/energy-conservation-program-energy-conservation-standards-for-commercial-packaged-boilers

DOE made mandatory changes to the boiler test procedures in 2023. The test procedures for residential boilers can be found in 10 CFR Part 430, Subpart B, Appendix N, Uniform Test Method for Measuring the Energy Consumption of Furnaces and Boilers. The new rules updated Appendix N to remove the items applicable only to consumer boilers and to rename the appendix “Uniform Test Method for Measuring the Energy Consumption of Furnaces.” Simultaneously, the final rule established a new test procedure specific to residential boilers in a new appendix EE. With modifications, the DOE includes all provisions currently included in Appendix N that relate to boilers. The final rule may be found here:

https://www.regulations.gov/document/EERE-2019-BT-TP-0037-0027

Water Heaters

The DOE is required by Congress under the 1987 National Appliance and Energy Conservation Act (NAECA) to set energy and performance standards for residential water heaters. The revised mandates, which were last completed in 2010 and took effect in April 2015, essentially set higher minimum efficiencies for electric, oil, and gas water heaters, as measured by the Energy Factor (EF) rating, to reduce home appliance energy use, conserve more natural resources, and generate substantial energy savings for homeowners.

The DOE has again updated residential water heater efficiency standards to double the efficiency for the most common sizes of electric storage water heaters. Finalized in May 2024 with a compliance date of May 6, 2029, the new requirements will effectively create a massive shift from the more common electric storage water heaters to electric heat pump water heaters—an estimated 50% of electric storage water heaters will utilize heat pump technology from just 3% today. The DOE estimates that units meeting the new standards will save consumers an average of $1,800 on their utility bills over the life of the appliance. Furthermore, they estimate that under the new rule, 30 years of energy savings will amount to 17.6 quadrillion BTUs—more energy used in a single year than the entire residential building sector. This will be the largest savings ever from a single DOE efficiency standard.

The final rule may be found here: https://www.regulations.gov/document/EERE-2017-BT-STD-0019-1426.

In August 2023, the US Department of Energy announced new efficiency standards for commercial water heaters to conserve energy and water while cutting harmful carbon pollution. The DOE is adopting a performance standard that will require condensing technology for new models of commercial gas-fired storage, instantaneous, and hot water supply boilers. The standards are set to start in 2026.

Furnaces

In 2023, furnace efficiencies increased from 80% AFUE to 81% across the country. In June 2022, President Biden proposed new energy efficiency standards for

residential gas furnaces with the objective of reducing greenhouse emissions and saving money for consumers. In September 2023, the U.S. Department of Energy (DOE) finalized the energy efficiency standards. The new standards are to come into effect in late 2028 and will require non-weatherized gas furnaces and those used in mobile homes to achieve an annual fuel utilization efficiency of 95%, leaving non-condensing gas furnaces practically out of the market. The US Senate passed a joint resolution to block the DOE rule in May 2024, which President Joe Biden has threatened to veto if brought before him.

In June 2011, the DOE established new energy efficiency standards for central air conditioners and furnaces, as well as strengthened the national standards for heat pumps. The new rules are based on a joint recommendation filed with the DOE by consumer, manufacturing, and environmental groups in 2009. The standards established were due to be applied starting May 1, 2013, for non-weatherized furnaces, and January 1, 2015, for weatherized furnaces, central air conditioners, and heat pumps. However, the AMERICAN PUBLIC GAS ASSOCIATION (APGA) filed an appeal of the regional standards rule in the US Court of Appeals for the DC Circuit. In March 2014, a settlement was reached in the lawsuit, which remands the efficiency standards for gas furnaces, forcing the DOE to reassess efficiency standards in a more transparent manner. The increase and regionalization in the central air conditioner standard remain, but the DOE allowed for a "sellthrough' period of 18 months and agreed to not penalize HVACR distribution as part of enforcement of the regional standard.

On June 25, 2014, the DOE issued new efficiency standards for residential furnace fans manufactured on or after July 3, 2019. The standards do not apply to furnace fans that are integrated into hydronic air handlers, SDHV modular blowers, SDHV electric furnaces, or CAC/HP indoor units. To learn the specifics about the new furnace fan standards, visit the US DOE website at:

www1.eere.energy.gov/buildings/appliance_standards/product.aspx/productid/42.

Central Air Conditioners and Heat Pumps

Manufacturers have been required to comply with the DOE's energy conservation standards for residential central air conditioners and heat pumps since 1992. A federal mandate applying to all residential central-cooling systems came into effect on January 23, 2006. The new standard of 13 SEER represented a 30% increase in efficiency over the previous standard of 10 SEER.

On January 1, 2010, in accordance with the US Clean Air Act Amendments of 1990, R22 refrigerants were phased out. In all systems, ozone-friendly R-410A-type refrigerants are now mandatory.

In April 2014, the U.S. DOE made changes in HVAC efficiency standards for residential and commercial equipment. Starting in 2015, new Seasonal Energy Efficiency Ratio (SEER) standards have been enforced. The new regulations significantly changed all split-system heat pumps. All regions of the US moved from

a minimum of 13 SEER and 7.7 HSPF (Heating Seasonal Performance Factor) to the new national heat pump efficiency minimum of 14 SEER and 8.2 HSPF for units installed on or after January 1, 2015.

In 2023, all new residential central air-conditioning and air-source heat pump systems sold in the United States were required to meet new minimum energy efficiency standards. The new standards required a SEER of no less than 14 SEER for residential systems in the northern part of the United States and 15 SEER in the southern part of the United States, where cooling loads are a larger share of home energy use. In addition, the standards required an increase in the heating efficiency of air-source heat pumps, measured by the equipment's heating seasonal performance factor (HSPF). The minimum HSPF is 8.8 HSPF, compared with the 8.2 HSPF required by the previous standard that went into effect in 2015.

One of the most recent changes is the phase-out of R-404A and R410-A refrigerants, as well as the adoption of new refrigerant standards in 2023. Given the mild flammability of the new refrigerants, the industry now faces new challenges in properly utilizing them. By 2024, HFCs will be phased out by 40%, and by 2025, A2L refrigerants like R-32 or R-454B will replace HFCs as the industry norm. All mini-splits and VRF systems will have their global warming potential decreased because of this adjustment. Residential and light commercial air conditioners and heat pumps manufactured after January 1, 2025, must use the new refrigerant. The EPA has set a one-year sell-through period for equipment manufactured before this deadline. Products like ductless multi and mini splits, PTACs, and VTACs will have a three-year sell-through date (2027).

Room Air Conditioners

Room air conditioners were regulated by several states in the 1970s and 1980s, and became federally regulated in 1987. The standards vary as a function of cooling capacity and other features, but for the most common type of unit (an 8,000–13,999 BTU/hour unit with side vents), the 1987 law required an efficiency of 9.0 EER (energy efficiency ratio), effective in 1990. EER is cooling capacity (in BTU/hour) divided by power input (in watts). In 1997, the DOE published amended standards for room air conditioners, which became effective in October 2000. For the most common unit, the EER must be at least 9.8.

In July 2010, efficiency proponents and manufacturers agreed to jointly support new room AC standards that would reduce energy consumption by about 11% for the most common units. The DOE published a direct final rule in 2011 that reflects the standard levels in the consensus agreement. The standards are based on a new metric called the combined energy efficiency ratio (CEER), which incorporates standby energy consumption; the standard for the most common unit is 10.9 CEER. The DOE estimates that the new standards will save 0.31 quadrillion BTUs (quads) over 30 years and yield a net present value of $1.5 billion at a 3% discount rate. Updated DOE standards are due soon.

After a lawsuit, the DOE published the first new national appliance efficiency standards since 2017 on January 10, 2020. The new standards, finalized under the Obama administration in December 2016 but withheld from official publication by the Trump administration, are to cut energy waste for four product categories, including portable air conditioners. The effective date of this rule is March 10, 2020. Compliance with the standards established for all single-duct portable ACs and dual-duct portable ACs that are manufactured in, or imported into, the US is required on and after January 10, 2025:

https://www.regulations.gov/document?D=EERE-2013-BT-STD-0033-0053

The standards will reduce portable air conditioning energy consumption by more than 20%. In 2018 and 2019, California, Colorado, Vermont, and Washington adopted the final national standards as state standards. The CA standards took effect between 02/01/2020 and 02/01/2022.

PTACs

In October 2008, the DOE published a final rule establishing new energy efficiency standards for packaged terminal air conditioners (PTACs) and packaged terminal heat pumps (PTHPs). DOE agreed to follow ASHRAE 90.1's energy efficiency rules for non-standard-size PTACs and PTHPs, along with AHRI's definition of these products. They also released an addition to ASHRAE 90.1-2007 for standard and non-standard-size package terminal equipment. However, contrary to AHRI’s recommendation, the DOE adopted energy efficiency standards for standard-size equipment that are significantly more stringent than the levels established in ASHRAE 90.1. The new standards became effective in October 2010 for non-standard-size equipment and became effective in October 2012 for standard-size products.

Residential Wood Heaters

The US Environmental Protection Agency’s (EPA) New Source Performance Standards (NSPS) were entered into the Federal Register on March 16, 2015, after nearly a decade of development going back to the US EPA’s Phase I and Phase II Voluntary Programs adopted by several of the North-Eastern States starting in 2008. These new standards are the first-ever federal standards for hydronic heaters, wood-fired forced air furnaces, pellet stoves, and a previously unregulated type of wood stove called a single-burn rate stove.

For wood boilers or residential wood hydronic heaters, per the EPA's language, a 2-step approach was taken in the final ruling:

Step 1: Units manufactured after May 15, 2015 must meet particulate emissions limits of 0.32 lbs/MMBtu heat output, with a cap of 18 g/hr for each individual test run.

Step 2: Units manufactured after May 15, 2020, must meet particulate emissions limits of 0.10 lbs/MMBtu or 0.15 lbs/MMBtu when tested with cordwood.

To summarize, only EPA ‘Step-1’ certified units are now allowed to be sold in the US (since December 31, 2015). These standards apply to each wood boiler manufactured after May 15, 2015. The EPA stressed that the new rule does not affect current heaters already in use in homes today and does not replace state or local requirements governing wood heater use. Rather, the agency said the new rules will ensure that consumers buying wood heaters in the US in the future will be able to choose from cleaner-burning models.

Energy Efficient Programs and Organizations

There are several programs that promote higher energy efficiency in new housing and commercial buildings through voluntary guidelines. The major programs include:

The non-profit US GREEN BUILDING COUNCIL advocates for a voluntary, consensus-based national rating system known as Leadership in Energy and Environmental Design (LEED).

- BUILDING AMERICA is a collaborative effort between the US Department of Energy (DOE) and industry leaders to promote the ENERGY AND ENVIRONMENTAL BUILDING ASSOCIATION. A Builder's Guide for Home Builders

ENERGY STAR is a joint program by the DOE and the US Environmental Protection Agency, helping save money and protecting the environment through energy-efficient products and practices.

Working with these organizations is ASHRAE. As a technical organization, ASHRAE prescribes methods that detail how to achieve certain levels of performance.

Federal Government Incentives

Tax credits for energy-efficient home improvements (including biomass stoves, HVAC appliances, and non-solar water heaters) were reinstated as part of the American Taxpayer Relief Act in January 2013 for purchases made in both 2012 and 2013, but these tax credits expired at the end of 2013 and no longer apply.

In 2015, the 30% tax credit for solar thermal systems was extended to the end of 2019. The credit decreases to 26% for 2020, then to 22% for 2021. The same percentage credits are available for both residential and non-residential solar thermal installations through 2021. The only difference is that the federal Business Energy Investment Tax Credit (ITC), which was passed in 2018, continues a 10% tax credit on solar thermal installations for all years after 2021 for non-residential installations.

The Inflation Reduction Act of 2022 provides federal tax credits and deductions to help Americans make homes and buildings more energy efficient. Homeowners can save up to $3,200 annually on taxes for energy-efficient upgrades through 2032. This includes saving up to $2,000 on the costs of upgrading to heat pump technology. Homeowners can also take advantage of a modified and extended Residential Clean Energy Credit that provides a 30% income tax credit through 2032 (20% in 2033 and 2043) for the installation of clean energy equipment such as rooftop solar, wind energy, geothermal heat pumps, and battery storage.

At the beginning of 2018, the 30% tax credit for residential geothermal systems was extended to the end of 2019. The tax credit for the commercial geothermal heat pumps was 10%. Like the solar thermal tax credit, the geothermal credit went down to 26% for systems placed in service after December 31, 2019, and before December 31, 2022 (extended from the initial deadline of January 1, 2021); then a decrease to 22% is to take place for systems placed in service in 2023. The Business Energy Investment Tax Credit (ITC) allows a 10% tax credit on geothermal installation costs for all years, back to 2008, for the following types of non-residential buildings: commercial, industrial, investor-owned utility, cooperative utility, and agricultural. The 10% commercial investment tax credit remains for commercial systems if construction is started before January 1, 2024.

State and Local Incentives

There are several other state- or local-level initiatives promoting energy efficiency. For more information, visit www.dsireusa.org.

A.2 Key Demographic Figures

US - DEMOGRAPHICS, HOUSEHOLDS AND CLIMATE

Segmentation	Category	2023
Households	Total ('000)	130 152.26
	Average Size (Persons h/h)	2.57
Demographics	Population ('000)	335 893.00
	Population (% Urban)	82.90
	Population (% Rural)	17.10

Source: BRG based on Country Statistics

A.3 Macroeconomic Trends and Forecasts

US - MACROECONOMIC KEY INDICATORS

Category	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028
Consumer Price Index (% change)	3.82	- 0.32	1.64	3.14	2.07	1.47	1.62	0.12	1.27	2.13	2.44	1.81	1.25	4.68	7.99	4.13	2.91	2.00	2.06	2.10	2.12
GDP (% change)	0.11	- 2.58	2.70	1.56	2.29	2.12	2.52	2.95	1.82	2.46	2.97	2.47	- 2.21	5.80	1.94	2.53	2.73	1.88	2.03	2.12	2.12
Government Balance (% of GDP)	- 6.61	- 13.18	- 10.99	- 9.72	- 8.11	- 4.56	- 4.03	- 3.53	- 4.36	- 4.79	- 5.34	- 5.81	- 13.95	- 11.09	- 4.07	- 8.79	- 6.54	- 7.05	- 6.56	- 6.25	- 6.41
Household Saving Rate (% of Disposable Income)	4.30	5.94	6.14	6.76	8.12	5.13	5.66	6.04	5.54	5.96	6.68	7.63	15.86	11.67	3.43	4.44	3.49	4.01	0.00	0.00	0.00
Private Consumption (% change)	0.15	- 1.31	1.91	1.70	1.37	1.74	2.83	3.37	2.45	2.64	2.74	2.00	- 2.52	8.41	2.53	2.19	1.50	1.61	0.00	0.00	0.00
Unemployment Rate (%)	5.80	9.28	9.61	8.93	8.08	7.36	6.16	5.28	4.88	4.36	3.89	3.68	8.09	5.35	3.63	3.63	3.99	4.22	4.26	4.21	4.16

Source: IMF, SCOTIABANK and BRG projection. N.B. Household Saving Rate and Private Consumption not available for all years.

A.4 US Historical Annual Construction Spending

Annual Value of Construction Put in Place 2012 - 2023

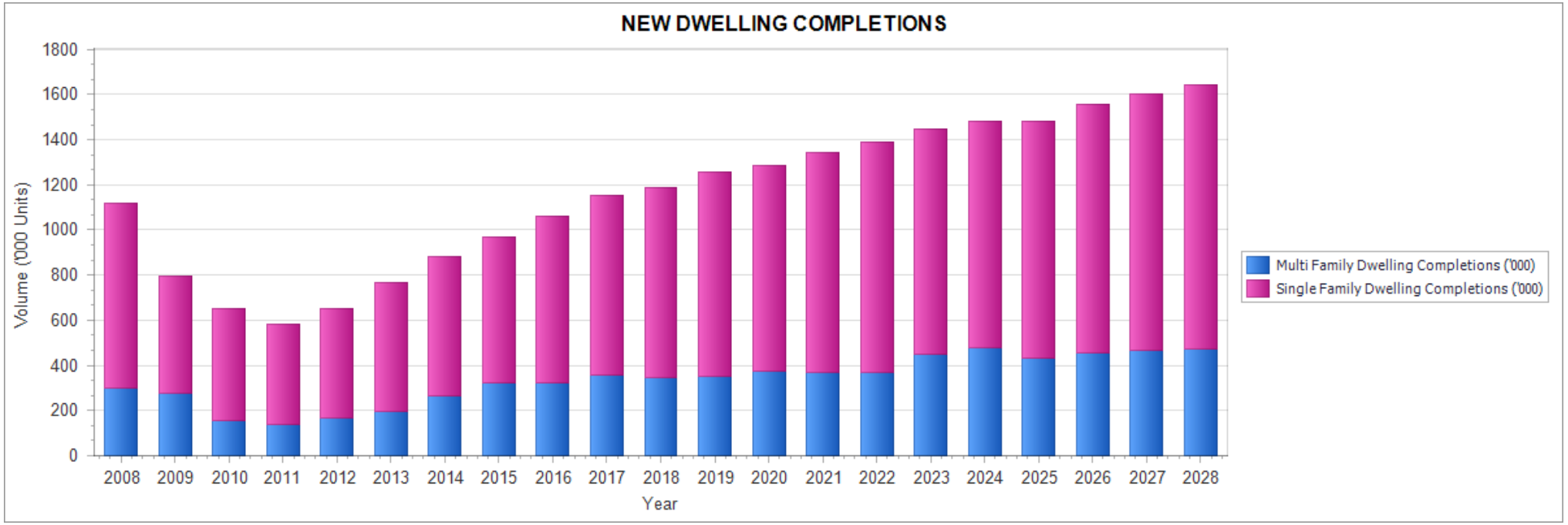
(Millions of dollars. Details may not add to totals due to rounding.)

Type of Construction:	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	% to Non-Residential
Total Construction	**854,401**	**914,582**	**1,015,292**	**1,132,116**	**1,213,144**	**1,279,841**	**1,333,183**	**1,391,039**	**1,499,570**	**1,653,375**	**1,848,676**	**1,979,667**	
% to last year:		7.0%	11.0%	11.5%	7.2%	5.5%	4.2%	4.3%	7.8%	10.3%	11.8%	19.7%	
Residential	**280,003**	**335,476**	**382,528**	**438,695**	**485,965**	**545,753**	**563,877**	**553,442**	**644,257**	**808,968**	**927,439**	**872,859**	
% to last year:		19.8%	14.0%	14.7%	10.8%	12.3%	3.3%	-1.9%	16.4%	25.6%	14.6%	7.9%	
Nonresidential	**574,399**	**579,106**	**632,764**	**693,421**	**727,179**	**734,088**	**769,306**	**837,597**	**855,313**	**844,408**	**921,237**	**1,106,807**	**100.0%**
% to last year:		0.8%	9.3%	9.6%	4.9%	1.0%	4.8%	8.9%	2.1%	-1.3%	9.1%	31.1%	
Amusement and recreation	15,480	15,207	16,773	20,527	23,652	26,569	28,068	30,416	28,288	27,102	30,001	32,569	**3.3%**
% to last year:		-1.8%	10.3%	22.4%	15.2%	12.3%	5.6%	8.4%	-7.0%	-4.2%	10.7%	20.2%	
Commercial	47,335	53,159	62,841	66,977	78,696	87,626	86,422	84,345	89,714	97,394	121,349	132,092	**13.2%**
% to last year:		12.3%	18.2%	6.6%	17.5%	11.3%	-1.4%	-2.4%	6.4%	8.6%	24.6%	35.6%	
Communication	16,165	17,783	17,298	21,691	22,179	23,696	24,502	22,184	23,876	23,091	24,306	24,883	**2.6%**
% to last year:		10.0%	-2.7%	25.4%	2.2%	6.8%	3.4%	-9.5%	7.6%	-3.3%	5.3%	7.8%	
Conservation and development	6,228	5,967	7,310	7,735	7,745	7,464	8,229	9,207	8,903	7,911	9,434	11,693	**1.0%**
% to last year:		-4.2%	22.5%	5.8%	0.1%	-3.6%	10.2%	11.9%	-3.3%	-11.1%	19.3%	47.8%	
Educational	84,672	79,060	79,681	85,346	91,629	96,685	101,210	108,952	110,692	100,988	102,086	116,377	**11.1%**
% to last year:		-6.6%	0.8%	7.1%	7.4%	5.5%	4.7%	7.6%	1.6%	-8.8%	1.1%	15.2%	
Health care	42,544	40,689	38,647	39,665	40,574	43,120	43,450	46,263	48,599	50,327	54,757	63,222	**5.9%**
% to last year:		-4.4%	-5.0%	2.6%	2.3%	6.3%	0.8%	6.5%	5.0%	3.6%	8.8%	25.6%	
Highway and street	80,546	81,364	84,743	91,701	93,187	89,620	91,745	99,402	102,321	103,381	114,107	134,693	**12.4%**
% to last year:		1.0%	4.2%	8.2%	1.6%	-3.8%	2.4%	8.3%	2.9%	1.0%	10.4%	30.3%	
Lodging	10,836	13,484	16,738	22,012	27,077	28,660	31,464	33,461	28,483	19,082	19,745	23,612	**2.1%**
% to last year:		24.4%	24.1%	31.5%	23.0%	5.8%	9.8%	6.3%	-14.9%	-33.0%	3.5%	23.7%	
Manufacturing	47,741	52,520	60,645	83,143	79,633	70,682	72,508	80,978	75,425	82,030	114,706	196,447	**12.5%**
% to last year:		10.0%	15.5%	37.1%	-4.2%	-11.2%	2.6%	11.7%	-6.9%	8.8%	39.8%	139.5%	
Office	37,800	37,979	46,582	56,013	67,907	68,685	76,662	88,724	92,831	89,902	91,623	98,888	**9.9%**
% to last year:		0.5%	22.7%	20.2%	21.2%	1.1%	11.6%	15.7%	4.6%	-3.2%	1.9%	10.0%	
Power	97,434	93,317	110,089	103,449	101,540	95,951	99,569	117,960	118,168	119,108	109,816	122,957	**11.9%**
% to last year:		-4.2%	18.0%	-6.0%	-1.8%	-5.5%	3.8%	18.5%	0.2%	0.8%	-7.8%	3.2%	
Public safety	10,431	9,506	9,437	8,505	8,177	8,539	9,353	12,012	17,667	12,826	11,574	13,323	**1.3%**
% to last year:		-8.9%	-0.7%	-9.9%	-3.9%	4.4%	9.5%	28.4%	47.1%	-27.4%	-9.8%	3.9%	
Religious	3,846	3,590	3,386	3,598	3,752	3,586	3,499	3,730	3,472	3,096	2,946	3,454	**0.3%**
% to last year:		-6.7%	-5.7%	6.3%	4.3%	-4.4%	-2.4%	6.6%	-6.9%	-10.8%	-4.8%	11.6%	
Sewage and waste disposal	22,261	22,425	23,173	24,599	24,151	22,901	23,931	26,119	27,189	28,811	32,660	41,382	**3.5%**
% to last year:		0.7%	3.3%	6.2%	-1.8%	-5.2%	4.5%	9.1%	4.1%	6.0%	13.4%	43.6%	
Transportation	37,862	39,459	42,043	45,106	43,339	46,137	53,219	57,448	60,734	59,075	58,725	63,776	**6.4%**
% to last year:		4.2%	6.5%	7.3%	-3.9%	6.5%	15.3%	7.9%	5.7%	-2.7%	-0.6%	8.0%	
Water supply	13,218	13,597	13,380	13,353	13,940	14,168	15,477	16,397	18,952	20,284	23,402	27,439	**2.5%**
% to last year:		2.9%	-1.6%	-0.2%	4.4%	1.6%	9.2%	5.9%	15.6%	7.0%	15.4%	35.3%	

SOURCE: U.S. Census Bureau Construction Spending

A.5 New Dwelling Construction

US - NEW DWELLING COMPLETIONS ('000 UNITS)



Category	Data	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028
Multi Family Dwelling Completions ('000)	Value	300.90	274.30	155.40	138.30	166.30	195.30	264.30	320.30	321.35	357.60	344.70	351.90	375.20	371.40	368.50	450.00	480.00	430.00	455.00	465.00	475.00
	% change	-	-8.84%	-43.35%	-11.00%	20.25%	17.44%	35.33%	21.19%	0.33%	11.28%	-3.61%	2.09%	6.62%	-1.01%	-0.78%	22.12%	6.67%	-10.42%	5.81%	2.20%	2.15%
Single Family Dwelling Completions ('000)	Value	818.80	520.10	496.30	446.60	483.00	569.10	619.50	647.90	738.35	795.30	840.20	903.20	911.70	969.70	1 022.00	998.90	1 000.00	1 050.00	1 100.00	1 140.00	1 165.00
	% change	-	-36.48%	-4.58%	-10.01%	8.15%	17.83%	8.86%	4.58%	13.96%	7.71%	5.65%	7.50%	0.94%	6.36%	5.39%	-2.26%	0.11%	5.00%	4.76%	3.64%	2.19%
Grand Total	Value	1 119.70	794.40	651.70	584.90	649.30	764.40	883.80	968.20	1 059.70	1 152.90	1 184.90	1 255.10	1 286.90	1 341.10	1 390.50	1 448.90	1 480.00	1 480.00	1 555.00	1 605.00	1 640.00
	% change Total	-	-29.05%	-17.96%	-10.25%	11.01%	17.73%	15.62%	9.55%	9.45%	8.79%	2.78%	5.92%	2.53%	4.21%	3.68%	4.20%	2.15%	0.00%	5.07%	3.22%	2.18%

Source: BRG based on Country Statistics.

A.5.1 US CUMULATIVE ESTIMATES OF HOUSING UNIT CHANGE FOR THE USA AND STATES, AND STATE RANKINGS 2020-2023

Cumulative Estimates of Housing Unit Change for the United States, States, District of Columbia, and State Rankings: April 1, 2020 to July 1, 2023

Geographic Area	April 1, 2020 Estimates Base	July 1, 2023 Estimate	Change, 2020 to 2023		State Ranking			
			Number	Percent	April 1, 2020 Estimates Base	July 1, 2023 Estimate	Change, 2020 to 2023	
							Number	Percent
United States	140,498,736	145,344,636	4,845,900	3.4	(X)	(X)	(X)	(X)
Alabama	2,288,335	2,360,135	71,800	3.1	24	24	19	21
Alaska	326,199	329,699	3,500	1.1	50	50	51	47
Arizona	3,081,997	3,239,581	157,584	5.1	14	14	8	11
Arkansas	1,365,265	1,409,134	43,869	3.2	31	31	28	20
California	14,392,140	14,763,237	371,097	2.6	1	1	3	29
Colorado	2,491,403	2,638,724	147,321	5.9	21	20	9	7
Connecticut	1,530,194	1,546,032	15,838	1.0	29	29	43	48
Delaware	448,736	471,458	22,722	5.1	45	45	39	12
District of Columbia	350,365	367,147	16,782	4.8	48	48	42	14
Florida	9,865,356	10,451,818	586,462	5.9	3	3	2	6
Georgia	4,410,953	4,610,567	199,614	4.5	10	10	5	15
Hawaii	561,065	572,063	10,998	2.0	42	42	45	39
Idaho	751,858	815,293	63,435	8.4	39	39	22	2
Illinois	5,426,429	5,471,027	44,598	0.8	6	6	27	50
Indiana	2,923,176	3,002,605	79,429	2.7	17	17	16	25
Iowa	1,412,789	1,449,340	36,551	2.6	30	30	33	28
Kansas	1,275,684	1,300,712	25,028	2.0	34	34	36	38
Kentucky	1,994,315	2,036,728	42,413	2.1	26	26	30	35
Louisiana	2,073,197	2,124,961	51,764	2.5	25	25	25	30
Maine	739,072	758,815	19,743	2.7	40	40	41	26
Maryland	2,530,846	2,572,412	41,566	1.6	20	22	32	42
Massachusetts	2,998,548	3,045,948	47,400	1.6	16	16	26	44
Michigan	4,570,170	4,643,473	73,303	1.6	9	9	18	43
Minnesota	2,485,558	2,575,411	89,853	3.6	22	21	15	18
Mississippi	1,319,942	1,350,642	30,700	2.3	32	32	34	32
Missouri	2,786,618	2,844,346	57,728	2.1	18	18	24	37
Montana	514,799	535,500	20,701	4.0	43	43	40	16
Nebraska	844,275	873,672	29,397	3.5	38	37	35	19
Nevada	1,281,017	1,348,211	67,194	5.2	33	33	20	10
New Hampshire	638,799	653,069	14,270	2.2	41	41	44	33
New Jersey	3,761,240	3,804,012	42,772	1.1	11	11	29	46
New Mexico	940,859	965,399	24,540	2.6	36	36	37	27
New York	8,488,064	8,631,525	143,461	1.7	4	4	11	41
North Carolina	4,708,715	4,979,204	270,489	5.7	8	8	4	8
North Dakota	370,642	380,841	10,199	2.8	47	47	46	23
Ohio	5,242,526	5,317,298	74,772	1.4	7	7	17	45
Oklahoma	1,746,807	1,788,972	42,165	2.4	28	28	31	31
Oregon	1,813,746	1,879,586	65,840	3.6	27	27	21	17
Pennsylvania	5,742,837	5,839,963	97,126	1.7	5	5	14	40
Rhode Island	483,468	487,132	3,664	0.8	44	44	50	51
South Carolina	2,344,965	2,490,344	145,379	6.2	23	23	10	4
South Dakota	393,371	417,220	23,849	6.1	46	46	38	5
Tennessee	3,031,606	3,195,304	163,698	5.4	15	15	6	9
Texas	11,589,315	12,395,364	806,049	7.0	2	2	1	3
Utah	1,151,414	1,258,623	107,209	9.3	35	35	12	1
Vermont	334,318	341,375	7,057	2.1	49	49	49	36
Virginia	3,618,242	3,717,677	99,435	2.7	12	12	13	24
Washington	3,202,240	3,361,615	159,375	5.0	13	13	7	13
West Virginia	855,647	863,745	8,098	0.9	37	38	48	49
Wisconsin	2,727,726	2,787,556	59,830	2.2	19	19	23	34
Wyoming	271,888	280,121	8,233	3.0	51	51	47	22

(X) Not applicable.

Dash (-) represents zero or rounds to zero.

Vintage 2023 data products are associated with Data Management System projects P-6000042, P-7501659, and P-7527355. The U.S. Census Bureau reviewed these data products for unauthorized disclosure of confidential information and approved the disclosure avoidance practices applied to this release (CBDRB-FY24-0085).

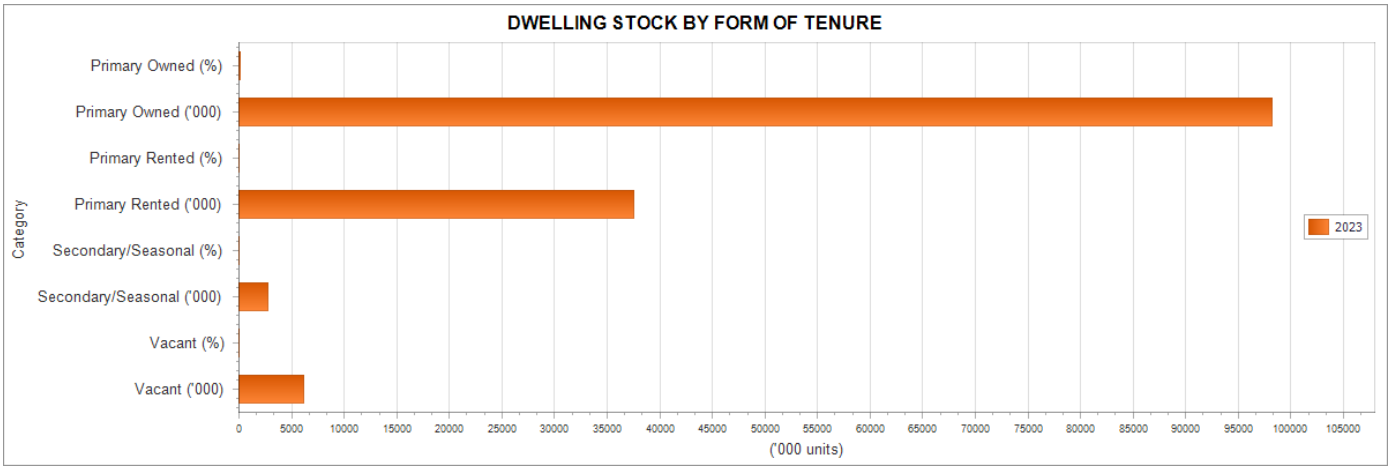
Cumulative Estimates of Housing Unit Change for the United States, States, District of Columbia, and State Rankings: April 1, 2020 to July 1, 2023 (NST-EST2023-HU-CUMCHG)

Source: U.S. Census Bureau, Population Division

Release Date: May 2024

A.5.2 Dwelling Stock by Type and Tenure

US - DWELLING STOCK BY TYPE AND TENURE ('000 UNITS)



Category	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Primary Owned (%)	57.27	56.76	56.43	56.23	55.85	55.46	55.33	55.77	56.49	56.77	59.58	62.38	65.14	67.80
Primary Owned ('000)	74 791.00	75 091.00	74 929.00	74 668.00	74 427.00	74 737.00	75 016.00	76 170.00	78 211.00	79 323.00	83 884.00	88 640.00	93 396.00	98 152.00
Primary Rented (%)	28.38	29.06	29.81	30.12	30.79	31.66	31.91	31.56	31.24	31.17	29.85	28.54	27.24	26.00
Primary Rented ('000)	37 069.00	38 443.00	39 583.00	40 004.00	41 034.00	42 658.00	43 259.00	43 102.00	43 247.00	43 558.00	42 036.00	40 549.00	39 062.00	37 575.00
Secondary/Seasonal (%)	3.40	3.42	3.36	3.34	3.32	3.25	3.20	3.04	2.88	2.76	2.54	2.33	2.12	1.90
Secondary/Seasonal ('000)	4 444.00	4 519.00	4 466.00	4 432.00	4 430.00	4 381.00	4 342.00	4 151.00	3 993.00	3 857.00	3 583.00	3 309.00	3 035.00	2 761.00
Vacant (%)	10.95	10.76	10.39	10.31	10.04	9.63	9.56	9.63	9.39	9.30	8.03	6.75	5.50	4.30
Vacant ('000)	14 294.00	14 239.00	13 800.00	13 695.00	13 379.00	12 983.00	12 961.00	13 147.00	12 997.00	12 997.00	11 301.00	9 592.00	7 883.00	6 174.00

Source: BRG based on Country Statistics.

A.6.1.a - DOMESTIC SPACE HEATING PARK, BY HOUSING UNIT TYPE 2020 (LATEST AVAILABLE DATA)

Number of Housing Units (million)

		Housing Unit Type				
		Single-Family Units		Apartments in Buildings With		
Water Heating	Total U.S.[1] (millions)	Detached	Attached	2 to 4 Units	5 or More Units	Mobile Homes
All homes	123.5	77.1	7.5	9.3	22.8	6.8
Space heating equipment						
Use space heating equipment	117.4	74.7	7.0	8.7	20.5	6.6
Have space heating equipment but do not use it	4.1	1.5	0.4	0.4	1.7	Q
Do not have space heating equipment	2.0	0.9	0.1	0.2	0.7	0.1
Main heating fuel and equipment						
Natural gas	56.3	39.7	4.1	4.0	7.0	1.4
Central warm-air furnace	47.4	35.9	3.4	2.5	4.3	1.3
Steam or hot water system	6.4	2.6	0.5	1.2	2.0	Q
Built-in room heater	2.3	1.1	0.2	0.4	0.7	Q
Some other equipment	0.2	0.1	Q	Q	Q	Q
Electricity	48.9	25.3	2.6	4.3	12.7	4.1
Central warm-air furnace	17.5	9.5	1.2	1.4	4.0	1.5
Heat pump	17.8	11.3	0.8	1.0	3.0	1.6
Ductless heat pump (mini-split)	1.1	0.6	0.1	Q	0.3	Q
Built-in electric units	8.3	2.1	0.4	1.2	4.3	0.3
Portable electric heaters	3.2	1.4	0.1	0.4	0.8	0.6
Some other equipment	1.0	0.4	Q	0.2	0.4	Q
Fuel oil/kerosene	5.0	3.7	0.2	0.4	0.5	0.2
Central warm-air furnace	3.1	2.5	0.1	0.2	0.1	0.2
Steam or hot water system	1.7	1.1	0.0	0.2	0.4	Q
Some other equipment	0.2	0.1	Q	Q	Q	Q
Propane	5.1	4.0	0.1	0.1	0.2	0.7
Central warm-air furnace	3.9	3.2	0.1	Q	0.1	0.5
Some other equipment	1.2	0.9	0.0	Q	0.1	0.1
Wood	2.2	1.9	Q	Q	Q	0.2
Heating stove	2.0	1.7	Q	Q	Q	0.2
Some other equipment	0.2	0.2	N	N	Q	Q
Some other fuel[3]	Q	Q	N	N	Q	Q
Do not use heating equipment	6.1	2.4	0.5	0.6	2.4	0.2
Main heating equipment (including all fuels)						
Central warm-air furnace	72.0	51.2	4.8	4.0	8.5	3.5
Heat pump	17.8	11.3	0.8	1.0	3.0	1.6
Steam or hot water system	9.3	4.3	0.6	1.6	2.8	0.1
Ductless heat pump (mini-split)	1.1	0.6	0.1	Q	0.3	Q
Built-in electric units	8.3	2.1	0.4	1.2	4.3	0.3
Built-in oil or gas room heater	3.0	1.5	0.2	0.4	0.8	0.2
Portable electric heaters	3.2	1.4	0.1	0.4	0.8	0.6
Heating stove burning wood	2.0	1.7	Q	Q	Q	0.2
Some other equipment	0.7	0.5	Q	Q	0.1	0.1
Do not use heating equipment	6.1	2.4	0.5	0.6	2.4	0.2

[1]Housing characteristics data were collected in late 2020 and early 2021

[2]Total U.S. includes all primary occupied housing units in the 50 states and the District of Columbia. Vacant housing units, seasonal units, second homes, military houses, and group quarters are excluded.

Q = Data withheld because either the Relative Standard Error (RSE) was greater than 50% or fewer than 10 households

N = No cases responded.

Notes: Because of rounding, data may not sum to totals. See RECS Terminology for definition of terms used in these tables.

Source: U.S. Energy Information Administration, Office of Energy Demand and Integrated Statistics, Form EIA-457A of the 2020 Residential Energy Consumption Survey.

A.6.1.b - DOMESTIC SPACE HEATING PARK, BY HOUSING UNIT TYPE 2020 (LATEST AVAILABLE DATA) - CONTINUED

Number of Housing Units (million)

		Housing Unit Type				
		Single-Family Units		Apartments in Buildings With		
Water Heating	Total U.S.[1] (millions)	Detached	Attached	2 to 4 Units	5 or More Units	Mobile Homes
All homes	123.5	77.1	7.5	9.3	22.8	6.8
Main heating equipment age						
Less than 2 years old	13.3	9.3	0.6	0.7	1.2	1.5
2 to 4 years old	17.2	11.8	1.0	1.1	2.1	1.2
5 to 9 years old	26.6	17.7	1.7	2.1	3.8	1.3
10 to 14 years old	22.3	14.1	1.7	2.0	3.5	1.0
15 to 19 years old	14.7	9.0	0.8	1.2	3.0	0.6
20 or more years old	23.4	12.7	1.2	1.7	6.9	1.0
Do not use heating equipment	6.1	2.4	0.5	0.6	2.4	0.2
Has thermostat						
Yes	109.4	72.5	6.8	7.3	17.2	5.6
Smart or internet-connected thermostat	12.8	10.6	0.7	0.3	0.9	0.3
Programmable thermostat	52.5	37.2	3.5	2.9	6.5	2.4
Non-programmable thermostat	44.1	24.7	2.6	4.1	9.8	2.9
No	12.7	4.0	0.5	1.9	5.1	1.2
Does not use heating or air-conditioning equipment	1.5	0.6	0.1	0.2	0.5	Q
Main heating equipment control						
Sets one temperature and leaves it there most of the time	50.5	32.7	3.0	3.6	7.9	3.3
Manually adjusts the temperature	30.9	18.9	2.0	2.6	5.8	1.7
Programmable or smart thermostat automatically adjusts the temperature	19.8	16.5	1.1	0.7	1.1	0.4
Turns equipment on or off as needed	12.1	5.7	0.7	1.2	3.6	0.9
Household does not have control over the equipment	3.9	0.8	0.1	0.7	2.0	0.3
Controls heating equipment in some other way	0.2	0.1	Q	Q	Q	Q
Does not use heating equipment	6.1	2.4	0.5	0.6	2.4	0.2
Secondary heating fuel and equipment						
Uses secondary heating equipment	51.3	39.0	2.7	2.1	4.7	2.8
Natural gas	7.6	6.6	0.4	Q	0.4	0.1
Fireplace	6.9	6.1	0.4	Q	0.3	0.1
Some other equipment	0.6	0.5	Q	Q	Q	Q
Electricity	31.5	21.7	2.0	1.9	3.9	2.0
Portable electric heaters	25.0	16.8	1.6	1.6	3.3	1.7
Ductless heat pump (mini-split)	0.9	0.8	0.1	Q	Q	Q
Built-in electric units	3.2	2.3	0.2	0.1	0.4	0.1
Fireplace	1.8	1.3	0.1	0.1	0.1	0.1
Some other equipment	0.6	0.5	Q	Q	Q	Q
Wood	8.6	7.5	0.2	0.1	0.4	0.4
Heating stove	3.1	2.9	0.0	Q	Q	0.1
Fireplace	5.5	4.6	0.2	0.1	0.3	0.3
Some other equipment	Q	Q	Q	N	N	N
Some other fuelb	3.6	3.2	0.1	Q	Q	0.3
Does not use secondary heating equipment	66.2	35.7	4.3	6.6	15.8	3.8
Does not use any heating equipment	6.1	2.4	0.5	0.6	2.4	0.2

[1]Housing characteristics data were collected in late 2020 and early 2021

[2]Total U.S. includes all primary occupied housing units in the 50 states and the District of Columbia. Vacant housing units, seasonal units, second homes, military houses, and group quarters are excluded.

Q = Data withheld because either the Relative Standard Error (RSE) was greater than 50% or fewer than 10 households

N = No cases responded.

Notes: Because of rounding, data may not sum to totals. See RECS Terminology for definition of terms used in these tables.

Source: U.S. Energy Information Administration, Office of Energy Demand and Integrated Statistics, Form EIA-457A of the 2020 Residential Energy Consumption Survey

A.6.2 - DOMESTIC WATER HEATING PARK, BY HOUSING UNIT TYPE 2020 (LATEST AVAILABLE DATA)

Number of Housing Units (million)

		Housing Unit Type				
		Single-Family Units		Apartments in Buildings With		
Water Heating	Total U.S.[1] (millions)	Detached	Attached	2 to 4 Units	5 or More Units	Mobile Homes
All homes	123.5	77.1	7.5	9.3	22.8	6.8
Fuel used by main water heater						
Natural gas	58.0	39.8	4.2	4.3	8.3	1.3
Electricity	58.2	31.7	3.1	4.6	13.7	5.2
Propane	4.3	3.4	0.1	0.1	0.3	0.4
Fuel oil or kerosene	2.6	1.9	0.1	0.2	0.5	Q
Some other fuel	0.4	0.3	0.0	Q	0.1	Q
Size of main water heater						
Small (30 gallons or less)	13.3	5.0	0.8	1.8	4.1	1.7
Medium (31 to 49 gallons)	55.4	34.8	3.9	4.7	8.2	3.7
Large (50 gallons or more)	46.8	31.4	2.4	2.6	9.2	1.3
Tankless	8.1	5.9	0.4	0.3	1.4	0.2
Age of main water heater						
Less than 2 years old	17.7	12.6	0.8	1.0	1.7	1.4
2 to 4 years old	22.5	15.1	1.6	1.8	2.9	1.2
5 to 9 years old	39.0	22.9	2.6	4.0	7.7	1.8
10 to 14 years old	25.1	14.9	1.4	1.7	5.8	1.3
15 to 19 years old	10.4	6.4	0.6	0.5	2.3	0.7
20 years or more	9.0	5.1	0.5	0.4	2.5	0.5
Fuel used by secondary water heater						
Natural gas	2.4	1.8	0.1	0.1	0.3	Q
Electricity	2.9	2.3	0.1	0.1	0.4	Q
Some other fuel	0.5	0.4	Q	Q	Q	N
Does not have a secondary water heater	117.8	72.6	7.2	9.1	22.1	6.8

[1]Housing characteristics data were collected in late 2020 and early 2021.

[2]Total U.S. includes all primary occupied housing units in the 50 states and the District of Columbia. Vacant housing units, seasonal units, second homes, military houses, and group quarters are excluded.

Q = Data withheld because either the Relative Standard Error (RSE) was greater than 50% or fewer than 10 households

N = No cases responded.

Notes: Because of rounding, data may not sum to totals. See RECS Terminology for definition of terms used in these tables.

Source: U.S. Energy Information Administration, Office of Energy Demand and Integrated Statistics, Form EIA-457A of the 2020

A.6.3 - AIR CONDITIONING PARK, BY HOUSING UNIT TYPE 2020 (LATEST AVAILABLE DATA)

Number of Housing Units (million)

		Housing Unit Type				
		Single-Family Units		Apartments in Buildings With		
Water Heating	Total U.S.[1] (millions)	Detached	Attached	2 to 4 Units	5 or More Units	Mobile Homes
All homes	123.5	77.1	7.5	9.3	22.8	6.8
Air-conditioning equipment						
Uses air-conditioning equipment	108.9	69.5	6.5	7.4	19.5	6.0
Does not use air-conditioning equipment	14.6	7.6	0.9	1.9	3.4	0.8
Main type of air-conditioning equipment used						
Central air-conditioning equipment (including central heat pump)	82.1	57.5	5.1	3.7	12.0	3.9
Ductless heat pump (mini-split)	1.8	1.2	0.1	Q	0.4	0.1
Window or wall air conditioner	21.4	8.9	1.1	3.4	6.2	1.8
Portable air conditioner	2.7	1.3	0.2	0.3	0.8	0.1
Evaporative or swamp cooler	1.0	0.7	Q	Q	0.1	0.1
Does not use air-conditioning equipment	14.6	7.6	0.9	1.9	3.4	0.8
Main air-conditioning equipment age						
Less than 2 years old	15.6	10.4	0.8	1.0	2.0	1.4
2 to 4 years old	21.8	13.8	1.3	1.9	3.3	1.5
5 to 9 years old	29.5	19.0	1.8	2.2	5.1	1.5
10 to 14 years old	20.5	12.9	1.4	1.6	3.7	0.8
15 to 19 years old	11.0	6.9	0.7	0.5	2.6	0.4
20 or more years old	10.6	6.5	0.6	0.3	2.8	0.5
Does not use air-conditioning equipment	14.6	7.6	0.9	1.9	3.4	0.8
Has thermostat						
Yes	109.4	72.5	6.8	7.3	17.2	5.6
Smart or internet-connected thermostat	12.8	10.6	0.7	0.3	0.9	0.3
Programmable thermostat	52.5	37.2	3.5	2.9	6.5	2.4
Non-programmable thermostat	44.1	24.7	2.6	4.1	9.8	2.9
No	12.7	4.0	0.5	1.9	5.1	1.2
Does not use heating or air-conditioning equipment	1.5	0.6	0.1	0.2	0.5	Q

[1]Housing characteristics data were collected in late 2020 and early 2021.

[2]Total U.S. includes all primary occupied housing units in the 50 states and the District of Columbia. Vacant housing units, seasonal units, second homes, military houses, and group quarters are excluded.

Q = Data withheld because either the Relative Standard Error (RSE) was greater than 50% or fewer than 10 households

N = No cases responded.

Notes: Because of rounding, data may not sum to totals. See RECS Terminology for definition of terms used in these tables.

U.S. Energy Information Administration, Office of Energy Demand and Integrated Statistics, Form EIA-457A of the 2020 Residential

A.7.1 Top 10 U.S. Builders

RANK 2024	RANK 2023	COMPANY	TOTAL 2023 REV ($ MIL.)	2023 CLOSINGS
1	1	D.R. HORTON; Arlington, TX	32,295	90,777
2	2	LENNAR CORP.; Miami, FL	32,459	73,087
3	3	PULTE GROUP; Atlanta, GA	15,598	28,603
4	4	NVR; Reston, VA	9,314	20,662
5	5	MERITAGE HOMES CORP.; Scottsdale, AZ	6,056	13,976
6	6	KB HOME; Los Angeles, CA	6,381	13,236
7	7	TAYLOR MORRISON; Scottsdale, AZ	7,200	11,495
8	8	CLAYTON PROPERTIES GROUP; Maryville, TN	3,993	9,957
9	10	TOLL BROTHERS; Fort Washington, PA	9,866	9,597
10	9	CENTURY COMMUNITIES; Greenwood Village, CO	3,604	9,568

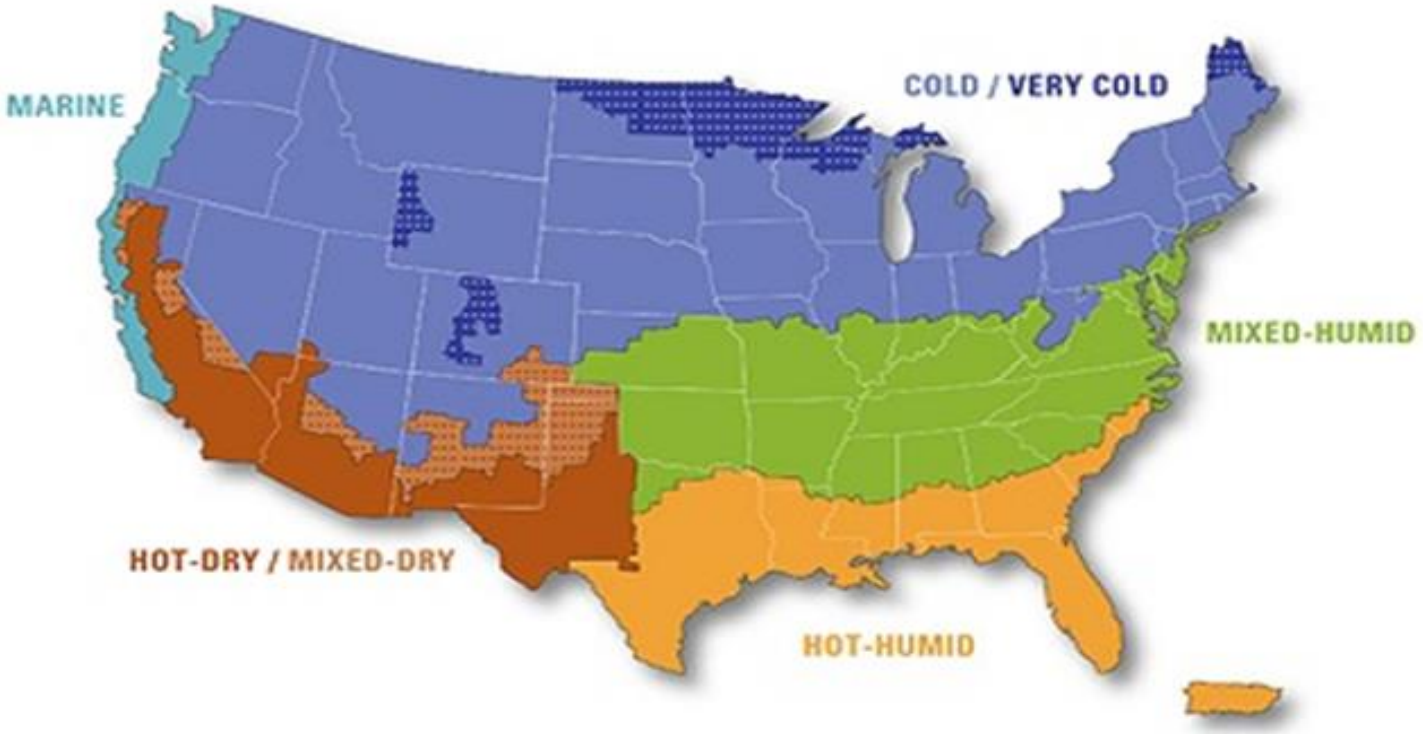
Source: Builder Online

A.7.2 Top 10 U.S. Contractors

RANK 2024	RANK 2023	FIRM
1	1	The Turner Corp., New York, N.Y.
2	4	Kiewit Corp., Omaha, Neb.
3	2	Bechtel, Reston, Va.
4	3	MasTec Inc., Coral Gables, Fla.
5	7	The Whiting-Turner Contracting Co., Baltimore, Md.
6	5	STO Building Group, New York, N.Y.
7	6	DPR Construction, Redwood City, Calif.
8	8	Fluor, Irving, Texas
9	--	McDermott International, Houston, Texas
10	13	PCL Construction Enterprises, Denver, Colo.

Source: Engineering News-Record

A.7.3 US Climate Map



A.8.1 Boiler Standards (as of September 1, 2012)

Prescribed Standards for Residential Boilers	**Current Efficiency (since 2012)**	**Jan. 15, 2021 Efficiency Requirements**	**Jan. 15, 2021 New Maxiumum Standby and Off-Mode Consumption Standards**
Boiler Type	AFUE (%)	AFUE (%)	Design Requirements
Gas Hot Water	82	84	9 W
Gas Steam	80	82	8 W
Oil Hot Water	84	86	11 W
Oil Steam	82	85	11 W
Electric Hot Water	None	None	8 W

Source: www.appliance-standards.org

A.8.2 Water Heater Standards (as of April 16, 2015)

Product Class	**Rated Storage Volume**	**NAECA Updated Energy Factor Requirements**
Gas-fired Storage Water Heater	≥ 20 gal and ≤ 55 gal	0.675 – (0.0015 x Vs)
Gas-fired Storage Water Heater	> 55 gal and ≤ 100 gal	0.8012 – (0.0015 x Vs)
Oil-fired Storage Water Heater	≤ 50 gal	0.68 – (0.0019 x Vs)
Electric Storage Water Heater	≥ 20 gal and ≤ 55 gal	0.960 – (0.0003 x Vs)
Electric Storage Water Heater	> 55 gal and ≤ 120 gal	2.057 – (0.00113 x Vs)
Gas-fired Instantaneous Water Heater	< 2 gal	0.82 – (0.0019 x Vs)
Electric Instantaneous Water Heater*	< 2 gal	0.93 – (0.00132 x Vs)

Vs: Rated Storage Volume - (in gallons)
*Rated Storage Volume
Source: DOE

A.8.3 Residential Central Air Conditioners and Heat Pumps Standards

Federal Standards for Units Manufactured on or after January 1, 2023

Product Class	Seasonal energy efficiency ratio (SEER2)	Heating seasonal performance factor (HSPF2)
(i) Split system air conditioners	13.4	
(ii) Split system heat pumps	14.3	7.5
(iii) Single package air conditioners	13.4	6.7
(iv) Single package heat pumps	13.4	6.7
(v) Small duct, high velocity systems	12	6.1
(vi)(A) Space constrained products- air conditioners	11.7	
(vi)(B) Space constrained products- heat pumps	11.9	6.3

**In addition to meeting the applicable Federal requirements in the table above, split-system air conditioners installed on or after January 1, 2023 in the following states shall have a SEER2 of not less than 14.3 for systems with a cooling capacity less than 45,000 BTU and 13.8 for systems with a cooling capacity greater than or equal to 45,000: Alabama, Arkansas, Delaware, Florida, Georgia, Hawaii, Kentucky, Louisiana, Maryland, Mississippi, North Carolina, Oklahoma, South Carolina, Tennessee, Texas, Virginia, or in the
Source: U.S. Department of Energy

Additional Energy Efficiency Ratio State Standards for Arizona, California, Nevada, and New Mexico
Manufactured on or after January 1, 2023

Product class	Energy efficiency ratio (EER2)
(i) Split-system rated cooling capacity less than 45,000 Btu/hr	11.7
(ii) Split-system rated cooling capacity equal to or greater than 45,000 Btu/hr	11.2
(iii) Single-package systems	10.6

Source: U.S. Department of Energy

Federal Standard for Off Mode Electrical Power Consumption
Manufactured on or after January 1, 2015

Product class	Average off mode power consumption PW,OFF (watts)
(i) Split-system air conditioners	30
(ii) Split-system heat pumps	33
(iii) Single-package air conditioners	30
(iv) Sing-package heat pumps	33
(v) Small-duct, high-velocity systems	30
(vi) Space-constrained air conditioners	30
(vii) Space-constrained heat pumps	33

Source: U.S. Department of Energy

A.8.4 Commercial Central Air Conditioners and Heat Pumps Standards

Table 1. Minimum Cooling Efficiency Levels for Commercial Package Air Conditioners and Heat Pumps (including Variable Refrigerant Flow Systems)

Equipment Type	Cooling Capacity	Sub-Category	Heating Type	Efficiency Level	Compliance Date
Small Commercial Packaged Air-Conditioning and Heating Equipment (Air-Cooled, 3 Phase)	<65,000 Btu/h	AC	All	SEER = 13	6/16/2008
		HP	All	SEER = 13	6/16/2008
Small Commercial Packaged Air-Conditioning and Heating Equipment (Air-Cooled)	≥ 65,000 Btu/h and <135,000 Btu/h	AC	No Heating or Electric Resistance Heating	EER = 11.2	1/1/2010
			All Other Types of Heating	EER = 11.0	1/1/2010
		HP	No Heating or Electric Resistance Heating	EER = 11.0	1/1/2010
			All Other Types of Heating	EER = 10.8	1/1/2010
Large Commercial Packaged Air-Conditioning and Heating Equipment (Air-Cooled)	≥135,000 Btu/h and <240,000 Btu/h	AC	No Heating or Electric Resistance Heating	EER = 11.0	1/1/2010
			All Other Types of Heating	EER = 10.8	1/1/2010
		HP	No Heating or Electric Resistance Heating	EER = 10.6	1/1/2010
			All Other Types of Heating	EER = 10.4	1/1/2010
Very Large Commercial Packaged Air-Conditioning and Heating Equipment (Air-Cooled)	≥ 240,000 Btu/h and <760,000 Btu/h	AC	No Heating or Electric Resistance Heating	EER = 10.0	1/1/2010
			All Other Types of Heating	EER = 9.8	1/1/2010
		HP	No Heating or Electric Resistance Heating	EER = 9.5	1/1/2010
			All Other Types of Heating	EER = 9.3	1/1/2010
Small Commercial Packaged Air-Conditioning and Heating Equipment (Water-Cooled, Evaporatively-Cooled, and Water-Source)	<17,000 Btu/h	AC	All	EER = 12.1	10/29/2003
		HP	All	EER = 11.2	10/29/2003
	≥17,000 and <65,000	AC	All	EER = 12.1	10/29/2003
		HP	All	EER = 12.0	10/29/2003
	≥65,000 Btu/h and <135,000 Btu/h	AC	No Heating or Electric Resistance Heating	EER = 11.5	10/29/2003
			All Other Types of Heating	EER = 11.3	10/29/2003
		HP	All	EER = 12.0	10/29/2003
Large Commercial Packaged Air-Conditioning and Heating Equipment (Water-Cooled, Evaporatively-Cooled, and Water-Source)	≥135,000 Btu/h and <240,000 Btu/h	AC	All	EER = 11.0	10/29/2004
		HP	All	EER = 11.0	10/29/2004
Very Large Commercial Packaged Air-Conditioning and Heating Equipment (Water-Cooled, Evaporatively-Cooled, and Water-Source)	≥240,000 Btu/h and <760,000 Btu/h	AC	No Heating or Electric Resistance Heating	EER = 11.0	1/10/2011
			All Other Types of Heating	EER = 10.8	1/10/2011
		HP	No Heating or Electric Resistance Heating	EER = 11.0	1/10/2011
			All Other Types of Heating	EER = 10.8	1/10/2011

Table 2. Minimum Heating Efficiency Levels for Commercial Package Heat Pumps (including Variable Refrigerant Flow Systems)

Equipment Type	Cooling Capacity	Efficiency Level	Compliance Date
Small Commercial Packaged Air-Conditioning and Heating Equipment (Air-Cooled, 3 Phase)	<65,000 Btu/h	HSPF = 7.7	6/16/2008
Small Commercial Packaged Air-Conditioning and Heating Equipment (Air-Cooled)	≥65,000 Btu/h and <135,000 Btu/h	COP = 3.3	1/1/2010
Large Commercial Packaged Air-Conditioning and Heating Equipment (Air-Cooled)	≥135,000 Btu/h and <240,000 Btu/h	COP = 3.2	1/1/2010
Very Large Commercial Packaged Air-Conditioning and Heating Equipment (Air-Cooled)	≥240,000 Btu/h and <760,000 Btu/h	COP = 3.2	1/1/2010
Small Commercial Packaged Air-Conditioning and Heating Equipment (Water-Source)	<135,000 Btu/h	COP = 4.2	10/29/2003

Table 3. Amended Energy Conservation Standards for Water-Cooled and Evaporatively-Cooled Commercial Packaged Air-Conditioning and Heating Equipment

Product	Cooling Capacity	Heating Type	Efficiency Level	Compliance Date
Small Commercial Package Air-Conditioning and Heating Equipment (Water-Cooled)	≥65,000 and <135,000 Btu/h	No Heating or Electric Resistance Heating	EER = 12.1	6/1/2013
		All Other Types of Heating	EER = 11.9	6/1/2013
Large Commercial Package Air-Conditioning and Heating Equipment (Water-Cooled)	=135,000 and <240,000 Btu/h	No Heating or Electric Resistance Heating	EER = 12.5	6/1/2014
		All Other Types of Heating	EER = 12.3	6/1/2014
Very Large Commercial Package Air-Conditioning and Heating Equipment (Water-Cooled)	≥240,000 and <760,000 Btu/h	No Heating or Electric Resistance Heating	EER = 12.4	6/1/2014
		All Other Types of Heating	EER = 12.2	6/1/2014
Small Commercial Package Air-Conditioning and Heating Equipment (Evaporatively-Cooled)	≥65,000 and <135,000 Btu/h	No Heating or Electric Resistance Heating	EER = 12.1	6/1/2013
		All Other Types of Heating	EER = 11.9	6/1/2013
Large Commercial Package Air-Conditioning and Heating Equipment (Evaporatively-Cooled)	≥135,000 and <240,000 Btu/h	No Heating or Electric Resistance Heating	EER = 12.0	6/1/2014
		All Other Types of Heating	EER = 11.8	6/1/2014
Very Large Commercial Package Air-Conditioning and Heating Equipment (Evaporatively-Cooled)	=240,000 and <760,000 Btu/h	No Heating or Electric Resistance Heating	EER = 11.9	6/1/2014
		All Other Types of Heating	EER = 11.7	6/1/2014

Table 4. Energy Conservation Standards for Variable Refrigerant Flow Multi-Split Air Conditioners and Heat Pumps*

Equipment Type	Cooling Capacity	Heating Type	Efficiency Level	Compliance Date
VRF Multi-Split Air Conditioners (Air-cooled)	<65,000 Btu/h	All	SEER = 13.0	6/16/2008
	≥65,000 Btu/h and <135,000 Btu/h	No Heating or Electric Resistance Heating	EER = 11.2	1/1/2010
		All Other Types of Heating	EER = 11.0	1/1/2010
	≥135,000 Btu/h and < 240,000 Btu/h	No Heating or Electric Resistance Heating	EER = 11.0	1/1/2010
		All Other Types of Heating	EER =10.8	1/1/2010
	≥240,000 Btu/h and < 760,000 Btu/h	No Heating or Electric Resistance Heating	EER = 10.0	1/1/2010
		All Other Types of Heating	EER = 9.8	1/1/2010
VRF Multi-Split Heat Pumps (Air-Cooled)	<65,000 Btu/h	All	SEER = 13.0	6/16/2008
			HSPF = 7.7	
	≥65,000 Btu/h and <135,000 Btu/h	No Heating or Electric Resistance Heating	EER = 11.0	1/1/2010
			COP = 3.3	
		All Other Types of Heating	EER = 10.8	1/1/2010
			COP = 3.3	
	≥135,000 Btu/h and <240,000 Btu/h	No Heating or Electric Resistance Heating	EER = 10.6	1/1/2010
			COP = 3.2	
		All Other Types of Heating	EER = 10.4	1/1/2010
			COP = 3.2	
	≥240,000 Btu/h and <760,000 Btu/h	No Heating or Electric Resistance Heating	EER = 9.5	1/1/2010
			COP = 3.2	
		All Other Types of Heating	EER = 9.3	1/1/2010
			COP = 3.2	
VRF Multi-Split Heat Pumps (Water-Source)	<17,000 Btu/h	Without Heat Recovery	EER = 12.0	10/29/2012 (EER Level)
			COP = 4.2	10/29/2003 (COP Level)
		With Heat Recovery	EER = 11.8	10/29/2012 (EER Level)
			COP = 4.2	10/29/2003 (COP Level)
	≥17,000 Btu/h and <65,000 Btu/h	All	EER =12.0	10/29/2013
			COP = 4.2	
	≥65,000 Btu/h and <135,000 Btu/h	All	EER =12.0	10/29/2013
			COP = 4.2	
	≥135,000 Btu/h and < 760,000 Btu/h	Without Heat Recovery	EER = 10.0	10/29/2013
			COP = 3.9	
		With Heat Recovery	EER = 9.8	10/29/2013
			COP = 3.9	

Source: DOE

A.9.1 US Status of State Energy Code Adoption Map Summary - Residential

State	State Current Code (As of 3/31/2024)
Alabama	2015 IECC with amendments
Alaska	None statewide
Arizona	Home rule
Arkansas	2009 IECC with amendments
California	2019 Bldg. Energy Efficiency Standards
Colorado	Home rule
Connecticut	2021 IECC with amendments
Delaware	2018 IECC
District of Columbia	2015 IECC with amendments
Florida	2021 IECC with amendments
Georgia	2015 IECC with amendments
Hawaii	Home rule
Idaho	2018 IECC with amendments
Illinois	2021 IECC with amendments
Indiana	2018 IECC with amendments
Iowa	2012 IECC with amendments
Kansas	Home rule
Kentucky	2009 IECC
Louisiana	2021 IECC with amendments
Maine	2015 IECC with amendments
Maryland	2021 IECC with amendments
Massachusetts	2018 IECC with amendments
Michigan	2015 IECC with amendments
Minnesota	2012 IECC with amendments
Mississippi	None statewide
Missouri	Home rule
Montana	2021 IECC with amendments
Nebraska	2018 IECC
Nevada	2018 IECC with amendments
New Hampshire	2018 IECC
New Jersey	2021 IECC
New Mexico	2021 IECC with amendments
New York	2018 IECC
North Carolina	2015 IECC with amendments
North Dakota	Home rule
Ohio	2018 IECC with amendments
Oklahoma	2009 IECC with amendments
Oregon	2021 IECC with amendments
Pennsylvania	2018 IECC with amendments
Rhode Island	2018 IECC with amendments
South Carolina	2009 IECC
South Dakota	Home rule
Tennessee	2018 IECC with amendments
Texas	2015 IECC
Utah	2015 IECC with amendments
Vermont	2018 IECC with amendments
Virginia	2021 IECC with amendments
Washington	Custom
West Virginia	2015 IECC with amendments
Wisconsin	2009 IECC with amendments
Wyoming	Home rule

Source: US Department of Energy (www.energycodes.gov)

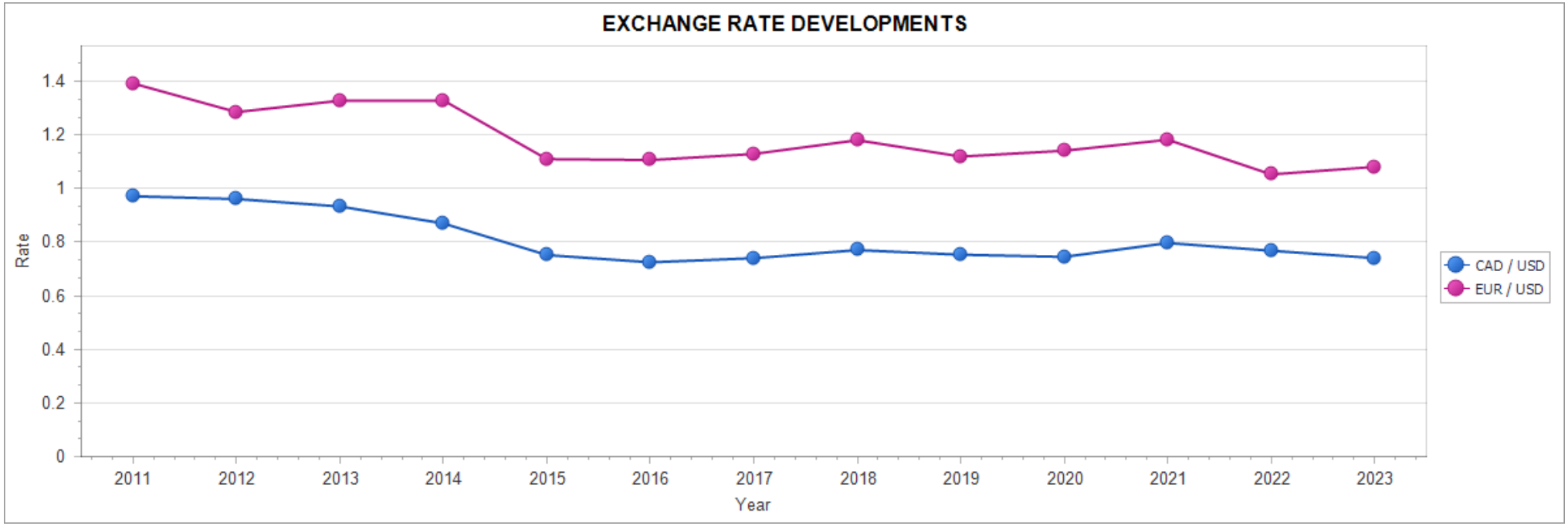
A.9.2 US Status of State Energy Code Adoption Map Summary - Commercial

State	State Current Code (As of 3/31/2024)
Alabama	90.1-2013
Alaska	None statewide
Arizona	Home rule
Arkansas	2009 IECC and 90.1-2007
California	2022 Building Energy Efficiency Standards
Colorado	Home rule
Connecticut	2021 IECC and 90.1-2019
Delaware	2018 IECC and 90.1-2016
District of Columbia	90.1-2013
Florida	2021 IECC and 90.1-2019
Georgia	2015 IECC and 90.1-2013
Hawaii	Home rule
Idaho	2018 IECC and 90.1-2016
Illinois	2021 IECC and 90.1-2019
Indiana	90.1-2007
Iowa	2012 IECC and 90.1-2010
Kansas	Home rule
Kentucky	2012 IECC and 90.1-2010
Louisiana	2021-IECC and 90.1-2019
Maine	2015 IECC and 90.1-2013
Maryland	2021 IECC and 90.1-2019
Massachusetts	2018 IECC and 90.1-2016
Michigan	2015 IECC and 90.1-2013
Minnesota	90.1-2019
Mississippi	None statewide
Missouri	Home rule
Montana	2021 IECC and 90.1-2019
Nebraska	2018 IECC and 90.1-2016
Nevada	2018 IECC and 90.1-2016
New Hampshire	2018 IECC and 90.1-2016
New Jersey	90.1-2019
New Mexico	2021 IECC and 90.1-2019
New York	2018 IECC and 90.1-2016
North Carolina	2015 IECC and 90.1-2013
North Dakota	Home rule
Ohio	2021 IECC and 90.1-2019
Oklahoma	2006 IECC and 90.1-2004
Oregon	90.1-2019
Pennsylvania	2018 IECC and 90.1-2016
Rhode Island	2018 IECC and 90.1-2016
South Carolina	2009 IECC and 90.1-2007
South Dakota	Home rule
Tennessee	2012 IECC and 90.1-2010
Texas	2015 IECC and 90.1-2013
Utah	2021-IECC and 90.1-2019
Vermont	2018 IECC and 90.1-2016
Virginia	2021 IECC and 90.1-2019
Washington	2018 Washington State Energy Code
West Virginia	90.1-2013
Wisconsin	2015 IECC and 90.1-2013
Wyoming	Home rule

Source: US Department of Energy (www.energycodes.gov)

A.10 Historical and Current Exchange Rates

US - EXCHANGE RATES : HISTORICAL AND CURRENT



Category	Data	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
CAD / USD	Value	0.97	0.96	0.93	0.87	0.75	0.73	0.74	0.77	0.75	0.75	0.80	0.77	0.74
	% change	-	-1.06%	-2.89%	-6.79%	-13.55%	-3.59%	2.11%	4.09%	-2.26%	-1.05%	6.93%	-3.61%	-3.59%
EUR / USD	Value	1.39	1.29	1.33	1.33	1.11	1.11	1.13	1.18	1.12	1.14	1.18	1.05	1.08
	% change	-	-7.67%	3.31%	0.06%	-16.46%	-0.32%	2.07%	4.54%	-5.17%	1.95%	3.57%	-10.94%	2.68%

Source: OANDA. The US dollar (USD) is the currency of USA. The exchange rates shown are the EUR to the USD.

USA

DISTRIBUTION

THE NORTH AMERICAN HEATING & COOLING PRODUCT MARKETS

BRG BUILDING SOLUTIONS

B.1 Distribution : Structural Overview

Overview

In the US, the distribution of heating and cooling equipment is characterized by the number of channels, fragmented ownership structure, and dependence on the cyclical and seasonal construction industry.

Distribution generally concentrates on supplying either service-oriented professional contractors (wholesale) or price-oriented retail consumers (home center retail). However, the most prominent feature of the North American distribution industry may be how quickly it is evolving. Technology is influencing every aspect of a successful distributor's business, and growth in e-commerce is changing the playing field.

Wholesale HVAC distributors are the traditional and largest distribution channel for HVAC products in the United States. Because of the size and complexity of most HVAC products, manufacturers are generally loyal to the two-step wholesale distribution model. They rely on the local wholesalers to provide additional sales support, solve warranty issues, provide installer training, stock repair parts, do repairs, and answer questions.

Distribution channels focused on the professional contractor are characterized by a few large, national wholesalers and a vast number of privately owned, small local or regional distribution companies, with the contractor typically using the same supplier for all project materials. Customer service and product availability are very important. Typically, professional contractors work within 20 miles of their local wholesaler and may visit it several times per week or even multiple times a day. Therefore, professional customers tend to be quite loyal to their distributors, forming personal relationships with the sales personnel working at the order desk. This channel is highly fragmented, with no markets dominated by a single distributor, but there has been merger and acquisition activity.

In recent years, consolidation has occurred throughout the HVAC distribution industry. Companies like FERGUSON and WATSCO are expanding their territories by buying smaller local and regional wholesalers that already have an established and loyal clientele. In addition, e-commerce routes to market, such as AMAZON BUSINESS, are causing disruptions as well.

Manufacturers of central air conditioning and forced air heating equipment typically sell their products under multiple brand names and, to some extent, limit the number and territory of wholesalers that may distribute their brands, but exclusivity is rare. Some manufacturers distribute their products via factory-owned branches or exclusive sales representatives. Most residential boiler manufacturers sell to wholesalers. However, it is very rare for a manufacturer to enter into national agreements; partnerships tend to be regional or local.

In residential and commercial remodeling situations, the end user will often research replacement products or request the recommendation of a repair technician before ordering through a nearby wholesaler. For residential sales, larger HVAC manufacturers, such as CARRIER and TRANE, typically sell directly to installation dealers. These dealerships may be factory-owned and company-branded, or they may sell and install products under an exclusivity agreement.

A three-step process is the most common distribution method for large, often custom-made, HVAC equipment in a ‘plan and spec’ bid situation. A three-step distribution model involves a commission-based sales representative and a traditional wholesale distributor. Like a buy-sell rep, commission-based sales reps target specific geographies, types of jobs, contractors, builders, and/or architects to promote their product lines; these lines of products are non-competitive but complimentary brands. However, the actual product sale will typically still happen through a commercial wholesale distributor after the price and delivery terms have been agreed upon through a request for proposal process.

A master distributor, or redistributor, is another type of business-to-business distribution relationship that qualifies as a three-step model. A master distributor purchases products from a manufacturer in order to supply smaller HVAC wholesalers or installers. Master distributors have found a niche handling small or custom orders, quick delivery times, and other value-added services. In the future, the trend seems to point to fewer steps in the distribution model, which can reduce prices for the end user and increase profits for the manufacturer.

With an aging contractor workforce and increasing adoption of technology, there is an opportunity for new methods to evolve that deliver HVAC products to the end user. New and younger contractors may not adhere to traditional distribution methods. In addition, with fewer distributors to choose from due to consolidation, buyers may look for alternative purchasing options.

Eliminating intermediaries and leveraging the Internet could potentially transform the distribution of HVAC products. It is possible now to buy a furnace (for example) online, but in general, it seems that manufacturers frown on this method of purchase and may refuse to honor the product’s warranty, stating that units purchased this way have higher return rates, possibly because they were improperly installed. In addition, manufacturers depend on their HVAC contractors to recommend their brands to buyers. Thus, they tend to favor policies that keep their contractors happy. In the future, it may be possible to find a win-win solution to this dilemma, at least for some segments of the market.

Home product retailers target the mass consumer market, which consists primarily of homeowners. This industry is highly competitive, with competition based primarily on price, store location, appearance, product quality, customer service, availability, and merchandise assortment. A customer-friendly, easily navigable website with multiple pick-up and quick delivery options has also become an extremely important element for success in the retail channel. Home center retailers, which have historically concentrated their sales efforts on retail

consumers and remodelers, are increasingly selling directly to professional contractors and installers. This has an adverse effect on some manufacturers' profit margins on products that contractors and installers would otherwise buy through our dealers and wholesalers.

The Internet has become a factor in only a few product categories. Room air conditioners, tankless water heaters, and water heating heat pumps are among the products readily available from both large home centers and some smaller manufacturers' websites. However, e-commerce is the fastest-growing sales channel in today's HVAC marketplace. Even products that would not sell well online are reportedly starting to sell through this channel. With that said, there is also a concerted effort being made by other manufacturers to prevent sales of their HVAC products online. Often, warranties will be invalidated if equipment is purchased through any type of non-approved dealer.

Wholesale/Distributor Trade

As stated earlier, wholesale HVAC distributors are the largest sales channel for HVAC products in the United States. In addition to supplying residential and commercial equipment, parts, and supplies, today's successful distributors also provide their customers with engineering support, plans and quotes, project management help, technical services, training, and networking events. Distributors' customers aren't always the homeowner, builder, contractor, or installer; many companies also work with independent commission-based sales reps and agencies in bid and spec processes.

According to the 2024 edition of the SUPPLY HOUSE TIMES Top 150 Market Leaders, the top five wholesale leaders in the HVACR supply segment are:

1. WATSCO
2. FERGUSON
3. WINSUPPLY
4. R.E. MICHEL CO.
5. US AIR CONDITIONING DISTRIBUTORS

WATSCO, INC. describes itself as the largest distribution network for HVACR products in the estimated $60 billion North American distribution market. In FY2023, the company generated $7.3 billion in revenue. It employs more than 40,000 people and has 692 locations in the US, Canada, and Mexico, but also exports to Latin America and the Caribbean. WATSCO dominates the Sun Belt, with the highest concentration of stores in Florida and Texas. The company has a 38.1% ownership interest in RUSSELL SIGLER, a notable HVAC distributor in North America with annual sales of approximately $1 billion serving customers from 34 locations throughout Arizona, California, Idaho, New Mexico, Nevada, and portions of Texas.

FERGUSON was founded in 1953 and is headquartered in Newport News, VA. It was formerly a wholly owned subsidiary of British FERGUSON PLC but recently underwent a demerger. Early in 2021, the UK operations of WOLSELEY were sold to the American private equity firm CLAYTON, DUBILIER &RICE.

FERGUSON's net sales were $29.7 billion in FY2023, a 4.1% increase over 2022, driven by price inflation of about 8% along with a 2.5% increase in sales via acquisitions, both partially offsetting lower sales volumes. Its US operations, which represented 95% of sales, were the primary driver of growth, rising 4.5% mainly due to non-residential markets and RMI sales within the residential segment. As of July 2023, the company operated 1,549 branches and 10 distribution centers in the US, employing 32,000 associates. More than 95% of the products sold in the United States are sourced from US-based suppliers.

In 2023, FERGUSON's sales among end-use markets in the US were 60% RMI versus 40% new construction and 52% residential versus 48% non-residential (68% commercial, 17% civil/infrastructure, 14% industrial). Nonresidential sales jumped 8.5% (commercial up 7.1% and largest in absolute value, civil/infrastructure 8.4%, and industrial 16.2%), while residential sales grew 1.1%. The company estimates it holds the leading North American market position in the residential building and remodeling market, with 13% of the market. FERGUSON calculates it is the number one commercial and mechanical provider in North America, with an estimated 22% market share. It also estimates that it holds a 17% market share in the North American residential trade plumbing segment, making it the second-largest supplier behind THE HOME DEPOT. As of April 2022, FERGUSON is the US's number one residential showroom brand, with an estimated 12% market share and 247 showrooms. E-commerce generates approximately 21% of all FERGUSON sales. The company owns the business-to-consumer website BUILD.COM, which uses the same distribution network and generates about 10% of US revenue. FERGUSON has been aggressively expanding its US business by buying smaller competitors. From fiscal 2021 to 2023, the company completed a total of 8, 17, and 7 acquisitions, respectively. All 2023 acquisitions were USA-based, and almost all of those were made in 2022.

WINSUPPLY is a privately held company based in Dayton, Ohio. It predominantly sells plumbing supplies in a shared ownership model; the parent company is WINSUPPLY, but each location may be owned up to 40% by the local “president” and team of employees. Winsupply has acquired several companies in the last few years; its most recent acquisition was HYDROLOGIC DISTRIBUTION COMPANY in October 2022. There are more than 660 locations across the country.

R.E. MICHEL Co. is an HVAC business with over 1,950 employees and 300 branches across the United States. It has been a family-owned operation since 1935, starting in Baltimore, MD. The company sells over 35,000 items from over 1,000 leading industry manufacturers. R.E. MICHEL’s main HVAC line is AIRTEMP, for which it is an exclusive marketer.

US AIR CONDITIONING DISTRIBUTORS, founded in 1990, has grown to 52 locations in California, Nevada, Utah, Idaho, and Arizona. In these states, the company is the exclusive YORK distributor. In 2019, it became the exclusive distributor of Samsung HVAC products in its territories.

SID HARVEY'S is an HVACR manufacturer and distributor operating a network of 76 branches from Maine to South Carolina on the East Coast, moving west throughout the Midwest to the Rocky Mountain region. The company manufactures SID HARVEY brand products in South Carolina and provides more than 50,000 SKUs from its Pennsylvania distribution center.

F.W. WEBB CO. is a family-owned business out of Bedford, MA, that has been in business for 150 years and has a strong reputation in the Northeast. The company, called FRANK WEBB HOME, operates 106 branches and 48 showrooms. About 55% of sales come from plumbing and hydronics. It belongs to the AFFILIATED DISTRIBUTORS buying group.

CREGGER COMPANY is a wholesale distributor founded in 1978 serving contractors in the plumbing, HVAC, and electrical industries through 39 locations in South Carolina, North Carolina, and Georgia.

W.W. GRAINGER, INC. was established in 1928. It is an Illinois-based distributor that offers a broad selection of maintenance, repair, and operating (MRO) supplies, as well as other related products and services. It conducts business via local branches, sales representatives, catalogs, and e-commerce, including ZORO.COM. GRAINGER sales reached $16.5 billion in 2023, an 8.2% increase. The company estimated its share of the 2022 US MRO market (latest available) at 6%, and globally at 4%. More than 75% of 2021 orders (the latest available) originated through digital channels. Roughly 20% of US sales are private label MRO items bearing GRAINGER's registered trademarks, including the GRAINGER APPROVED brand in plumbing.

REECE is a provider of HVAC, bathroom, kitchen, plumbing, and waterworks products. The Australian REECE GROUP acquired MORSCO INC. in July 2018 and recently combined the MORSCO brands with several other companies under the REECE name. REECE GROUP reported US revenue of $5.0 billion for FY2023, a 21% increase. In 2023, the organization added 27 new branches, totaling 231 across the US, primarily in the Sun Belt region. The company recently opened its first REECE BATH+KITCHEN USA showroom in Houston.

THE HOME DEPOT completed its acquisition of HD SUPPLY in late 2020. HD SUPPLY offered a full selection of products to the MRO market sector across multiple end uses, such as multi-family housing, hospitality, healthcare, and institutional facilities. The company operated more than 40 distribution centers across 25 states and over a dozen retail outlets in California.

HAJOCA (www.hajoca.com) claims to be the nation's largest privately held wholesale distributor of plumbing, heating, and industrial supplies. Founded in

1858, the company has been in business for over 160 years and owns Canada's EMCO CORPORATION. It operates 430 branches and employs 5,500.

Exclusivity arrangements are the norm for America's largest HVAC companies. As mentioned above, CARRIER jointly owns stores with WATSCO. LENNOX has over 200 LENNOX PARTSPLUS stores that provide contractors and independent dealers (who sell to homeowners) with a place to obtain universal service and replacement parts, LENNOX equipment, and OEM parts. TRANE utilizes locally owned and operated, independent HVAC wholesalers as exclusive TRANE dealerships that offer both residential and commercial equipment. YORK (JOHNSON CONTROL) utilizes CCE (Certified Comfort Expert—a YORK designation) contractors to install and service its products.

Buying Groups

In addition to the large wholesaler chains mentioned above, there are large buying groups that provide products to thousands of regional and local independent wholesale plumbing distributors. Wholesale buying groups are typically defined as a group of non-competing wholesalers pooling their purchasing power for select vendors and categories. The accumulated buying power far exceeds the purchasing influence of any one member and may consequently earn larger volume discounts, resulting in an overall lower net price for the members.

The largest wholesale buying groups are: affiliated distributors, which serve 3,800 branches in North America through 550 different owners; THE COMMONWEALTH GROUP, with almost 1,600 member-owned branches; EQUITY PLUMBING, with 400 branches; and OMNI, which buys products for 240 branches. The wholesalers supplied by a plumbing buying group are all independently branded and do not carry the purchasing group's banner name.

AFFILIATED DISTRIBUTORS, a $75 billion wholesale "marketing/buying group for construction and industrial supplies," boasts over 850 distributor members worldwide. AD Plumbing is North America's #1 buying and marketing group for top plumbing distributors and leading suppliers. AD HVAC is North America's premier buying and marketing group for mid-sized to large independent HVAC distributors.

The COMMONWEALTH GROUP was established in January 2020 when the EMBASSY GROUP LTD, with 800 member-owned branches, joined WIT, with 670 member-owned branches. These two long-standing, premier buying groups now boast combined revenue of more than $9 billion via 1,800 branches across the continental United States. This group is, in terms of revenue, the largest plumbing group in the country.

In 2019, OMNI CORPORATE SERVICES, a member-owned US plumbing and HVAC distributor, and EQUITY PLUMBING, the largest US plumbing group in terms of membership, merged to form IMARK PLUMBING, the plumbing vertical of the IMARK GROUP. The IMARK GROUP is a member-owned and member-governed organization of independent distributors that generates more than $28

billion in annual sales. The Luxury Products Group caters to the needs of independent decorative plumbing and lighting showrooms.

Also, in May of 2019, BLUE HAWK joined forces with the IMARK GROUP. Blue Hawk is an HVAC co-op distributor with 220 member owners and more than 1,700 locations in 49 states, the U.S. Virgin Islands, and Puerto Rico. Its members distribute a wide range of HVACR-related products to the residential and commercial construction trades. In combination, IMARK Electrical (850+ members), IMARK Plumbing (650+ members), and BLUE HAWK have created the largest member-owned and member-controlled B2B wholesale distributor network in the world, with cumulative revenue exceeding $25 billion.

Hardware Stores

In the United States, there are also several significant buying groups that include independent hardware stores as members. These transactions, in the BRG distribution flow, are recorded as sales to specialty retailers. Buying groups, also known as co-operatives or co-ops for short, primarily serve locally owned, independently operated hardware stores that cater to DIY customers. These stores are typically branded under the buying group's name and carry a wide variety of categories, with a limited selection in each category that a DIY customer can install, such as faucets and shower accessories. In the United States, the co-op's largest buying groups are ACE HARDWARE and DO IT BEST CORPORATION.

ACE HARDWARE is the largest retailer-owned hardware cooperative in the world, with consolidated revenues of $9.1 billion in fiscal year 2023, a 0.4% decrease from the prior year. Omni's website generated $1.3 billion, or just over 14% of revenue. The global count was 5,870 at the end of FY2023, of which 4,994 were in the US.

DO IT BEST, formerly known as HARDWARE WHOLESALERS, INC., is a member-owned hardware, lumber, and building materials cooperative based in Fort Wayne, Indiana. The company had gross sales of $4.8 billion in FY2023, down 13.8% from the $5.5 billion posted in 2022.

TRUE VALUE, an American hardware store cooperative, has 30 independently owned stores in Canada. In 2018, ACON INVESTMENTS acquired a 70% stake in TRUE VALUE, while existing dealers retained 30% ownership in the new TRUE VALUE COMPANY. TRUE VALUE abandoned its 70-year-old co-op business model in favor of becoming a wholesale supplier to its independent retailers. According to their website, they now serve 4,500 stores worldwide through 13 regional distribution centers. The company has sales of about $10 billion per year and employs 2,500 associates.

ORGILL is an independent hardline distributor based in Memphis, Tennessee, with annual sales of $3.6 billion in 2021. The company serves more than 12,000 retail hardware stores, home centers, pro lumber dealers, and farm stores in over 50 countries; 700+ TRUE VALUE retailers, 600+ ACE retailers, and 250+ DO IT BEST retailers use ORGILL as a secondary supplier. The company has seven distribution

centers and 400 field reps located throughout the country.

Retail Trade

The North American Hardware and Paint Association (NHPA) collects sales data in mostly homeowner-driven retail channels. The NHPA tracks sales through the following channels: hardware, home centers, lumber and building material outlets, and paint and decorating outlets. The Association estimates North American retail home improvement sales will hit $545 billion in 2023, down 3.8% from $566 billion in 2022. Home centers are believed to have accounted for 62.0% of the total, lumberyards 25.3%, and hardware stores 12.6%. NHPA estimates 17,850 hardware stores, 9,540 home centers, and 9,470 lumberyards were active during the year.

The North American retail industry is supported by many different types of outlets. BRG categorizes them as two channels. First, the DIY channel includes home improvement centers as well as independent and chain hardware stores. Second, specialist retailers, such as general retailers or mass merchants, are included.

DIY Retailers / Home Centers

The largest home centers in the United States are THE HOME DEPOT, LOWE'S, and MENARDS.

As of the end of 2023, THE HOME DEPOT is the largest home improvement retailer in the US, with 2,015 stores and 411,200 associates across the country and its territories. In its March 2023 earnings release, the company estimated it holds a 17% share of the ~$1 trillion addressable market. FY2023 sales hit $152.7 billion, a decline of 3.0% compared to the previous year. In its Q4 2023 earnings call, the company stated that nearly half of online orders were fulfilled through stores.

LOWE'S COMPANIES INC is the second-largest home improvement retailer in the US, operating 1,746 stores at the end of FY2023, employing 284,000 associates. The company generated net sales of $86.4 billion for the year, an 11% drop versus 2022. Pro customers represented about 25% of total sales. Online sales accounted for about 11% of total sales.

MENARDS is a privately held company headquartered in Eau Claire, WI. Because it is a Midwest home store, it has not enjoyed the growth levels of its two larger competitors. Customers perceive MENARDS as having lower prices than HOME DEPOT, but the latter has higher-quality goods.

These stores began as a place where the do-it-yourself homeowner could find anything they needed for home improvement and repair projects, stealing market share from the local hardware store because of their large selection across seemingly every category. However, these DIY Home Center stores have evolved into destinations for the professional customer, who has traditionally been very loyal to their local wholesaler, because of better pricing and product availability.

Online ordering capabilities through user-friendly apps, along with in-store pickup, also save professionals' time, which for many is even more important than saving money. THE HOME DEPOT said that pro, contractor, and builder accounted for about 45% of total sales as of July 2021. Pro customers represent about 25% of LOWE'S total sales in 2021, with growth to pro customers outpacing the company's overall sales growth.

Specialist / Other Retailers

Specialists include all types of reps and segment-focused retailers. Other retailers, such as mass merchants (WALMART) and membership warehouse distributors (COSTCO and SAM'S CLUB) will generally offer a selection of portable and window air conditioners for purchase on their websites and sometimes in their stores in the summer. Heating and cooling equipment sales in this channel are very low.

Two of the top three largest retailers in North America are general merchandise chains. WALMART STORES (including SAM'S CLUB) and COSTCO. WALMART did $642.6 billion in sales in FY2024, up 6.1% from the prior year; WALMART US revenue hit $441.8 billion, up 5.1%.

COSTCO Global revenue for FY2023 was $238 billion. The company had 604 locations in the US, with 206,000 full-time and part-time employees.

Internet Retailers ("E-Commerce")

The growing e-commerce channel is not without its perils for manufacturers; it eases entry for new brands and allows greater pricing transparency for consumers. Each of these factors puts downward pressure on already shrinking profit margins. It also becomes more difficult to manage long-standing relationships with existing brick-and-mortar distribution channels that are seeing their in-store traffic decline, yet their profit motives are as strong as ever. As manufacturers search for ways to increase margins, some are eyeing distributor profit pools and developing their own direct relationships with customers. The rise of new digital technologies makes it easier than ever for manufacturers to skip a level of profit-sharing and sell directly to end users.

AMAZON is the largest Internet marketplace in the world and is also the largest in Canada. For FY2023, net sales increased 12% to $574.8 billion USD, compared to $514.0 billion in 2021. North America accounted for 61% of net sales, growing 12% year over year. It currently has more than 110 active fulfillment centers in the US and 46 in Canada, which has enabled the company to deliver most products to Prime members in one day.

Supplyhouse.com, ACWHolesaler.com, AlpineHomeAirProducts.com, eComfort.com, and HVACDirect.com are a few other Internet-only stores that sell heating and cooling products.

In the Internet channel, BRG does not account for brick-and-mortar wholesalers or retailers such as HOMEDEPOT.COM and FERGUSON.COM's internet sales. The BRG distribution flow counts these sales in the wholesaler, retailer, or other respective channel since their primary means of selling is through brick-and-mortar locations.

B.2 US - PROFILES OF THE LEADING DISTRIBUTORS

Holding	Trading Name	Distributor Type	Turnover		Home Market Coverage		Specific Product Orientation				
			Year End	US$ million	# of Stores/ Branches	Geographical Coverage	Heat & Cool	K&B	Pipe & Fittings	Vent	AIP
FERGUSON PLC	FERGUSON, JONES STEPHENS, BUILD.COM	Wholesaler, E-Commerce	FY 2023	28,291	1,549	50 US States	XX	XXX	XXX	XX	X
WATSCO INC.	WATSCO, CARRIER ENTERPRISE, BAKER DISTRIBUTING, and others	Wholesaler	FY 2022	$7,274	673	42 US States, Canada, Mexico and Puerto Rico	XXX			XX	
WINSUPPLY	WINSUPPLY	Wholesaler	2021	$5,000	600	45 States	XX	XXX	XX		X
F.W. WEBB CO.	F.W. WEBB, FRANK WEBB HOME	Wholesaler	2020	$1,400	100+	9 States in the NE	XXX	XXX	XXX	XX	X
JOHNSTONE SUPPLY	JOHNSTONE SUPPLY LLC	Co-Op Wholesaler	2021	$3,000	450	47 States	XXX		X	XXX	
W.W. GRAINGER, INC.	GRAINGER, ZORO.COM	Wholesaler (MRO), E-Commerce	2023	$13,389	397	245 US Stores	XX	X	X	XX	
THE HOME DEPOT	HD SUPPLY	Wholesaler (MRO)	FY 2019	$6,047	270	40+ Distribution Centers across 25 US states and 2 Canadian Provinces	XX	XXX	X	XX	X
HAJOCA	HAJOCA CORPORATION	Wholesaler	2020	$5,390	400+	32 States	XX	XXX	XX	X	X
REECE LIMITED	MORSCO, MORRISON SUPPLY, FARNSWORTH WHOLESALE	Wholesaler, K&B Showrooms	FY 2023	$4,987	231	16 States in the SE and SW	XX	XXX	XX		X
RUSSELL SIGLER, INC.	SIGLER	Wholesaler	2019	$960	35	AZ, CA, ID, NM, NV, TX	XXX			XX	
R.E. MICHEL CO.	R.E. MICHEL CO.	Wholesaler	2020	$982	300+	USA	XXX		XX	XX	
KELLER SUPPLY	KELLER SUPPLY COMPANY	Wholesaler, K&B Showrooms	2020	$1,770	70	9 Western States	XX	XXX		XX	X
THE HOME DEPOT INC.	THE HOME DEPOT, INTERLINE BRANDS, BARNETT	DIY Retailer	FY 2023	$140,083	2,335	2015 in USA; 182 in CA; 138 in Mexico	X	XXX	XX	X	X
LOWE'S COMPANIES INC	LOWE'S	DIY Retailer	FY 2023	$86,377 (Worldwide)	1,746	50 US States	X	XXX	XX	X	X
MENARDS	MENARDS INC	DIY Retailer	2022	$13,000	336	15 Midwestern States	X	XXX	XX	X	X
AFFILIATED DISTRIBUTORS	AD	Buying Group - Wholesale	2021	$58,500	5,000+	National	X	X	X	X	X
THE COMMONWEALTH GROUP	fka The EMBASSY GROUP LTD and WIT & COMPANY LTD	Buying Group - Wholesale	2021	$8,500	800+	National	X	X	X	X	X
IMARK GROUP.	BLUE HAWK	Buying Group - Wholesale Co-Op	2020	$28,000	2,300	National	X	X	X	X	X
ORGILL, INC.	ORGILL	Master Distributor	FY 2022	$3,600	6,000+	US and Canada	X	X	X	X	X
ACE HARDWARE	ACE HARDWARE	Buying Group - Hardware Co-Op	2023	$9,131 (Worldwide)	5,870 Worldwide	4,538 stores in 50 US States	X	X	X	X	X
DO IT BEST	DO IT BEST CORP	Buying Group - Hardware Co-Op	FY 2023	$4,638 (Worldwide)	2900+	50 US States + 52 other countries	X	X	X	X	X
TRUE VALUE	TRUE VALUE	Buying Group - Hardware Co-Op	2018	$5,341	4,500	58 Countries	X	X	X	X	X
WALMART	WALMART, SAM'S CLUB	General Retailer, Warehouse Club	FY 2023	$441,817	5214	4,615 Walmart locations, 599 Sam's Club locations	X	X		X	X
COSTCO	COSTCO	Warehouse Club	FY 2023	$176,630	600	All 50 States & Puerto Rico	X	X		X	
AMAZON	AMAZON.COM	Internet Retailer	FY 2023	$352,828	NA	US, Canada and Mexico	X	X	X	X	X

Source: BRG based on secondary information

x = marginal activity
xx = substantial activity
xxx = core activity

USA

BOILERS

THE NORTH AMERICAN HEATING & COOLING PRODUCT MARKETS

BRG BUILDING SOLUTIONS

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Market Characteristics and Technological Trends

The hot water and steam boiler markets in the US are categorized into floor-standing and wall-hung products, condensing and non-condensing types, and using gas, oil, or electricity as a fuel source. A very small number of boilers on the market utilize wood chips, pellets, or another biofuel. In comparison to other heating options in the US, boilers effectively represent a niche residential home heating solution, with only 1% of new single-family dwellings and 6% of new multi-family dwellings built in 2022 using hot water or steam as the primary heat source. Multi-family units rarely install an individual boiler per living unit and will more likely connect to a collective system.

Between September 2012 and January 2021, the minimum boiler efficiency requirement in the US was 82% AFUE for gas hot water and 84% for oil hot water boilers (80% for gas steam and 82% for oil steam boilers). In 2016, the Department of Energy (DOE) finalized new standards for residential boilers that raised minimum efficiency levels from 82% to 84% AFUE for gas-fired boilers and 84% to 86% for oil-fired hot water boilers beginning January 15, 2021.

A second regulation was published by the DOE in January 2020, raising minimum efficiency levels and cutting energy waste for four product categories, including commercial boilers. Compliance with these amended standards has been ongoing since January 10, 2023. The documentation can be found here:

https://www.federalregister.gov/documents/2020/01/10/2019-26356/energy-conservation-program-energy-conservation-standards-for-commercial-packaged-boilers

Condensing gas boilers have seen a steady rise in popularity since appearing in the US market at the end of the 1990s, driven by growing awareness of better energy efficiency versus noncondensing types and the available incentive programs. The penetration of condensing oil boilers is very low, however, with sales of just a couple of hundred units a year and a very limited number of market players. This is reported to be at least partially due to the difference in oil quality between North America and Europe (higher sulfur content).

A significant proportion of boilers are sold in the Northeast, while sales to regions with warmer climates are limited. Condensing gas wall-hung boilers pair well with hydronic radiant heating (underfloor heating), which has a higher penetration in the West. In recent years, boilers have increasingly been installed with an air handler to create a ducted air system. This pairing has several benefits, including better humidity, which can eliminate the need for a humidifier. However, for this type of system, a tankless water heater can also be installed, creating competition.

In the US, there is a large emphasis on the material used to make a boiler, including the heat exchanger. Aluminum condensing boilers became popular

thanks to WEIL-MCLAIN and its ULTRA boilers, BOSCH/BUDERUS, TRIANGLE TUBE, and SPX's PATTERSON-KELLEY. However, stainless steel is the vastly dominant heat exchanger material for condensing boilers in North America today, and it is used by virtually all players.

In the non-condensing boiler market, manufacturers such as AO SMITH and its subsidiary LOCHINVAR, BRADFORD WHITE's LAARS, PALOMA/RHEEM's RAYPAK, MESTEK's RBI, BURNHAM HOLDINGS' THERMAL SOLUTIONS, and HARBOUR GROUP's CAMUS HYDRONICS promote commercial copper boilers.

Copper boilers are marketed as light, modern, and more efficient but will typically be replaced more frequently than alternatives. In contrast, marketing for cast iron boilers positions them as very reliable and require minimal maintenance. Heat exchangers for oil and dual fuel boilers are generally made of cast iron.

When discussing condensing boilers, the North American industry refers to boiler inputs, while for non-condensing boilers such as cast iron, steel water tube, and fire tube, they refer to either inputs or outputs. In terms of input, copper boilers are more commonly advertised.

The extreme price spike of oil in 2008 and the elevated prices of 2011–2014 contributed to a decrease in the number of residential oil boilers installed since that time. Since then, the image of oil as a home heating fuel has trended negatively, especially following incidents such as the 2010 BP oil spill in the Gulf of Mexico. However, like-for-like replacements continue to occur because some customers still hold the belief that oil is superior, or they simply prefer not to rely on gas companies, despite the ongoing growth in gas penetration. Oil prices had remained low since late 2014, even going negative briefly in 2020, which slowed the rate of decline in oil boiler sales. According to Trading Economics, in 2022, crude oil prices sharply increased in the first half of the year, hitting a five-year high of $116 per barrel at the end of May 2022. However, in the second half of the year, prices leveled out, ending the year at $79 per barrel.

The US market for electric boilers has historically been small, but it did rise in 2008 when oil prices spiked. An electric boiler is typically installed only if no other fuel can be used because the price of electricity paid by households in the US is about 2.5 times higher than natural gas. However, prices do vary greatly by region. For example, according to the EIA, the 2023 residential price of natural gas in Florida was roughly 70% higher than in New York. Further, the outdated US electric infrastructure lacks the robustness needed to support mass use of electric boilers because these units draw a lot of power. Even on a commercial scale, it is not feasible in many cases to run an electric boiler.

Some multi-family buildings are heated by steam boilers, although this represents a low percentage. Such installations include Boston row homes and some apartment blocks in New York City and Chicago. Steam boilers are sold primarily into the like-for-like replacement market in the Northeast, as it is very costly to replace a steam system with a hot water system.

Recent Market Trends

After most boiler types saw sales growth in 2022, the boiler market in the United States decreased heavily in 2023. The percentages of decline varied; nonetheless, most product sales saw a double-digit decrease compared to the previous year.

Demand started to slow down considerably in the second half of 2022, with distributors reporting unusually high levels of inventory following a somewhat artificially strong market in 2021, the year following the beginning of the COVID-19 pandemic. Even though the overall boiler market experienced positive development in 2022, supply chain issues, high prices due to general inflation, and a lower purchasing power among Americans were observed to have a greater influence on market development. In 2023, many distributors continued to report higher-than-normal inventory levels; hence, manufacturers shipped fewer boilers than the previous year.

The market for electric boilers remains niche; however, interest in electrification and regulations are expected to help the market grow further. In some areas of the country, such as New York, Oregon, Washington, and California, municipalities are making it more difficult to install gas products in new construction, which should help electric boiler sales in the long term because they are seen as more environmentally friendly.

Wall-hung boilers can be both heating-only and combi (space heating and domestic hot water heating) configured, with the latter option gaining healthy momentum within the last couple of years, following in Europe’s footsteps. The replacement market remains the main driver for both wall-hung and floor-standing condensing models.

Condensing boilers continue to be pushed by utility companies, and with $1,500 rebates available for installing a combi boiler in parts of the country, they can cost as little as $1,200 in some areas.

The non-condensing wall-hung boiler market has been contracting, with only a couple of players left supplying niche markets, mainly in the New York City area. However, because the market is so small, one larger project can significantly increase its size.

Condensing floor-standing boiler sales have increased in volume over time, with around half going to commercial and light commercial applications. There are more incentives for boilers in the commercial sector than in the residential sector.

The solid-fuel boiler market continues to struggle to find a niche for itself. Sales are limited to projects where there is an abundance of fuel available, like greenhouses, or for homes located in rural areas that are off the grid with no access to natural gas.

The pellet boiler market has continued to see a steady decrease over the past several years. This decrease is likely due, in part, to the demographic of consumers purchasing these boilers. The baby boomer generation is the most likely group to use a pellet boiler, but as this generation continues to age, it becomes less practical to carry the wood used for the boiler. This has led to the aging population preferring different heating technologies, like heat pumps, that do not involve physical activity.

It was reported that a 2020 EPA rule resulted in some solid fuel boiler models slipping off the list of certified boilers, thus no longer being eligible for sale in the US. The tougher emission regulations have seemingly put a lot of players out of business. A common frustration among manufacturers, however, stems from discrepancies with EU efficiency standards, which are said to be higher than the EPA's. Nonetheless, the cost of getting boilers certified can be prohibitive. Fortunately, a number of major insurance companies, such as STATE FARM and GEIKO, accept European boiler standards when insuring a home, provided that their efficiency standards exceed the EPA's. This has helped the market stay viable.

According to The Daily Journal of the United States Government, in 2022, the EPA finalized amendments to the national emission standards for hazardous air pollutants (NESHAP). This action applies to new and existing industrial, commercial, and institutional boilers. The action outlines amendments to numerous numeric emission limits for new and existing boilers, along with a compliance date of December 5, 2022, for the new emission limits.

With fuel prices reaching a high in 2022 and prevalent uncertainties due to the war in Ukraine, some US consumers preferred installing solid fuel boilers to circumvent the expensive cost of oil and natural gas. As a result, the overall solid fuel boiler market ended 2022 at a slightly higher level than the previous year. As we moved into 2023, fuel prices began to normalize, causing the initial surge in solid-fuel boiler installations to taper off. As a result, the solid fuel boiler market experienced a decline in growth, ending 2023 at a slightly lower level than the previous year.

Prices and Market Values

In 2021, after the COVID-19 pandemic, the average price of boilers spiked; in 2022, the average price of boilers leveled off, but the increases observed were still high compared to pre-pandemic levels. In 2023, however, prices continued to settle even further with the highest price increase being in the mid-single digits. Furthermore, some products saw a slight price drop for the first time since 2019.

Although price increases cooled off, boiler sales in 2023 did not reap the benefits. Boilers remain an expensive investment for most consumers, and high inflation and interest rates continued to affect sales in the market. Sales in the boiler market spiked after the pandemic in 2021, when distributors overpurchased for fear or ran out of products. Consumers were more willing to invest their discretionary income

into a boiler purchase. In turn, the influx of sales in 2021-2022 caused the boiler market to level out and return to more normal levels in 2023.

New Housing

The number of boilers installed in new homes or apartments is small in comparison to the US housing stock. It is a segment that could potentially grow as more manufacturers introduce wall-hung models to the market, which can be paired with an air handler for use in a ducted system (a popular system in new construction).

Condensing wall-hung boilers account for the majority of boiler sales in new single-family housing. In the multi-family segment, almost all boilers installed in individual apartments are wall-hung with a small number of electric boilers.

Historically, boilers installed in apartment blocks were part of a centralized system, with all apartments having radiators or baseboards. These boilers are usually floor-standing models, but in some instances, cascades of wall-hung models have been used. With the advent of smaller wall-hung models, each apartment can now have individual space heating and hot water, giving the occupant much more flexibility over their heating bills.

First-Time Installation

Most dwellings in the United States use forced air or hydronic central heating. Therefore, the first-time installation segment is residual and declines annually. There are a couple of reasons for this: the wide variety of climates in the US, with many regions having limited demand for heating; and the history of installer habits. American builders build houses to accommodate forced-air heating with ducts and crawl spaces because this is the method they are familiar with, and most Americans have grown up with and understand these systems.

Housing RMI

Boilers are predominantly sold for replacement purposes. Virtually all non-condensing boilers make their way into this segment. Some smaller solid-fuel boilers are used as an add-on to a different main heating system using another fuel source, and they tend to be installed in large houses surrounded by considerable land with readily available wood to use.

Condensing boiler sales into RMI is driven by a variety of regional and state utility rebate programs and other financial incentives. There is a large replacement market for older steam boilers used for collective heating in urban areas, like New York and Chicago. Most combi boilers are sold for renovation in the Northeast, replacing an oil or gas boiler with a tankless coil.

Oil boilers and steam boilers are most often sold as like-for-like replacements. For example, if a steam boiler needs replacing, it will typically be replaced by a steam boiler, because switching to a hot water system is very costly. In the gas floor-

standing market, there is a gradual shift from non-condensing to condensing-type boilers.

Electric boilers can be installed as a replacement for oil- or gas-fired boilers or in new commercial construction, such as hospitals and schools. However, in 2023, replacement demand for boilers in the US decreased significantly, falling by double digits. The significant decline in replacement installations can be partially attributed to the surge of boiler replacements that took place in 2021 and high inventories at the distributor level in 2022 and 2023.

Non-Housing

Floor-standing boilers are the most prevalent boiler type in the non-residential segment. Due to their large size, they are suitable for a wide range of applications, from small, light commercial to very large industrial. Most of these boilers are over 300,000 BTU in size.

There are some condensing wall-hung boilers with higher BTU ranges (representing 2% of total wall-hung boiler sales), but smaller sizes will also be used in light commercial applications, even though they are also suitable for residential applications.

Supply Structure

WEIL-MCLAIN remains the leader of the overall US boiler market, followed by NAVIEN, and BURNHAM. NAVIEN continues to sell an impressive number of condensing wall-hung boilers. WEIL-MCLAIN and BURNHAM have a strong non-condensing floor-standing boiler offering, but both companies continue to invest in high-efficiency technology. BURNHAM is also multi-branded, with US BOILER remaining the largest brand by volume.

In August 2021, Canadian building product manufacturer, GRANBY INDUSTRIES LP, a division of TERRAVEST INDUSTRIES INC., acquired ECR INTERNATIONAL from BDR THERMEA (which had owned the company since 2015). ECR's brand names remain (UTICA, DUNKIRK, PENNCO and ARGO), and the company now has the added benefit of GRANBY's North American manufacturing facilities. BAXI boilers continue to be distributed by ECR INTERNATIONAL, but quantities are negligible.

BOSCH THERMOTECHNOLOGY, now called BOSCH HOME COMFORT GROUP, sold its remaining inventory of commercial cast iron boilers in 2020 and concentrates on condensing only boilers within the commercial market.

VIESSMANN launched additional wall-hung condensing boilers in North America in June 2021, including the VITODENS 100-W, B1HE/B1KE and VITODENS 200-W, B2HE series. The company launched a new floor standing commercial condensing boiler series in 2022, the VITOCROSSAL CI2. In April 2023, it was announced that CARRIER GLOBAL CORP. would buy VIESSMANN in a 12 billion euros ($13.17

billion) cash-and-stock-deal, and by early January 2024 the acquisition was completed.

SPX, the parent company of WEIL-MCLAIN, acquired PATTERSON-KELLEY from HARSCO CORPORATION in November 2019.

RHEEM acquired INTERGAS and IBC TECHNOLOGIES in June 2019. RHEEM also purchased FRIEDRICH, an air conditioning manufacturer, in September 2021.

LAARS launched commercial electric boilers in 2019, sourced from ELECTRO INDUSTRIES. In January 2024, LAARS acquired ELECTRO INDUSTRIES, manufacturer of electric boilers, air-to-water heat pumps, electric duct heaters and controls.

MESTEK's RBI launched its TORUS water-tube boiler in 2020 and its ENCORE model in 2023 (replacing the INFINITE ENERGY II model). MESTEK's EMBASSY INDUSTRIES stopped sourcing from the Italian COSMOGAS and started offering a product made by PB HEAT which also got discontinued due to limited demand (mostly NY and northern NJ). Like BOSCH, MESTEK no longer sells residential cast iron non-condensing boilers under SMITH. SMITH was essentially out of business during 2023 other than for parts sales. The Boyertown foundry closed in September 2023 and SMITH now sources the castings for its 19-Series and 28-Series.

In 2019, the Japanese NORITZ acquired the US boiler manufacturer PB HEAT LLC, owner of the PEERLESS boiler brand.

ACV and TRIANGLE TUBE joined the French GROUPE ATLANTIC in 2018. TRIANGLE TUBE launched the UK-made combi and heating only INSTINCT boiler models in 2019, sourced from British sister company IDEAL HEATING. The IDEAL brand was launched in December 2019 and at first exclusively distributed in the US by F.W. WEBB, with new distributors in the West added more recently. TRIANGLE TUBE launched its INSTINCT floor-standing combi boiler at the end of 2021 and its MONARCH indirect tank in October of 2023. TRIANGLE TUBE also launched its new EXCALIBUR wall-hung and floor-standing models at the AHR Expo in January 2024, sourced from the Dutch company ECO HEATING SYSTEMS.

To grow its North American portfolio, ARISTON acquired NTI in 2016 followed by HTP (as well as Israeli electric water heater manufacturer ATMOR) in 2017, and CALOREX from the Mexican conglomerate GIS in 2019. CALOREX sells its water heaters under AMERICAN STANDARD in the USA, however the brand ARISTON has now also been launched into the North American market. The US-based business, located in Massachusetts, was renamed ARISTON USA.

CALORITECH was acquired by THERMON GROUP in 2017. Electric boilers are only a small part of the company's business. VAPOR POWER (which owns PRECISION BOILERS) was also acquired by THERMON GROUP in December 2023.

In 2021, IBC released its new IFIRE-B commercial condensing boiler sourced from sister company RAYPAK.

On the wholesale side, continued consolidation is evident. This is expected to continue to impact manufacturers, as it provides or takes away access to certain distribution channels.

In July 2022, DHT purchased SUSSMAN ELECTRIC BOILERS. SUSSMAN continues its current business operation in its New York manufacturing facility.

HAMILTON ENGINEERING released its new EVO XL boiler and water heater system at AHR Expo 2023. In early 2024, it was reported that HAMILTON's floor standing boiler model was recently launched and is starting production.

NEW HORIZON CORP is expected to soon be adding water-based heat pumps to its product portfolio.

In 2024, DDR AMERICAS is expected to launch its ABSOLUTE/SPIRE stainless boiler model.

ACE HEATERS is due to launch the AJAX E-SERIES boiler in 2024, the company stopped selling non-condensing boilers in 2021.

RINNAI is expected to launch a new line of boilers called the ISERIES Plus, which is 60,000 to 199,000 BTUs, and started selling its new RCB301 and 399 commercial boilers in early 2024.

ALTERNATIVE HEATING OF NORTH AMERICA discontinued selling VIADRUS solid fuel boilers in 2019.

VARMEBARONEN intends to exit the US logwood and pellet markets due to recent EPA standards. The company plans to continue to sell the existing inventory as well as spare parts.

In 2020, the Italian boiler company BIASI will stop supplying wall-hung boilers to the US and Canadian markets. It still sells cast iron boilers through QHT, a New Hampshire-based exclusive distributor.

Forecasts

Economic factors, including legislation, inflation, interest rates, and still-high inventory levels, are expected to continue to weigh heavily on the boiler market in 2024. The market's decline is closely linked to the economic pressures of inflation and interest rates, which have put a significant strain on consumer spending. It is anticipated that high interest rates will remain stagnant in the 5.25%–5.50% range for the first half of 2024, further impeding investment activity and growth.

Due to monetary tightening, inflation levels have moderately recovered, although they remain higher than the 2% target rate. Nevertheless, the high interest rate environment continues to be one of the headlines for 2024, making the outlook for the year less positive.

Moving into the second half of 2024, the forecast remains negative, although only by a marginal percentage, which may come as somewhat of a relief after the significant decline observed during the year prior. The non-condensing market will continue to dissolve as legislation and bans against the products continue to take effect, and in return, this is expected to bring growth to the wall-hung and floor-standing gas condensing markets in 2024. In addition, the electric boiler market is expected to continue its growth in 2024 and beyond, as electrification continues to be a hot topic across the US.

Indeed, electrification has recently become a trend-setting buzzword in the HVAC universe. Certain areas of the United States are experiencing a change in attitude towards electricity as a source of energy because more people are learning that electric-based appliances consume less energy and thus reduce carbon emissions compared with gas-based heating systems, even if the electricity is produced using fossil fuels. Several states have jumped on board with helpful incentives.

Most of the United States has been slow to adopt electrification compared to the rest of the world. California, however, is one of America's leading electric states. According to the Sierra Club, as of May 2023, 77 cities in California have embraced electrification by adopting building codes to reduce their reliance on gas. Most of these new codes require new residential and commercial buildings to be all-electric; many cities do allow exemptions; however, for certain types of buildings, affordable housing projects, laboratories, and commercial kitchens are popular exemptions to the all-electric rule.

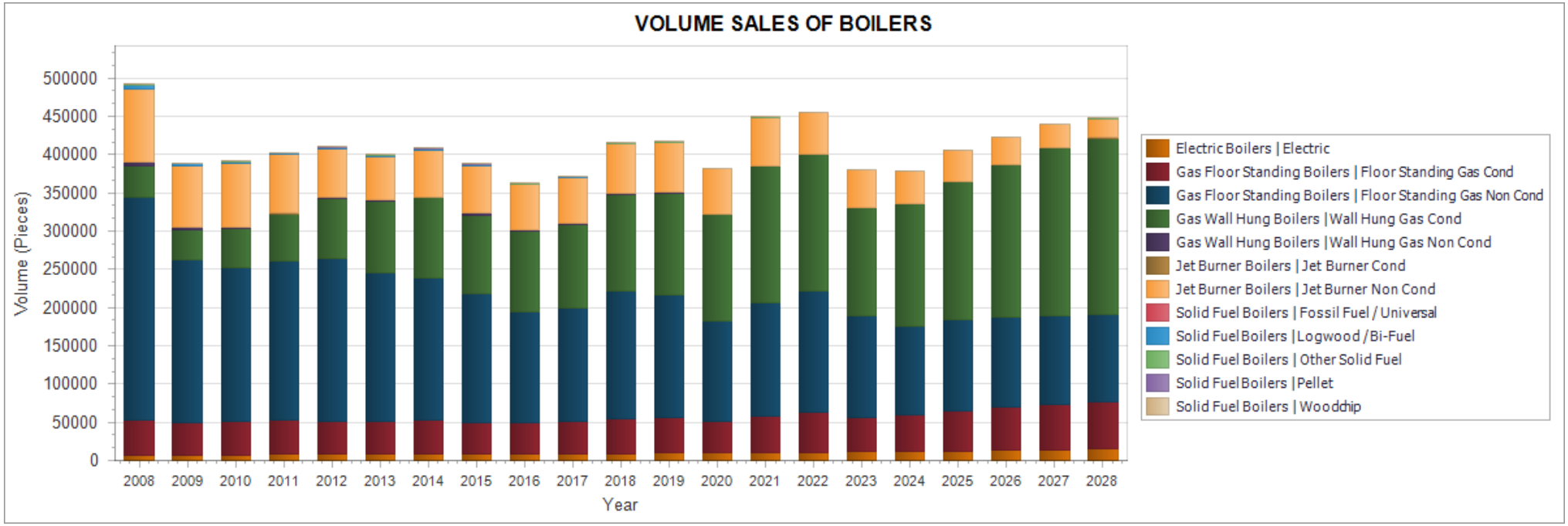
New York is yet another trailblazer in America's electrification. New York Governor Andrew Cuomo and the Public Service Commission approved an energy-efficiency program in December 2018 that requires utilities to install some 83,000 heat pump systems in five years. The New York State Public Service Commission approved an additional $2 billion for building electrification and energy efficiency programs in January 2020. In addition, it also reauthorized $1.3 billion for existing programs. New York's Clean Heat Program is run by homeowners' utilities, each of which has a unique incentive and is an "opt-in system," so customers can choose to become electric. In total, the state’s earmarked funds for electrification and efficiency are a whopping US $6.8 billion through 2025. In December 2021, the New York City Council banned gas and oil hookups on new heating, hot water, and kitchen stove construction starting in 2024. There are exceptions based on the size of the building and end use: commercial kitchens, manufacturing, labs, hospitals, laundromats, crematoria, waste treatment facilities operated by the Department of Environmental Conservation (DEC), and buildings used for electric power and steam generation regulated by the Public Service Commission (PSC). The deadline for building compliance is December 2027.

Because of the volatile nature of oil and gas prices, price spikes are a real possibility during our forecast period. The continued increase in the cost of these fuels could be positive for solid fuel boilers, as they would be comparatively cheaper energy solutions. If tax credits remain and energy prices continue to increase, demand for logwood boiler sales could return to growth. However, many market players have not been able to pass the necessary EPA certifications. In some cases, the time and money it takes to obtain the certifications do not outweigh the benefits of receiving them. It reportedly costs roughly $20,000 to test each boiler model for EPA certification; therefore, many players are moving away from the US solid fuel market. BRG anticipates that the solid fuel market will decline in 2024 and beyond.

After the expected decline in 2024, it is anticipated that the overall boiler market will experience a recovery with a healthy increase in the mid-single digits in 2025. After 2025, the market is expected to continue growing, but by a lesser percentage, year over year, through 2028.

1.2.1 Historical Trends and Forecasts : All Boilers

US - SUMMARY OF TRENDS IN THE BOILER MARKET : HISTORICAL AND FORECASTS

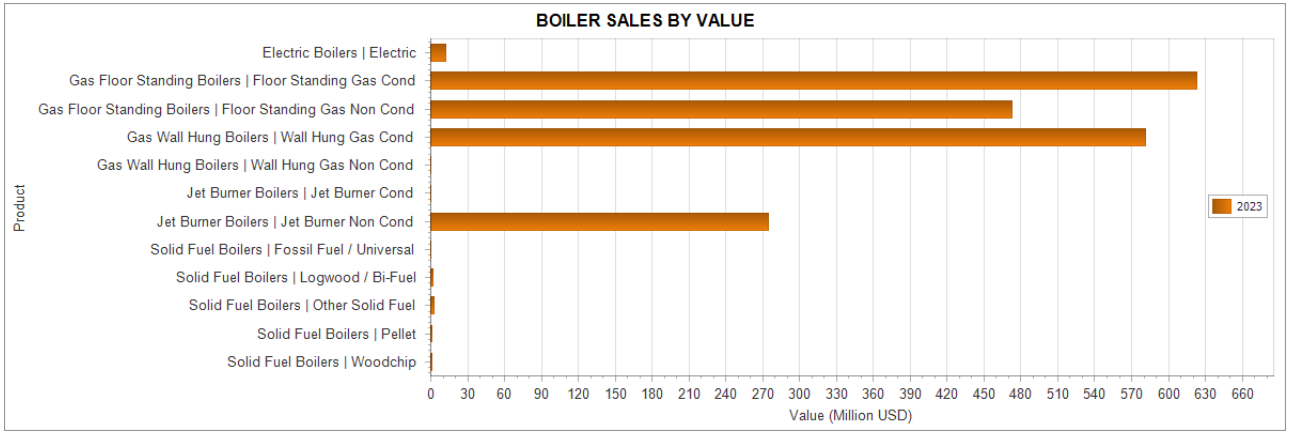


Product Category	Product	Data	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028
Electric Boilers	Electric		6 800	7 400	6 700	9 300	8 900	8 300	8 800	7 850	8 300	8 600	9 100	9 700	9 600	10 500	10 500	11 300	12 000	12 600	13 300	14 000	14 700
Gas Floor Standing Boilers	Floor Standing Gas Cond		45 800	42 200	45 000	43 300	42 900	43 100	43 300	42 000	41 500	42 200	46 000	47 300	41 500	48 300	53 300	44 200	48 000	52 000	56 000	59 000	62 000
	Floor Standing Gas Non Cond		292 000	213 000	200 000	208 200	211 700	194 100	187 000	168 900	145 300	149 300	166 000	160 000	131 000	148 000	158 000	134 500	115 000	120 000	118 000	116 000	115 000
Gas Wall Hung Boilers	Wall Hung Gas Cond		40 000	39 600	51 700	60 600	79 000	93 400	104 400	102 250	104 400	108 550	127 400	132 600	139 800	178 300	178 800	141 000	160 000	180 000	200 000	220 000	230 000
	Wall Hung Gas Non Cond		5 000	2 500	1 600	1 600	1 500	1 200	1 230	2 700	1 870	1 100	1 150	1 000	700	350	300	200	190	180	170	160	150
Jet Burner Boilers	Jet Burner Cond		140	450	260	260	260	125	100	125	120	110	120	150	130	110	100	80	80	80	80	80	80
	Jet Burner Non Cond		95 700	80 300	83 500	76 500	63 000	57 600	60 700	61 900	59 400	60 800	64 300	65 000	58 600	63 000	53 500	48 300	43 000	40 000	35 000	30 000	25 000
Solid Fuel Boilers	Fossil Fuel / Universal		250	170	150	165	125	100	100	75	55	50	40	40	45	45	75	60	50	40	30	25	20
	Logwood / Bi-Fuel		5 600	2 500	1 900	1 900	1 800	1 600	1 575	1 200	750	660	610	550	500	450	410	315	305	300	290	280	265
	Other Solid Fuel		800	700	650	700	650	700	750	700	550	575	550	525	450	400	420	350	340	330	320	310	300
	Pellet		570	290	280	420	380	420	450	350	260	250	220	200	140	135	130	110	100	95	90	85	80
	Wooddhip		100	65	50	50	50	35	35	30	35	30	30	20	10	10	10	10	10	10	10	10	10
Grand Total		Trends	492 760	389 175	391 790	402 995	410 265	400 680	408 440	388 080	362 540	372 225	415 520	417 085	382 475	449 600	455 545	380 425	379 075	405 635	423 290	439 950	447 605
		% change Total		-21.02%	0.67%	2.86%	1.80%	-2.34%	1.94%	-4.98%	-6.58%	2.67%	11.63%	0.38%	-8.30%	17.55%	1.32%	-16.49%	-0.35%	7.01%	4.35%	3.94%	1.74%

Source: BRG based on industry interviews

1.3.1 Prices and Market Values : Boilers

US - ESTIMATED VALUES OF THE BOILER MARKET

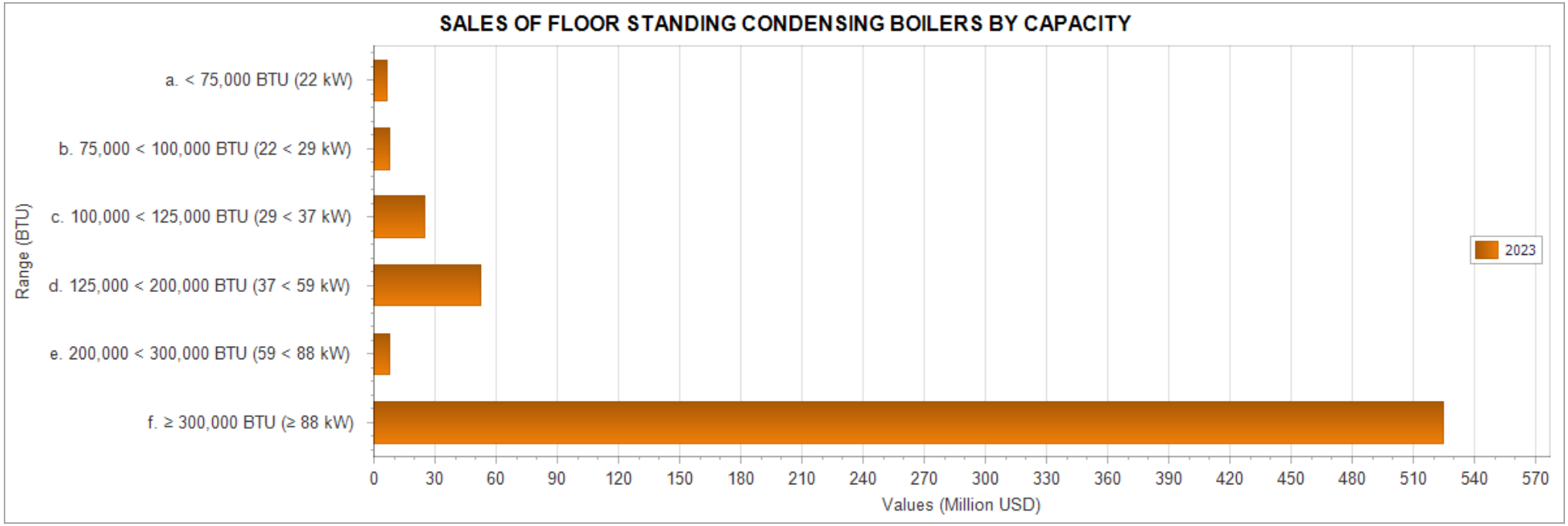


		2023			2022		
Product Category	Product	Volumes	MSP (Local)	Value (MSP Local)	Volumes	MSP (Local)	Value (MSP Local)
Electric Boilers	Electric	11 300	1,079.00	12.19	10 500	1,030.00	10.82
Gas Floor Standing Boilers	Floor Standing Gas Cond	44 200	14,099.93	623.22	53 300	13,446.68	716.71
	Floor Standing Gas Non Cond	134 500	3,515.16	472.79	158 000	3,310.51	523.06
Gas Wall Hung Boilers	Wall Hung Gas Cond	141 000	4,121.68	581.16	178 800	3,960.34	708.11
	Wall Hung Gas Non Cond	200	1,755.71	0.35	300	1,694.00	0.51
Jet Burner Boilers	Jet Burner Cond	80	4,247.25	0.34	100	4,144.00	0.41
	Jet Burner Non Cond	48 300	5,685.34	274.60	53 500	5,639.23	301.70
Solid Fuel Boilers	Fossil Fuel / Universal	60	3,131.84	0.19	75	3,171.05	0.24
	Logwood / Bi-Fuel	315	5,452.60	1.72	410	5,506.19	2.26
	Other Solid Fuel	350	7,882.04	2.76	420	7,975.00	3.35
	Pellet	110	8,569.19	0.94	130	8,619.32	1.12
	Woodchip	10	63,306.57	0.63	10	64,100.00	0.64
Grand Total		380 425	5,180.75	1 970.89	455 545	4,980.67	2 268.92

Source: BRG based on industry interviews. Local Prices Shown in USD. Values Shown in Millions (at MSP = Manufacturers'Selling Price).

1.3.2 Prices & Market Values by Capacity: GFSB Cond

US - ESTIMATED VALUE OF THE FLOOR STANDING CONDENSING BOILER MARKET BY CAPACITY (BTU)

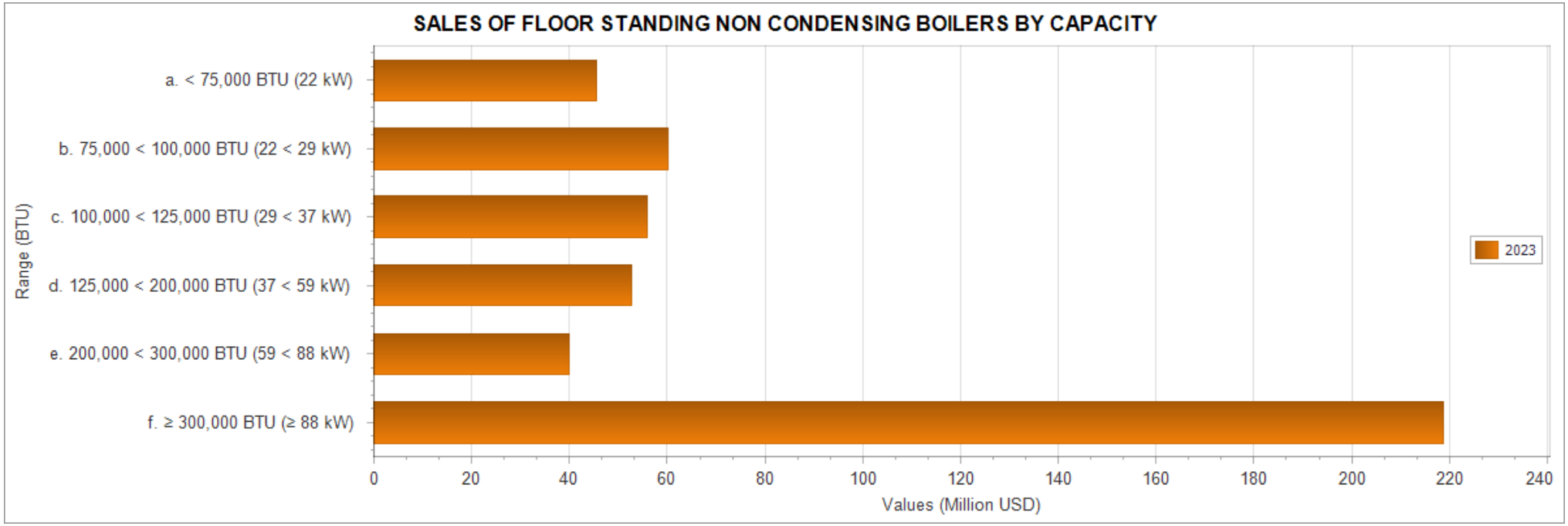


	2023					2022				
Category	Attributed Volume	% (Col Vol)	MSP (Local)	Value (MSP Local)	% (Col Val)	Attributed Volume	% (Col Vol)	MSP (Local)	Value (MSP Local)	% (Col Val)
a. < 75,000 BTU (22 kW)	2.21	5.00%	2,814.65	6.22	1.00%	2.83	5.30%	2,710.00	7.66	1.07%
b. 75,000 < 100,000 BTU (22 < 29 kW)	2.12	4.80%	3,704.59	7.86	1.26%	2.72	5.10%	3,560.00	9.68	1.35%
c. 100,000 < 125,000 BTU (29 < 37 kW)	5.57	12.60%	4,414.47	24.59	3.95%	7.09	13.30%	4,300.00	30.48	4.25%
d. 125,000 < 200,000 BTU (37 < 59 kW)	10.47	23.70%	4,999.75	52.37	8.40%	10.99	20.62%	4,800.00	52.76	7.36%
e. 200,000 < 300,000 BTU (59 < 88 kW)	1.23	2.77%	6,207.71	7.61	1.22%	3.13	5.87%	6,000.00	18.77	2.62%
f. ≥ 300,000 BTU (≥ 88 kW)	22.60	51.13%	23,211.42	524.58	84.17%	26.55	49.81%	22,500.00	597.40	83.35%
Grand Total	44.20	100.00%	14,099.50	623.23	100.00%	53.30	100.00%	13,446.53	716.74	100.00%

Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Local Prices Shown in USD. Volume Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price)

1.3.3 Prices & Market Values by Capacity: GFSB Non Cond

US - ESTIMATED VALUE OF THE FLOOR STANDING NON CONDENSING BOILER MARKET BY CAPACITY (BTU)

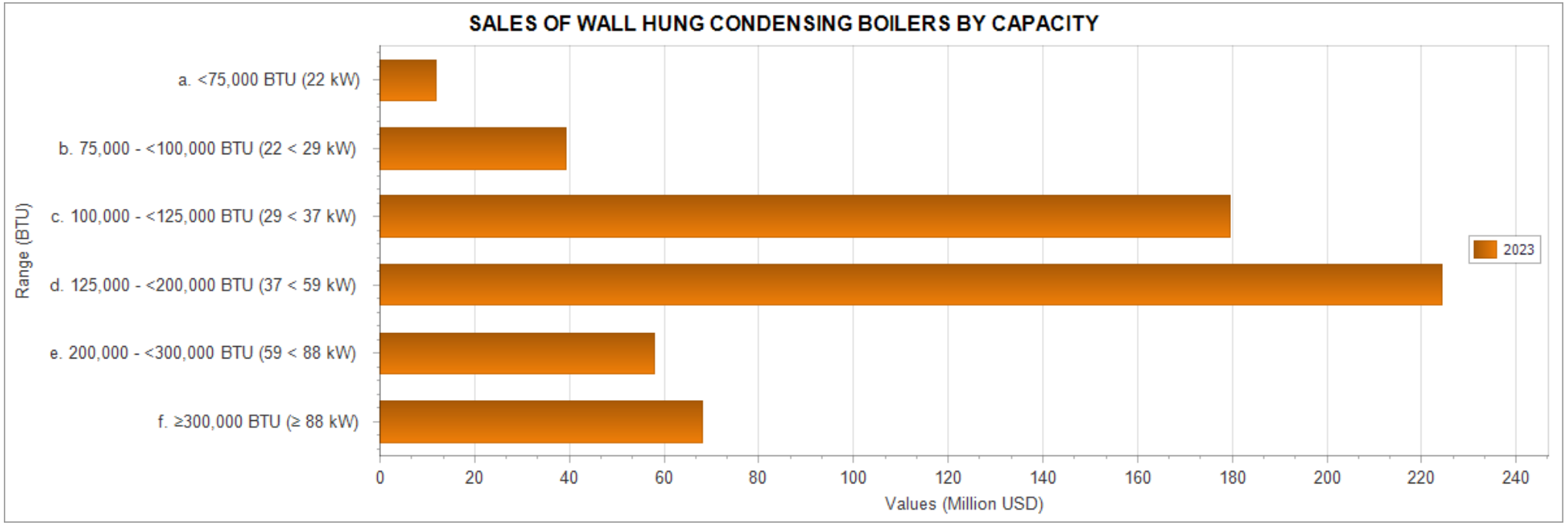


	2023					2022				
Category	Attributed Volume	% (Col Vol)	MSP (Local)	Value (MSP Local)	% (Col Val)	Attributed Volume	% (Col Vol)	MSP (Local)	Value (MSP Local)	% (Col Val)
a. < 75,000 BTU (22 kW)	30.94	23.00%	1,473.78	45.59	9.64%	36.47	23.08%	1,430.00	52.15	9.97%
b. 75,000 < 100,000 BTU (22 < 29 kW)	34.97	26.00%	1,717.79	60.07	12.71%	41.42	26.21%	1,670.00	69.17	13.22%
c. 100,000 < 125,000 BTU (29 < 37 kW)	26.63	19.80%	2,100.27	55.93	11.83%	31.44	19.90%	2,030.00	63.83	12.20%
d. 125,000 < 200,000 BTU (37 < 59 kW)	18.56	13.80%	2,837.59	52.67	11.14%	22.12	14.00%	2,740.00	60.61	11.59%
e. 200,000 < 300,000 BTU (59 < 88 kW)	9.20	6.84%	4,330.70	39.86	8.43%	11.21	7.09%	4,200.00	47.07	9.00%
f. ≥ 300,000 BTU (≥ 88 kW)	14.20	10.56%	15,399.30	218.69	46.25%	15.35	9.72%	15,000.00	230.26	44.02%
Grand Total	134.50	100.00%	3,515.25	472.81	100.00%	158.00	100.00%	3,310.57	523.08	100.00%

Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Local Prices Shown in USD. Volume Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price)

1.3.4 Prices & Market Values by Capacity: GWHB Cond

US - ESTIMATED VALUE OF THE WALL HUNG CONDENSING BOILER MARKET BY CAPACITY (BTU)

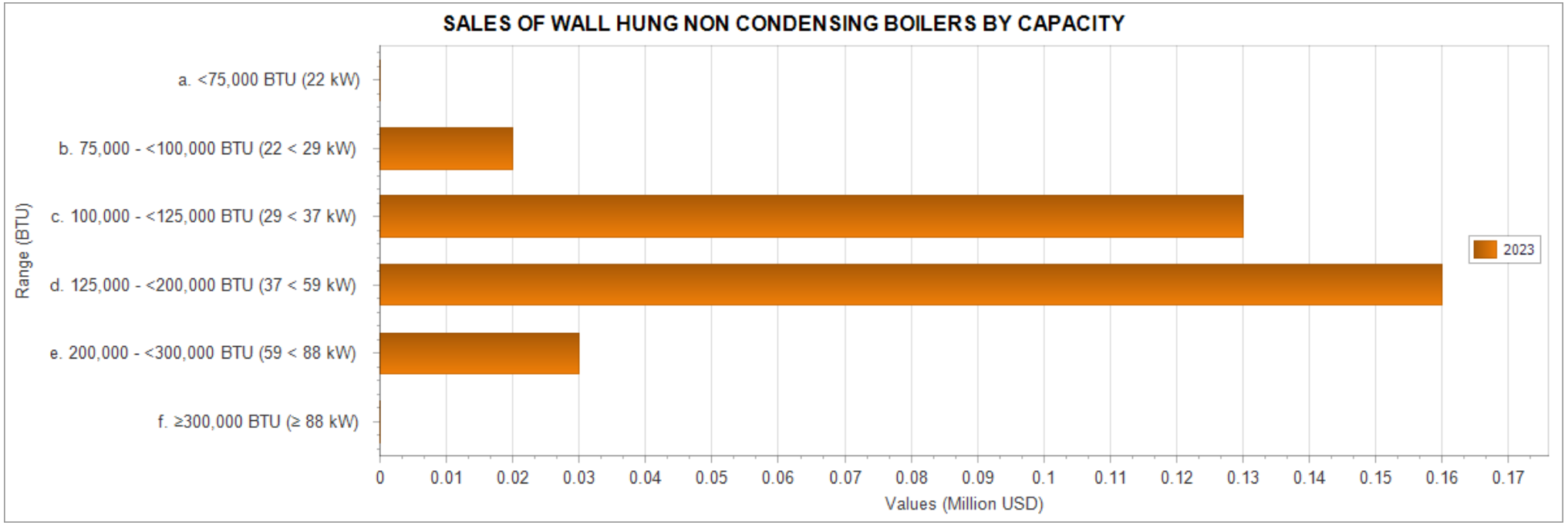


	2023					2022				
Category	Attributed Volume	% (Col Vol)	MSP (Local)	Value (MSP Local)	% (Col Val)	Attributed Volume	% (Col Vol)	MSP (Local)	Value (MSP Local)	% (Col Val)
a. <75,000 BTU (22 kW)	4.65	3.30%	2,548.49	11.86	2.04%	6.26	3.50%	2,480.00	15.52	2.19%
b. 75,000 - <100,000 BTU (22 < 29 kW)	14.10	10.00%	2,786.97	39.30	6.76%	18.42	10.30%	2,720.00	50.09	7.07%
c. 100,000 - <125,000 BTU (29 < 37 kW)	52.69	37.37%	3,408.04	179.56	30.90%	66.63	37.26%	3,310.00	220.55	31.15%
d. 125,000 - <200,000 BTU (37 < 59 kW)	54.43	38.60%	4,122.48	224.37	38.61%	69.38	38.80%	4,000.00	277.50	39.19%
e. 200,000 - <300,000 BTU (59 < 88 kW)	10.64	7.54%	5,456.98	58.04	9.99%	12.97	7.25%	5,300.00	68.75	9.71%
f. ≥300,000 BTU (≥ 88 kW)	4.50	3.19%	15,120.71	68.06	11.71%	5.15	2.88%	14,700.00	75.71	10.69%
Grand Total	141.00	100.00%	4,121.76	581.18	100.00%	178.80	100.00%	3,960.33	708.12	100.00%

Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Local Prices Shown in USD. Volume Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price)

1.3.5 Prices & Market Values by Capacity: GWHB Non Cond

US - ESTIMATED VALUE OF THE WALL HUNG NON CONDENSING BOILER MARKET BY CAPACITY (BTU)

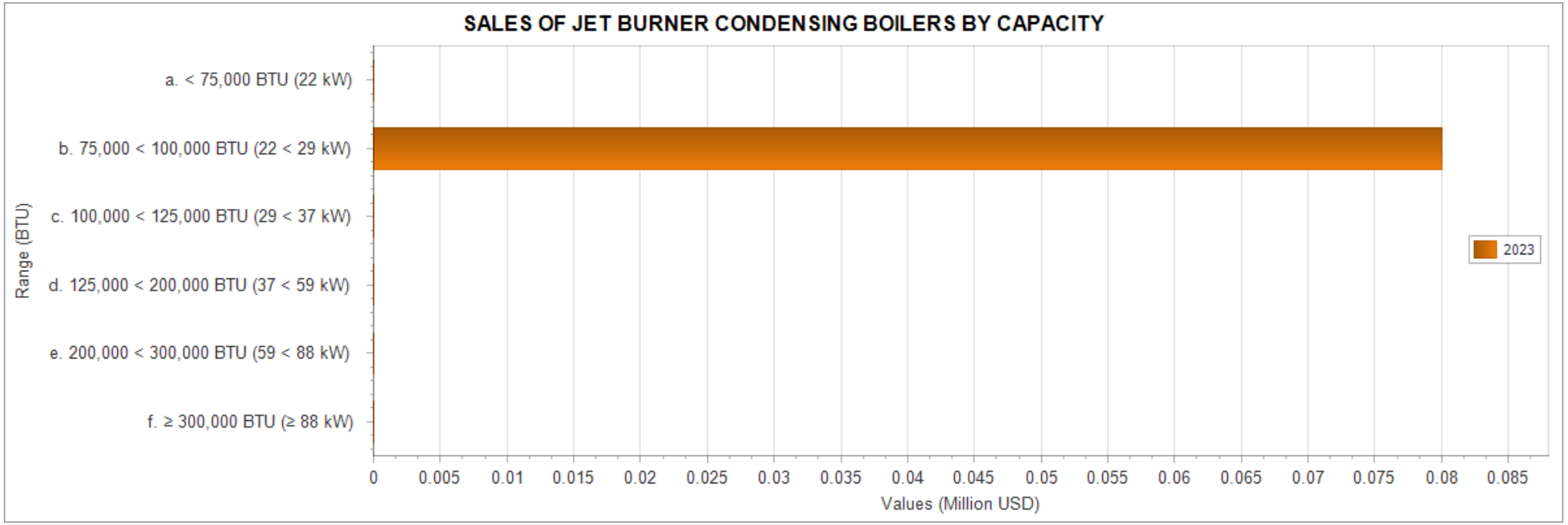


	2023					2022				
Category	Attributed Volume	% (Col Vol)	MSP (Local)	Value (MSP Local)	% (Col Val)	Attributed Volume	% (Col Vol)	MSP (Local)	Value (MSP Local)	% (Col Val)
a. <75,000 BTU (22 kW)	0.00	0.00%	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00%
b. 75,000 - <100,000 BTU (22 < 29 kW)	0.02	10.00%	1,157.65	0.02	6.59%	0.03	10.00%	1,120.00	0.03	6.61%
c. 100,000 - <125,000 BTU (29 < 37 kW)	0.09	46.00%	1,429.15	0.13	37.44%	0.14	46.00%	1,380.00	0.19	37.47%
d. 125,000 - <200,000 BTU (37 < 59 kW)	0.08	39.00%	2,108.39	0.16	46.83%	0.12	39.00%	2,030.00	0.24	46.74%
e. 200,000 - <300,000 BTU (59 < 88 kW)	0.01	5.00%	3,205.22	0.03	9.13%	0.02	5.00%	3,110.00	0.05	9.18%
f. ≥300,000 BTU (≥ 88 kW)	0.00	0.00%	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00%
Grand Total	0.20	100.00%	1,755.71	0.35	100.00%	0.30	100.00%	1,694.00	0.51	100.00%

Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Local Prices Shown in USD. Volume Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price)

1.3.6 Prices & Market Values by Capacity: JBB Cond

US - ESTIMATED VALUE OF THE JET BURNER CONDENSING BOILER MARKET BY CAPACITY (BTU)

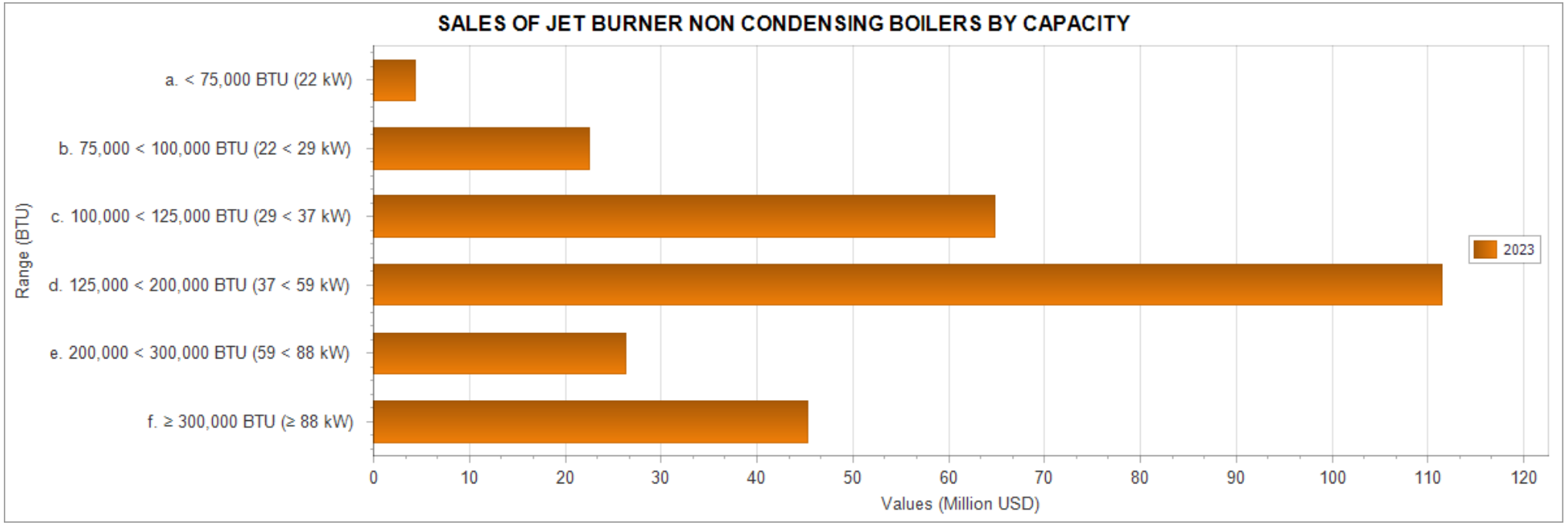


	2023					2022				
Category	Attributed Volume	% (Col Vol)	MSP (Local)	Value (MSP Local)	% (Col Val)	Attributed Volume	% (Col Vol)	MSP (Local)	Value (MSP Local)	% (Col Val)
a. < 75,000 BTU (22 kW)	0.00	5.00%	2,790.76	0.01	3.29%	0.01	5.00%	2,700.00	0.01	3.26%
b. 75,000 < 100,000 BTU (22 < 29 kW)	0.08	95.00%	4,323.90	0.33	96.71%	0.10	95.00%	4,220.00	0.40	96.74%
c. 100,000 < 125,000 BTU (29 < 37 kW)	0.00	0.00%	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00%
d. 125,000 < 200,000 BTU (37 < 59 kW)	0.00	0.00%	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00%
e. 200,000 < 300,000 BTU (59 < 88 kW)	0.00	0.00%	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00%
f. ≥ 300,000 BTU (≥ 88 kW)	0.00	0.00%	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00%
Grand Total	0.08	100.00%	4,247.24	0.34	100.00%	0.10	100.00%	4,144.00	0.41	100.00%

Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Local Prices Shown in USD. Volumes Shown in Thousands and Values in Millions (at MSP = Manufacturers' Selling Price)

1.3.7 Prices & Market Values by Capacity: JBB Non Cond

US - ESTIMATED VALUE OF THE JET BURNER NON CONDENSING BOILER MARKET BY CAPACITY (BTU)

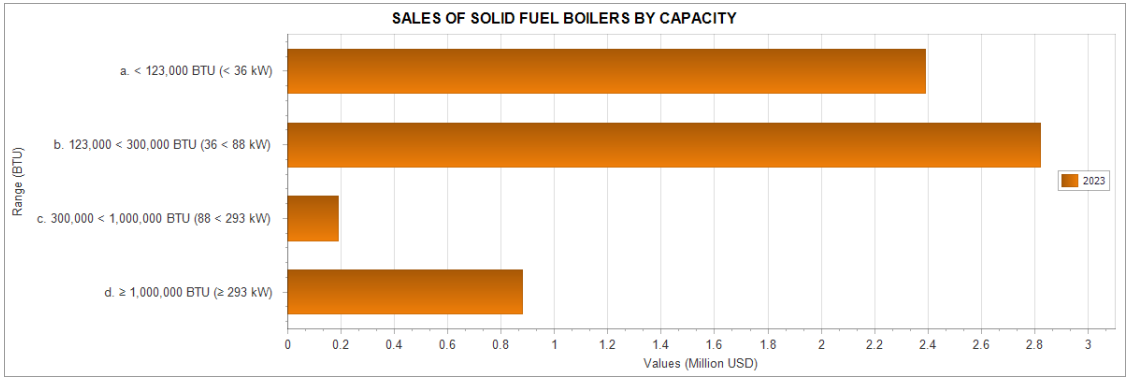


	2023					2022				
Category	Attributed Volume	% (Col Vol)	MSP (Local)	Value (MSP Local)	% (Col Val)	Attributed Volume	% (Col Vol)	MSP (Local)	Value (MSP Local)	% (Col Val)
a. < 75,000 BTU (22 kW)	1.98	4.10%	2,221.82	4.40	1.60%	2.14	4.00%	2,160.00	4.62	1.53%
b. 75,000 < 100,000 BTU (22 < 29 kW)	6.52	13.50%	3,452.98	22.52	8.20%	7.06	13.20%	3,370.00	23.80	7.89%
c. 100,000 < 125,000 BTU (29 < 37 kW)	13.81	28.60%	4,689.32	64.78	23.59%	15.14	28.30%	4,550.00	68.89	22.83%
d. 125,000 < 200,000 BTU (37 < 59 kW)	19.08	39.50%	5,840.34	111.43	40.57%	21.13	39.50%	5,700.00	120.46	39.92%
e. 200,000 < 300,000 BTU (59 < 88 kW)	3.71	7.67%	7,083.68	26.26	9.56%	3.83	7.15%	6,900.00	26.39	8.75%
f. ≥ 300,000 BTU (≥ 88 kW)	3.20	6.63%	14,133.17	45.24	16.47%	4.20	7.85%	13,700.00	57.55	19.08%
Grand Total	48.30	100.00%	5,685.44	274.62	100.00%	53.50	100.00%	5,639.36	301.72	100.00%

Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Local Prices Shown in USD. Volume Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price)

1.3.8 Prices & Market Values by Capacity: Solid Fuel

US - ESTIMATED VALUE OF THE SOLID FUEL BOILER MARKET BY CAPACITY (BTU)

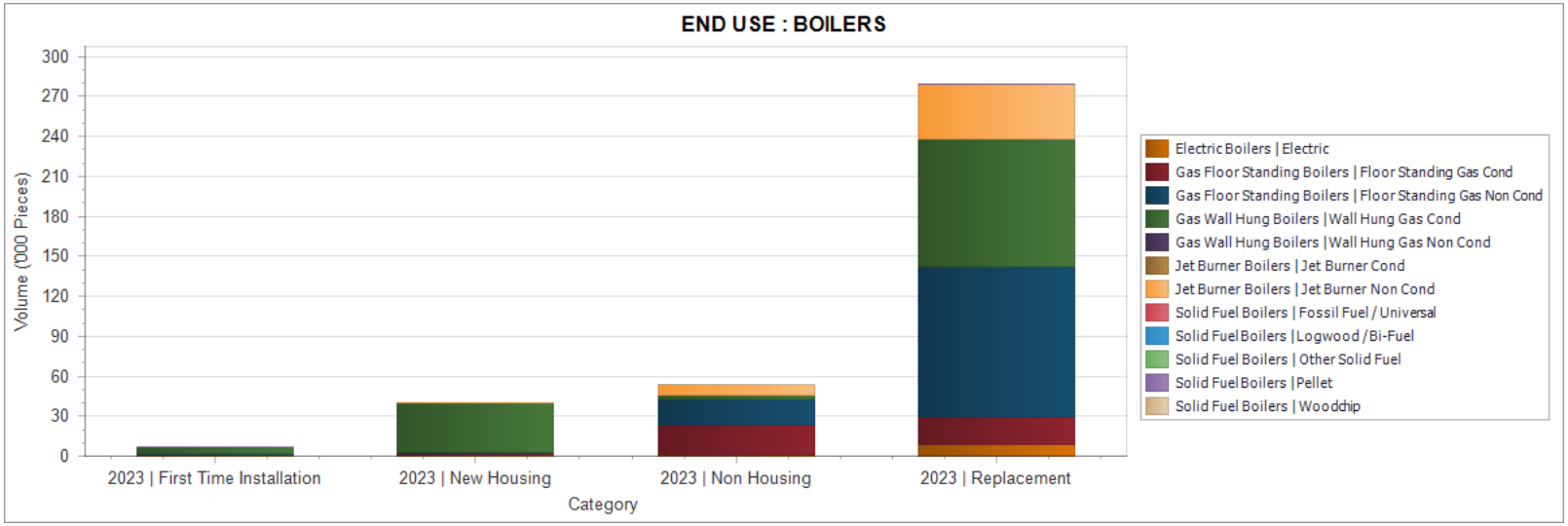


		2023					2022				
Product	Category	Attributed Volume	% (Col Vol)	MSP (Local)	Value (MSP Local)	% (Col Val)	Attributed Volume	% (Col Vol)	MSP (Local)	Value (MSP Local)	% (Col Val)
Fossil Fuel / Universal	a. < 123,000 BTU (< 36 kW)	0.05	80.00%	2,923.92	0.14	74.69%	0.06	78.95%	2,950.00	0.18	73.44%
	b. 123,000 < 300,000 BTU (36 < 88 kW)	0.01	10.00%	3,528.70	0.02	11.27%	0.01	10.53%	3,550.00	0.03	11.78%
	c. 300,000 < 1,000,000 BTU (88 < 293 kW)	0.01	10.00%	4,398.37	0.03	14.04%	0.01	10.53%	4,450.00	0.04	14.77%
	d. ≥ 1,000,000 BTU (≥ 293 kW)	0.00	0.00%	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00%
Fossil Fuel / Universal Total		0.06	7.06%	3,131.84	0.19	2.99%	0.08	7.22%	3,171.05	0.24	3.15%
Logwood / Bi-Fuel	a. < 123,000 BTU (< 36 kW)	0.17	52.85%	4,708.26	0.79	45.63%	0.22	52.91%	4,750.00	1.04	45.65%
	b. 123,000 < 300,000 BTU (36 < 88 kW)	0.13	39.87%	5,887.82	0.74	43.06%	0.16	39.81%	5,950.00	0.98	43.01%
	c. 300,000 < 1,000,000 BTU (88 < 293 kW)	0.01	3.16%	7,207.33	0.07	4.18%	0.01	3.16%	7,300.00	0.09	4.18%
	d. ≥ 1,000,000 BTU (≥ 293 kW)	0.01	4.11%	9,446.50	0.12	7.13%	0.02	4.13%	9,550.00	0.16	7.16%
Logwood / Bi-Fuel Total		0.32	37.18%	5,452.60	1.72	27.42%	0.41	39.16%	5,506.19	2.27	29.62%
Other Solid Fuel	a. < 123,000 BTU (< 36 kW)	0.15	42.90%	6,932.44	1.05	37.66%	0.18	42.89%	7,000.00	1.27	37.59%
	b. 123,000 < 300,000 BTU (36 < 88 kW)	0.19	53.69%	8,436.88	1.59	57.37%	0.23	53.79%	8,550.00	1.94	57.59%
	c. 300,000 < 1,000,000 BTU (88 < 293 kW)	0.01	1.70%	9,846.36	0.06	2.13%	0.01	1.66%	9,950.00	0.07	2.07%
	d. ≥ 1,000,000 BTU (≥ 293 kW)	0.01	1.70%	13,165.13	0.08	2.84%	0.01	1.66%	13,250.00	0.09	2.75%
Other Solid Fuel Total		0.35	41.41%	7,896.13	2.78	44.24%	0.42	40.11%	7,986.37	3.37	44.01%
Pellet	a. < 123,000 BTU (< 36 kW)	0.05	48.21%	7,762.13	0.42	43.67%	0.06	48.48%	7,850.00	0.50	44.16%
	b. 123,000 < 300,000 BTU (36 < 88 kW)	0.05	46.43%	8,981.68	0.47	48.66%	0.06	46.97%	9,050.00	0.56	49.32%
	c. 300,000 < 1,000,000 BTU (88 < 293 kW)	0.00	2.68%	11,100.89	0.03	3.47%	0.00	2.27%	11,250.00	0.03	2.97%
	d. ≥ 1,000,000 BTU (≥ 293 kW)	0.00	2.68%	13,415.04	0.04	4.19%	0.00	2.27%	13,500.00	0.04	3.56%
Pellet Total		0.11	13.18%	8,569.19	0.96	15.27%	0.13	12.55%	8,619.32	1.14	14.86%
Woodchip	a. < 123,000 BTU (< 36 kW)	0.00	0.00%	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00%
	b. 123,000 < 300,000 BTU (36 < 88 kW)	0.00	0.00%	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00%
	c. 300,000 < 1,000,000 BTU (88 < 293 kW)	0.00	0.00%	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00%
	d. ≥ 1,000,000 BTU (≥ 293 kW)	0.01	100.00%	63,306.58	0.63	100.00%	0.01	100.00%	64,100.00	0.64	100.00%
Woodchip Total		0.01	1.18%	63,306.58	0.63	10.08%	0.01	0.95%	64,100.00	0.64	8.37%
Grand Total		0.85	100.00%	7,391.98	6.28	100.00%	1.05	100.00%	7,279.99	7.66	100.00%

Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Local Prices Shown in USD. Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price)

1.4.1 End Use : Base Year

US - SALES OF BOILERS BY TYPE AND END USE SEGMENT ('000 PIECES)

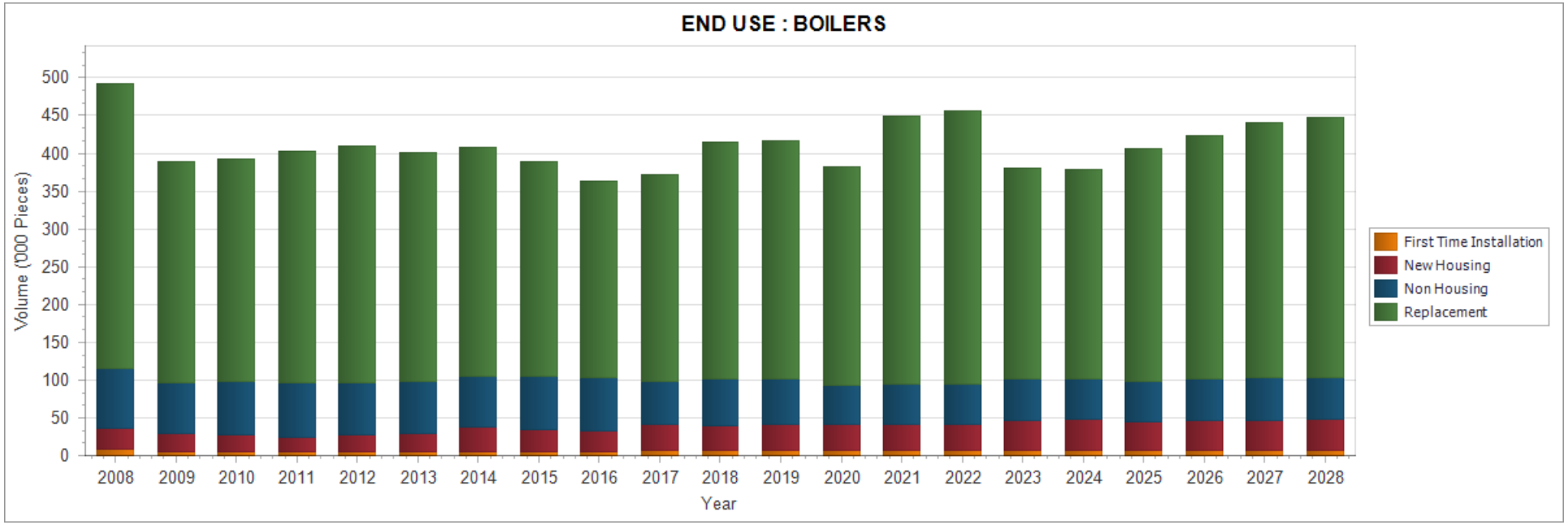


		2023									2023 Total		
		First Time Installation			New Housing			Non Housing			Replacement					
Product Category	Product	Attributed Volume	% (Col Total)	% (Row)	Attributed Volume	% (Col Total)	% (Row)	Attributed Volume	% (Col Total)	% (Row)	Attributed Volume	% (Col Total)	% (Row)	Attributed Volume	% (Col Total)	% (Row)
Electric Boilers	Electric	0.52	7.74%	4.59%	0.95	2.35%	8.37%	0.79	1.47%	7.02%	9.04	3.23%	80.02%	11.30	2.97%	100.00%
Gas Floor Standing Boilers	Floor Standing Gas Cond	0.59	8.82%	1.34%	0.62	1.53%	1.40%	22.39	41.58%	50.64%	20.61	7.37%	46.62%	44.20	11.62%	100.00%
	Floor Standing Gas Non Cond	0.62	9.30%	0.46%	1.05	2.61%	0.78%	19.89	36.94%	14.79%	112.94	40.39%	83.97%	134.50	35.35%	100.00%
Gas Wall Hung Boilers	Wall Hung Gas Cond	4.92	73.29%	3.49%	37.33	92.64%	26.48%	3.23	6.00%	2.29%	95.53	34.17%	67.75%	141.00	37.06%	100.00%
	Wall Hung Gas Non Cond	0.01	0.10%	3.47%	0.09	0.22%	43.56%	0.01	0.02%	5.45%	0.10	0.03%	47.52%	0.20	0.05%	100.00%
Jet Burner Boilers	Jet Burner Cond	0.00	0.00%	0.00%	0.01	0.03%	13.58%	0.00	0.00%	0.00%	0.07	0.03%	86.42%	0.08	0.02%	100.00%
	Jet Burner Non Cond	0.00	0.00%	0.00%	0.25	0.63%	0.52%	7.29	13.54%	15.09%	40.76	14.58%	84.39%	48.30	12.70%	100.00%
Solid Fuel Boilers	Fossil Fuel / Universal	0.00	0.00%	0.00%	0.00	0.00%	0.00%	0.01	0.01%	13.11%	0.05	0.02%	86.89%	0.06	0.02%	100.00%
	Logwood / Bi-Fuel	0.03	0.42%	8.83%	0.00	0.00%	0.00%	0.16	0.29%	49.84%	0.13	0.05%	41.32%	0.32	0.08%	100.00%
	Other Solid Fuel	0.01	0.19%	3.70%	0.00	0.00%	0.00%	0.03	0.05%	7.41%	0.31	0.11%	88.89%	0.35	0.09%	100.00%
	Pellet	0.01	0.13%	8.04%	0.00	0.00%	0.00%	0.05	0.09%	41.07%	0.06	0.02%	50.89%	0.11	0.03%	100.00%
	Woodchip	0.00	0.00%	0.00%	0.00	0.00%	0.00%	0.01	0.02%	90.91%	0.00	0.00%	9.09%	0.01	0.00%	100.00%
Grand Total		6.71	100.00%	1.76%	40.30	100.00%	10.59%	53.84	100.00%	14.15%	279.60	100.00%	73.49%	380.45	100.00%	100.00%

Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

1.4.2 End Use : Historical and Forecasts

US - SALES OF BOILERS BY END USE SEGMENT ('000 PIECES)



Category	Data	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028
First Time Installation	Attributed Volume	9.02	4.38	4.35	4.64	4.92	4.96	5.03	5.00	5.48	6.03	6.53	6.80	7.17	7.03	6.81	6.71	6.65	6.60	6.55	6.50	6.44
	% (Col)	1.83%	1.13%	1.11%	1.15%	1.20%	1.24%	1.23%	1.29%	1.51%	1.62%	1.57%	1.63%	1.87%	1.56%	1.49%	1.76%	1.76%	1.63%	1.55%	1.48%	1.44%
New Housing	Attributed Volume	27.51	24.57	22.54	19.60	22.55	24.93	32.44	29.98	27.65	35.87	33.76	35.14	33.51	34.30	34.46	40.30	41.41	37.66	39.37	40.21	41.00
	% (Col)	5.58%	6.31%	5.75%	4.86%	5.50%	6.22%	7.94%	7.72%	7.63%	9.64%	8.12%	8.42%	8.76%	7.63%	7.56%	10.59%	10.92%	9.28%	9.30%	9.14%	9.16%
Non Housing	Attributed Volume	78.65	67.20	71.65	72.69	68.51	67.89	67.88	69.86	70.08	55.53	61.37	59.63	52.54	53.34	53.28	53.84	53.99	54.28	55.11	56.09	55.91
	% (Col)	15.96%	17.27%	18.29%	18.04%	16.70%	16.94%	16.62%	18.00%	19.33%	14.92%	14.77%	14.30%	13.74%	11.86%	11.70%	14.15%	14.24%	13.38%	13.02%	12.75%	12.49%
Replacement	Attributed Volume	377.60	293.03	293.27	306.08	314.30	302.92	303.12	283.26	259.35	274.81	313.88	315.54	289.28	354.95	361.02	279.60	277.04	307.11	322.28	337.17	344.27
	% (Col)	76.63%	75.29%	74.85%	75.95%	76.61%	75.60%	74.21%	72.99%	71.53%	73.83%	75.54%	75.65%	75.63%	78.95%	79.25%	73.49%	73.08%	75.71%	76.13%	76.63%	76.91%
Grand Total	Attributed Volume	492.78	389.19	391.81	403.01	410.28	400.70	408.46	388.10	362.56	372.24	415.54	417.10	382.49	449.62	455.57	380.45	379.09	405.66	423.31	439.97	447.63
	% (Col)	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%

Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

1.4.3 Technical Segmentation : Gas FS Boilers

US - FLOOR STANDING GAS BOILERS : TECHNICAL SEGMENTATION ('000 PIECES)

		2023				2023 Total	
		Floor Standing Gas Cond		Floor Standing Gas Non Cond			
Segmentation	Category	Attributed Volume	% (Col Vol)	Attributed Volume	% (Col Vol)	Attributed Volume	% (Col Vol)
Burner Type	Conventional	0.00	0.00%	0.00	0.00%	0.00	0.00%
	Other Low NOx	0.00	0.00%	134.50	100.00%	134.50	75.27%
	Premix	44.20	100.00%	0.00	0.00%	44.20	24.73%
Flue Type	Open	0.00	0.00%	56.49	42.00%	56.49	31.61%
	Room Sealed (Balanced)	0.00	0.00%	0.00	0.00%	0.00	0.00%
	Room Sealed (Fanned)	44.20	100.00%	78.01	58.00%	122.21	68.39%
Gas Type	LPG	0.00	0.00%	20.18	15.00%	20.18	11.29%
	Natural	44.20	100.00%	114.33	85.00%	158.53	88.71%
Heat Exchanger	Aluminium	4.42	10.00%	0.00	0.00%	4.42	2.47%
	Cast Iron	1.55	3.50%	117.72	87.52%	119.27	66.74%
	Copper	2.03	4.60%	12.75	9.48%	14.78	8.27%
	Stainless Steel	36.20	81.90%	0.00	0.00%	36.20	20.26%
	Steel	0.00	0.00%	4.04	3.00%	4.04	2.26%
Hot Water Production	Combi	1.67	3.78%	0.00	0.00%	1.67	0.93%
	Heating Only With External Cylinder	8.94	20.22%	26.63	19.80%	35.57	19.90%
	Heating Only With Integrated Cylinder	0.00	0.00%	0.00	0.00%	0.00	0.00%
	Heating Only Without Cylinder	33.59	76.00%	107.87	80.20%	141.46	79.16%
Ignition Type	Electronic	44.20	100.00%	134.50	100.00%	178.70	100.00%
	Pilot	0.00	0.00%	0.00	0.00%	0.00	0.00%

Source: BRG based on industry interviews. Totals may show slight variance due to rounding.

1.4.4 Technical Segmentation : Gas Wall Hung Boilers

US - WALL HUNG GAS BOILERS : TECHNICAL SEGMENTATION ('000 PIECES)

		2023				2023 Total	
		Wall Hung Gas Cond		Wall Hung Gas Non Cond			
Segmentation	Category	Attributed Volume	% (Col Vol)	Attributed Volume	% (Col Vol)	Attributed Volume	% (Col Vol)
Burner Type	Conventional	0.00	0.00%	0.00	0.00%	0.00	0.00%
	Other Low NOx	0.00	0.00%	0.20	100.00%	0.20	0.14%
	Premix	141.00	100.00%	0.00	0.00%	141.00	99.86%
Flue Type	Open	0.00	0.00%	0.00	0.00%	0.00	0.00%
	Room Sealed (Balanced)	0.00	0.00%	0.00	0.00%	0.00	0.00%
	Room Sealed (Fanned)	141.00	100.00%	0.20	100.00%	141.20	100.00%
Gas Type	LPG	14.10	10.00%	0.04	19.00%	14.14	10.01%
	Natural	126.90	90.00%	0.16	81.00%	127.06	89.99%
Heat Exchanger	Aluminium	14.24	10.10%	0.00	0.00%	14.24	10.09%
	Cast Iron	0.00	0.00%	0.00	0.00%	0.00	0.00%
	Copper	0.00	0.00%	0.20	100.00%	0.20	0.14%
	Stainless Steel	126.76	89.90%	0.00	0.00%	126.76	89.77%
	Steel	0.00	0.00%	0.00	0.00%	0.00	0.00%
Hot Water Production	Combi Conventional	72.40	51.35%	0.14	70.00%	72.54	51.37%
	Combi With Pre-Heat	0.00	0.00%	0.00	0.00%	0.00	0.00%
	Combi With Storage > 40L	2.80	1.99%	0.00	0.00%	2.80	1.98%
	Heating Only With External Cylinder	45.62	32.36%	0.04	21.00%	45.67	32.34%
	Heating Only Without Cylinder	20.18	14.31%	0.02	9.00%	20.20	14.30%
Ignition Type	Electronic	141.00	100.00%	0.20	100.00%	141.20	100.00%
	Pilot	0.00	0.00%	0.00	0.00%	0.00	0.00%

Source: BRG based on industry interviews. Totals may show slight variance due to rounding.

1.4.5 Technical Segmentation : Jet Burner Boilers

US - OIL (JET BURNER) BOILERS : TECHNICAL SEGMENTATION ('000 PIECES)

		2023				2023 Total	
		Jet Burner Cond		Jet Burner Non Cond			
Segmentation	Category	Attributed Volume	% (Col Vol)	Attributed Volume	% (Col Vol)	Attributed Volume	% (Col Vol)
Flue Type	Open	0.00	0.00%	19.80	41.00%	19.80	40.93%
	Room Sealed (Balanced)	0.00	0.00%	0.00	0.00%	0.00	0.00%
	Room Sealed (Fanned)	0.08	100.00%	28.50	59.00%	28.58	59.07%
Fuel Type	Bi-Fuel Units	0.02	30.00%	3.86	8.00%	3.89	8.04%
	Gas Units	0.00	0.00%	0.00	0.00%	0.00	0.00%
	Oil Units	0.06	70.00%	44.44	92.00%	44.49	91.96%
	Oil/Gas Units	0.00	0.00%	0.00	0.00%	0.00	0.00%
Heat Exchanger	Cast Iron	0.00	0.00%	46.37	96.00%	46.37	95.84%
	Other	0.00	0.00%	0.00	0.00%	0.00	0.00%
	Steel, Mixed, Stainless Steel	0.08	100.00%	1.93	4.00%	2.01	4.16%
Hot Water Production	Heating Only With External Cylinder	0.01	10.00%	4.83	10.00%	4.84	10.00%
	Heating Only With Integrated Cylinder	0.00	0.00%	0.00	0.00%	0.00	0.00%
	Heating Only Without Cylinder	0.07	90.00%	43.47	90.00%	43.54	90.00%

Source: BRG based on industry interviews. Totals may show slight variance due to rounding.

1.4.6 Technical Segmentation : Solid Fuel Boilers

US - SOLID FUEL BOILERS : TECHNICAL SEGMENTATION ('000 PIECES)

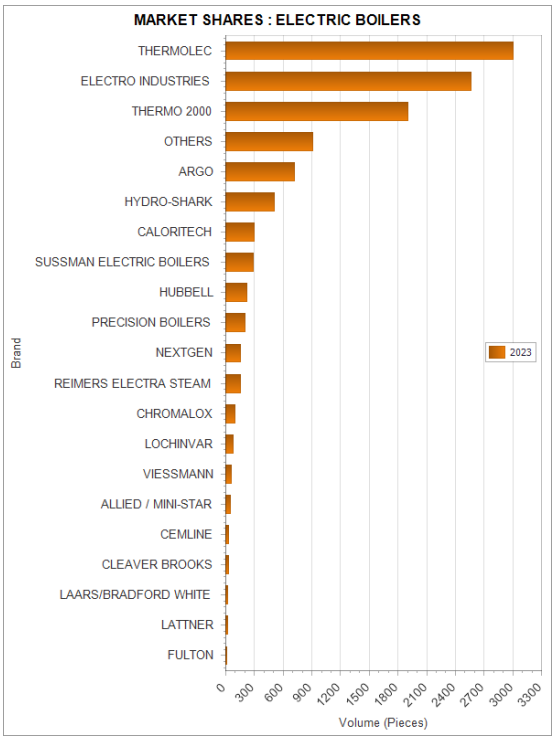
		2023										2023 Total	
		Fossil Fuel / Universal		Logwood / Bi-Fuel		Other Solid Fuel		Pellet		Woodchip			
Segmentation	Category	Attributed Volume	% (Col Vol)	Attributed Volume	% (Col Vol)	Attributed Volume	% (Col Vol)	Attributed Volume	% (Col Vol)	Attributed Volume	% (Col Vol)	Attributed Volume	% (Col Vol)
Hot Water Production	Heating Only with External Cylinder (Non-Solar)	0.00	0.00%	0.00	0.00%	0.00	0.00%	0.00	0.00%	0.00	0.00%	0.00	0.00%
	Heating Only with External Cylinder (Solar)	0.00	0.00%	0.00	0.00%	0.00	0.00%	0.00	0.00%	0.00	0.00%	0.00	0.00%
	Heating Only Without Cylinder	0.06	100.00%	0.32	100.00%	0.35	100.00%	0.11	100.00%	0.01	100.00%	0.85	100.00%

Source: BRG based on industry interviews. Totals may show slight variance due to rounding.

1.5.1 Market Shares : Electric

US - ESTIMATED SHARES : ELECTRIC BOILERS

	2023		2022	
Holding Company (Brand)	Shares	% of total	Shares	% of total
THERMOLEC (THERMOLEC)	3 000	26.55%	2 750	26.19%
BRADFORD WHITE (ELECTRO INDUSTRIES)	2 560	22.65%	2 520	24.00%
THERMO 2000 (THERMO 2000)	1 900	16.81%	1 800	17.14%
OTHERS (OTHERS)	913	8.08%	834	7.94%
TERRAVEST INDUSTRIES (ARGO)	720	6.37%	545	5.19%
STIEBEL ELTRON (HYDRO-SHARK)	505	4.47%	465	4.43%
THERMON GROUP (CALORITECH)	300	2.65%	290	2.76%
DHT (SUSSMAN ELECTRIC BOILERS)	285	2.52%	290	2.76%
NUDYNE GROUP (HUBBELL)	220	1.95%	215	2.05%
THERMON GROUP (PRECISION BOILERS)	200	1.77%	180	1.71%
NEXTGEN (NEXTGEN)	155	1.37%	120	1.14%
REIMERS (REIMERS ELECTRA STEAM)	150	1.33%	145	1.38%
SPIRAX SARCO (CHROMALOX)	100	0.88%	90	0.86%
AO SMITH (LOCHINVAR)	80	0.71%	50	0.48%
CARRIER GLOBAL (VIESSMANN)	60	0.53%	45	0.43%
E-Z RECT MFG (ALLIED / MINI-STAR)	50	0.44%	60	0.57%
HARBOUR GROUP (CLEAVER BROOKS)	30	0.27%	20	0.19%
CEMLINE (CEMLINE)	30	0.27%	35	0.33%
BRADFORD WHITE (LAARS/BRADFORD WHITE)	20	0.18%	20	0.19%
LATTNER BOILER (LATTNER)	15	0.13%	15	0.14%
THE FULTON COMPANIES (FULTON)	7	0.06%	11	0.10%
Grand Total	11 300	100.00%	10 500	100.00%

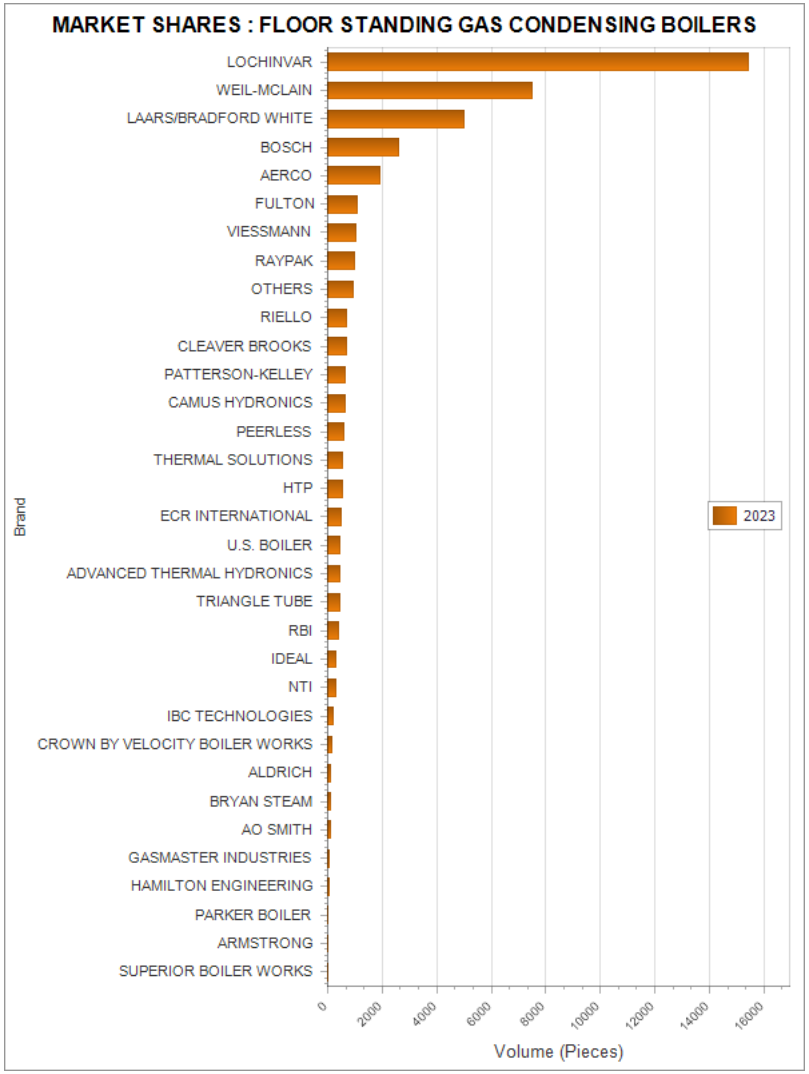


Source: BRG based on industry interviews

1.5.2 Market Shares : Floor Standing Gas Condensing

US - ESTIMATED SHARES : FLOOR STANDING GAS CONDENSING BOILERS

	2023		2022	
Holding Company (Brand)	Shares	% of total	Shares	% of total
AO SMITH (LOCHINVAR)	15 400	34.84%	18 450	34.62%
SPX (WEIL-MCLAIN)	7 490	16.95%	10 495	19.69%
BRADFORD WHITE (LAARS/BRADFORD WHITE)	5 000	11.31%	4 335	8.13%
BOSCH (BOSCH)	2 580	5.84%	3 205	6.01%
WATTS WATER TECHNOLOGIES (AERCO)	1 900	4.30%	2 250	4.22%
THE FULTON COMPANIES (FULTON)	1 070	2.42%	1 075	2.02%
CARRIER GLOBAL (VIESSMANN)	1 040	2.35%	1 105	2.07%
PALOMA (RAYPAK)	975	2.21%	1 150	2.16%
OTHERS (OTHERS)	950	2.15%	1 082	2.03%
CARRIER GLOBAL (RIELLO)	685	1.55%	760	1.43%
HARBOUR GROUP (CLEAVER BROOKS)	675	1.53%	725	1.36%
SPX (PATTERSON-KELLEY)	630	1.43%	530	0.99%
HARBOUR GROUP (CAMUS HYDRONICS)	620	1.40%	600	1.13%
NORITZ (PEERLESS)	600	1.36%	635	1.19%
BURNHAM (THERMAL SOLUTIONS)	540	1.22%	525	0.98%
ARISTON GROUP (HTP)	515	1.17%	940	1.76%
TERRAVEST INDUSTRIES (ECR INTERNATIONAL)	485	1.10%	700	1.31%
BURNHAM (U.S.BOILER)	460	1.04%	420	0.79%
MESTEK (ADVANCED THERMAL HYDRONICS)	450	1.02%	785	1.47%
GROUPE ATLANTIC (TRIANGLE TUBE)	450	1.02%	750	1.41%
MESTEK (RBI)	370	0.84%	525	0.98%
GROUPE ATLANTIC (IDEAL)	300	0.68%	700	1.31%
ARISTON GROUP (NTI)	295	0.67%	775	1.45%
PALOMA (IBC TECHNOLOGIES)	180	0.41%	200	0.38%
BURNHAM (CROWN BY VELOCITY BOILER WORKS)	155	0.35%	275	0.52%
ALDRICH (ALDRICH)	100	0.23%	75	0.14%
BURNHAM (BRYAN STEAM)	90	0.20%	55	0.10%
AO SMITH (AO SMITH)	90	0.20%	100	0.19%
GASMASTER INDUSTRIES (GASMASTER INDUSTRIES)	45	0.10%	50	0.09%
ARMSTRONG INTERNATIONAL (HAMILTON ENGINEERING)	30	0.07%		0.00%
PARKER BOILER (PARKER BOILER)	20	0.05%	25	0.05%
SUPERIOR BOILER (SUPERIOR BOILER WORKS)	5	0.01%	3	0.01%
ARMSTRONG INTERNATIONAL (ARMSTRONG)	5	0.01%		0.00%
Grand Total	44 200	100.00%	53 300	100.00%

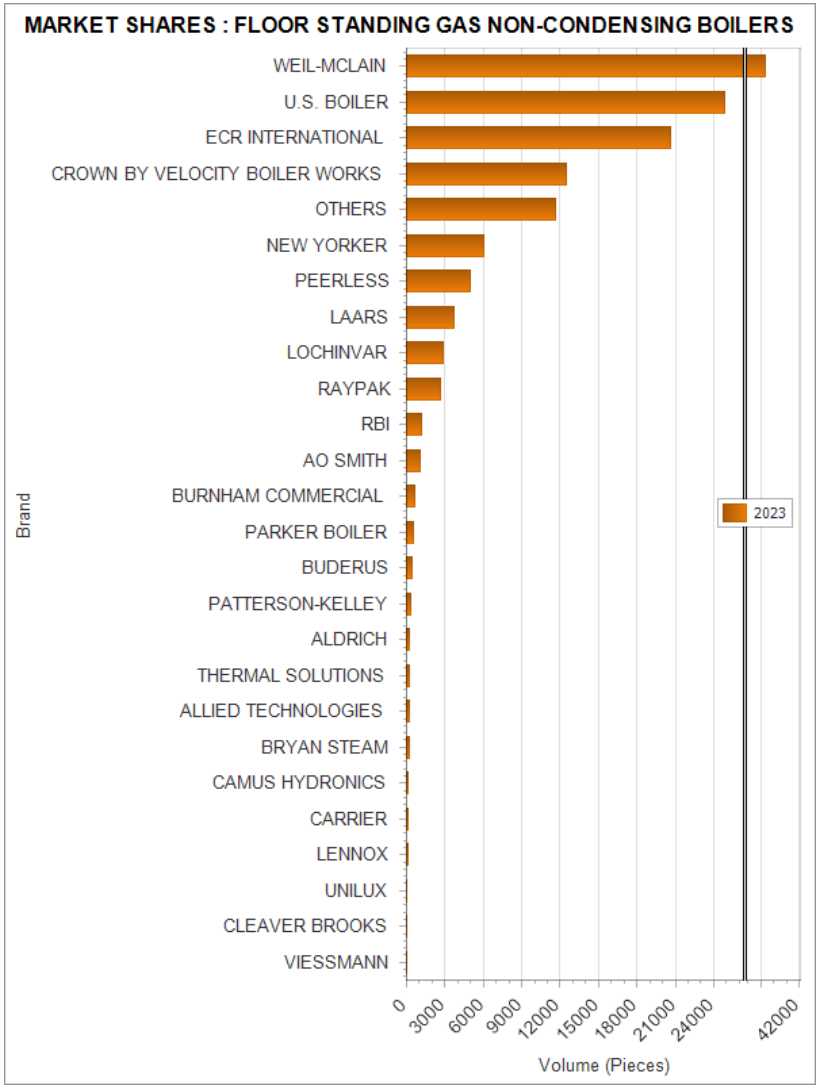


Source: BRG based on industry interviews

1.5.3 Market Shares : Floor Standing Gas Non-Condensing

US - ESTIMATED SHARES : FLOOR STANDING GAS NON-CONDENSING BOILERS

	2023		2022	
Holding Company (Brand)	Shares	% of total	Shares	% of total
SPX (WEIL-MCLAIN)	39 295	29.22%	51 570	32.64%
BURNHAM (U.S.BOILER)	24 900	18.51%	21 870	13.84%
TERRAVEST INDUSTRIES (ECR INTERNATIONAL)	20 645	15.35%	27 170	17.20%
BURNHAM (CROWN BY VELOCITY BOILER WORKS)	12 465	9.27%	13 555	8.58%
OTHERS (OTHERS)	11 660	8.67%	14 205	8.99%
BURNHAM (NEWYORKER)	6 000	4.46%	5 390	3.41%
NORITZ (PEERLESS)	4 975	3.70%	5 950	3.77%
BRADFORD WHITE (LAARS)	3 685	2.74%	5 305	3.36%
AO SMITH (LOCHINVAR)	2 900	2.16%	3 395	2.15%
PALOMA (RAYPAK)	2 600	1.93%	3 150	1.99%
MESTEK (RBI)	1 120	0.83%	1 250	0.79%
AO SMITH (AO SMITH)	1 100	0.82%	1 125	0.71%
BURNHAM (BURNHAM COMMERCIAL)	680	0.51%	1 020	0.65%
PARKER BOILER (PARKER BOILER)	525	0.39%	620	0.39%
BOSCH (BUDERUS)	400	0.30%	875	0.55%
SPX (PATTERSON-KELLEY)	370	0.28%	280	0.18%
ALDRICH (ALDRICH)	250	0.19%	225	0.14%
BURNHAM (THERMAL SOLUTIONS)	200	0.15%	230	0.15%
E-Z RECT MFG (ALLIED TECHNOLOGIES)	190	0.14%	255	0.16%
BURNHAM (BRYAN STEAM)	180	0.13%	175	0.11%
HARBOUR GROUP (CAMUS HYDRONICS)	140	0.10%	135	0.09%
CARRIER GLOBAL (CARRIER)	75	0.06%	90	0.06%
LENNOX INTERNATIONAL (LENNOX)	70	0.05%	85	0.05%
UNILUX (UNILUX)	30	0.02%	35	0.02%
HARBOUR GROUP (CLEAVER BROOKS)	30	0.02%	30	0.02%
CARRIER GLOBAL (VIESSMANN)	15	0.01%	10	0.01%
Grand Total	134 500	100.00%	158 000	100.00%

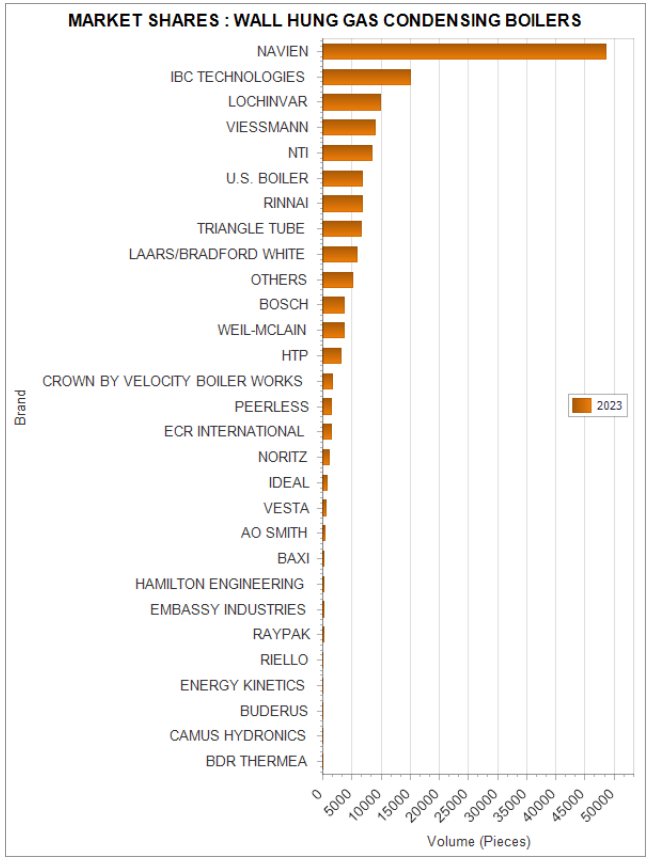


Source: BRG based on industry interviews

1.5.4 Market Shares : Wall Hung Gas Condensing

US - ESTIMATED SHARES : WALL HUNG GAS CONDENSING BOILERS

	2023		2022	
Holding Company (Brand)	Shares	% of total	Shares	% of total
KD NAVIEN (NAVIEN)	48 520	34.41%	59 590	33.33%
PALOMA (IBC TECHNOLOGIES)	15 100	10.71%	18 800	10.51%
AO SMITH (LOCHINVAR)	9 900	7.02%	16 750	9.37%
CARRIER GLOBAL (VIESSMANN)	9 025	6.40%	14 650	8.19%
ARISTON GROUP (NTI)	8 440	5.99%	11 195	6.26%
BURNHAM (U.S. BOILER)	6 740	4.78%	6 250	3.50%
RINNAI (RINNAI)	6 705	4.76%	9 760	5.46%
GROUPE ATLANTIC (TRIANGLE TUBE)	6 650	4.72%	8 250	4.61%
BRADFORD WHITE (LAARS/BRADFORD WHITE)	5 920	4.20%	5 680	3.18%
OTHERS (OTHERS)	5 195	3.68%	5 765	3.22%
BOSCH (BOSCH)	3 665	2.60%	4 275	2.39%
SPX (WEIL-MCLAIN)	3 615	2.56%	3 465	1.94%
ARISTON GROUP (HTP)	3 160	2.24%	4 005	2.24%
BURNHAM (CROWN BY VELOCITY BOILER WORKS)	1 560	1.11%	1 310	0.73%
NORITZ (PEERLESS)	1 500	1.06%	1 800	1.01%
TERRAVEST INDUSTRIES (ECR INTERNATIONAL)	1 450	1.03%	2 075	1.16%
NORITZ (NORITZ)	1 140	0.81%	1 465	0.82%
GROUPE ATLANTIC (IDEAL)	650	0.46%	950	0.53%
DAESUNG (VESTA)	550	0.39%	350	0.20%
AO SMITH (AO SMITH)	400	0.28%	775	0.43%
BDR THERMEA (BAXI)	270	0.19%	430	0.24%
ARMSTRONG INTERNATIONAL (HAMILTON ENGINEERING)	250	0.18%	235	0.13%
MESTEK (EMBASSY INDUSTRIES)	170	0.12%	200	0.11%
PALOMA (RAYPAK)	140	0.10%	165	0.09%
CARRIER GLOBAL (RIELLO)	85	0.06%	260	0.15%
ENERGY KINETICS (ENERGY KINETICS)	80	0.06%	110	0.06%
BOSCH (BUDERUS)	80	0.06%	195	0.11%
HARBOUR GROUP (CAMUS HYDRONICS)	35	0.02%	20	0.01%
BDR THERMEA (BDR THERMEA)	5	0.00%		0.00%
SPX (PATTERSON-KELLEY)		0.00%	25	0.01%
Grand Total	141 000	100.00%	178 800	100.00%



Source: BRG based on industry interviews

1.5.5 Market Shares: Wall Hung Gas Condensing - Combi vs. Heating Only

US - ESTIMATED MARKET SHARES FOR GAS WALL-HUNG CONDENSING BOILERS: COMBI vs HEATING ONLY

Holding Group	Company/Brand	Total Units				Combi Boilers, Units				Heating Only Boilers, Units			
		2021	2022	2023	2023 Share	2021	2022	2023	2023 Share	2021	2022	2023	2023 Share
KD NAVIEN	NAVIEN	69,645	59,590	**48,520**	**34.4%**	50,875	42,225	**34,490**	**45.9%**	18,770	17,365	**14,030**	**21.3%**
PALOMA	IBC TECHNOLOGIES	13,465	18,800	**15,100**	**10.7%**	5,340	7,950	**6,345**	**8.4%**	8,125	10,850	**8,755**	**13.3%**
AO SMITH	LOCHINVAR	15,530	16,750	**9,900**	**7.0%**	6,480	6,300	**4,500**	**6.0%**	9,050	10,450	**5,400**	**8.2%**
VIESSMANN	VIESSMANN	12,040	14,650	**9,025**	**6.4%**	4,850	5,005	**3,560**	**4.7%**	7,190	9,645	**5,465**	**8.3%**
ARISTON GROUP	NTI	10,575	11,195	**8,440**	**6.0%**	3,780	3,605	**1,660**	**2.2%**	6,795	7,590	**6,780**	**10.3%**
BURNHAM	U.S. BOILER	4,925	6,250	**6,740**	**4.8%**	1,330	2,425	**2,805**	**3.7%**	3,595	3,825	**3,935**	**6.0%**
RINNAI	RINNAI	7,465	9,760	**6,705**	**4.8%**	5,260	7,655	**4,880**	**6.5%**	2,205	2,105	**1,825**	**2.8%**
ATLANTIC	TRIANGLE TUBE	8,000	8,250	**6,650**	**4.7%**	2,400	2,475	**2,000**	**2.7%**	5,600	5,775	**4,650**	**7.1%**
BRADFORD WHITE	LAARS/BRADFORD WHITE	5,640	5,680	**5,920**	**4.2%**	2,990	3,180	**3,015**	**4.0%**	2,650	2,500	**2,905**	**4.4%**
BOSCH	BOSCH	4,050	4,275	**3,665**	**2.6%**	3,050	3,375	**2,965**	**3.9%**	1,000	900	**700**	**1.1%**
SPX	WEIL-MCLAIN	4,510	3,465	**3,615**	**2.6%**	2,490	2,265	**1,800**	**2.4%**	2,020	1,200	**1,815**	**2.8%**
ARISTON GROUP	HTP	5,185	4,005	**3,160**	**2.2%**	2,380	1,550	**1,300**	**1.7%**	2,805	2,455	**1,860**	**2.8%**
BURNHAM	CROWN BY VELOCITY BOILER WORKS	1,270	1,310	**1,560**	**1.1%**	360	405	**180**	**0.2%**	910	905	**1,380**	**2.1%**
NORITZ	PEERLESS	1,750	1,800	**1,500**	**1.1%**	690	720	**680**	**0.9%**	1,060	1,080	**820**	**1.2%**
TERRAVEST INDUSTRIES	ECR INTERNATIONAL	2,625	2,075	**1,450**	**1.0%**	1,300	935	**905**	**1.2%**	1,325	1,140	**545**	**0.8%**
NORITZ	NORITZ	2,340	1,465	**1,140**	**0.8%**	2,340	1,465	**1,140**	**1.5%**	0	0	**0**	**0.0%**
GROUPE ATLANTIC	IDEAL	800	950	**650**	**0.5%**	520	625	**425**	**0.6%**	280	325	**225**	**0.3%**
DAESUNG	VESTA	480	350	**550**	**0.4%**	335	250	**500**	**0.7%**	145	100	**50**	**0.1%**
AO SMITH	AO SMITH	730	775	**400**	**0.3%**	300	300	**155**	**0.2%**	430	475	**245**	**0.4%**
BDR THERMEA	BAXI	275	430	**270**	**0.2%**	245	0	**0**	**0.0%**	30	430	**270**	**0.4%**
ARMSTRONG INTERNATIONAL	HAMILTON ENGINEERING	300	235	**250**	**0.2%**	0	0	**0**	**0.0%**	300	235	**250**	**0.4%**
MESTEK	EMBASSY INDUSTRIES	50	200	**170**	**0.1%**	50	200	**170**	**0.2%**	0	0	**0**	**0.0%**
PALOMA	RAYPAK	160	165	**140**	**0.1%**	0	0	**0**	**0.0%**	160	165	**140**	**0.2%**
CARRIER GLOBAL	RIELLO	70	260	**85**	**0.1%**	1	0	**0**	**0.0%**	69	260	**85**	**0.1%**
BOSCH	BUDERUS	170	195	**80**	**0.1%**	0	0	**0**	**0.0%**	170	195	**80**	**0.1%**
ENERGY KINETICS	ENERGY KINETICS	105	110	**80**	**0.1%**	0	0	**0**	**0.0%**	105	110	**80**	**0.1%**
HARBOUR GROUP	CAMUS HYDRONICS	25	20	**35**	**0.0%**	13	5	**5**	**0.0%**	12	15	**30**	**0.0%**
BDR THERMEA	DDR AMERICAS	0	0	**5**	**0.0%**	0	0	**0**	**0.0%**	0	0	**5**	**0.0%**
SPX	PATTERSON-KELLEY	20	25	**0**	**0.0%**	0	0	**0**	**0.0%**	20	25	**0**	**0.0%**
Others	Others	6,100	5,765	**5,195**	**3.7%**	1,621	1,785	**1,720**	**2.3%**	4,479	3,980	**3,475**	**5.3%**
TOTAL		178,300	178,800	**141,000**	**100.0%**	99,000	94,700	**75,200**	**100.0%**	79,300	84,100	**65,800**	**100.0%**
% of Combi and Heating Only Boilers						55.5%	53.0%	**53.3%**		44.5%	47.0%	**46.7%**	

Source: BRG based on industry interviews

1.5.6 Market Shares : Residential vs Commercial Condensing Boilers

US - ESTIMATED MARKET SHARES FOR BOILERS: TOTAL GAS CONDENSING (RESIDENTIAL vs COMMERCIAL)

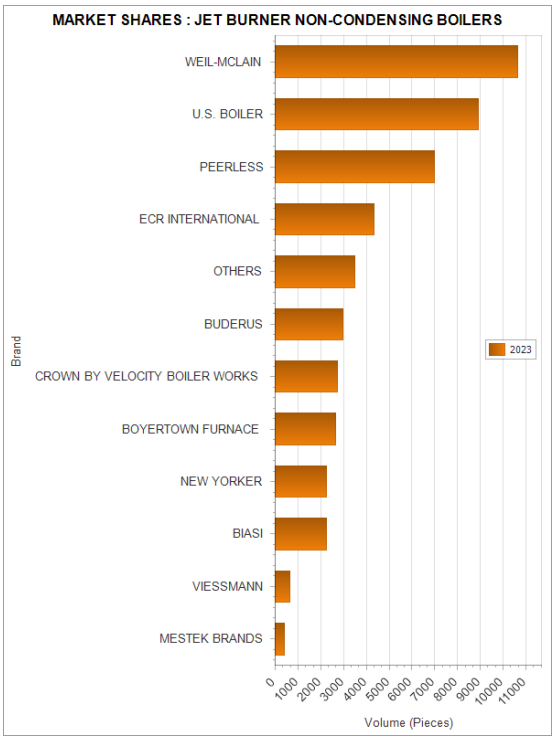
Holding Group	Company/Brand	Residential (<300k BTU) Units				Commercial (≥300k BTU) Units				Total Units				Of Which Mounted on:			
		2021	2022	2023	2023 Share	2021	2022	2023	2023 Share	2021	2022	2023	2023 Share	Wall (2023)	%	Floor (2023)	%
KD NAVIEN	*NAVIEN*	68,885	58,180	**47,560**	***30.1%***	760	1,410	**960**	***3.5%***	69,645	59,590	**48,520**	***26.2%***	48,520	***34.4%***	-	***0.0%***
AO SMITH	*LOCHINVAR*	21,580	22,550	**14,400**	***9.1%***	10,700	12,650	**10,900**	***40.2%***	32,280	35,200	**25,300**	***13.7%***	9,900	***7.0%***	15,400	***34.8%***
PALOMA	*IBC TECHNOLOGIES*	13,315	18,600	**14,955**	***9.5%***	320	400	**325**	***1.2%***	13,635	19,000	**15,280**	***8.3%***	15,100	***10.7%***	180	***0.4%***
SPX	*WEIL-MCLAIN*	12,270	12,890	**10,050**	***6.4%***	930	1,070	**1,055**	***3.9%***	13,200	13,960	**11,105**	***6.0%***	3,615	***2.6%***	7,490	***16.9%***
BRADFORD WHITE	*LAARS/BRADFORD WHITE*	7,520	8,930	**9,805**	***6.2%***	1,050	1,085	**1,115**	***4.1%***	8,570	10,015	**10,920**	***5.9%***	5,920	***4.2%***	5,000	***11.3%***
VIESSMANN	*VIESSMANN*	12,235	14,535	**8,930**	***5.6%***	1,065	1,220	**1,135**	***4.2%***	13,300	15,755	**10,065**	***5.4%***	9,025	***6.4%***	1,040	***2.4%***
ARISTON GROUP	*NTI*	10,360	10,795	**8,090**	***5.1%***	840	1,175	**645**	***2.4%***	11,200	11,970	**8,735**	***4.7%***	8,440	***6.0%***	295	***0.7%***
BURNHAM	*U.S. BOILER*	5,120	6,380	**6,920**	***4.4%***	325	290	**280**	***1.0%***	5,445	6,670	**7,200**	***3.9%***	6,740	***4.8%***	460	***1.0%***
ATLANTIC	*TRIANGLE TUBE*	8,345	8,740	**6,900**	***4.4%***	255	260	**200**	***0.7%***	8,600	9,000	**7,100**	***3.8%***	6,650	***4.7%***	450	***1.0%***
RINNAI	*RINNAI*	7,465	9,760	**6,705**	***4.2%***	-	-	**-**	***0.0%***	7,465	9,760	**6,705**	***3.6%***	6,705	***4.8%***	-	***0.0%***
BOSCH	*BOSCH*	7,600	7,425	**6,215**	***3.9%***	70	55	**30**	***0.1%***	7,670	7,480	**6,245**	***3.4%***	3,665	***2.6%***	2,580	***5.8%***
ARISTON GROUP	*HTP*	5,510	4,315	**3,285**	***2.1%***	570	630	**390**	***1.4%***	6,080	4,945	**3,675**	***2.0%***	3,160	***2.2%***	515	***1.2%***
NORITZ	*PEERLESS*	2,370	2,300	**1,980**	***1.3%***	130	135	**120**	***0.4%***	2,500	2,435	**2,100**	***1.1%***	1,500	***1.1%***	600	***1.4%***
TERRAVEST INDUSTRIES	*ECR INTERNATIONAL*	3,665	2,720	**1,895**	***1.2%***	40	55	**40**	***0.1%***	3,705	2,775	**1,935**	***1.0%***	1,450	***1.0%***	485	***1.1%***
WATTS WATER TECHNOLOGIES	*AERCO*	-	-	**-**	***0.0%***	2,200	2,250	**1,900**	***7.0%***	2,200	2,250	**1,900**	***1.0%***	-	***0.0%***	1,900	***4.3%***
BURNHAM	*CROWN BY VELOCITY BOILER WORKS*	1,345	1,450	**1,590**	***1.0%***	60	135	**125**	***0.5%***	1,405	1,585	**1,715**	***0.9%***	1,560	***1.1%***	155	***0.4%***
NORITZ	*NORITZ*	2,340	1,465	**1,140**	***0.7%***	-	-	**-**	***0.0%***	2,340	1,465	**1,140**	***0.6%***	1,140	***0.8%***	-	***0.0%***
PALOMA	*RAYPAK*	30	15	**15**	***0.0%***	1,300	1,300	**1,100**	***4.1%***	1,330	1,315	**1,115**	***0.6%***	140	***0.1%***	975	***2.2%***
THE FULTON COMPANIES	*FULTON*	-	-	**20**	***0.0%***	1,010	1,075	**1,050**	***3.9%***	1,010	1,075	**1,070**	***0.6%***	-	***0.0%***	1,070	***2.4%***
GROUPE ATLANTIC	*IDEAL*	1,070	1,510	**870**	***0.6%***	130	140	**80**	***0.3%***	1,200	1,650	**950**	***0.5%***	650	***0.5%***	300	***0.7%***
CARRIER GLOBAL	*RIELLO*	5	190	**85**	***0.1%***	750	830	**685**	***2.5%***	755	1,020	**770**	***0.4%***	85	***0.1%***	685	***1.5%***
HARBOUR GROUP	*CLEAVER BROOKS*	-	-	**-**	***0.0%***	680	725	**675**	***2.5%***	680	725	**675**	***0.4%***	-	***0.0%***	675	***1.5%***
HARBOUR GROUP	*CAMUS HYDRONICS*	30	20	**-**	***0.0%***	485	600	**655**	***2.4%***	515	620	**655**	***0.4%***	35	***0.0%***	620	***1.4%***
SPX	*PATTERSON-KELLEY*	-	-	**-**	***0.0%***	590	555	**630**	***2.3%***	590	555	**630**	***0.3%***	-	***0.0%***	630	***1.4%***
DAESUNG	*VESTA*	480	350	**550**	***0.3%***	-	-	**-**	***0.0%***	480	350	**550**	***0.3%***	550	***0.4%***	-	***0.0%***
BURNHAM	*THERMAL SOLUTIONS*	-	-	**-**	***0.0%***	650	525	**540**	***2.0%***	650	525	**540**	***0.3%***	-	***0.0%***	540	***1.2%***
AO SMITH	*AO SMITH*	730	875	**490**	***0.3%***	-	-	**-**	***0.0%***	730	875	**490**	***0.3%***	400	***0.3%***	90	***0.2%***
MESTEK	*ADVANCED THERMAL HYDRONICS*	25	45	**30**	***0.0%***	575	740	**420**	***1.5%***	600	785	**450**	***0.2%***	-	***0.0%***	450	***1.0%***
MESTEK	*RBI*	-	-	**-**	***0.0%***	400	525	**370**	***1.4%***	400	525	**370**	***0.2%***	-	***0.0%***	370	***0.8%***
ARMSTRONG INTERNATIONAL	*HAMILTON ENGINEERING*	50	35	**40**	***0.0%***	250	200	**240**	***0.9%***	300	235	**280**	***0.2%***	250	***0.2%***	30	***0.1%***
BDR THERMEA	*BAXI*	260	420	**260**	***0.2%***	15	10	**10**	***0.0%***	275	430	**270**	***0.1%***	270	***0.2%***	-	***0.0%***
MESTEK	*EMBASSY INDUSTRIES*	50	200	**170**	***0.1%***	-	-	**-**	***0.0%***	50	200	**170**	***0.1%***	170	***0.1%***	-	***0.0%***
ALDRICH	*ALDRICH*	-	-	**-**	***0.0%***	50	75	**100**	***0.4%***	50	75	**100**	***0.1%***	-	***0.0%***	100	***0.2%***
BURNHAM	*BRYAN STEAM*	-	-	**-**	***0.0%***	120	55	**90**	***0.3%***	120	55	**90**	***0.0%***	-	***0.0%***	90	***0.2%***
BOSCH	*BUDERUS*	80	90	**20**	***0.0%***	90	105	**60**	***0.2%***	170	195	**80**	***0.0%***	80	***0.1%***	-	***0.0%***
ENERGY KINETICS	*ENERGY KINETICS*	105	110	**80**	***0.1%***	-	-	**-**	***0.0%***	105	110	**80**	***0.0%***	80	***0.1%***	-	***0.0%***
GASMASTER INDUSTRIES	*GASMASTER INDUSTRIES*	-	-	**5**	***0.0%***	50	50	**40**	***0.1%***	50	50	**45**	***0.0%***	-	***0.0%***	45	***0.1%***
PARKER BOILER	*PARKER BOILER*	-	-	**-**	***0.0%***	25	25	**20**	***0.1%***	25	25	**20**	***0.0%***	-	***0.0%***	20	***0.0%***
SUPERIOR BOILER	*SUPERIOR BOILER WORKS*	-	-	**-**	***0.0%***	-	3	**5**	***0.0%***	-	3	**5**	***0.0%***	-	***0.0%***	5	***0.0%***
ARMSTRONG INTERNATIONAL	*ARMSTRONG*	-	-	**-**	***0.0%***	-	-	**5**	***0.0%***	-	-	**5**	***0.0%***	-	***0.0%***	5	***0.0%***
BDR THERMEA	*DDR AMERICAS*	-	-	**5**	***0.0%***	-	-	**-**	***0.0%***	-	-	**5**	***0.0%***	5	***0.0%***	-	***0.0%***
NUDYNE GROUP	*ACE HEATERS*	-	-	**-**	***0.0%***	12	-	**-**	***0.0%***	12	-	**-**	***0.0%***	-	***0.0%***	-	***0.0%***
Others	*Others*	5,060	5,505	**5,040**	***3.2%***	2,203	1,342	**1,105**	***4.1%***	7,263	6,847	**6,145**	***3.3%***	5,195	***3.7%***	950	***2.1%***
TOTAL		**197,900**	**200,400**	**158,100**	***100.0%***	**28,700**	**31,700**	**27,100**	***100.0%***	**226,600**	**232,100**	**185,200**	***100.0%***	**141,000**	***100.0%***	**44,200**	***100.0%***

Source: BRG based on industry interviews

1.5.7 Market Shares : Oil Non-Condensing

US - ESTIMATED SHARES : OIL (JET BURNER) NON-CONDENSING BOILERS

	2023		2022	
Holding Company (Brand)	Shares	% of total	Shares	% of total
SPX (WEIL-MCLAIN)	10 620	21.99%	15 055	28.14%
BURNHAM (U.S.BOILER)	8 900	18.43%	7 725	14.44%
NORITZ (PEERLESS)	7 000	14.49%	7 400	13.83%
TERRAVEST INDUSTRIES (ECR INTERNATIONAL)	4 325	8.95%	4 620	8.64%
OTHERS (OTHERS)	3 505	7.26%	3 565	6.66%
BOSCH (BUDERUS)	2 990	6.19%	3 265	6.10%
BURNHAM (CROWN BY VELOCITY BOILER WORKS)	2 730	5.65%	2 860	5.35%
BOYERTOWN FURNACE COMPANY (BOYERTOWN FURNACE)	2 650	5.49%	2 730	5.10%
BURNHAM (NEW YORKER)	2 250	4.66%	2 540	4.75%
BIASI (BIASI)	2 250	4.66%	2 300	4.30%
CARRIER GLOBAL (VIESSMANN)	660	1.37%	990	1.85%
MESTEK (MESTEK BRANDS)	420	0.87%	450	0.84%
Grand Total	48 300	100.00%	53 500	100.00%



Source: BRG based on industry interviews

1.6.1 US - IMPORTS OF BOILERS 2023

Source Country	2023				2022			
	Volume		Value 000		Volume		Value 000	
	Pieces	%	USD	%	Pieces	%	USD	%
Korea, Republic of	62 440	31.2%	74 541	22.8%	91 741	35.5%	101 709	26.5%
Canada	60 541	30.2%	110 836	33.8%	58 366	22.6%	111 738	29.1%
Germany	30 011	15.0%	37 936	11.6%	34 466	13.3%	49 942	13.0%
Japan	12 793	6.4%	10 512	3.2%	14 310	5.5%	13 120	3.4%
Netherlands	9 227	4.6%	8 215	2.5%	15 822	6.1%	14 605	3.8%
United Kingdom	7 595	3.8%	10 506	3.2%	8 904	3.4%	13 383	3.5%
Italy	6 183	3.1%	19 481	5.9%	15 246	5.9%	27 992	7.3%
France	2 622	1.3%	1 777	0.5%	1 056	0.4%	2 303	0.6%
Belgium	2 328	1.2%	5 804	1.8%	6 334	2.5%	11 232	2.9%
China	2 045	1.0%	15 295	4.7%	1 891	0.7%	11 900	3.1%
OTHERS	4 516	2.3%	32 555	9.9%	10 089	3.9%	25 476	6.6%
Total	**200 301**	**100.0%**	**327 458**	**100.0%**	**258 225**	**100.0%**	**383 400**	**100.0%**

Source : Trademap

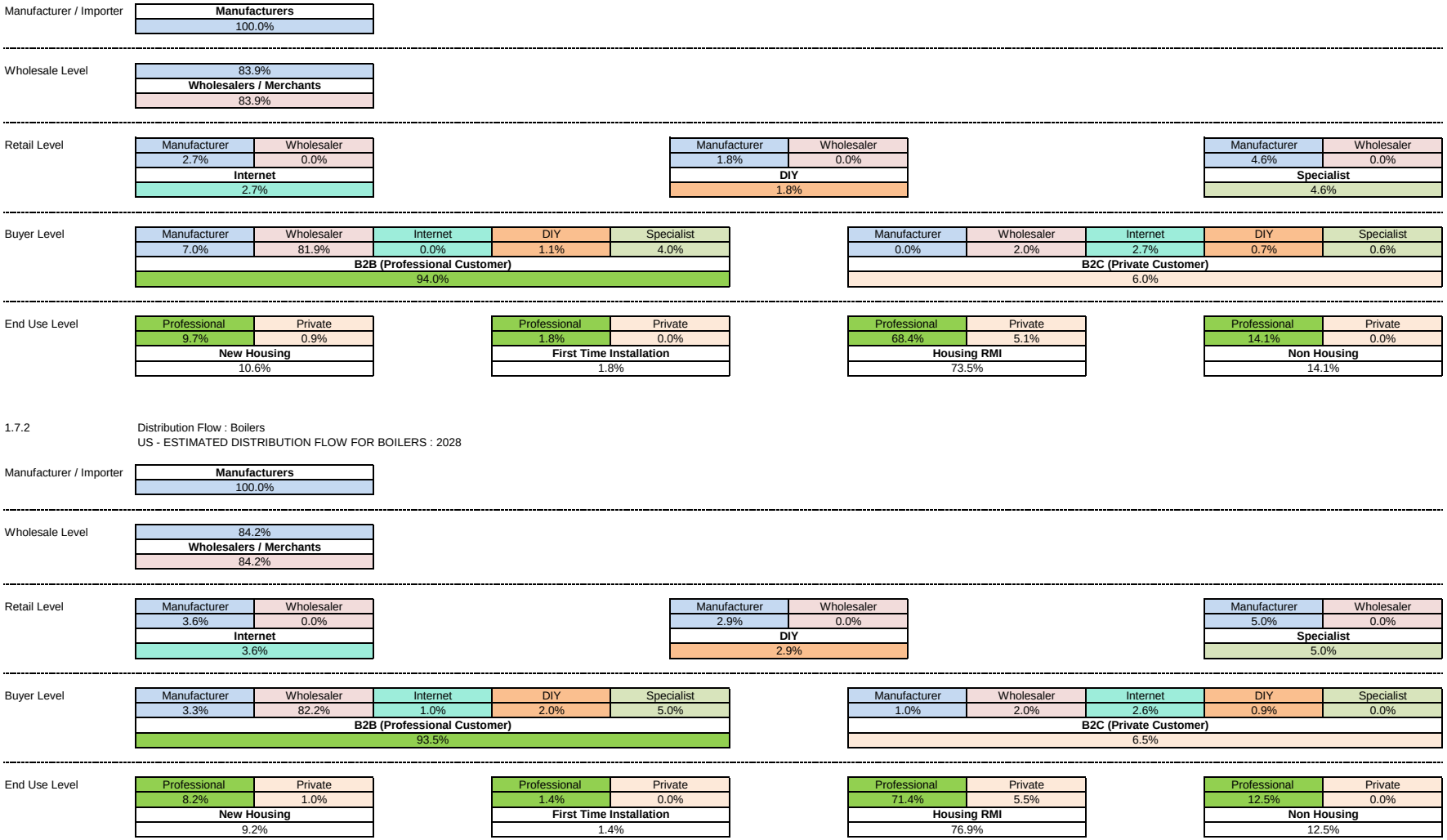
1.6.2 US - EXPORTS OF BOILERS 2023

Destination	2023				2022			
	Volume		Value 000		Volume		Value 000	
	Pieces	%	USD	%	Pieces	%	USD	%
Canada	22 763	89.7%	74 662	88.7%	33 590	92.0%	94 264	90.2%
Mexico	808	3.2%	2 506	3.0%	688	1.9%	1 172	1.1%
Korea, Republic of	440	1.7%	2 069	2.5%	378	1.0%	1 476	1.4%
OTHERS	1 376	5.4%	4 975	5.9%	1 860	5.1%	7 639	7.3%
Total	**25 387**	**100.0%**	**84 212**	**100.0%**	**36 516**	**100.0%**	**104 551**	**100.0%**

Source : Trademap

1.7.1 Distribution Flow : Boilers

US - ESTIMATED DISTRIBUTION FLOW FOR BOILERS : 2023



1.7.2 Distribution Flow : Boilers

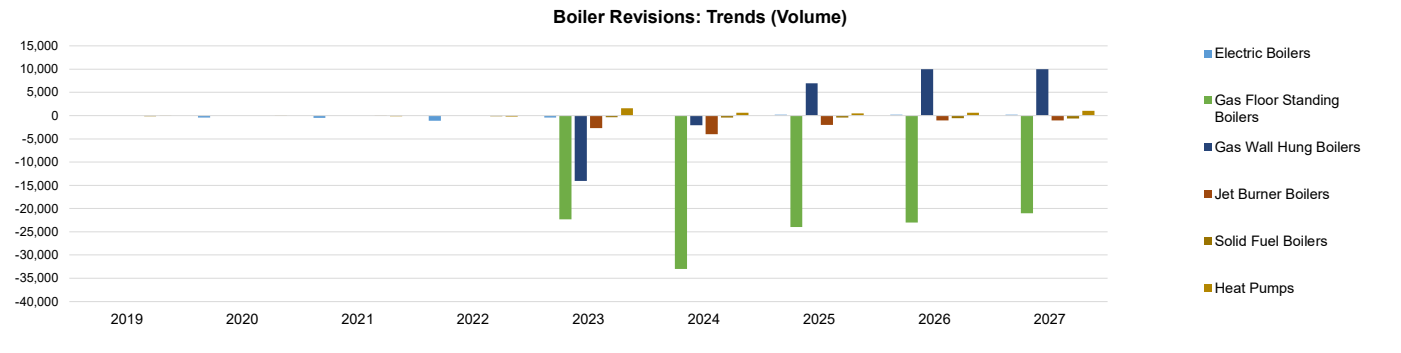
US - ESTIMATED DISTRIBUTION FLOW FOR BOILERS : 2028

Source: BRG based on Industry Interviews.

1.8 Revisions in the US Boiler Market (May 2024)

US - REVISIONS IN THE US BOILER MARKET

Data Comparison: May 2023



Product Category	Alt Product	Latest Data	2019	2020	2021	2022	2023	2024	2025	2026	2027
Electric Boilers	Electric	*Vol*	9,700	9,600	10,500	10,500	11,300	12,000	12,600	13,300	14,000
Electric Boilers	Electric	*Δ Change*	*0*	*-400*	*-500*	*-1,100*	*-400*	*0*	*200*	*200*	*200*
Gas Floor Standing Boilers	Floor Standing Gas Cond	*Vol*	47,300	41,500	48,300	53,300	44,200	48,000	52,000	56,000	59,000
Gas Floor Standing Boilers	Floor Standing Gas Cond	*Δ Change*	*0*	*0*	*0*	*0*	*-5,800*	*-1,000*	*-1,000*	*-1,000*	*-2,000*
Gas Floor Standing Boilers	Floor Standing Gas Non Cond	*Vol*	160,000	131,000	148,000	158,000	134,500	115,000	120,000	118,000	116,000
Gas Floor Standing Boilers	Floor Standing Gas Non Cond	*Δ Change*	*0*	*0*	*0*	*0*	*-16,500*	*-32,000*	*-23,000*	*-22,000*	*-19,000*
Gas Wall Hung Boilers	Wall Hung Gas Cond	*Vol*	132,600	139,800	178,300	178,800	141,000	160,000	180,000	200,000	220,000
Gas Wall Hung Boilers	Wall Hung Gas Cond	*Δ Change*	*0*	*0*	*0*	*0*	*-14,000*	*-2,000*	*7,000*	*10,000*	*10,000*
Gas Wall Hung Boilers	Wall Hung Gas Non Cond	*Vol*	1,000	700	350	300	200	190	180	170	160
Gas Wall Hung Boilers	Wall Hung Gas Non Cond	*Δ Change*	*0*	*0*	*0*	*0*	*-80*	*-85*	*-70*	*-50*	*-40*
Jet Burner Boilers	Jet Burner Cond	*Vol*	150	130	110	100	80	80	80	80	80
Jet Burner Boilers	Jet Burner Cond	*Δ Change*	*0*	*0*	*0*	*0*	*-20*	*-20*	*-20*	*-20*	*-20*
Jet Burner Boilers	Jet Burner Non Cond	*Vol*	65,000	58,600	63,000	53,500	48,300	43,000	40,000	35,000	30,000
Jet Burner Boilers	Jet Burner Non Cond	*Δ Change*	*0*	*0*	*0*	*0*	*-2,700*	*-4,000*	*-2,000*	*-1,000*	*-1,000*
Solid Fuel Boilers	Fossil Fuel / Universal	*Vol*	40	45	45	75	60	50	40	30	25
Solid Fuel Boilers	Fossil Fuel / Universal	*Δ Change*	*0*	*0*	*5*	*0*	*0*	*0*	*0*	*0*	*-5*
Solid Fuel Boilers	Logwood / Bi-Fuel	*Vol*	550	500	450	410	315	305	300	290	280
Solid Fuel Boilers	Logwood / Bi-Fuel	*Δ Change*	*-110*	*0*	*-100*	*-170*	*-245*	*-285*	*-320*	*-360*	*-400*
Solid Fuel Boilers	Other Solid Fuel	*Vol*	525	450	400	420	350	340	330	320	310
Solid Fuel Boilers	Other Solid Fuel	*Δ Change*	*0*	*0*	*0*	*0*	*-70*	*-60*	*-50*	*-50*	*-60*
Solid Fuel Boilers	Pellet	*Vol*	200	140	135	130	110	100	95	90	85
Solid Fuel Boilers	Pellet	*Δ Change*	*0*	*0*	*15*	*5*	*-25*	*-50*	*-85*	*-120*	*-155*
Solid Fuel Boilers	Woodchip	*Vol*	20	10	10	10	10	10	10	10	10
Solid Fuel Boilers	Woodchip	*Δ Change*	*0*	*0*	*0*	*0*	*0*	*0*	*0*	*0*	*0*
Heat Pumps	Heat Pumps	*Vol*	74,425	62,415	65,320	68,450	72,950	75,800	80,500	85,000	90,000
Heat Pumps	Heat Pumps	*Δ Change*	*-75*	*-85*	*-130*	*-200*	*1,550*	*600*	*500*	*600*	*1,000*
Total	**Total**	**Vol**	**491,510**	**444,890**	**514,920**	**523,995**	**453,375**	**454,875**	**486,135**	**508,290**	**529,950**
Total	**Total**	***Δ Change***	***-185***	***-485***	***-710***	***-1,465***	***-38,290***	***-38,900***	***-18,845***	***-13,800***	***-11,480***

The total market size for electric boilers from 2020 to 2022 was adjusted downward due to updated data obtained from an important market player.

The total market size for logwood boilers was revised downwards for 2019, 2021 and 2022 due to updated sales data provided by an important market player.



USA

WATER HEATERS

THE NORTH AMERICAN HEATING & COOLING PRODUCT MARKETS

BRG BUILDING SOLUTIONS

Market Characteristics and Technological Trends

Historically, the US market has been dominated by dedicated tank-type (storage) water heaters, a product category that has had little competition from other water heater types in the US. The tankless water heater and combi boiler markets, although growing, remain smaller in comparison. The market share of electric vs. gas tanks is almost equal. In the past, any shift from one to the other was considered minor. Some regions of the country, namely the South, prefer electric over gas, while other areas, like the West, largely favor gas over electric. The US water heater market is mature. Every home in the country has a water heater of some sort, so market fluctuations are closely tied to the overall health of the economy, but particularly the state of the consumer and housing markets.

Manufacturers have been required to comply with the U.S. Department of Energy's (DOE) energy conservation standards for consumer water heaters since 1990. In the past, the DOE focused on safety regulations. However, the market is currently in an efficiency cycle, with legislation and technological advancements driving a rapid shift towards energy-efficient products. Condensing gas storage, tankless, and HP water heaters are expected to be the technologies of the future, although growth has not been as fast as expected. Stricter water heater efficiency standards that took effect in April 2015 have started a shift in the market towards high-efficiency units.

As part of the National Appliance and Energy Conservation Act (NAECA), the DOE made its new energy efficiency mandates effective on April 16, 2015 (known as NAECA 3). For almost all residential water heaters, NAECA requires higher energy efficiencies, as measured by EF ratings. This reduces home appliance energy consumption, conserves more natural resources, and generates significant energy savings for homeowners. The NAECA 3 standards required nearly all electric, oil, and gas water heaters to carry higher Energy Factor (EF) ratings. Electric and gas water heaters were grouped into two size categories: 20 to 55 gallons and 55 gallons and larger. NAECA also affects some light-duty commercial water heaters. These new efficiencies are boosting sales of condensing water heaters, tankless water heaters, and HP water heaters.

In July 2023, the US Department of Energy (DOE) proposed new energy efficiency standards for water heaters that aim to save American homeowners more than $11 billion on annual utility bills. It was determined that the amended energy conservation standards would result in Americans saving roughly $200 billion and reducing 501 million metric tons of harmful carbon dioxide emissions cumulatively over 30 years.

Under this final rule, the DOE is finalizing standards for all consumer water heaters, except for gas-fired instantaneous water heaters. The proposal requires the most common-sized electric water heaters to achieve efficiency gains with heat pump technology. It requires higher efficiency standards for electric storage water heaters

between 20 and 120 gallons and gas-fired storage water heaters between 20 and 100 gallons. The target effective date of the rule is July 5, 2024, with a compliance date set for May 6, 2029.

Compliance flexibilities may be available in some cases; the Energy Policy and Conservation Act (EPCA) states that a manufacturer whose annual gross revenue from all operations does not exceed $8 million may apply for full or partial exemption for a period no longer than 24 months after the effective date of a final rule. In addition, manufacturers subject to the DOE's energy efficiency standards may apply to the DOE's Office of Hearings and Appeals for exception relief under certain circumstances.

In August 2023, the US Department of Energy announced new efficiency standards for commercial water heaters to conserve energy and water while cutting harmful carbon pollution. The DOE is adopting a performance standard that will require condensing technology for new models of commercial gas-fired storage, instantaneous, and hot water supply boilers; the standards are set to start in 2026.

Although the uptake of heat pump water heaters was initially slow, sales are now improving and expected to take off with the new government regulations. Manufacturers are offering more models as an alternative to a standard storage water heater.

ENERGY STAR water heaters have created more product and efficiency awareness in the market. It is still an endorsement that customers look for, as they associate it with a 'good purchase'. However, the payback period on qualifying tankless gas water heaters is long. When consumers are concerned about saving cash, such as during economic contractions, they will prioritize a lower up-front cost. In May of 2024, the EPA revealed the launch of ENERGY STAR NextGen Certification for new homes and apartments. It is a voluntary certification program for the US residential new construction sector that aims to increase national energy and emissions savings by accelerating the building industry's adoption of advanced, energy efficient technologies. The certification will recognize homes and apartments with increased energy efficiency through reduced on-site emissions from heat pumps and heat pump water heaters.

Water heater storage capacity is measured in US gallons (1 US gallon = 3.785 liters), ranging from 2.5 to over 119-gallon units. Over time, the share of LPG-fueled water heaters versus natural gas has reportedly remained static, with electric-to-LPG shifts (and LPG-to-natural gas) being limited.

Gas residential storage water heaters have a lifespan of 12–15 years (the life expectancy of water heaters varies regionally due to water quality). In parts of Texas with very hard water, the lifecycle is reduced by 20% compared to the average. Only a small percentage of water heaters sold in the US are mechanically vented (including power vents and power-assisted direct vents). The rest are products with traditional, atmospheric, or direct venting. Concentric vents are already used with tankless gas water heaters, and they are expected to become

standard for atmospheric venting in storage tank water heaters. Sales are most important in the Midwest, followed by the Pacific West, Mid-Atlantic, Southwest, and Southeast. Most condensing residential storage water heaters are sold to high-end homes.

The commercial electric storage market is more evenly spread across the country; however, the Southeast is estimated to be the largest region in terms of sales. The average life expectancy of commercial storage water heaters is estimated at 10 years (electric lifespans are slightly longer than gas). Based on the type of construction, both fuels can be found in the same building; for instance, a commercial gas-fired water heater can supply water on the top 4 floors of a building with a penthouse, while the lower 4 floors will be supplied by an electric water heater located in the basement (for economics of construction, it is more economical for builders to use gas fired in concentrated areas). The commercial gas condensing market is expected to overtake its non-condensing counterpart during the forecast period. Restaurants, for example, require a different temperature of water to kill bacteria. The vast majority (>90%) are made of glass.

The bulk of the gas tankless market is comprised of indoor units. Most of the outdoor tankless water heater market is estimated to be in the West (CA, AZ, NV, etc.), due to the climate and because it frees up living space (e.g., as real estate in California is expensive). The life expectancy of tankless water heaters is estimated at 12 to 20 years. Unlike storage water heaters, where units need to be replaced when there is a leak or hole, tankless water heaters can be repaired, so it is much harder to determine the exact lifespan of the product. In the condensing part of the market, the most popular BTU is 199,000 BTU.

Residential electric water heaters have a life expectancy of 13–17 years. Water heaters with a 2.5- to 6-gallon storage capacity are referred to as'mini-tanks’. In terms of popularity, most electric storage water heater sales are between 35 and 55 gallons (132.5 and 208.2 liters). Sales tend to be most important in the Southeast, followed by the Midwest, Southwest, Mid-Atlantic, and Pacific West. The vast majority (>90%) are made of glass line.

Point-of-use (<13 kW) electric tankless water heaters are used when the hot water requirement comes from only one sink, one shower, or where water is needed somewhere but not everywhere. Single-point-of-use models are sold throughout the United States. Applications include warehouses, factory floors, fire stations, and churches. Outputs used in the market for whole-house units (providing hot water to the whole house) range between 13 and 36 kW, with the trend being towards larger outputs. Another small category within the electric instantaneous/tankless market is the one for commercial and industrial units (>36 kW). Commercial units consume a lot of electricity, but efforts are being made to make them more efficient.

Water heating boilers (also referred to as hot water supply boilers), commercial HP water heaters, semi-instantaneous, and commercial indirect water heaters are covered in the BRG North American Commercial Boiler and Water Heater Report.

Most indirect water heaters are connected to a gas or oil boiler, have a stainless-steel heat exchanger, and range between 20 and 80 gallons. The indirect water heater market is losing share to combi boilers.

Recent Market Trends

After the market correction of 2022, the US water heater market experienced a mid-single digit increase in volume for 2023. While 2022 was marked by an overall economic slowdown, dwindling demand, and high inventories from unusually high sales in pandemic years 2020 and 2021, 2023 proved to be resilient against market constraints. The US water heater market has seen an increase in sales three out of the past four years, despite numerous challenges. Non-condensing residential and commercial gas storage water heater sales underwent mid-single digit growth in 2023. These increases can be attributed to the looming ban on non-condensing technologies, which pushes manufacturers to make the product while they still can and customers to buy it before it is gone.

Data from the BLS Consumer Price Index shows that inflation fell from 6.5% in 2022 to a 3.4% in 2023. The US Federal Reserve voted to raise the interest rate paid on reserve balances in March 2023. This decision saw the Federal Funds Effective Rate rise from 4.1% in December 2022 to 5.3% in December 2023. In 2023, the Federal Reserve announced a series of interest rate hikes. By the end of the year, these hikes had led to an increase in benchmark borrowing costs to their highest level in over 22 years, between 5.25% and 5.5%, making all forms of consumer credit more expensive. High interest rates are expected to remain stagnant in the 5.25%-5.50% range for the first half of 2024, which may impede growth.

Compared to the strong decrease in the 2022 market, the dedicated water heater market flourished in 2023 due to a combination of factors. Sales volume in this market can often be closely linked to prices, legislation, and overall economic health. Inflation falling 3.1% from the previous year, was a significant factor in water heater growth where the average cost of a residential product ranges roughly between $400 and $1,700 depending on the product, while the average cost of a commercial product is roughly between $2,200 and $35,000. Higher efficiency standards were also driving growth for specific products like heat pump water heaters and condensing water heaters. Although the US water heater market is a predominately mature market, many homeowners jumped at the opportunity to buy a water heater in 2023 while price increases levelled out, incentives were up, and legislation changes were coming to the forefront.

Linked water heaters saw a decline for the second year in a row, with an overall double-digit decline in 2023, in line with the drop seen in the boiler market due to unusually high levels of inventories at the distributor level.

In states with milder climates, HP water heaters are becoming more popular. It seems to be the technology of the future, with electrification and energy efficiency

standards as a tailwind, although most customers have not made the switch yet due to the large price tag. So far, customers have preferred to install a smaller storage water heater or a tankless water heater. This technology still only represents just over 1% of the US water heater market, but it is growing at a faster pace than all other residential water heating products, seeing growth every year since 2017.

HP water heaters are the most exciting product in the water heating market due to their high efficiency and environmentally friendly features. In 2023, HP water heater sales experienced strong double digit growth once again due to increased awareness from customers, a push by manufacturers, and available incentives from local and state governments as well as utility companies in selected parts of the country. In December 2022, Congress passed the Inflation Reduction Act, making federal income tax credits available for ENERGY STAR-certified tankless water heater, heating boiler, and HP water heater purchases in 2023. A provision in this new law gave customers up to a $2,000 federal tax credit and a $1,750 rebate based on income, starting January 1, 2023. Additionally, the updated Title 24 Building Energy Efficiency Standards in California continue to encourage contractors to install electric HP water heaters at the expense of gas products.

Sales of condensing gas tankless models are now almost double than that of their non-condensing counterparts. Non-condensing models are typically favored in the milder climate of the southern states, in like-for-like replacement, or with price-conscious consumers such as builders looking for the most cost-effective solution.

Different energy initiatives remain in place in several states, for instance PACE (Property Assessed Clean Energy) programs (www.pacenation.us), which help property owners finance renewable energy and energy efficiency improvements on residential and commercial properties, thus acting as a market driver in the water heater market (the program can, for instance, finance water heater upgrades). Differences are seen in each region, state, and even at the metropolitan area level.

Prices and Market Values

In 2023, overall prices for water heaters remained somewhat stable compared to the previous year. In 2022, each individual water heater had seen a price increase of at least 15%. In 2023, however, the highest price increase remained in the low-single digits. Heat pump water heaters were the only product that decreased in price due to growing retail sales.

These small rises were a welcome relief after the record-high price increases experienced in 2022. They can be attributed to a multitude of factors, including a cooling down of the economy, lower freight and container costs, and supply chain improvement.

New Housing

It is estimated that there is just slightly more than one water heater per dwelling (1 water heater per multi-family housing unit and around 1.2 water heaters per single-family unit). Single-family housing continues to be the largest sector where instantaneous/tankless water heaters are installed, primarily for retrofit purposes but increasingly for new construction as well, especially on the gas non-condensing side, where they are preferred (over more expensive condensing models) by builders. The new construction market continues to develop, and tankless water heaters over tanks are growing in some areas, with homeowners seeing them as more worthwhile and the price easily included with their mortgage. The tankless water heater's location in the house (installed outdoors in some states), as well as the limited amount of space the unit takes up, appeal to many contractors and homeowners. In the southern states, mid-efficiency products are often preferred because the payback period is shorter than that of a high-efficiency water heater.

First-Time Installation

In the United States, the penetration of water heaters is very high. In fact, only an estimated 0.4 million (or 0.35% of) housing units do not use hot water, so the share of first-time installation sales is negligible in the US.

Housing RMI

The market's largest share of water heaters is on the replacement/retrofit side. Historically, around 80% of sales were for RMI (replacement, maintenance, and improvement), both for tank and tankless models. Normally, some estimated 25–30% of replacements are proactive, but in times of economic slowdown, this percentage goes down considerably as homeowners have less disposable income.

In 2023, replacement sales increased by mid-single digits overall, in part due to the stabilization of water heater prices. After a difficult year in 2022, the reduction of inflationary pressure made consumers more willing to spend their discretionary income on water heaters.

The number of condensing residential gas storage water heaters sold for replacement is currently relatively small due to low consumer awareness. Typically, with emergency replacements, there is no time to contemplate the installation of a condensing residential storage water heater, as this entails a venting upgrade, among other things.

Oil-water heaters are used for like-for-like replacement. Installations in the replacement segment continued to decrease in 2023, making it the fourth consecutive year the market has seen a double-digit decrease in installations.

Non-Housing

Electric tankless units and electric storage (tank-type) water heaters dominate the non-housing market in terms of sales. Sales in the electric tankless sector reflect the high proportion of ‘point-of-use’ units (<12 kW), commonly installed in factories,

manufacturing plants, warehouses, churches, fire stations, or any other commercial place where one sink or shower with hot water is needed.

The commercial tankless gas market, if referred to as 200,000 BTU and over, is estimated to account for less than 1% of the market. However, the threshold for commercial water heaters is not clear-cut because light commercial applications will often have two smaller water heaters.

Supply Structure

Over 85% of the overall water heater market remains in the hands of three large manufacturing groups: AO SMITH, PALOMA (RHEEM's parent company), and BRADFORD WHITE.

In 2021, HISENSE HOME APPLIANCE GROUP, a subsidiary of HISENSE GROUP, acquired 75% of SANDEN GROUP. Furthermore, SANDEN exited the North American HP water heater market during 2020. In early 2021, a new company called ECO2 SYSTEMS was created to sell the SANCO2 HP water heaters, sourced from CHOFU.

HTP began operating as a brand of parent company ARISTON THERMO USA in mid-2020, following the acquisition of HTP by ARISTON THERMO GROUP in 2017. HTP offers stainless steel electric water heaters as opposed to the more commonly found glass-lined models. Stainless steel electric water heaters are preferred in some pockets of the country.

GE APPLIANCES launched its Air &Water Solutions division in January 2021. The company's portfolio is made up of over 350 products across 15 HVAC categories, including water heaters. GE continues to sell standard electric tanks sourced from parent company HAIER in China, initially distributed mostly through retail. It launched its 'flexible capacity' electric tanks (which have a mixing valve on top and act like an 80-gallon tank) in sizes of 50, 40, and 30g in April 2020. GE also offers tankless electric water heaters, but sales have remained negligible up until now. In January 2022, the company launched residential atmospheric gas tanks, which have been made in South Carolina since May 2022. GE exited the HP water heater market in 2016 due to disappointing sales, but the company now aims to re-enter the US HPWH market in the next couple of years. In early 2023, GE launched a power vent water heater, and it plans to launch the RealMAX Ultra-Low NOx atmospheric gas water heater in late 2024. GE is investing in innovation with its $60 million, 259,000 square foot, high volume water heater production facility in Camden, South Carolina. The plant opened in May 2022, and it was the first new tank-style water heater facility built in the US in over 30 years. This plant will manufacture RealMAX water heaters and applied products.

The French company ESSENCY, part of GT (GENERAL TRANSMISSION), launched an on-demand 55-gallon electric tank water heater (superheated water in the storage tank is utilized as a heat source for drawn water) in Texas in 2021. The company has since entered other states (FL, GA, AZ, and CO), and sales more

than doubled in 2023.

RINNAI makes and assembles residential and commercial water heating solutions in its new state-of-the-art manufacturing facility in Griffin, Georgia. Some of its models are still coming from its parent company in Japan. At AHR 2024, RINNAI unveiled the RX tankless condensing water heater with Bluetooth and adapter technology. RINNAI also released a heat pump water heater, the REHP Series, as well as its new RCB commercial boiler, in the first half of 2024.

NAVIEN announced in 2020 that it was in the process of establishing its first US manufacturing and assembly operation in Virginia. The 300,000-square-foot facility was set to be used as a distribution center for the first two years, and then began the process of transitioning into manufacturing operations in 2022. In early 2021, the company launched its NPE-2 series condensing tankless water heater, and in October 2023, it began shipping its NHW-A non-condensing tankless water heater with ComfortFlow. NAVIEN is expected to launch a heat pump water heater by the end of 2024.

In July 2017, ARISTON THERMO GROUP added HTP to its growing North America portfolio, having acquired NTI in 2016. ARISTON also acquired the Israeli electric tankless water heater manufacturer ATMOR in September 2017 and the Mexican water heater giant CALOREX from the conglomerate GRUPO INDUSTRIAL SALTILLO (GIS) in January 2019. In 2019, HTP began selling ATMOR-sourced electric tankless water heaters.

BRADFORD WHITE purchased the heat pump and water heater manufacturing assets of GE in 2017 and started manufacturing heat pump water heaters at its Michigan facility in 2018. BRADFORD WHITE also acquired KELTECH from BRADLEY CORPORATION in December 2020. More recently, BRADFORD WHITE moved its commercial tankless electric manufacturing from Delton, MI, to its LAARS manufacturing facility in Rochester, NH. In 2023, BRADFORD WHITE launched the Infinity G circulating and non-circulating tankless water heater models.

VESTA (owned by DAESUNG) entered the US market for tankless and hybrid water heaters in 2017 as well as Canada in 2021.

In 2017, STIEBEL ELTRON acquired TANKLESS INC. TANKLESS INC. is a Florida-based company that manufactures commercial and industrial units designed to work with trench showers. The TANKLESS INC. (TI) name disappeared during 2019, and all products are now branded STIEBEL ELTRON.

AO SMITH acquired SEISCO in 2018 and has been selling electric tankless water heaters under AO SMITH in retail and wholesale since 2019. In 2021, the company acquired GIANT FACTORIES, a Canada-based manufacturer of residential and commercial water heaters. AO SMITH is in the process of phasing out its TAKAGI tankless water heaters and launched its new ADAPT condensing tankless water heater in the first half of 2024. The ADAPT water heaters are currently made in

China, but production is expected to be transferred to Mexico in 2025.

In 2022, HUBBELL, VAUGHN, RECO, and ACE HEATERS' parent company, HEH HOLDINGS, was renamed THE NUDYNE GROUP.

TRIANGLE TUBE launched its MONARCH indirect water heater in October 2023, as well as the EXCALIBUR wall-hung and floor-standing boilers at AHR Expo 2024, which are used for both space and water heating. According to reports, the EXCALIBUR line was sourced from ECO HEATING SYSTEMS in the Netherlands and shipped to the UK first.

BOSCH is expected to release the GREENTRONIC, a heat pump water heater, in the summer of 2024.

In April 2023, CARRIER GLOBAL CORP. announced it would buy the German indirect tank supplier (and manufacturer of other heating equipment) VIESSMANN in a 12-billion-euro ($13.17 billion) cash-and-stock deal. By January 2024, the acquisition was completed.

Forecasts

The headlines for 2024 will continue to be higher-for-longer interest rates, which are likely to remain in the 5.25%–5.50% range for at least the first six months of the year and then possibly decline to 4.6%, which is still considerably higher than pre-pandemic records. Due to monetary tightening, inflation levels have moderately recovered, although they remain higher than the 2% target rate. Nevertheless, the high interest rate environment continues to hinder investment activity and slow growth.

Residential investment is one of the most affected sectors. Multifamily completions for 2024 are already showing debilitating signs as construction starts declined 14% in 2023. High-rise construction is projected to continue the downhill trajectory with a 20% decrease in 2024, according to the National Association of Home Builders. High costs are not the only challenge delaying projects for residential construction; the constant lack of skilled labor keeps debilitating the potential growth. According to the NAHB, the industry is short more than 400,000 workers, an issue that will only exacerbate once home builders resume the pace to catch up with the shortage of houses in the country.

The high cost of borrowing and the constant increase in a home’s sales price continue to discourage people from taking out a mortgage. According to the National Association of Realtors, the average sales price for a single-family home increased by 42% from the end of 2019 to the end of 2023.

Non-residential construction relies on loans from smaller regional banks, which have tightened their credit conditions because of the high interest rates and current challenges facing the banking sector. Investors in the non-housing sector appear to be pushing the breaks, as they face a sustained increase in development loans,

tightening credit conditions, and higher construction costs. According to the American Institute of Architects, overall non-residential construction is expected to have more modest single-digit growth in the next two years.

Meanwhile, remodeling activity is likely to have another difficult year. As specified by the Leading Indicator of Remodeling Activity (LIRA), expenses in repair and remodeling projects for owner-occupied homes are expected to decrease in 2024 and then rebound slightly in the second quarter of 2025.

The market for water heaters is predicted to grow at a slow and steady pace. The economic situation, global geopolitical tensions, high interest rates, and a decline in private consumption may prevent the market from experiencing higher growth. Overall sales of residential heaters are expected to increase slightly in 2024.

Sales of commercial water heaters are expected to be slightly higher than residential products as there continues to be a backlog of orders from prior years. HP water heaters are expected to experience another year of double-digit percentage growth thanks to government incentives and increased customer awareness about the products.

2025 is predicted to be a consistent year for water heater sales, and the market is expected to grow by a single-digit percentage. After that, the market should continue to experience single-digit percentage growth, fueled by new efficiency regulations and government incentives for more efficient products such as condensing gas products and HP water heaters.

Recently, the term "electrification" has taken center stage in the HVAC industry. Because more people are becoming aware that electric appliances use less energy and produce fewer carbon emissions than gas-based heating systems - even when the electricity is generated using fossil fuels - some parts of the United States are witnessing a shift in attitudes toward electricity as a source of energy. Numerous states have adopted beneficial incentives.

In contrast to the rest of the world, most of the United States has been sluggish to embrace electrification. However, more states are embracing electrification; California is among the nation's top states in this regard. Contractors in California will have substantial incentives to select electric heating products over gas-fired ones for both new residential and light commercial construction, according to the new Title 24 Building Energy Efficiency Standards. For new buildings, electric heating appliances are therefore the preferred option. Most of these new rules mandate that all new residential and business structures be electrified. Nonetheless, a lot of communities do grant exceptions for specific kinds of structures, such as commercial kitchens, affordable housing complexes, and labs.

Another notable example is Washington, which was the first state to require all-electric space and water heating systems to be installed in most newly constructed commercial buildings and big multifamily homes. Future sales of HP water heaters are anticipated to benefit from these modifications to building codes.

Due to electrification projects that are sprouting up across the nation, a gradual move toward more electric products in the residential water heater market is anticipated in the medium to long term. Many prefer gas installations, whereas commercial solutions are more complex. The US ban on non-condensing commercial water heaters is scheduled to take effect in 2026, as announced by the DOE in July 2023. As a result, BRG expects that these items will disappear from the market starting in 2027.

On the residential side, it is anticipated that non-condensing tankless technology will be phased out of the market beginning in 2029, with water heating heat pumps essentially taking the place of electric storage water heaters.

Condensing gas water heaters that satisfy the federal requirements for a residential unit are still rare in the North American market. These units have been used for years in the commercial sector. The domestic market should continue to recognize and welcome new products.

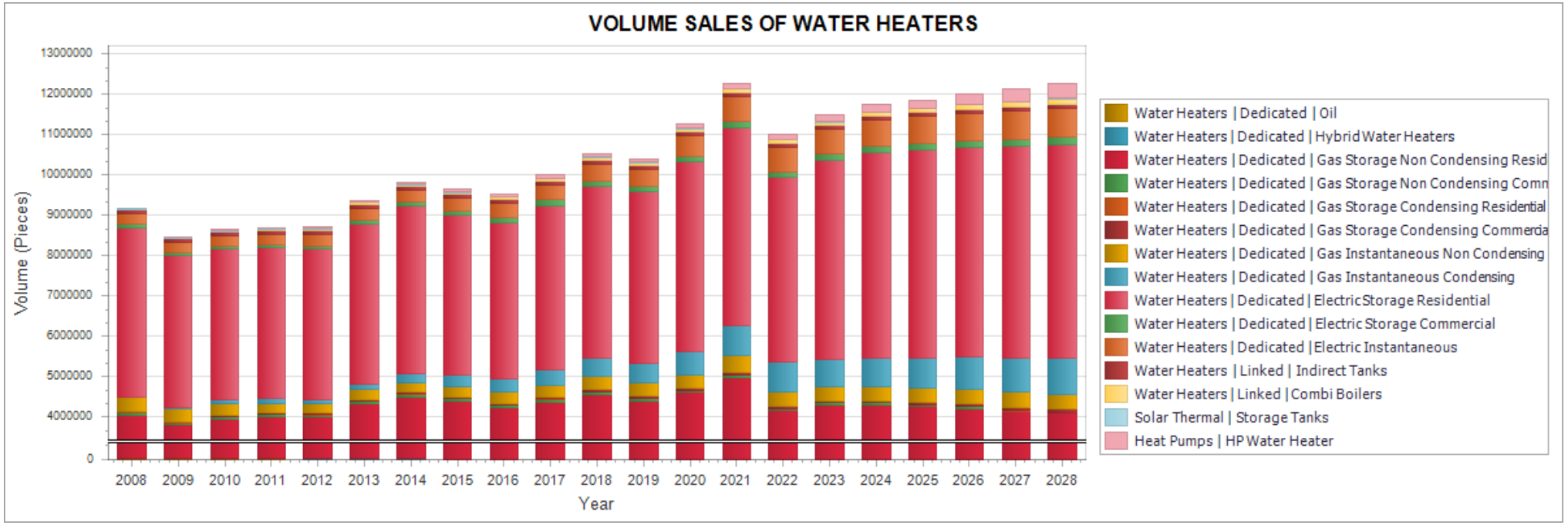
2023 saw the publication of an update to the National Appliance Energy Conservation Act (NAECA). The new NAECA 4 regulations require higher minimum efficiencies, which many believe will drastically alter the market's makeup by favoring heat pump and condensing gas water heater shipments over traditional electric tanks. In the next five years, it is expected that the mix of water heaters will continue to shift in favor of more energy-efficient models.

The trade typically adopts new technologies slowly. The main cause of the delayed launch is the installation requirements related to retrofitting a condensing appliance. Despite obstacles like higher startup costs and condensate drainage when replacing, these two technologies outperform standard-efficiency appliances in terms of long-term return on investment. High-efficiency technology is part of the product lines of most large manufacturers. These high-efficiency technologies should be pushed further in new build, where installation is very simple, and homeowners may factor the additional cost of a high-efficiency unit into their mortgage. During the projection period, opportunities to accelerate HP water heater sales are expected to become increasingly apparent.

In general, low single-digit growth is anticipated in the US water heater market in 2024, with modest growth continuing throughout the forecast period. As in many other businesses, the state of the economy is a major factor contributing to the strength of the HVAC and water heating industries, as it is in a variety of other businesses. Going forward, one of the main factors influencing the adoption rate of high-efficiency water heating equipment will continue to be the availability of financial incentives from federal, state, and utility sources for the purchase of such equipment.

3.2 Historical Trends and Forecasts : Water Heaters

US - SUMMARY OF TRENDS IN THE WATER HEATER MARKET : HISTORICAL AND FORECASTS

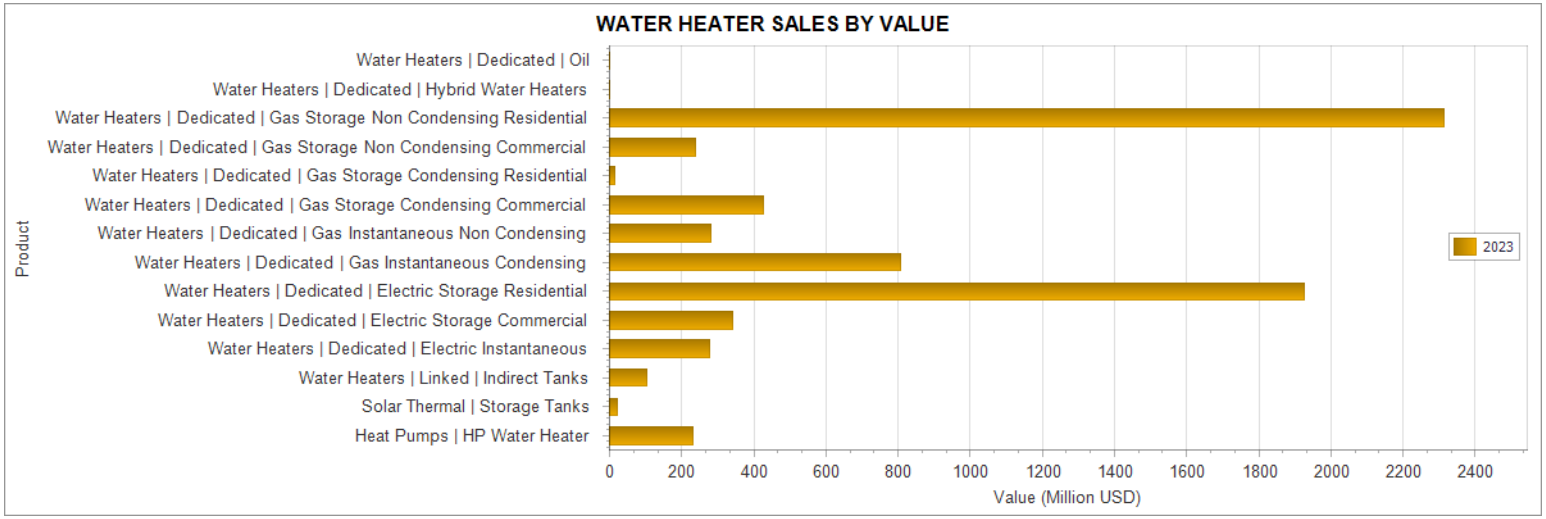


Product Type	Product Category	Product	Data	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028
Water Heaters	Dedicated	Oil		29 000	22 000	18 300	16 000	12 100	11 900	9 800	7 600	5 250	4 100	3 850	2 900	2 400	1 950	1 650	1 400	700	600	500	400	300
		Hybrid Water Heaters		900	2 800	11 000	14 700	15 300	11 500	9 200	1 000	2 050	3 000	3 200	1 850	2 200	1 075	1 400	550	500	500	500	500	500
		Gas Storage Non Condensing Residential		4 018 000	3 779 000	3 922 000	3 964 000	3 965 000	4 292 300	4 480 000	4 382 000	4 220 000	4 370 000	4 550 000	4 400 000	4 600 000	4 980 000	4 150 000	4 300 000	4 300 000	4 250 000	4 200 000	4 150 000	4 100 000
		Gas Storage Non Condensing Commercial		67 000	57 200	60 650	63 200	57 500	62 400	64 800	61 600	59 100	58 500	59 300	55 000	46 000	51 500	45 400	48 100	49 000	50 000	54 000		
		Gas Storage Condensing Residential		6 000	7 500	8 000	9 000	9 500	10 000	10 500	11 000	11 400	11 900	12 400	14 000	16 000	16 900	17 700	7 300	10 000	12 000	13 500	15 000	16 500
		Gas Storage Condensing Commercial		27 500	25 600	24 650	26 700	31 500	32 900	36 800	39 400	41 500	42 400	43 400	40 500	36 700	40 000	39 400	40 400	43 000	44 300	45 000	70 000	71 400
		Gas Instantaneous Non Condensing		338 000	300 000	291 400	250 000	235 000	260 000	247 000	256 000	273 000	288 000	320 000	322 000	351 000	416 000	368 000	357 000	359 000	363 000	368 000	372 000	380 000
		Gas Instantaneous Condensing		16 200	47 800	82 700	110 000	113 200	147 000	201 000	273 000	331 000	406 000	464 000	504 000	574 000	758 000	725 000	671 000	698 000	745 000	794 000	846 000	889 000
		Electric Storage Residential		4 196 000	3 760 000	3 745 000	3 747 000	3 740 000	3 968 000	4 181 000	3 972 000	3 885 000	4 055 000	4 265 000	4 234 000	4 686 000	4 913 000	4 600 000	4 950 000	5 100 000	5 150 000	5 200 000	5 250 000	5 300 000
		Electric Storage Commercial		69 000	56 500	59 000	61 100	68 000	70 000	74 350	88 800	127 500	152 500	139 000	152 000	142 000	155 500	121 200	149 300	160 000	165 000	170 000	175 000	170 000
		Electric Instantaneous		265 900	268 300	276 600	272 000	271 000	300 000	295 000	330 000	349 000	365 000	399 000	410 000	519 000	598 000	600 000	625 000	650 000	670 000	690 000	710 000	725 000
	Linked	Indirect Tanks		104 250	89 900	90 300	92 500	101 000	106 000	98 000	95 800	96 000	95 600	101 000	97 300	92 700	102 000	97 700	84 500	86 000	90 000	92 000	94 000	96 000
		Combi Boilers		8 500	7 100	13 000	20 400	31 900	42 500	49 100	52 400	55 500	61 000	74 800	78 100	78 900	101 000	96 900	77 000	91 500	104 800	118 500	132 400	140 700
Solar Thermal	Storage Tanks			26 000	25 000	26 300	26 500	21 750	17 500	14 000	10 200	9 400	10 150	9 300	7 400	7 400	6 600	6 700	6 100	5 700	5 400	5 250	5 000	4 700
Heat Pumps	HP Water Heater				5 000	12 200	23 000	34 000	39 700	46 900	55 600	52 000	72 000	67 000	85 000	105 000	113 000	141 000	160 000	180 000	200 000	240 000	300 000	380 000
Grand Total			Trends	9 172 250	8 453 700	8 641 100	8 696 100	8 706 750	9 371 700	9 817 450	9 636 400	9 517 700	9 995 150	10 511 250	10 404 050	11 259 300	12 254 525	11 012 050	11 477 650	11 733 400	11 850 600	11 991 250	12 120 300	12 274 100
			% change Total		-7.83%	2.22%	0.64%	0.12%	7.64%	4.76%	-1.84%	-1.23%	5.02%	5.16%	-1.02%	8.22%	8.84%	-10.14%	4.23%	2.23%	1.00%	1.19%	1.08%	1.27%

Source: BRG based on industry interviews

3.3.1 Prices and Market Values : Water Heaters

US - ESTIMATED VALUES OF THE WATER HEATER MARKET

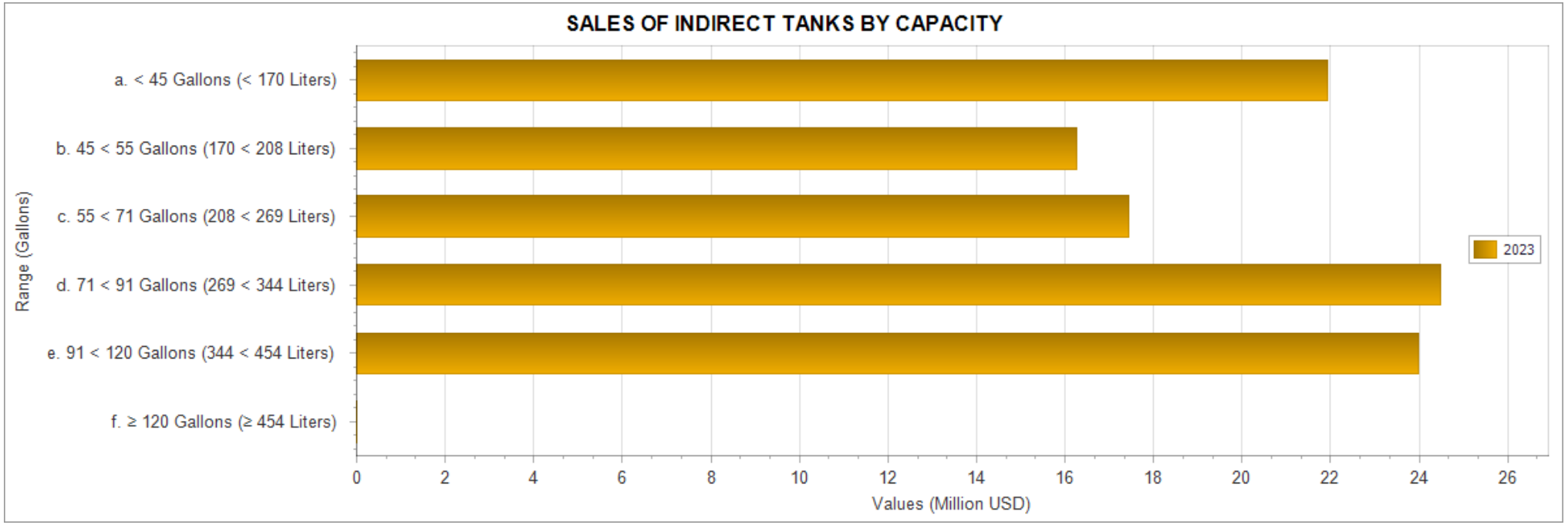


			2023			2022		
Product Type	Product Category	Product	Volumes	MSP (Local)	Value (MSP Local)	Volumes	MSP (Local)	Value (MSP Local)
Water Heaters	Dedicated	Oil	1 400	1,044.61	1.46	1 650	974.31	1.61
		Hybrid Water Heaters	550	1,799.34	0.99	1 400	1,705.00	2.39
		Gas Storage Non Condensing Residential	4 300 000	537.95	2 313.18	4 150 000	521.62	2 164.74
		Gas Storage Non Condensing Commercial	48 100	4,913.45	236.34	45 400	4,802.27	218.02
		Gas Storage Condensing Residential	7 300	1,797.53	13.12	17 700	1,744.64	30.88
		Gas Storage Condensing Commercial	40 400	10,560.04	426.63	39 400	10,323.50	406.75
		Gas Instantaneous Non Condensing	357 000	782.87	279.48	368 000	781.12	287.45
		Gas Instantaneous Condensing	671 000	1,204.05	807.92	725 000	1,199.22	869.43
		Electric Storage Residential	4 950 000	389.45	1 927.79	4 600 000	385.94	1 775.30
		Electric Storage Commercial	149 300	2,276.11	339.82	121 200	2,233.34	270.68
		Electric Instantaneous	625 000	446.07	278.80	600 000	439.92	263.95
	Linked	Indirect Tanks	84 500	1,231.94	104.10	97 700	1,228.30	120.00
Solar Thermal	Storage Tanks	Storage Tanks	6 100	3,453.83	21.07	6 700	3,300.00	22.11
Heat Pumps	HP Water Heater	HP Water Heater	160 000	1,439.10	230.26	141 000	1,466.56	206.78
Grand Total			11 400 650	612.33	6 980.95	10 915 150	608.34	6 640.11

Source: BRG based on industry interviews. Local Prices Shown in USD. Values Shown in Millions (at MSP = Manufacturers' Selling Price). Excludes Combi Boilers.

3.3.2 Prices & Market Values by Capacity: Indirect Tanks

US - ESTIMATED VALUE OF THE INDIRECT TANK MARKET BY CAPACITY (GALLONS)

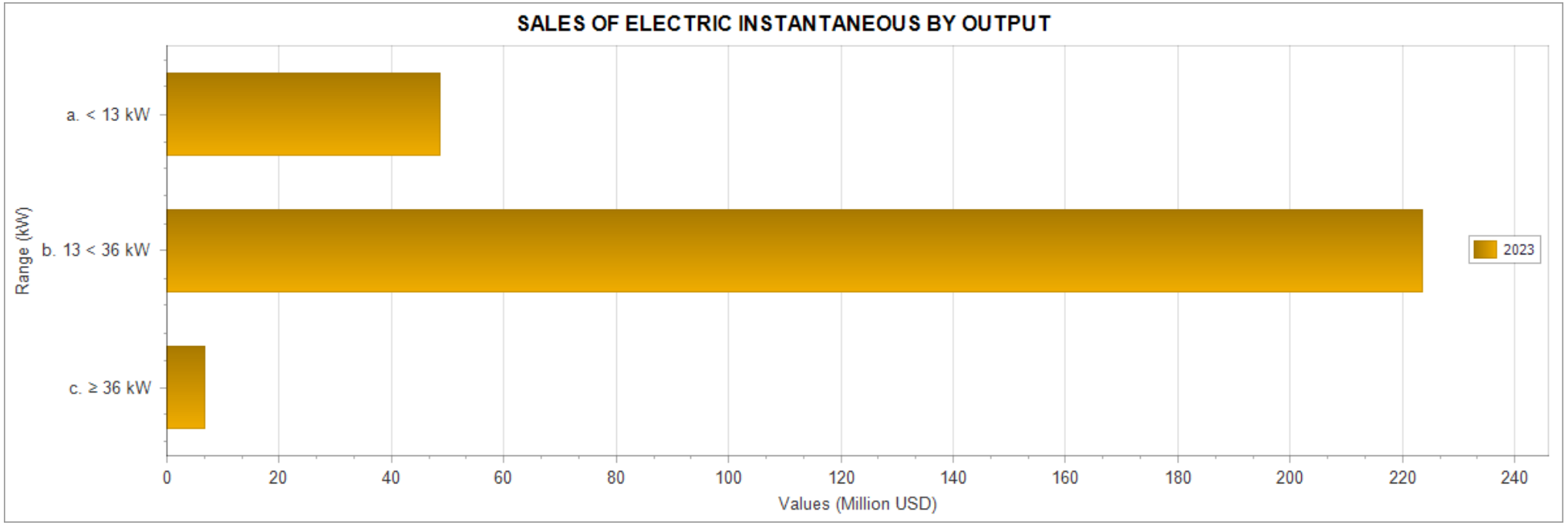


	2023					2022				
Category	Attributed Volume	% (Col Vol)	MSP (Local)	Value (MSP Local)	% (Col Val)	Attributed Volume	% (Col Vol)	MSP (Local)	Value (MSP Local)	% (Col Val)
a. < 45 Gallons (< 170 Liters)	25.35	30.00%	864.68	21.92	21.06%	29.31	30.00%	853.98	25.03	20.86%
b. 45 < 55 Gallons (170 < 208 Liters)	15.21	18.00%	1,069.61	16.27	15.63%	17.59	18.00%	1,060.05	18.64	15.53%
c. 55 < 71 Gallons (208 < 269 Liters)	14.37	17.00%	1,214.55	17.45	16.76%	16.61	17.00%	1,257.17	20.88	17.40%
d. 71 < 91 Gallons (269 < 344 Liters)	16.06	19.00%	1,524.44	24.47	23.51%	18.56	19.00%	1,507.82	27.99	23.32%
e. 91 < 120 Gallons (344 < 454 Liters)	13.52	16.00%	1,774.34	23.99	23.04%	15.63	16.00%	1,756.84	27.46	22.88%
f. ≥ 120 Gallons (≥ 454 Liters)	0.00	0.00%	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00%
Grand Total	84.50	100.00%	1,231.94	104.10	100.00%	97.70	100.00%	1,228.30	120.01	100.00%

Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Local Prices Shown in USD. Volume Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price)

3.3.3 Prices & Market Values by Output: Elec. Tankless

US - ESTIMATED VALUE OF THE ELECTRIC INSTANTANEOUS MARKET BY OUTPUT (KW)

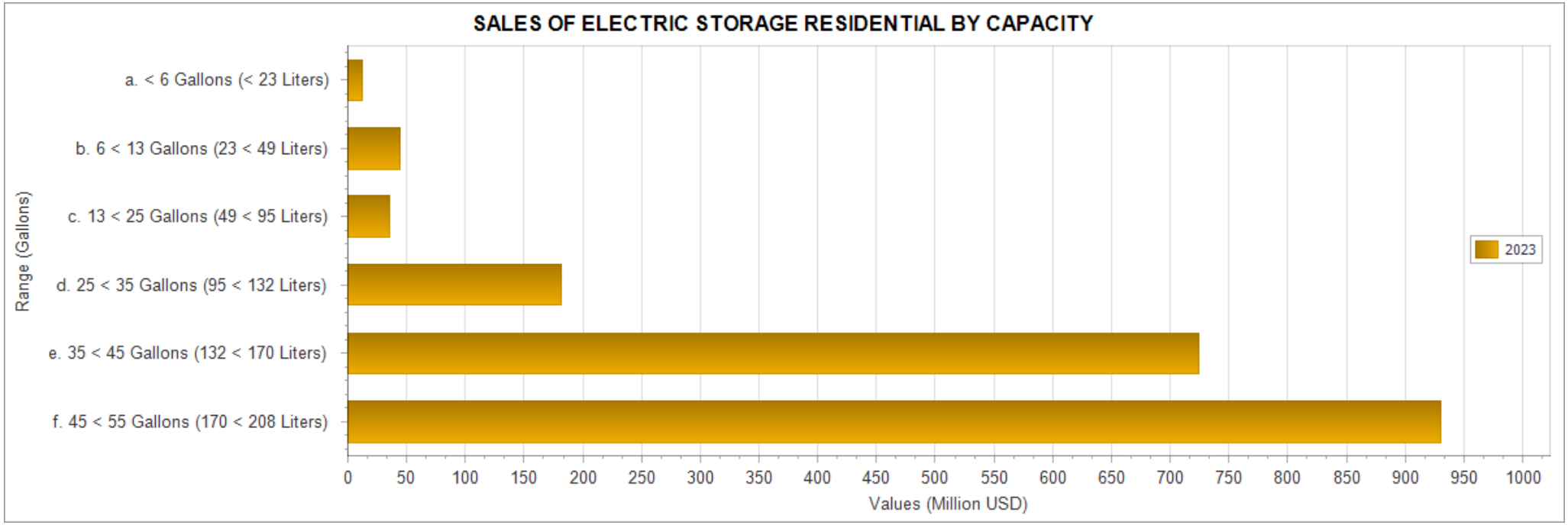


	2023					2022				
Category	Attributed Volume	% (Col Vol)	MSP (Local)	Value (MSP Local)	% (Col Val)	Attributed Volume	% (Col Vol)	MSP (Local)	Value (MSP Local)	% (Col Val)
a. < 13 kW	335.10	53.62%	144.95	48.57	17.42%	327.65	54.61%	145.09	47.54	18.01%
b. 13 < 36 kW	284.60	45.54%	785.71	223.61	80.21%	267.10	44.52%	785.71	209.86	79.51%
c. ≥ 36 kW	5.30	0.85%	1,247.54	6.61	2.37%	5.25	0.88%	1,248.11	6.55	2.48%
Grand Total	625.00	100.00%	446.07	278.80	100.00%	600.00	100.00%	439.93	263.96	100.00%

Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Local Prices Shown in USD. Volume Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price)

3.3.4 Prices & Market Values by Capacity: ESWH Res.

US - ESTIMATED VALUE OF THE ELECTRIC STORAGE RESIDENTIAL MARKET BY CAPACITY (GALLONS)

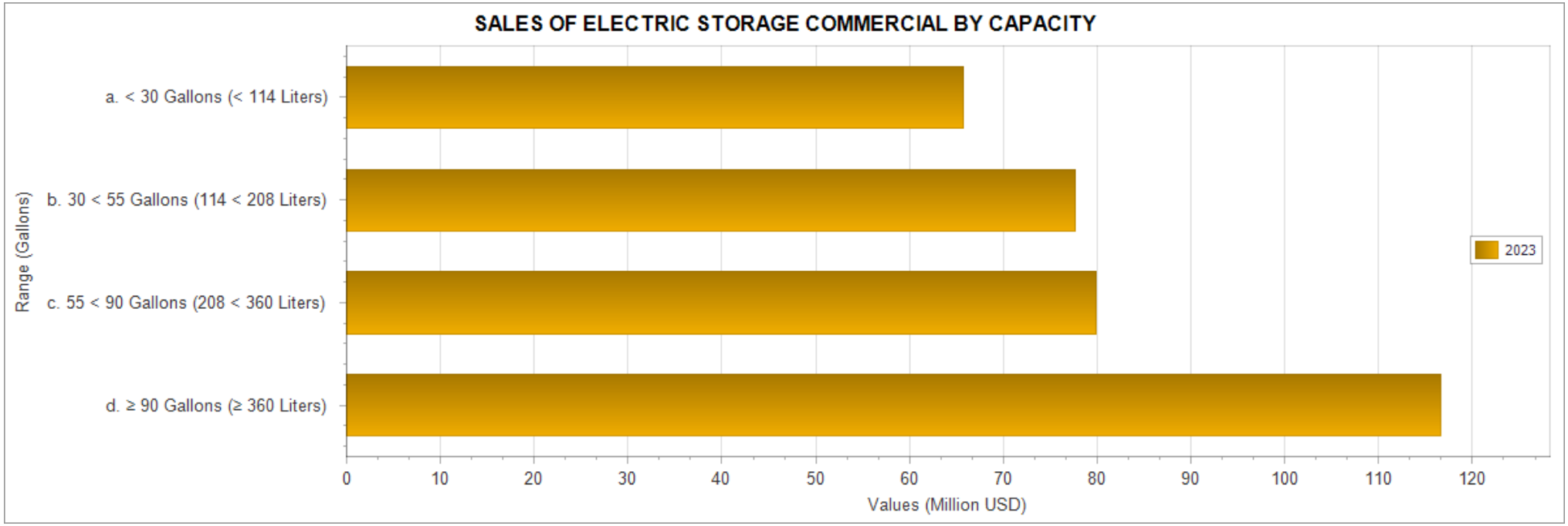


	2023					2022				
Category	Attributed Volume	% (Col Vol)	MSP (Local)	Value (MSP Local)	% (Col Val)	Attributed Volume	% (Col Vol)	MSP (Local)	Value (MSP Local)	% (Col Val)
a. < 6 Gallons (< 23 Liters)	99.00	2.00%	124.95	12.37	0.64%	92.00	2.00%	123.35	11.35	0.64%
b. 6 < 13 Gallons (23 < 49 Liters)	247.50	5.00%	179.93	44.53	2.31%	230.00	5.00%	182.08	41.88	2.36%
c. 13 < 25 Gallons (49 < 95 Liters)	148.50	3.00%	239.91	35.63	1.85%	138.00	3.00%	238.95	32.98	1.86%
d. 25 < 35 Gallons (95 < 132 Liters)	594.00	12.00%	304.89	181.10	9.39%	552.00	12.00%	298.28	164.65	9.27%
e. 35 < 45 Gallons (132 < 170 Liters)	1 881.00	38.00%	384.86	723.92	37.55%	1 748.00	38.00%	384.65	672.36	37.87%
f. 45 < 55 Gallons (170 < 208 Liters)	1 980.00	40.00%	469.83	930.26	48.25%	1 840.00	40.00%	463.09	852.08	48.00%
Grand Total	4 950.00	100.00%	389.46	1 927.81	100.00%	4 600.00	100.00%	385.93	1 775.29	100.00%

Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Local Prices Shown in USD. Volume Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price)

3.3.5 Prices & Market Values by Capacity: ESWH Com.

US - ESTIMATED VALUE OF THE ELECTRIC STORAGE COMMERCIAL MARKET BY CAPACITY (GALLONS)

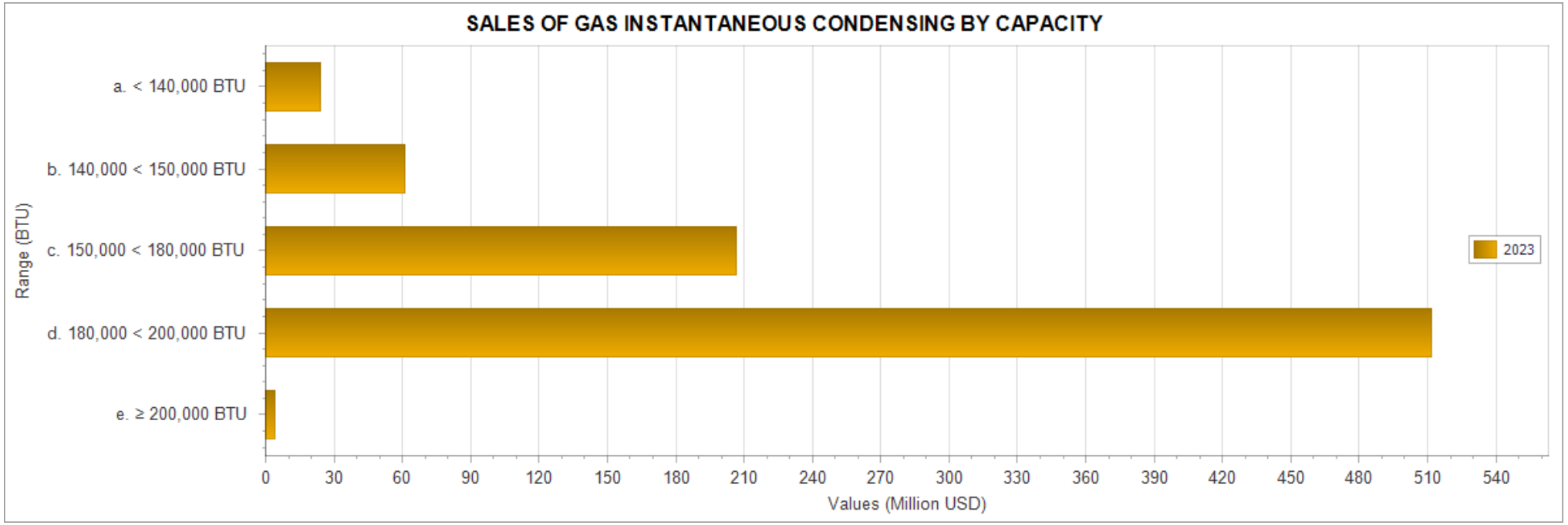


	2023					2022				
Category	Attributed Volume	% (Col Vol)	MSP (Local)	Value (MSP Local)	% (Col Val)	Attributed Volume	% (Col Vol)	MSP (Local)	Value (MSP Local)	% (Col Val)
a. < 30 Gallons (< 114 Liters)	43.30	29.00%	1,519.44	65.79	19.36%	35.15	29.00%	1,492.56	52.46	19.38%
b. 30 < 55 Gallons (114 < 208 Liters)	38.82	26.00%	1,999.26	77.61	22.84%	31.51	26.00%	1,966.94	61.98	22.90%
c. 55 < 90 Gallons (208 < 360 Liters)	28.37	19.00%	2,813.96	79.82	23.49%	23.03	19.00%	2,755.07	63.44	23.44%
d. ≥ 90 Gallons (≥ 360 Liters)	38.82	26.00%	3,003.89	116.60	34.31%	31.51	26.00%	2,944.72	92.79	34.28%
Grand Total	149.30	100.00%	2,276.11	339.82	100.00%	121.20	100.00%	2,233.34	270.68	100.00%

Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Local Prices Shown in USD. Volume Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price)

3.3.6 Prices & Market Values by Capacity: GIWH Cond

US - ESTIMATED VALUE OF THE GAS INSTANTANEOUS CONDENSING MARKET BY CAPACITY (BTU)

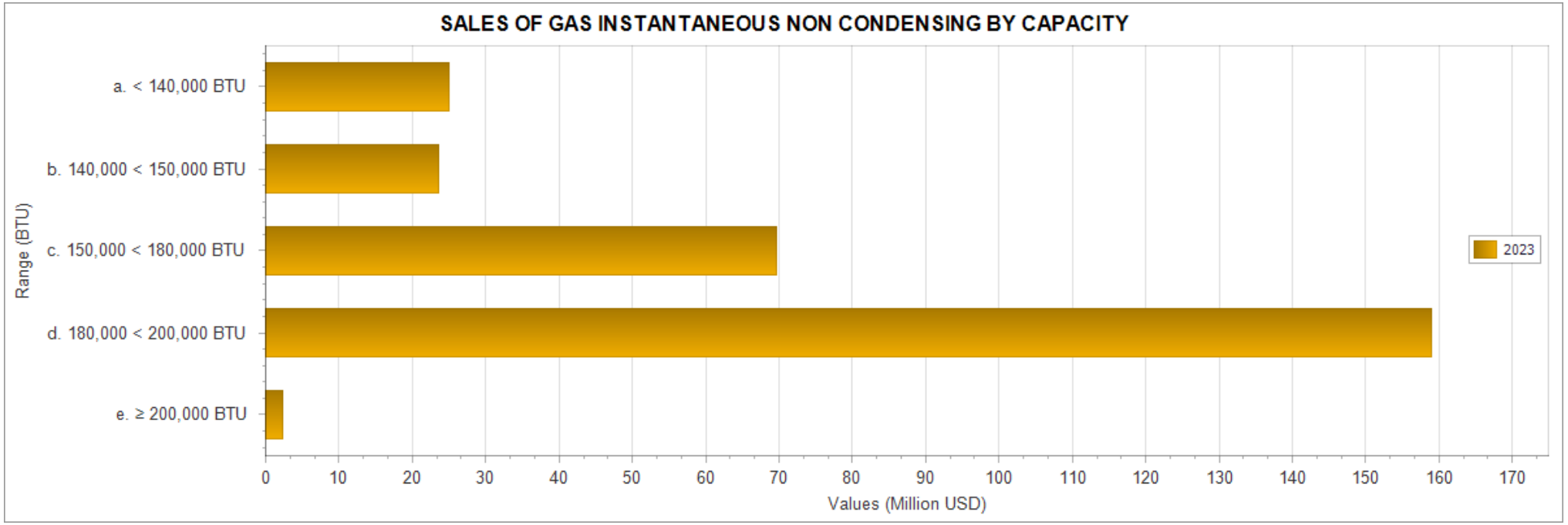


	2023					2022				
Category	Attributed Volume	% (Col Vol)	MSP (Local)	Value (MSP Local)	% (Col Val)	Attributed Volume	% (Col Vol)	MSP (Local)	Value (MSP Local)	% (Col Val)
a. < 140,000 BTU	33.55	5.00%	719.73	24.15	2.99%	36.25	5.00%	715.00	25.92	2.98%
b. 140,000 < 150,000 BTU	67.10	10.00%	909.66	61.04	7.56%	72.50	10.00%	920.00	66.70	7.67%
c. 150,000 < 180,000 BTU	197.01	29.36%	1,049.61	206.79	25.60%	212.87	29.36%	1,030.00	219.25	25.22%
d. 180,000 < 200,000 BTU	372.41	55.50%	1,374.49	511.87	63.36%	402.43	55.51%	1,375.00	553.34	63.64%
e. ≥ 200,000 BTU	0.91	0.14%	4,448.35	4.05	0.50%	0.95	0.13%	4,440.00	4.22	0.49%
Grand Total	670.98	100.00%	1,204.05	807.89	100.00%	725.00	100.00%	1,199.22	869.44	100.00%

Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Local Prices Shown in USD. Volume Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price)

3.3.7 Prices & Market Values by Capacity: GIWH N-Cond

US - ESTIMATED VALUE OF THE GAS INSTANTANEOUS NON CONDENSING MARKET BY CAPACITY (GALLONS)

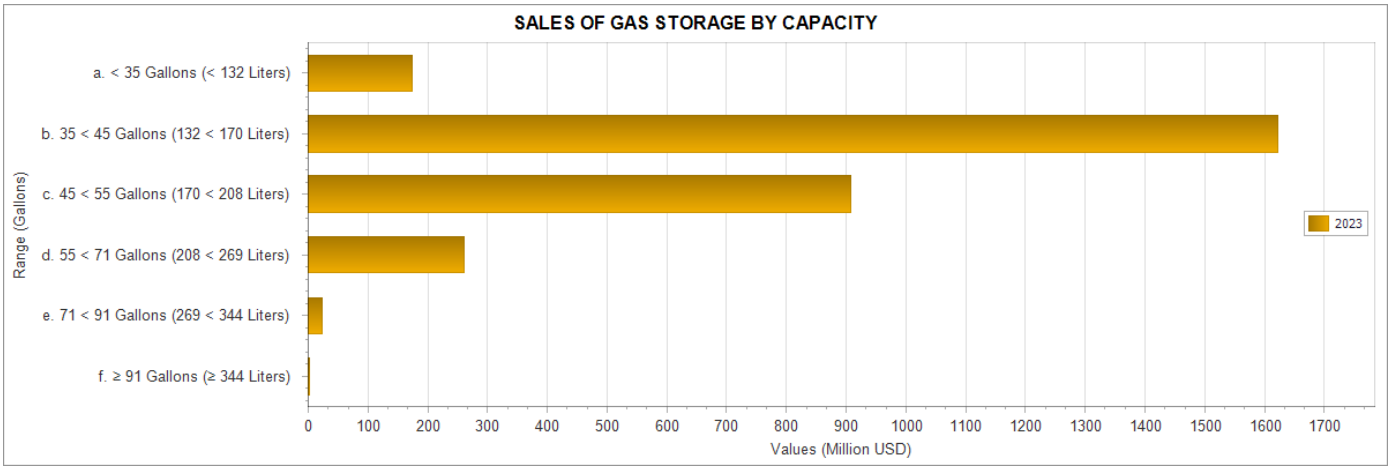


	2023					2022				
Category	Attributed Volume	% (Col Vol)	MSP (Local)	Value (MSP Local)	% (Col Val)	Attributed Volume	% (Col Vol)	MSP (Local)	Value (MSP Local)	% (Col Val)
a. < 140,000 BTU	46.41	13.00%	539.80	25.05	8.96%	49.68	13.50%	545.00	27.08	9.42%
b. 140,000 < 150,000 BTU	35.70	10.00%	659.76	23.55	8.43%	36.80	10.00%	653.00	24.03	8.36%
c. 150,000 < 180,000 BTU	89.25	25.00%	779.71	69.59	24.90%	92.00	25.00%	760.00	69.92	24.32%
d. 180,000 < 200,000 BTU	184.96	51.81%	859.68	159.01	56.89%	188.90	51.33%	870.00	164.34	57.17%
e. ≥ 200,000 BTU	0.68	0.19%	3,348.76	2.28	0.82%	0.63	0.17%	3,340.00	2.09	0.73%
Grand Total	357.00	100.00%	782.87	279.48	100.00%	368.00	100.00%	781.13	287.46	100.00%

Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Local Prices Shown in USD. Volume Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price)

3.3.8 Prices & Market Values by Capacity: Gas Storage

US - ESTIMATED VALUE OF THE GAS STORAGE MARKET BY CAPACITY (GALLONS)

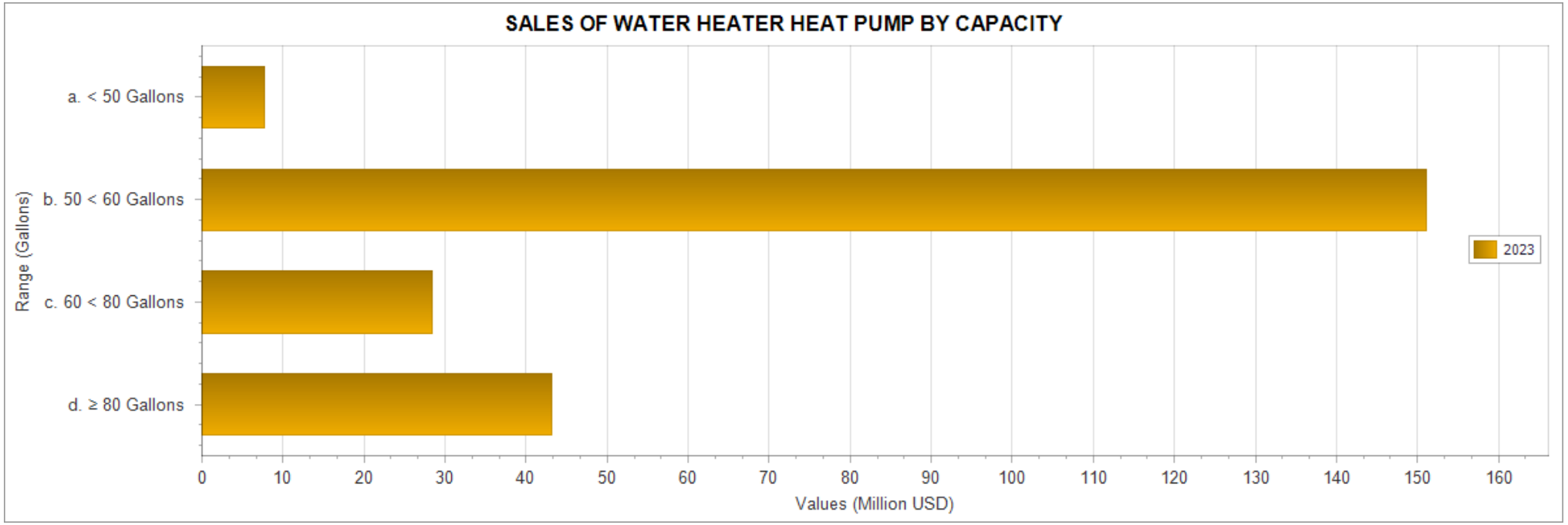


		2023					2022				
Product	Category	Attributed Volume	% (Col Vol)	MSP (Local)	Value (MSP Local)	% (Col Val)	Attributed Volume	% (Col Vol)	MSP (Local)	Value (MSP Local)	% (Col Val)
Gas Storage Condensing Commercial	a. < 35 Gallons (< 132 Liters)	6.46	16.00%	7,452.25	48.17	11.29%	6.30	16.00%	7,300.00	46.02	11.31%
	b. 35 < 45 Gallons (132 < 170 Liters)	25.45	63.00%	10,146.24	258.24	60.53%	24.82	63.00%	9,900.00	245.74	60.42%
	c. 45 < 55 Gallons (170 < 208 Liters)	3.84	9.50%	12,170.50	46.71	10.95%	3.74	9.50%	11,900.00	44.54	10.95%
	d. 55 < 71 Gallons (208 < 269 Liters)	4.04	10.00%	15,564.24	62.88	14.74%	3.94	10.00%	15,300.00	60.28	14.82%
	e. 71 < 91 Gallons (269 < 344 Liters)	0.61	1.50%	17,528.51	10.62	2.49%	0.59	1.50%	17,200.00	10.17	2.50%
	f. ≥ 91 Gallons (≥ 344 Liters)	0.00	0.00%	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00%
Gas Storage Condensing Commercial Total		40.40	0.92%	10,560.04	426.63	14.27%	39.40	0.93%	10,323.50	406.75	14.42%
Gas Storage Condensing Residential	a. < 35 Gallons (< 132 Liters)	0.00	0.00%	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00%
	b. 35 < 45 Gallons (132 < 170 Liters)	0.51	7.01%	1,419.47	0.73	5.54%	1.24	7.00%	1,376.81	1.71	5.53%
	c. 45 < 55 Gallons (170 < 208 Liters)	3.65	49.99%	1,674.38	6.11	46.56%	8.85	49.99%	1,627.79	14.41	46.65%
	d. 55 < 71 Gallons (208 < 269 Liters)	0.51	7.01%	1,709.37	0.88	6.67%	1.24	7.00%	1,660.04	2.06	6.67%
	e. 71 < 91 Gallons (269 < 344 Liters)	2.19	29.99%	2,024.25	4.43	33.78%	5.31	30.00%	1,961.56	10.42	33.73%
	f. ≥ 91 Gallons (≥ 344 Liters)	0.44	6.00%	2,234.17	0.98	7.46%	1.06	6.00%	2,161.70	2.30	7.43%
Gas Storage Condensing Residential Total		7.30	0.17%	1,797.47	13.13	0.44%	17.70	0.42%	1,744.62	30.88	1.09%
Gas Storage Non Condensing Commercial	a. < 35 Gallons (< 132 Liters)	6.98	14.50%	3,638.65	25.38	10.74%	6.77	14.90%	3,564.34	24.11	11.06%
	b. 35 < 45 Gallons (132 < 170 Liters)	29.82	62.00%	4,813.22	143.54	60.73%	28.15	62.00%	4,710.90	132.60	60.82%
	c. 45 < 55 Gallons (170 < 208 Liters)	4.71	9.80%	5,228.07	24.65	10.43%	4.36	9.60%	5,142.06	22.41	10.28%
	d. 55 < 71 Gallons (208 < 269 Liters)	5.39	11.20%	6,372.64	34.34	14.53%	4.99	11.00%	6,217.88	31.05	14.24%
	e. 71 < 91 Gallons (269 < 344 Liters)	1.20	2.50%	7,022.40	8.45	3.57%	1.14	2.50%	6,913.09	7.85	3.60%
	f. ≥ 91 Gallons (≥ 344 Liters)	0.00	0.00%	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00%
Gas Storage Non Condensing Commercial Total		48.10	1.09%	4,913.48	236.35	7.91%	45.40	1.07%	4,802.27	218.03	7.73%
Gas Storage Non Condensing Residential	a. < 35 Gallons (< 132 Liters)	301.00	7.00%	334.88	100.80	4.36%	290.50	7.00%	326.03	94.71	4.38%
	b. 35 < 45 Gallons (132 < 170 Liters)	2 515.50	58.50%	484.82	1 219.57	52.72%	2 427.75	58.50%	471.54	1 144.78	52.88%
	c. 45 < 55 Gallons (170 < 208 Liters)	1 268.50	29.50%	654.76	830.56	35.91%	1 224.25	29.50%	632.06	773.80	35.75%
	d. 55 < 71 Gallons (208 < 269 Liters)	215.00	5.00%	754.72	162.27	7.01%	207.50	5.00%	729.87	151.45	7.00%
	e. 71 < 91 Gallons (269 < 344 Liters)	0.00	0.00%	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00%
	f. ≥ 91 Gallons (≥ 344 Liters)	0.00	0.00%	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00%
Gas Storage Non Condensing Residential Total		4 300.00	97.82%	537.95	2 313.19	77.38%	4 150.00	97.59%	521.62	2 164.73	76.75%
Grand Total		4 395.80	100.00%	680.03	2 989.29	100.00%	4 252.50	100.00%	663.23	2 820.39	100.00%

Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Local Prices Shown in USD. Volume Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price)

3.3.9 Prices & Market Values by Capacity: HPWH

US - ESTIMATED VALUE OF THE WATER HEATER HEAT PUMP MARKET BY CAPACITY (GALLONS)

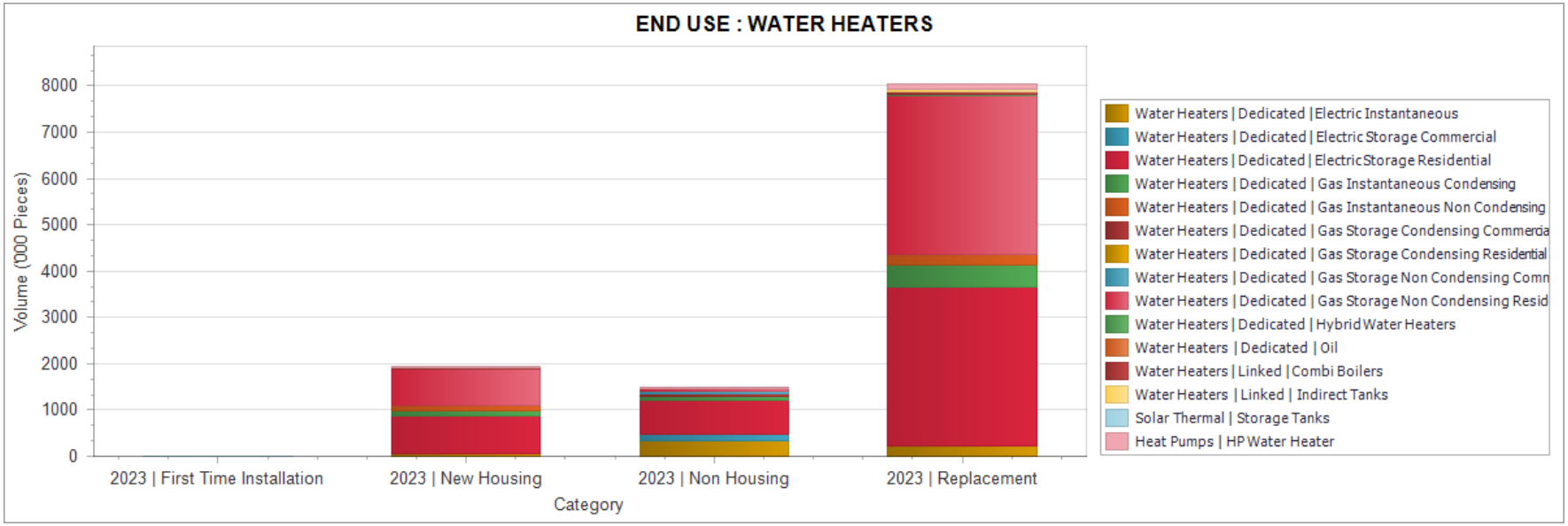


	2023					2022				
Category	Attributed Volume	% (Col Vol)	MSP (Local)	Value (MSP Local)	% (Col Val)	Attributed Volume	% (Col Vol)	MSP (Local)	Value (MSP Local)	% (Col Val)
a. < 50 Gallons	10.24	6.40%	749.72	7.68	3.33%	8.88	6.30%	760.00	6.75	3.26%
b. 50 < 60 Gallons	127.52	79.70%	1,184.56	151.06	65.60%	112.52	79.80%	1,210.00	136.15	65.84%
c. 60 < 80 Gallons	11.04	6.90%	2,564.05	28.31	12.29%	9.73	6.90%	2,600.00	25.30	12.23%
d. ≥ 80 Gallons	11.20	7.00%	3,858.57	43.22	18.77%	9.87	7.00%	3,910.00	38.60	18.66%
Grand Total	160.00	100.00%	1,439.12	230.26	100.00%	141.00	100.00%	1,466.58	206.79	100.00%

Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Local Prices Shown in USD. Volume Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price)

3.4.1 End Use : Base Year

US - SALES OF WATER HEATERS BY TYPE AND END USE SEGMENT ('000 PIECES)

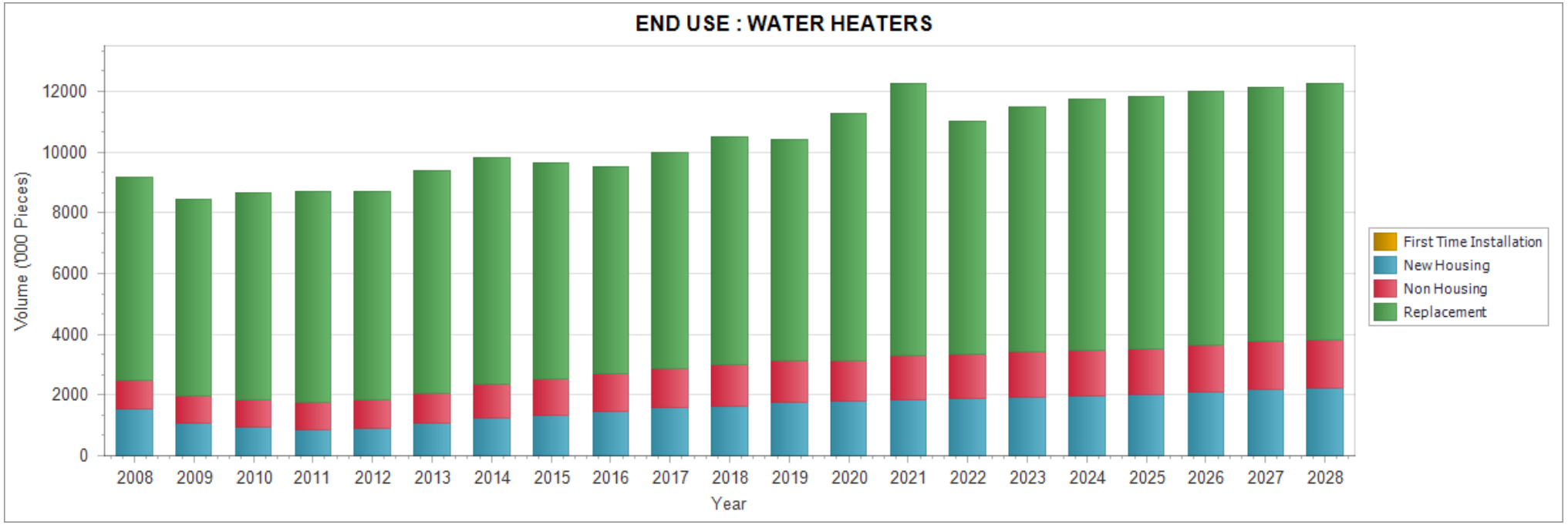


			2023												2023 Total		
			First Time Installation			New Housing			Non Housing			Replacement					
Product Type	Product Category	Product	Attributed Volume	% (Col Total)	% (Row)	Attributed Volume	% (Col Total)	% (Row)	Attributed Volume	% (Col Total)	% (Row)	Attributed Volume	% (Col Total)	% (Row)	Attributed Volume	% (Col Total)	% (Row)
Water Heaters	Dedicated	Electric Instantaneous	0.00	0.00%	0.00%	53.58	2.76%	8.57%	337.08	22.81%	53.93%	234.34	2.91%	37.49%	625.00	5.45%	100.00%
		Electric Storage Commercial	0.00	0.00%	0.00%	0.00	0.00%	0.00%	147.81	10.00%	99.00%	1.49	0.02%	1.00%	149.30	1.30%	100.00%
		Electric Storage Residential	0.00	0.00%	0.00%	812.87	41.87%	16.42%	712.33	48.21%	14.39%	3424.80	42.53%	69.19%	4950.00	43.13%	100.00%
		Gas Instantaneous Condensing	0.00	0.00%	0.00%	103.79	5.35%	15.47%	87.93	5.95%	13.10%	479.28	5.95%	71.43%	671.00	5.85%	100.00%
		Gas Instantaneous Non Condensing	0.00	0.00%	0.00%	127.90	6.59%	35.83%	19.05	1.29%	5.34%	210.05	2.61%	58.84%	357.00	3.11%	100.00%
		Gas Storage Condensing Commercial	0.00	0.00%	0.00%	0.00	0.00%	0.00%	38.84	2.63%	96.13%	1.56	0.02%	3.87%	40.40	0.35%	100.00%
		Gas Storage Condensing Residential	0.00	0.00%	0.00%	2.93	0.15%	40.08%	2.35	0.16%	32.12%	2.03	0.03%	27.80%	7.30	0.06%	100.00%
		Gas Storage Non Condensing Commercial	0.00	0.00%	0.00%	0.00	0.00%	0.00%	47.63	3.22%	99.02%	0.47	0.01%	0.98%	48.10	0.42%	100.00%
		Gas Storage Non Condensing Residential	0.00	0.00%	0.00%	792.95	40.84%	18.44%	58.62	3.97%	1.36%	3448.42	42.83%	80.20%	4300.00	37.46%	100.00%
		Hybrid Water Heaters	0.00	0.00%	0.00%	0.00	0.00%	0.00%	0.00	0.00%	0.00%	0.55	0.01%	100.00%	0.55	0.00%	100.00%
		Oil	0.00	0.00%	0.00%	0.00	0.00%	0.00%	0.00	0.00%	0.00%	1.40	0.02%	100.00%	1.40	0.01%	100.00%
	Linked	Combi Boilers	2.46	36.65%	3.20%	9.75	0.50%	12.66%	0.00	0.00%	0.00%	64.79	0.80%	84.14%	77.00	0.67%	100.00%
		Indirect Tanks	1.36	20.21%	1.61%	8.43	0.43%	9.98%	9.96	0.67%	11.79%	64.75	0.80%	76.63%	84.50	0.74%	100.00%
Solar Thermal	Storage Tanks		2.90	43.14%	47.49%	0.45	0.02%	7.31%	2.01	0.14%	32.99%	0.75	0.01%	12.21%	6.10	0.05%	100.00%
Heat Pumps	HP Water Heater		0.00	0.00%	0.00%	28.84	1.49%	18.03%	13.92	0.94%	8.70%	117.24	1.46%	73.27%	160.00	1.39%	100.00%
Grand Total			6.72	100.00%	0.06%	1941.49	100.00%	16.92%	1477.54	100.00%	12.87%	8051.92	100.00%	70.15%	11477.67	100.00%	100.00%

Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

3.4.2 End Use : Historical and Forecasts

US - SALES OF WATER HEATERS BY END USE SEGMENT ('000 PIECES)



Category	Data	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028
First Time Installation	Attributed Volume	19.61	18.15	19.52	19.61	16.37	13.73	11.41	8.45	8.26	8.93	8.68	7.69	7.84	7.30	7.15	6.72	6.48	6.30	6.20	6.04	5.86
	% (Col)	0.21%	0.21%	0.23%	0.23%	0.19%	0.15%	0.12%	0.09%	0.09%	0.09%	0.08%	0.07%	0.07%	0.06%	0.06%	0.06%	0.06%	0.05%	0.05%	0.05%	0.05%
New Housing	Attributed Volume	1535.48	1061.93	905.91	817.58	902.56	1062.84	1213.07	1316.04	1453.40	1577.16	1631.77	1733.30	1771.58	1854.01	1895.24	1941.49	1973.04	1998.33	2098.13	2167.85	2215.11
	% (Col)	16.74%	12.56%	10.48%	9.40%	10.37%	11.34%	12.36%	13.66%	15.27%	15.78%	15.52%	16.66%	15.73%	15.13%	17.21%	16.92%	16.82%	16.86%	17.50%	17.89%	18.05%
Non Housing	Attributed Volume	928.89	885.71	909.26	916.01	919.31	997.57	1124.90	1221.94	1253.49	1304.34	1344.25	1376.87	1340.93	1426.79	1447.53	1477.54	1499.17	1526.29	1557.71	1590.61	1590.51
	% (Col)	10.13%	10.48%	10.52%	10.53%	10.56%	10.64%	11.46%	12.68%	13.17%	13.05%	12.79%	13.23%	11.91%	11.64%	13.14%	12.87%	12.78%	12.88%	12.99%	13.12%	12.96%
Replacement	Attributed Volume	6688.28	6487.93	6806.42	6942.91	6868.53	7297.58	7468.09	7090.00	6802.57	7104.74	7526.57	7286.22	8138.96	8966.44	7662.15	8051.92	8254.73	8319.70	8329.24	8355.81	8462.63
	% (Col)	72.92%	76.75%	78.77%	79.84%	78.89%	77.87%	76.07%	73.58%	71.47%	71.08%	71.60%	70.03%	72.29%	73.17%	69.58%	70.15%	70.35%	70.20%	69.46%	68.94%	68.95%
Grand Total	Attributed Volume	9172.27	8453.72	8641.12	8696.12	8706.77	9371.73	9817.47	9636.42	9517.72	9995.17	10511.27	10404.07	11259.32	12254.54	11012.06	11477.67	11733.42	11850.62	11991.27	12120.31	12274.11
	% (Col)	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%

Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

3.4.3 Sales by Type : Water Heaters

US - WATER HEATER SALES BY TYPE OF PRODUCT ('000 PIECES)

				2023	
Product Type	Product Category	Product	Category	Attributed Volume	% (Col Total)
Water Heaters	Dedicated	Electric Instantaneous	a. < 13 kW	335.10	2.92%
			b. 13 < 36 kW	284.60	2.48%
			c. ≥ 36 kW	5.30	0.05%
		Electric Storage Commercial	Electric Storage Commercial	149.30	1.30%
		Electric Storage Residential	Electric Storage Residential	4 950.00	43.13%
		Gas Instantaneous Condensing	Gas Instantaneous Condensing	671.00	5.85%
		Gas Instantaneous Non Condensing	Gas Instantaneous Non Condensing	357.00	3.11%
		Gas Storage Condensing Commercial	Gas Storage Condensing Commercial	40.40	0.35%
		Gas Storage Condensing Residential	Gas Storage Condensing Residential	7.30	0.06%
		Gas Storage Non Condensing Commercial	Gas Storage Non Condensing Commercial	48.10	0.42%
		Gas Storage Non Condensing Residential	Gas Storage Non Condensing Residential	4 300.00	37.46%
		Hybrid Water Heaters	Hybrid	0.55	0.00%
		Oil	Oil	1.40	0.01%
	Linked	Combi Boilers	Combi Boilers	77.00	0.67%
		Indirect Tanks	Coiled	74.02	0.64%
			Plate-to-Plate	0.68	0.01%
			Tank-in-Tank	9.81	0.09%
Solar Thermal	Storage Tanks			6.10	0.05%
Heat Pumps	HP Water Heater			160.00	1.39%
Grand Total				11 477.65	100.00%

Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

3.4.4 Sales by Type of Flue and Type of Material : WH

US - WATER HEATER SALES BY TYPE OF FLUE AND MATERIAL : BASE YEAR & SPOT FORECAST ('000 PIECES)

			2023			2028		
Product Category	Product	Category	Attributed Volume	% (Col Total)	% (Col Vol)	Attributed Volume	% (Col Total)	% (Col Vol)
Dedicated	Gas Storage Condensing Residential	Mechanical Venting (Power and Power Direct Vent)	7.30	0.17%	100.00%	16.50	0.39%	100.00%
		Traditional Venting (Atmospheric and Direct Vent)	0.00	0.00%	0.00%	0.00	0.00%	0.00%
	Gas Storage Non Condensing Residential	Mechanical Venting (Power and Power Direct Vent)	817.00	18.60%	19.00%	840.50	19.95%	20.50%
		Traditional Venting (Atmospheric and Direct Vent)	3 483.00	79.31%	81.00%	3 259.50	77.38%	79.50%
Linked	Indirect Tanks	Enamelled	29.15	0.66%	34.50%	32.54	0.77%	33.90%
		Others	4.23	0.10%	5.00%	4.80	0.11%	5.00%
		Stainless Steel	51.12	1.16%	60.50%	58.66	1.39%	61.10%
Grand Total			4 391.80	100.00%	100.00%	4 212.50	100.00%	100.00%

Source: BRG based on industry interviews. Totals may show slight variance due to rounding. Excludes Gas Storage Commercial and Gas Tankless.

3.4.5 Sales by Storage Capacity : Electric Storage

US - ELECTRIC STORAGE WH SALES BY STORAGE CAPACITY : BASE YEAR & SPOT FORECAST ('000 PIECES)

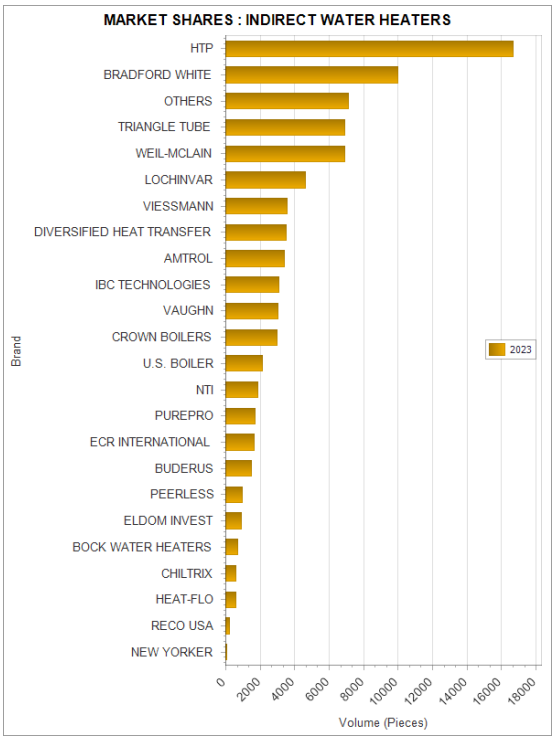
		2023		2028	
Product	Category	Attributed Volume	% (Col Vol)	Attributed Volume	% (Col Vol)
Electric Storage Commercial	a. < 30 Gallons (< 114 Liters)	43.30	29.00%	48.79	28.70%
	b. 30 < 55 Gallons (114 < 208 Liters)	38.82	26.00%	44.20	26.00%
	c. 55 < 90 Gallons (208 < 360 Liters)	28.37	19.00%	32.30	19.00%
	d. ≥ 90 Gallons (≥ 360 Liters)	38.82	26.00%	44.71	26.30%
Electric Storage Residential	a. < 6 Gallons (< 23 Liters)	99.00	2.00%	106.00	2.00%
	b. 6 < 13 Gallons (23 < 49 Liters)	247.50	5.00%	265.00	5.00%
	c. 13 < 25 Gallons (49 < 95 Liters)	148.50	3.00%	159.00	3.00%
	d. 25 < 35 Gallons (95 < 132 Liters)	594.00	12.00%	636.00	12.00%
	e. 35 < 45 Gallons (132 < 170 Liters)	1 881.00	38.00%	2 029.90	38.30%
	f. 45 < 55 Gallons (170 < 208 Liters)	1 980.00	40.00%	2 104.10	39.70%
Grand Total		5 099.30	100.00%	5 470.00	100.00%

Source: BRG based on industry interviews. Totals may show slight variance due to rounding.

3.5.1 Market Shares : Indirect Tanks

US - ESTIMATED SHARES : INDIRECT FIRED WATER HEATERS

	2023		2022	
Holding Company (Brand)	Shares	% of total	Shares	% of total
ARISTON GROUP (HTP)	16 650	19.70%	22 250	22.77%
BRADFORD WHITE (BRADFORD WHITE)	10 000	11.83%	10 500	10.75%
OTHERS (OTHERS)	7 095	8.40%	7 845	8.03%
GROUPE ATLANTIC (TRIANGLE TUBE)	6 925	8.20%	7 350	7.52%
SPX (WEIL-MCLAIN)	6 900	8.17%	7 620	7.80%
AO SMITH (LOCHINVAR)	4 625	5.47%	5 350	5.48%
CARRIER GLOBAL (VIESSMANN)	3 535	4.18%	4 480	4.59%
DHT (DIVERSIFIED HEAT TRANSFER)	3 500	4.14%	3 915	4.01%
WORTHINGTON INDUSTRIES (AMTROL)	3 400	4.02%	2 670	2.73%
PALOMA (IBC TECHNOLOGIES)	3 100	3.67%	3 300	3.38%
NUDYNE GROUP (VAUGHN)	3 000	3.55%	3 000	3.07%
BURNHAM (CROWN BOILERS)	2 980	3.53%	3 680	3.77%
BURNHAM (U.S. BOILER)	2 110	2.50%	2 615	2.68%
ARISTON GROUP (NTI)	1 835	2.17%	3 140	3.21%
F.W. WEBB (PUREPRO)	1 720	2.04%	1 685	1.72%
TERRAVEST INDUSTRIES (ECR INTERNATIONAL)	1 620	1.92%	2 435	2.49%
BOSCH (BUDERUS)	1 505	1.78%	1 550	1.59%
NORITZ (PEERLESS)	950	1.12%	1 000	1.02%
ELDOM INVEST (ELDOM INVEST)	900	1.07%	1 000	1.02%
BOCK WATER HEATERS (BOCK WATER HEATERS)	700	0.83%	720	0.74%
HEAT-FLO HEATING PRODUCTS (HEAT-FLO)	600	0.71%	670	0.69%
CHILTRIX (CHILTRIX)	600	0.71%	600	0.61%
NUDYNE GROUP (RECO USA)	205	0.24%	230	0.24%
BURNHAM (NEW YORKER)	45	0.05%	80	0.08%
BDR THERMEA (BAXI)		0.00%	15	0.02%
Grand Total	84 500	100.00%	97 700	100.00%

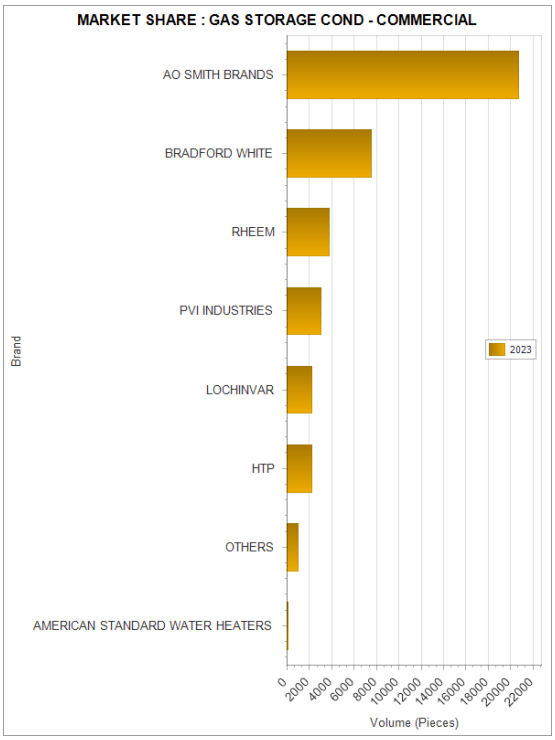


Source: BRG based on industry interviews

3.5.2 Market Shares : Gas Storage Condensing Commercial

US - ESTIMATED SHARES : COMMERCIAL GAS STORAGE COND WATER HEATERS

	2023		2022	
Holding Company (Brand)	Shares	% of total	Shares	% of total
AO SMITH (AO SMITH BRANDS)	20 700	51.24%	20 000	50.76%
BRADFORD WHITE (BRADFORD WHITE)	7 500	18.56%	6 500	16.50%
PALOMA (RHEEM)	3 800	9.41%	4 000	10.15%
WATTS WATER TECHNOLOGIES (PVI INDUSTRIES)	3 000	7.43%	3 000	7.61%
AO SMITH (LOCHINVAR)	2 200	5.45%	2 100	5.33%
ARISTON GROUP (HTP)	2 180	5.40%	2 765	7.02%
OTHERS (OTHERS)	970	2.40%	1 000	2.54%
ARISTON GROUP (AMERICAN STANDARD WATER HEATERS)	50	0.12%	35	0.09%
Grand Total	40 400	100.00%	39 400	100.00%

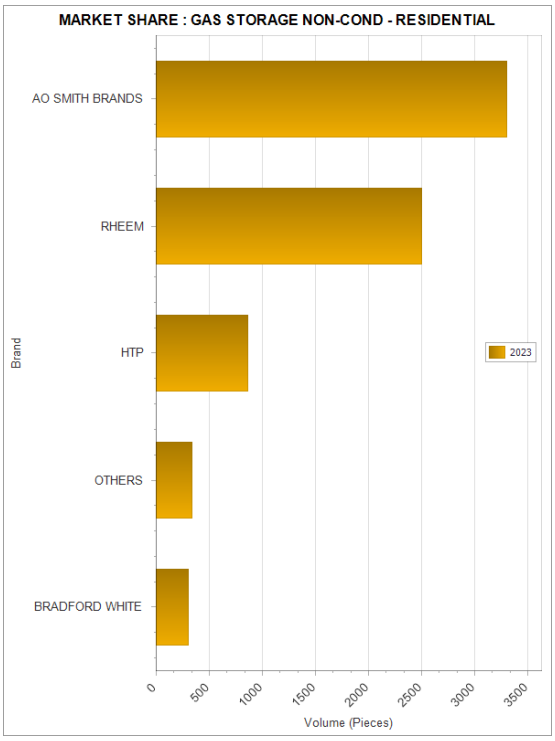


Source: BRG based on industry interviews

3.5.3 Market Shares : Gas Storage Condensing Residential

US - ESTIMATED SHARES : RESIDENTIAL GAS STORAGE CONDENSING WATER HEATERS

	2023		2022	
Holding Company (Brand)	Shares	% of total	Shares	% of total
AO SMITH (AO SMITH BRANDS)	3 300	45.21%	13 000	73.45%
PALOMA (RHEEM)	2 500	34.25%	2 550	14.41%
ARISTON GROUP (HTP)	860	11.78%	1 180	6.67%
OTHERS (OTHERS)	340	4.66%	270	1.53%
BRADFORD WHITE (BRADFORD WHITE)	300	4.11%	700	3.95%
Grand Total	7 300	100.00%	17 700	100.00%

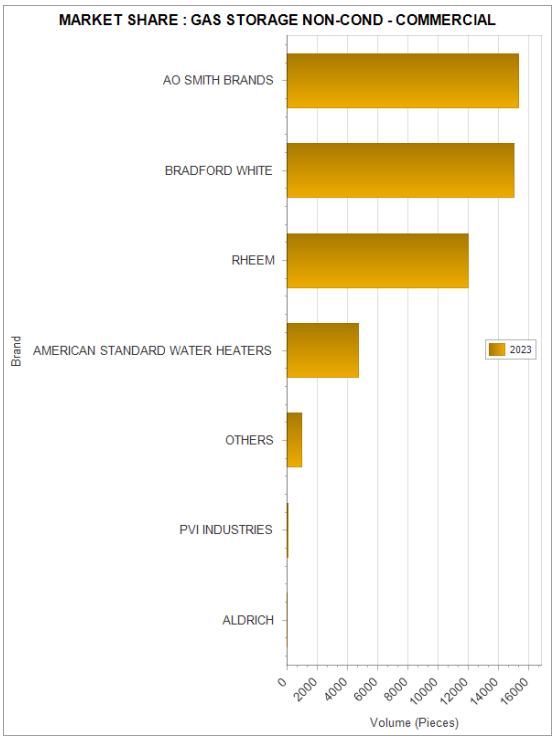


Source: BRG based on industry interviews

3.5.4 Market Shares : Gas Storage Non-Cond. Commercial

US - ESTIMATED SHARES : COMMERCIAL GAS STORAGE NON-CONDENSING WATER HEATERS

	2023		2022	
Holding Company (Brand)	Shares	% of total	Shares	% of total
AO SMITH (AO SMITH BRANDS)	15 300	31.81%	14 700	32.38%
BRADFORD WHITE (BRADFORD WHITE)	15 000	31.19%	13 500	29.74%
PALOMA (RHEEM)	12 000	24.95%	12 000	26.43%
ARISTON GROUP (AMERICAN STANDARD WATER HEATERS)	4 700	9.77%	4 000	8.81%
OTHERS (OTHERS)	990	2.06%	1 090	2.40%
WATTS WATER TECHNOLOGIES (PVI INDUSTRIES)	90	0.19%	90	0.20%
ALDRICH (ALDRICH)	20	0.04%	20	0.04%
Grand Total	48 100	100.00%	45 400	100.00%

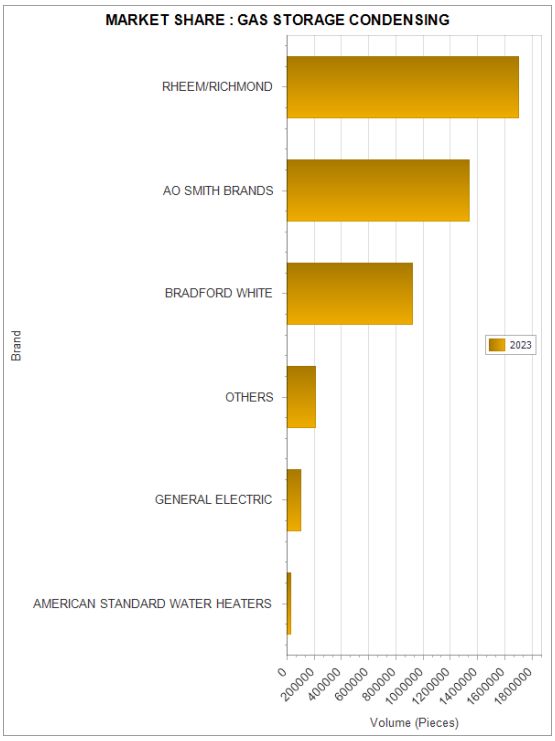


Source: BRG based on industry interviews

3.5.5 Market Shares : Gas Storage Non-Cond. Residential

US - ESTIMATED SHARES : RESIDENTIAL GAS STORAGE NON-CONDENSING WATER HEATERS

	2023		2022	
Holding Company (Brand)	Shares	% of total	Shares	% of total
PALOMA (RHEEM/RICHMOND)	1 700 000	39.53%	1 560 000	37.59%
AO SMITH (AO SMITH BRANDS)	1 340 000	31.16%	1 462 000	35.23%
BRADFORD WHITE (BRADFORD WHITE)	920 000	21.40%	855 000	20.60%
OTHERS (OTHERS)	210 900	4.90%	197 300	4.75%
HAIER (GENERAL ELECTRIC)	102 000	2.37%	45 000	1.08%
ARISTON GROUP (AMERICAN STANDARD WATER HEATERS)	27 100	0.63%	30 700	0.74%
Grand Total	4 300 000	100.00%	4 150 000	100.00%

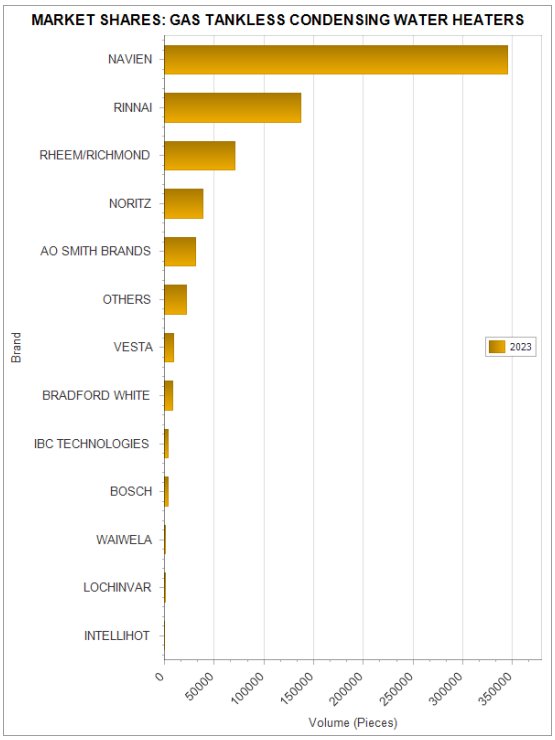


Source: BRG based on industry interviews

3.5.6 Market Shares : Gas Tankless Condensing

US - ESTIMATED SHARES : GAS TANKLESS CONDENSING WATER HEATERS

	2023		2022	
Holding Company (Brand)	Shares	% of total	Shares	% of total
KD NAVIEN (NAVIEN)	344 700	51.37%	350 000	48.28%
RINNAI (RINNAI)	137 100	20.43%	159 600	22.01%
PALOMA (RHEEM/RICHMOND)	70 700	10.54%	72 000	9.93%
NORITZ (NORITZ)	38 600	5.75%	50 350	6.94%
AO SMITH (AO SMITH BRANDS)	31 300	4.66%	43 700	6.03%
OTHERS (OTHERS)	22 045	3.29%	23 660	3.26%
DAESUNG (VESTA)	9 500	1.42%	8 000	1.10%
BRADFORD WHITE (BRADFORD WHITE)	8 000	1.19%	7 600	1.05%
PALOMA (IBC TECHNOLOGIES)	4 000	0.60%	4 580	0.63%
BOSCH (BOSCH)	3 425	0.51%	3 950	0.54%
EFFICIENT TECHNOLOGY SALES (WAIWELA)	700	0.10%	740	0.10%
AO SMITH (LOCHINVAR)	600	0.09%	500	0.07%
INTELLIHOT (INTELLIHOT)	330	0.05%	320	0.04%
Grand Total	671 000	100.00%	725 000	100.00%

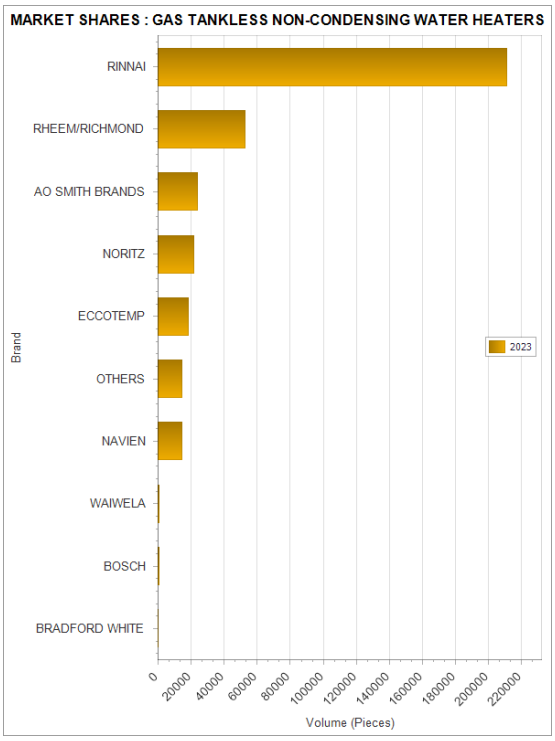


Source: BRG based on industry interviews

3.5.7 Market Shares : Gas Tankless Non-Condensing

US - ESTIMATED SHARES : GAS TANKLESS GAS NON-CONDENSING WATER HEATERS

	2023		2022	
Holding Company (Brand)	Shares	% of total	Shares	% of total
RINNAI (RINNAI)	210 900	59.08%	219 300	59.59%
PALOMA (RHEEM/RICHMOND)	52 800	14.79%	53 400	14.51%
AO SMITH (AO SMITH BRANDS)	24 000	6.72%	22 400	6.09%
NORITZ (NORITZ)	21 700	6.08%	21 800	5.92%
ECCOTEMP SYSTEMS (ECCOTEMP)	18 000	5.04%	18 400	5.00%
OTHERS (OTHERS)	14 620	4.10%	15 220	4.14%
KD NAVIEN (NAVIEN)	14 300	4.01%	16 700	4.54%
EFFICIENT TECHNOLOGY SALES (WAIWELA)	350	0.10%	370	0.10%
BOSCH (BOSCH)	280	0.08%	400	0.11%
BRADFORD WHITE (BRADFORD WHITE)	50	0.01%	10	0.00%
Grand Total	357 000	100.00%	368 000	100.00%

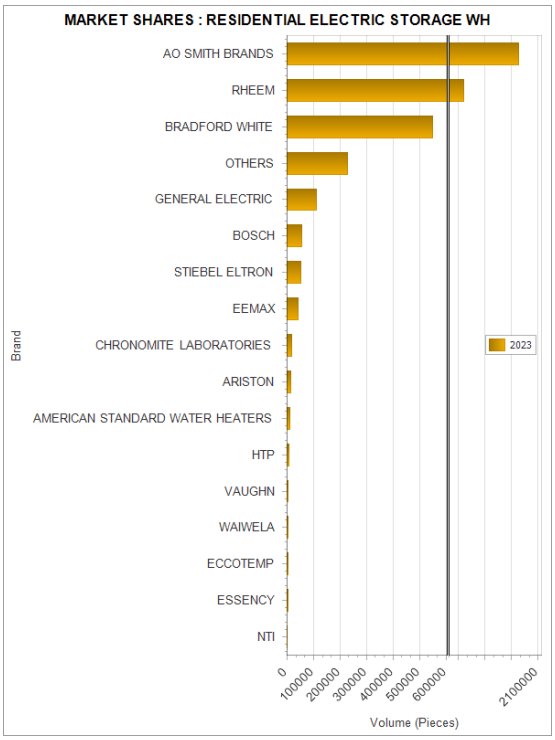


Source: BRG based on industry interviews

3.5.8 Market Shares : Electric Storage Residential

US - ESTIMATED SHARES : ELECTRIC STORAGE WATER HEATERS - RESIDENTIAL

	2023		2022	
Holding Company (Brand)	Shares	% of total	Shares	% of total
AO SMITH (AO SMITH BRANDS)	2 026 000	40.93%	1 800 000	39.13%
PALOMA (RHEEM)	1 820 000	36.77%	1 800 000	39.13%
BRADFORD WHITE (BRADFORD WHITE)	550 000	11.11%	456 000	9.91%
OTHERS (OTHERS)	229 405	4.63%	227 785	4.95%
HAIER (GENERAL ELECTRIC)	110 000	2.22%	100 000	2.17%
BOSCH (BOSCH)	55 800	1.13%	58 000	1.26%
STIEBEL ELTRON (STIEBEL ELTRON)	51 900	1.05%	50 600	1.10%
PALOMA (EEMAX)	43 000	0.87%	41 000	0.89%
MORRIS GROUP INTERNATIONAL (CHRONOMITELABORATORIES)	17 000	0.34%	16 000	0.35%
ARISTON GROUP (ARISTON)	12 500	0.25%	10 800	0.23%
ARISTON GROUP (AMERICAN STANDARD WATER HEATERS)	11 780	0.24%	18 800	0.41%
ARISTON GROUP (HTP)	6 420	0.13%	6 765	0.15%
NUDYNE GROUP (VAUGHN)	4 850	0.10%	4 550	0.10%
EFFICIENT TECHNOLOGY SALES (WAIWELA)	4 600	0.09%	4 500	0.10%
ECCOTEMP SYSTEMS (ECCOTEMP)	4 200	0.08%	4 000	0.09%
GT GROUP (ESSENCY)	2 500	0.05%	1 000	0.02%
ARISTON GROUP (NTI)	45	0.00%	200	0.00%
Grand Total	4 950 000	100.00%	4 600 000	100.00%

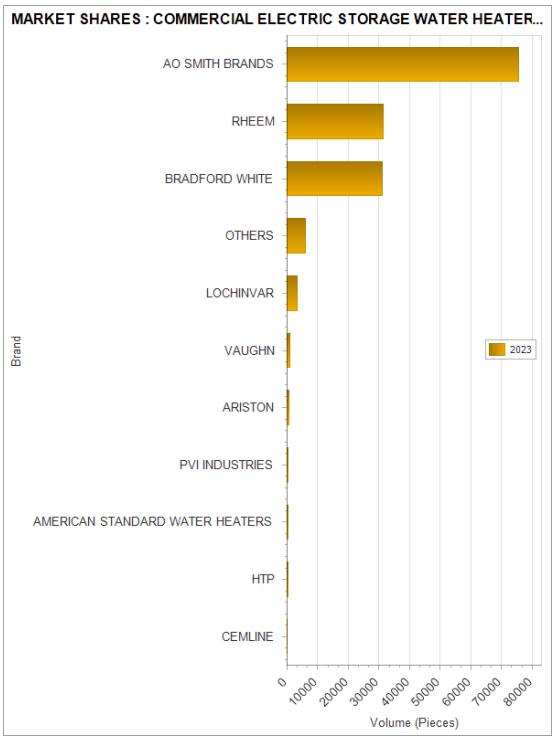


Source: BRG based on industry interviews

3.5.9 Market Shares : Electric Storage Commercial

US - ESTIMATED SHARES : ELECTRIC STORAGE WATER HEATERS - COMMERCIAL

	2023		2022	
Holding Company (Brand)	Shares	% of total	Shares	% of total
AO SMITH (AO SMITH BRANDS)	75 500	50.57%	61 000	50.33%
PALOMA (RHEEM)	31 300	20.96%	29 000	23.93%
BRADFORD WHITE (BRADFORD WHITE)	31 000	20.76%	21 000	17.33%
OTHERS (OTHERS)	6 055	4.06%	5 775	4.76%
AO SMITH (LOCHINVAR)	3 225	2.16%	2 700	2.23%
NUDYNE GROUP (VAUGHN)	880	0.59%	840	0.69%
ARISTON GROUP (ARISTON)	595	0.40%	315	0.26%
WATTS WATER TECHNOLOGIES (PVI INDUSTRIES)	285	0.19%	260	0.21%
ARISTON GROUP (AMERICAN STANDARD WATER HEATERS)	230	0.15%	125	0.10%
ARISTON GROUP (HTP)	215	0.14%	175	0.14%
CEMLINE (CEMLINE)	15	0.01%	10	0.01%
Grand Total	149 300	100.00%	121 200	100.00%

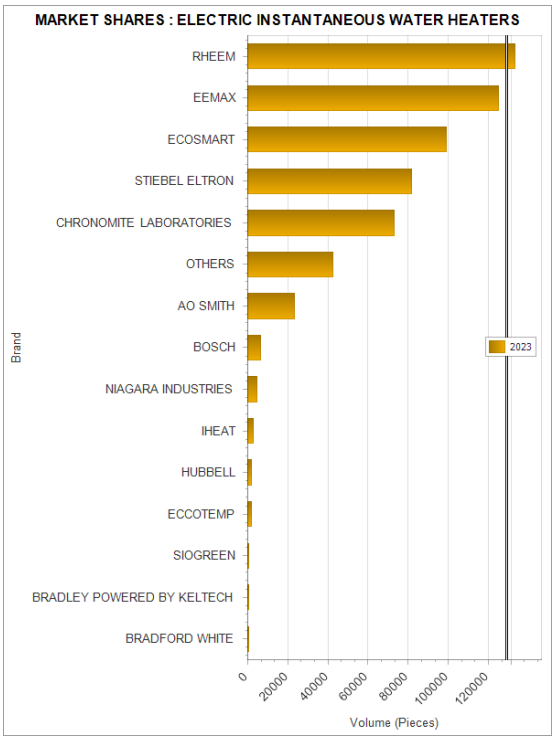


Source: BRG based on industry interviews

3.5.10 Market Shares : Electric Instantaneous

US - ESTIMATED SHARES : ELECTRIC TANKLESS WATER HEATERS

	2023		2022	
Holding Company (Brand)	Shares	% of total	Shares	% of total
PALOMA (RHEEM)	162 000	25.92%	156 000	26.00%
PALOMA (EEMAX)	125 000	20.00%	121 000	20.17%
PALOMA (ECOSMART)	99 000	15.84%	98 000	16.33%
STIEBEL ELTRON (STIEBEL ELTRON)	81 700	13.07%	75 850	12.64%
MORRIS GROUP INTERNATIONAL (CHRONOMITELABORATORIES)	73 000	11.68%	70 500	11.75%
OTHERS (OTHERS)	42 440	6.79%	41 150	6.86%
AO SMITH (AO SMITH)	23 200	3.71%	18 700	3.12%
BOSCH (BOSCH)	6 260	1.00%	5 700	0.95%
NIAGARA INDUSTRIES (NIAGARA INDUSTRIES)	4 700	0.75%	5 500	0.92%
DRAKKEN INDUSTRIES (IHEAT)	2 900	0.46%	2 900	0.48%
NUDYNE GROUP (HUBBELL)	1 900	0.30%	1 800	0.30%
ECCOTEMP SYSTEMS (ECCOTEMP)	1 700	0.27%	1 700	0.28%
SIOGREEN (SIOGREEN)	550	0.09%	550	0.09%
WATTS WATER TECHNOLOGIES (BRADLEY POWERED BY KELTECH)	400	0.06%	400	0.07%
BRADFORD WHITE (BRADFORD WHITE)	250	0.04%	250	0.04%
Grand Total	625 000	100.00%	600 000	100.00%



Source: BRG based on industry interviews

3.5.11 Market Shares : Electric Instantaneous (by type)

US - ESTIMATED MARKET SHARES FOR ELECTRIC TANKLESS: POINT OF USE, WHOLE HOUSE, COMMERCIAL/INDUSTRIAL

Holding group	Company/Brand	TANKLESS WATER HEATERS : Unit Sales (2021 - 2023)											
		Pont of Use (<13kW)				Whole House (13-36kW)				Commercial / Industrial (>36kW)			
		2021	2022	**2023**	***2023 Share***	2021	2022	**2023**	***2023 Share***	2021	2022	**2023**	***2023 Share***
PALOMA	RHEEM	2,000	2,000	2,000	***0.6%***	155,000	154,000	160,000	***56.2%***	-	-	-	***0.0%***
PALOMA	EEMAX	97,600	96,200	99,400	***29.7%***	21,100	20,800	21,500	***7.6%***	4,000	4,000	4,100	***77.4%***
PALOMA	ECOSMART	90,000	88,200	89,000	***26.6%***	10,000	9,800	10,000	***3.5%***	-	-	-	***0.0%***
STIEBEL ELTRON	STIEBEL ELTRON	52,530	45,730	45,420	***13.6%***	27,710	29,740	35,915	***12.6%***	310	380	365	***6.9%***
MORRIS GROUP INTERNATIONAL	CHRONOMITE LABORATORIES	65,790	64,870	67,300	***20.1%***	5,390	5,320	5,400	***1.9%***	320	310	300	***5.7%***
AO SMITH	AO SMITH	-	-	-	***0.0%***	17,500	18,700	23,200	***8.2%***	-	-	-	***0.0%***
BOSCH	BOSCH	6,500	4,250	5,185	***1.5%***	450	1,450	1,075	***0.4%***	-	-	-	***0.0%***
NIAGARA INDUSTRIES	NIAGARA INDUSTRIES	3,025	2,750	2,350	***0.7%***	2,975	2,750	2,350	***0.8%***	-	-	-	***0.0%***
DRAKKEN INDUSTRIES	IHEAT	2,970	2,600	2,600	***0.8%***	330	300	300	***0.1%***	-	-	-	***0.0%***
NUDYNE GROUP	HUBBELL	-	-	-	***0.0%***	1,970	1,750	1,850	***0.7%***	30	50	50	***0.9%***
ECCOTEMP SYSTEMS	ECCOTEMP	675	410	400	***0.1%***	2,125	1,290	1,300	***0.5%***	-	-	-	***0.0%***
SIOGREEN	SIOGREEN	65	55	50	***0.0%***	585	495	500	***0.2%***	-	-	-	***0.0%***
WATTS WATER TECHNOLOGIES	BRADLEY POWERED BY KELTECH	-	-	-	***0.0%***	-	-	-	***0.0%***	400	400	400	***7.5%***
BRADFORD WHITE	BRADFORD WHITE	285	240	240	***0.1%***	15	10	10	***0.0%***	-	-	-	***0.0%***
Others	Others	13,110	20,345	21,155	***6.3%***	13,100	20,695	21,200	***7.4%***	140	110	85	***1.6%***
TOTAL		**334,550**	**327,650**	**335,100**	***100.0%***	**258,250**	**267,100**	**284,600**	***100.0%***	**5,200**	**5,250**	**5,300**	***100.0%***
		55.9%	***54.6%***	***53.6%***		***43.2%***	***44.5%***	***45.5%***		***0.9%***	***0.9%***	***0.8%***	

Source: BRG based on industry interviews

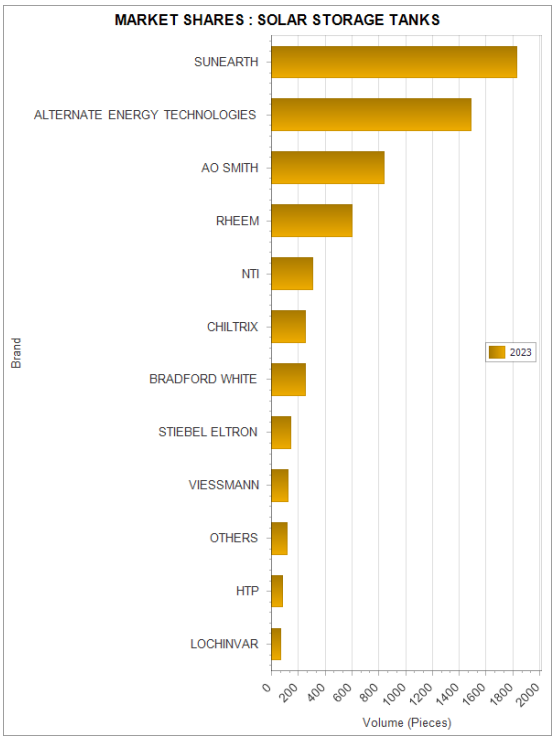
Holding group	Company/Brand	Total Unit Sales (2020 - 2023)			
		2021	2022	**2023**	***2023 Share***
PALOMA	RHEEM	157,000	156,000	162,000	***25.9%***
PALOMA	EEMAX	122,700	121,000	125,000	***20.0%***
PALOMA	ECOSMART	100,000	98,000	99,000	***15.8%***
STIEBEL ELTRON	STIEBEL ELTRON	80,550	75,850	81,700	***13.1%***
MORRIS GROUP INTERNATIONAL	CHRONOMITE LABORATORIES	71,500	70,500	73,000	***11.7%***
AO SMITH	AO SMITH	17,500	18,700	23,200	***3.7%***
BOSCH	BOSCH	6,950	5,700	6,260	***1.0%***
NIAGARA INDUSTRIES	NIAGARA INDUSTRIES	6,000	5,500	4,700	***0.8%***
DRAKKEN INDUSTRIES	IHEAT	3,300	2,900	2,900	***0.5%***
NUDYNE GROUP	HUBBELL	2,000	1,800	1,900	***0.3%***
ECCOTEMP SYSTEMS	ECCOTEMP	2,800	1,700	1,700	***0.3%***
SIOGREEN	SIOGREEN	650	550	550	***0.1%***
WATTS WATER TECHNOLOGIES	BRADLEY POWERED BY KELTECH	400	400	400	***0.1%***
BRADFORD WHITE	BRADFORD WHITE	300	250	250	***0.0%***
Others	Others	26,350	41,150	42,440	***6.8%***
TOTAL		**598,000**	**600,000**	**625,000**	***100.0%***

Source: BRG based on industry interviews

3.5.12 Market Shares : Solar Storage Tanks

US - ESTIMATED MARKET SHARES : SOLAR STORAGE TANKS

	2023		2022	
Holding Company (Brand)	Shares	% of total	Shares	% of total
THE SOLARAY CORPORATION (SUNEARTH)	1 830	30.00%	2 100	31.34%
THE SOLARAY CORPORATION (ALTERNATE ENERGY TECHNOLOGIES)	1 490	24.43%	1 570	23.43%
AO SMITH (AO SMITH)	840	13.77%	950	14.18%
PALOMA (RHEEM)	600	9.84%	660	9.85%
ARISTON GROUP (NTI)	305	5.00%		0.00%
CHILTRIX (CHILTRIX)	250	4.10%	260	3.88%
BRADFORD WHITE (BRADFORD WHITE)	250	4.10%	310	4.63%
STIEBEL ELTRON (STIEBEL ELTRON)	145	2.38%	140	2.09%
CARRIER GLOBAL (VIESSMANN)	125	2.05%	130	1.94%
OTHERS (OTHERS)	115	1.89%	160	2.39%
ARISTON GROUP (HTP)	80	1.31%	100	1.49%
AO SMITH (LOCHINVAR)	70	1.15%	100	1.49%
AO SMITH (STATE)		0.00%	220	3.28%
Grand Total	6 100	100.00%	6 700	100.00%

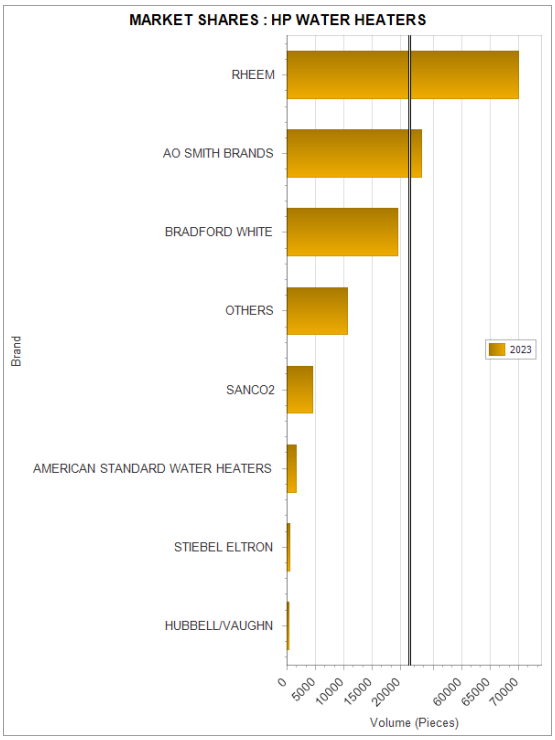


Source: BRG based on industry interviews

3.5.13 Market Shares : HP Water Heater

US - ESTIMATED MARKET SHARES : HP WATER HEATERS

	2023		2022	
Holding Company (Brand)	Shares	% of total	Shares	% of total
PALOMA (RHEEM)	70 000	43.75%	63 000	44.68%
AO SMITH (AO SMITH BRANDS)	53 000	33.13%	47 000	33.33%
BRADFORD WHITE (BRADFORD WHITE)	19 500	12.19%	16 500	11.70%
OTHERS (OTHERS)	10 590	6.62%	9 760	6.92%
ECO2 SYSTEMS (SANCO2)	4 500	2.81%	4 000	2.84%
ARISTON GROUP (AMERICAN STANDARD WATER HEATERS)	1 675	1.05%		0.00%
STIEBEL ELTRON (STIEBEL ELTRON)	425	0.27%	470	0.33%
NUDYNE GROUP (HUBBELL/VAUGHN)	310	0.19%	270	0.19%
Grand Total	160 000	100.00%	141 000	100.00%



Source: BRG based on industry interviews

3.6.1 US - IMPORTS OF WATER HEATERS - OTHER NON-ELECTRIC 2023

Source Country	2023				2022			
	Volume		Value 000		Volume		Value 000	
	Pieces	%	USD	%	Pieces	%	USD	%
Mexico	1 897 139	88.3%	455 306	86.8%	1 867 711	85.8%	475 888	85.7%
China	77 228	3.6%	11 859	2.3%	169 743	7.8%	25 575	4.6%
Italy	27 985	1.3%	8 017	1.5%	27 078	1.2%	9 470	1.7%
Japan	27 508	1.3%	6 253	1.2%	3 504	0.2%	1 142	0.2%
Canada	25 889	1.2%	8 585	1.6%	14 292	0.7%	4 228	0.8%
Germany	24 410	1.1%	7 340	1.4%	16 332	0.8%	4 444	0.8%
OTHERS	68 268	3.2%	26 942	5.1%	77 529	3.6%	34 771	6.3%
Total	**2 148 427**	**100.0%**	**524 302**	**100.0%**	**2 176 189**	**100.0%**	**555 518**	**100.0%**

Source : Trademap

3.6.2 US - EXPORTS OF WATER HEATERS - OTHER NON-ELECTRIC 2023

Destination	2023				2022			
	Volume		Value 000		Volume		Value 000	
	Pieces	%	USD	%	Pieces	%	USD	%
Canada	242 592	78.2%	166 791	85.0%	204 687	56.9%	142 603	75.3%
Mexico	22 862	7.4%	7 145	3.6%	79 523	22.1%	13 044	6.9%
Australia	11 009	3.5%	1 334	0.7%	20 451	5.7%	2 700	1.4%
OTHERS	33 915	10.9%	20 898	10.7%	54 790	15.2%	31 006	16.4%
Total	**310 378**	**100.0%**	**196 168**	**100.0%**	**359 451**	**100.0%**	**189 353**	**100.0%**

Source : Trademap

3.6.3 US - IMPORTS OF WATER HEATERS - GAS INSTANTANEOUS 2023

Source Country	2023				2022			
	Volume		Value 000		Volume		Value 000	
	Pieces	%	USD	%	Pieces	%	USD	%
Japan	382 159	42.0%	202 928	35.6%	811 809	54.7%	447 892	49.5%
Korea, Republic of	381 973	42.0%	291 178	51.0%	440 785	29.7%	336 755	37.2%
China	66 700	7.3%	6 866	1.2%	89 088	6.0%	9 002	1.0%
Mexico	45 174	5.0%	48 484	8.5%	92 135	6.2%	82 800	9.1%
Portugal	10 667	1.2%	7 332	1.3%	18 904	1.3%	11 135	1.2%
OTHERS	23 795	2.6%	13 828	2.4%	30 962	2.1%	17 949	2.0%
Total	**910 468**	**100.0%**	**570 616**	**100.0%**	**1 483 683**	**100.0%**	**905 533**	**100.0%**

Source : Trademap

3.6.4 US - EXPORTS OF WATER HEATERS - GAS INSTANTANEOUS 2023

Destination	2023				2022			
	Volume		Value 000		Volume		Value 000	
	Pieces	%	USD	%	Pieces	%	USD	%
Canada	61 063	56.9%	61 526	65.0%	105 233	23.4%	95 101	53.1%
United Kingdom	21 899	20.4%	3 920	4.1%	33	0.0%	124	0.1%
Mexico	16 330	15.2%	21 179	22.4%	322 528	71.7%	66 111	36.9%
Colombia	1 748	1.6%	1 224	1.3%	856	0.2%	1 116	0.6%
OTHERS	6 341	5.9%	6 824	7.2%	21 289	4.7%	16 544	9.2%
Total	**107 381**	**100.0%**	**94 673**	**100.0%**	**449 939**	**100.0%**	**178 996**	**100.0%**

Source : Trademap

3.6.5 US - IMPORTS OF WATER HEATERS - ELECTRIC STORAGE 2023

Source Country	2023				2022			
	Volume		Value 000		Volume		Value 000	
	Pieces	%	USD	%	Pieces	%	USD	%
Mexico	2 318 320	87.2%	459 300	91.1%	2 229 535	80.1%	441 077	89.8%
China	149 379	5.6%	18 288	3.6%	330 409	11.9%	24 036	4.9%
Germany	77 655	2.9%	10 889	2.2%	70 251	2.5%	9 902	2.0%
Viet Nam	48 602	1.8%	3 396	0.7%	52 383	1.9%	2 981	0.6%
OTHERS	65 255	2.5%	12 221	2.4%	99 847	3.6%	13 141	2.7%
Total	**2 659 211**	**100.0%**	**504 094**	**100.0%**	**2 782 425**	**100.0%**	**491 137**	**100.0%**

Source : Trademap

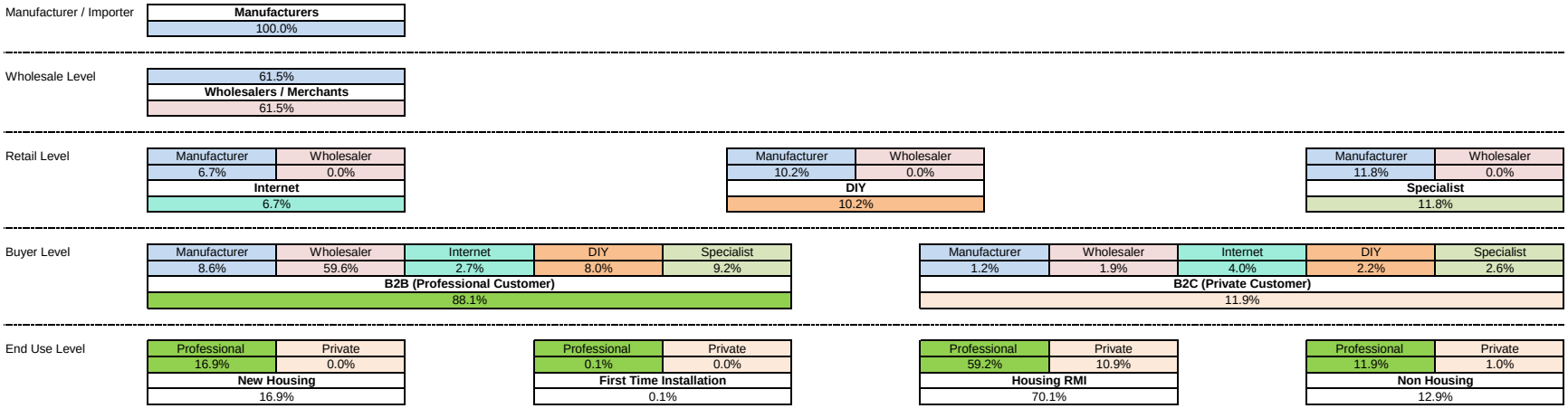
3.6.6 US - EXPORTS OF WATER HEATERS - ELECTRIC STORAGE 2023

Destination	2023				2022			
	Volume		Value 000		Volume		Value 000	
	Pieces	%	USD	%	Pieces	%	USD	%
Canada	288 493	52.9%	68 901	49.7%	267 535	52.7%	67 764	51.3%
Mexico	105 679	19.4%	29 105	21.0%	94 935	18.7%	22 307	16.9%
Saudi Arabia	30 405	5.6%	7 710	5.6%	24 279	4.8%	5 338	4.0%
Kuwait	11 477	2.1%	4 106	3.0%	11 890	2.3%	4 877	3.7%
Germany	9 843	1.8%	2 940	2.1%	6 653	1.3%	2 267	1.7%
Bahamas	7 385	1.4%	1 558	1.1%	6 129	1.2%	1 478	1.1%
China	6 498	1.2%	1 046	0.8%	2 684	0.5%	451	0.3%
OTHERS	85 126	15.6%	23 242	16.8%	93 908	18.5%	27 725	21.0%
Total	**544 906**	**100.0%**	**138 608**	**100.0%**	**508 013**	**100.0%**	**132 207**	**100.0%**

Source : Trademap

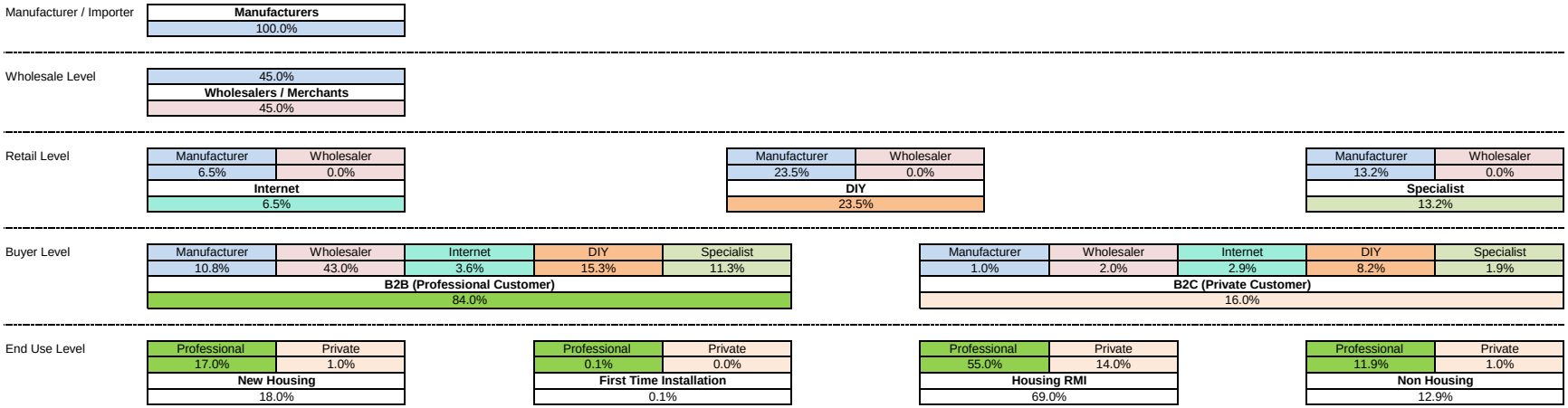
3.7.1 Distribution Flow : Water Heaters

US - ESTIMATED DISTRIBUTION FLOW FOR WATER HEATERS : 2023



3.7.2 Distribution Flow : Water Heaters

US - ESTIMATED DISTRIBUTION FLOW FOR WATER HEATERS : 2028

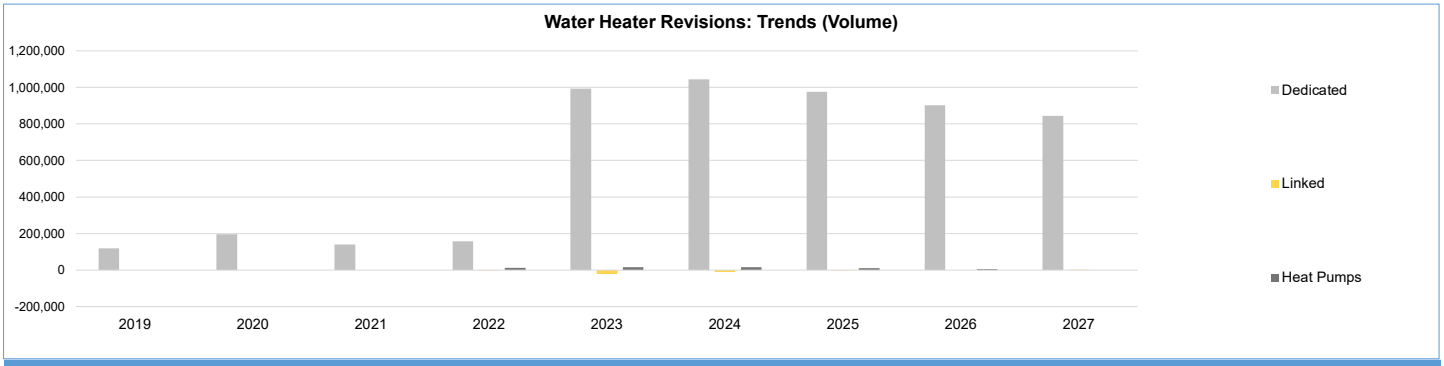


Source: BRG based on Industry Interviews.

3.8 Revisions in the US Water Heater Market (May 2024)

US - REVISIONS IN THE US WATER HEATER MARKET

Data Comparison: May 2023



Product Category	Alt Product	Latest Data	2019	2020	2021	2022	2023	2024	2025	2026	2027
Dedicated	Electric Instantaneous	*Vol*	410,000	519,000	598,000	600,000	625,000	650,000	670,000	690,000	710,000
Dedicated	Electric Instantaneous	*Δ Change*	*0*	*0*	*0*	*0*	*18,000*	*31,000*	*27,000*	*21,000*	*12,000*
Dedicated	Electric Storage Commercial	*Vol*	152,000	142,000	155,500	121,200	149,300	160,000	165,000	170,000	175,000
Dedicated	Electric Storage Commercial	*Δ Change*	*0*	*0*	*0*	*0*	*4,300*	*13,300*	*15,100*	*17,000*	*21,000*
Dedicated	Electric Storage Residential	*Vol*	4,234,000	4,686,000	4,913,000	4,600,000	4,950,000	5,100,000	5,150,000	5,200,000	5,250,000
Dedicated	Electric Storage Residential	*Δ Change*	*116,000*	*201,000*	*143,000*	*170,000*	*665,000*	*760,000*	*740,000*	*720,000*	*700,000*
Dedicated	Gas Instantaneous Condensing	*Vol*	504,000	574,000	758,000	725,000	671,000	698,000	745,000	794,000	846,000
Dedicated	Gas Instantaneous Condensing	*Δ Change*	*0*	*0*	*0*	*0*	*-34,000*	*-47,000*	*-50,000*	*-51,000*	*-34,000*
Dedicated	Gas Instantaneous Non Condensing	*Vol*	322,000	351,000	416,000	368,000	357,000	359,000	363,000	368,000	372,000
Dedicated	Gas Instantaneous Non Condensing	*Δ Change*	*0*	*0*	*0*	*-9,000*	*2,000*	*-6,000*	*-12,000*	*-22,000*	*-28,000*
Dedicated	Gas Storage Condensing Commercial	*Vol*	40,500	36,700	40,000	39,400	40,400	43,000	44,300	45,000	70,000
Dedicated	Gas Storage Condensing Commercial	*Δ Change*	*0*	*0*	*0*	*900*	*-600*	*-1,000*	*-2,700*	*-5,000*	*0*
Dedicated	Gas Storage Condensing Residential	*Vol*	14,000	16,000	16,900	17,700	7,300	10,000	12,000	13,500	15,000
Dedicated	Gas Storage Condensing Residential	*Δ Change*	*0*	*0*	*0*	*0*	*-10,900*	*-8,800*	*-7,200*	*-6,000*	*-4,800*
Dedicated	Gas Storage Non Condensing Commercial	*Vol*	55,000	46,000	51,500	45,400	48,100	49,000	50,000	54,000	0
Dedicated	Gas Storage Non Condensing Commercial	*Δ Change*	*0*	*-6,000*	*-5,000*	*-5,300*	*-900*	*3,000*	*7,000*	*9,000*	*0*
Dedicated	Gas Storage Non Condensing Residential	*Vol*	4,400,000	4,600,000	4,980,000	4,150,000	4,300,000	4,300,000	4,250,000	4,200,000	4,150,000
Dedicated	Gas Storage Non Condensing Residential	*Δ Change*	*0*	*0*	*0*	*0*	*350,000*	*300,000*	*260,000*	*220,000*	*180,000*
Dedicated	Hybrid Water Heaters	*Vol*	1,850	2,200	1,075	1,400	550	500	500	500	500
Dedicated	Hybrid Water Heaters	*Δ Change*	*0*	*0*	*0*	*0*	*-750*	*-700*	*-700*	*-700*	*-700*
Dedicated	Oil	*Vol*	2,900	2,400	1,950	1,650	1,400	700	600	500	400
Dedicated	Oil	*Δ Change*	*1,700*	*1,400*	*1,050*	*900*	*670*	*-20*	*-110*	*-200*	*-290*
Linked	Combi Boilers	*Vol*	78,100	78,900	101,000	96,900	77,000	91,500	104,800	118,500	132,400
Linked	Combi Boilers	*Δ Change*	*0*	*0*	*-500*	*0*	*-10,300*	*-1,200*	*3,900*	*5,800*	*5,800*
Linked	Indirect Tanks	*Vol*	97,300	92,700	102,000	97,700	84,500	86,000	90,000	92,000	94,000
Linked	Indirect Tanks	*Δ Change*	*0*	*0*	*0*	*-2,300*	*-11,500*	*-9,500*	*-7,000*	*-6,100*	*-5,200*
Heat Pumps	Heat Pumps	*Vol*	85,000	105,000	113,000	141,000	160,000	180,000	200,000	240,000	300,000
Heat Pumps	Heat Pumps	*Δ Change*	*0*	*0*	*0*	*12,000*	*15,000*	*15,000*	*10,000*	*5,000*	*0*
Total	**Total**	**Vol**	**10,396,650**	**11,251,900**	**12,247,925**	**11,005,350**	**11,471,550**	**11,727,700**	**11,845,200**	**11,986,000**	**12,115,300**
Total	**Total**	***Δ Change***	***117,700***	***196,400***	***138,550***	***167,200***	***986,020***	***1,048,080***	***983,290***	***906,800***	***845,810***

The historical market size for residential electric storage was revised upward due to updated sales figures submitted by an important market player.

The total market size for gas instantaneous noncondensing was revised downward for 2021 due to update sales submission from an important market player.

The total market size for heat pump water heaters was revised upwards for 2022 due to an updated sales submission by a trusted market player.